Templates list

Application Build

- Backend/API (API)
- Background Jobs & Scheduling (JBS)
- Eventing & Webhooks (EVT)
- Rate Limits & Abuse Controls (RLIM)
- Feature Flags & Config (FFCFG)
- API Pagination/Filtering/Sorting (PFS)
- File Processing & Media Pipelines (FPMP)
- Admin & Internal Tools APIs (ADMIN)
- Frontend/UI Implementation (FE)
- State Management & Data Fetching (SMD)
- Client Performance & Rendering (CPR)
- Forms & Validation (FORM)
- Client Routing & Deep Links (ROUTE)
- UI Composition & Layout Patterns (UICP)
- Client Error Handling & Recovery (CER)
- Client Security (CSec)
- Mobile Implementation (MOB)
- Mobile Device Capabilities (MDC)
- Offline & Sync (OFS)
- Mobile Performance & Battery (MBAT)
- Mobile Deep Links & Universal Links (MDL)
- Push Notifications (MPUSH)
- App Store Release & Signing (SIGN)

Application Build (Definition)

What it is

The Application Build group is the **implementation-facing specification layer**: the concrete contracts, module plans, and build rules needed to implement the product in code across backend, frontend, mobile, and supporting runtime systems (jobs, events, flags, rate limits, media pipelines). It translates architecture + data + design into **build-ready specs**.

What it produces

- API and backend execution specs (endpoints, handlers, services, jobs, events)

- Client implementation specs (routes, state, rendering/perf, forms, error recovery, security)

- Mobile implementation specs (navigation, offline, device features, push, deep links, signing)
- Runtime controls specs (rate limits, feature flags, pagination/filter/sort, webhook/event contracts)
- Admin/internal tool specs (privileged APIs, moderation/support tooling boundaries)

What it is used for (exactly)

- Driving code generation and implementation planning (IMP slices) with deterministic mappings
- Enforcing consistent patterns across modules (platform rules, error models, authz checks)
- Ensuring cross-cutting runtime behaviors (rate limits, flags, retries, validation) are implemented consistently
- Producing testable, verifiable build outputs (contracts and acceptance hooks)

What it is not

- Not product definition (Group 1) and not design deliverables (Group 2)
- Not architecture boundary decisions (Group 3) — assumes ARC is locked
- Not data governance/lifecycle (Group 4) — consumes DATA/DLR/DGL/DQV outputs
- Not ops/reliability runbooks (Group 8) — though it references them

Scope rules

- Must reference upstream truth: PRD feature IDs, DMG terms, ARC boundaries, DATA entities, ERR reason codes, PMAD authz policies
- Must define enforcement points (where validation/authz/rate limits happen in code)
- Must specify “shape” not “implementation code” (code comes later)
- Must include unknown handling and completeness gates per template

Subcategory meaning (one-line each)

- **API**: backend endpoint catalogs/specs, auth/errors, jobs + WS contracts pointer
- **JBS**: background jobs definitions, schedules, retries/DLQ, idempotency
- **EVT**: event schemas, producer/consumer rules, webhooks, delivery guarantees
- **RLIM**: rate limiting + abuse controls across API/realtime/search
- **FFCFG**: feature flag model, config matrix, rollout + kill-switch behavior
- **PFS**: pagination/filtering/sorting contract rules for list/search endpoints
- **FPMP**: file upload/processing, media pipeline stages, storage/transcode rules
- **ADMIN**: privileged/admin/internal APIs, moderation/support tooling endpoints
- **FE**: web client route/layout/component implementation surfaces
- **SMD**: client state architecture + data fetching/caching rules
- **CPR**: client performance budgets, rendering strategy, profiling hooks
- **FORM**: forms, input schemas, validation UX + mapping to DQV/ERR
- **ROUTE**: routing + deep links, route params, navigation contracts
- **UICP**: layout/composition patterns, reusable page structures, UI assembly rules
- **CER**: client error recovery patterns, retries, offline/error UX integration
- **CSec**: client-side security rules (storage, tokens, anti-tamper, CSP where applicable)
- **MOB**: mobile app implementation surfaces (navigation + screens + platform glue)
- **MDC**: device capability usage rules (camera/files/gps/push)
- **OFS**: offline-first rules, sync engine, conflict resolution
- **MBAT**: mobile perf/battery constraints, background work rules
- **MDL**: deep link/universal link rules on mobile

- **MPUSH**: push notification types, routing, permissions, delivery rules
- **SIGN**: app signing, build/release artifacts, store submission constraints

Template Sets

Group 5 — Application Build — Template Set (Locked v1)

Backend/API (API)
API-01 Endpoint Catalog
API-02 Endpoint Spec (per endpoint: request/response/auth/errors)
API-03 Error & Status Code Policy
API-04 AuthZ Rules (RBAC/ABAC checks)
API-05 Rate Limit & Abuse Controls (per endpoint class)
API-06 Background Jobs Spec (API-triggered + system jobs)
API-07 WebSocket API Spec

Background Jobs & Scheduling (JBS)
JBS-01 Jobs Inventory (by job_id)
JBS-02 Job Spec (per job: trigger, inputs, outputs, retries)
JBS-03 Scheduling Rules (cron/event-driven, env enablement)
JBS-04 Retry/DLQ Policy (backoff, poison messages)
JBS-05 Idempotency & Concurrency for Jobs
JBS-06 Job Observability (metrics/logs/traces, alerts)

Eventing & Webhooks (EVT)
EVT-01 Event Catalog (by event_id)
EVT-02 Event Schema Spec (payload, versioning, pii rules)
EVT-03 Producer/Consumer Map (who emits/consumes, contracts)
EVT-04 Delivery Semantics (ordering, retries, dedupe)
EVT-05 Webhook Producer Spec (outbound webhooks)
EVT-06 Webhook Consumer Spec (inbound webhooks)
EVT-07 Event Failure Handling (DLQ, replay, backfill)
EVT-08 Event Observability (lag, success rate, tracing)

Rate Limits & Abuse Controls (RLIM)
RLIM-01 Rate Limit Policy (global rules, scopes)
RLIM-02 Rate Limit Catalog (limits by surface/endpoint)
RLIM-03 Abuse Signals & Detection Rules
RLIM-04 Enforcement Actions Matrix (throttle/ban/captcha)
RLIM-05 Exemptions & Allowlist Policy
RLIM-06 Rate Limit Observability (alerts, dashboards)

Feature Flags & Config (FFCFG)
FFCFG-01 Feature Flag Registry (by flag_id)
FFCFG-02 Flag Behavior Spec (default, targeting, kill-switch)
FFCFG-03 Rollout Plan (phased, canary, % ramp)
FFCFG-04 Config Matrix (env vars/settings by env)
FFCFG-05 Safe Degradation Rules (flag off behavior)
FFCFG-06 Audit & Change Control (who can flip, logging)

API Pagination/Filtering/Sorting (PFS)
PFS-01 Query Contract (filter/sort operators, syntax)
PFS-02 Pagination Rules (cursor/offset, stability guarantees)
PFS-03 Default Ordering & Tie-Break Rules
PFS-04 Validation & Error Mapping (bad query → reason codes)
PFS-05 Performance Constraints (limits, indexed fields policy)

File Processing & Media Pipelines (FPMP)
FPMP-01 Upload Contract (types, limits, auth, scanning)
FPMP-02 Storage Strategy (buckets, paths, access modes)
FPMP-03 Processing Pipeline Stages (transcode/resize/parse)
FPMP-04 Async Processing & Status Model (jobs, retries, DLQ)
FPMP-05 CDN/Delivery Rules (cache headers, variants)
FPMP-06 Security & Compliance for Files (PII, retention, malware)
FPMP-07 Media Observability (latency, failure rates, QoS)

Admin & Internal Tools APIs (ADMIN)
ADMIN-01 Admin Capabilities Matrix
ADMIN-02 Moderation/Support Tools Spec (queues, actions)
ADMIN-03 Audit Trail Spec (actions + retention)
ADMIN-04 Data Repair Procedures (safe backfills)
ADMIN-05 Privileged API Surface Catalog (endpoints + authz)
ADMIN-06 Admin Observability & Safeguards (rate limits, alerts)

Frontend/UI Implementation (FE)
FE-01 Route Map + Layout Spec
FE-02 Component Implementation Spec
FE-03 UI State Model (loading/empty/error/success)
FE-04 Data Binding Rules (what calls what, when)
FE-05 Accessibility Implementation Notes (a11y enforcement)
FE-06 Theming/Tokens Integration (DSYS binding)
FE-07 Error Handling UX Implementation (surface rules)

State Management & Data Fetching (SMD)
SMD-01 State Architecture (global vs local, stores)
SMD-02 Query/Cache Strategy (client cache rules, invalidation)
SMD-03 Mutation Patterns (optimistic updates, rollback)
SMD-04 Realtime Subscription Patterns (WS hooks, reconciliation)
SMD-05 Offline State Handling (queueing, retry)
SMD-06 Observability for Client Data Layer (logs/metrics)

Client Performance & Rendering (CPR)
CPR-01 Performance Budget (web + mobile targets)
CPR-02 Rendering Strategy (SSR/CSR, list virtualization)
CPR-03 Asset Loading Strategy (code split, lazy load)

CPR-04 Profiling Plan (what to measure, tools)
CPR-05 Perf Regression Gates (thresholds, CI checks)

Forms & Validation (FORM)
FORM-01 Forms Inventory (by form_id/screen_id)
FORM-02 Field Spec (per field: type, rules, copy)
FORM-03 Validation UX Rules (inline, focus, announce)
FORM-04 Schema Mapping (forms ↔ DATA-06/DQV-03)
FORM-05 Submission & Error Recovery Patterns
FORM-06 Anti-Abuse for Forms (spam, throttles, bot defense)

Client Routing & Deep Links (ROUTE)
ROUTE-01 Route Contract (paths, params, types)
ROUTE-02 Navigation Map (flows, entry points)
ROUTE-03 Deep Link Map (URLs → screens/actions)
ROUTE-04 Guard Rules (auth gating, role gating)
ROUTE-05 Back/History Rules (expected behavior)
ROUTE-06 Link Validation & Unknown Handling (fallback routes)

UI Composition & Layout Patterns (UICP)
UICP-01 Page Composition Patterns (shells, sections)
UICP-02 Component Composition Rules (props, slots, variants)
UICP-03 Responsive Layout Patterns (breakpoints usage)
UICP-04 Content Hierarchy Rules (typography, spacing)
UICP-05 Interaction Patterns (modals, drawers, toasts)

Client Error Handling & Recovery (CER)
CER-01 Error Boundary Strategy (global vs local)
CER-02 Retry & Recovery Patterns (per error class)
CER-03 Offline/Error Mode UX (degraded experiences)
CER-04 Session Expiry Handling (re-auth prompts, restore state)
CER-05 Client Logging & Crash Reporting (fields, redaction)

Client Security (CSec)
CSec-01 Token Storage Policy (secure storage, rotation)
CSec-02 Client Data Protection (PII redaction, screenshots, clipboard)
CSec-03 Web Security Headers/CSP (if web)
CSec-04 Anti-Tamper & Integrity Checks (basic)
CSec-05 Secure Deep Link Handling (validation, allowlists)

Mobile Implementation (MOB)
MOB-01 Navigation Map (stacks/tabs/modals)
MOB-02 Screen Implementation Spec (per screen binding)
MOB-03 Native Integration Map (bridges, permissions)

MOB-04 App Lifecycle + State (foreground/background)
MOB-05 Release & Signing (stores, builds)

Mobile Device Capabilities (MDC)
MDC-01 Capabilities Inventory (camera/gps/files/push)
MDC-02 Permissions UX Rules (prompts, denial handling)
MDC-03 Capability Security Rules (least privilege)
MDC-04 Capability Failure Handling (fallbacks)
MDC-05 Telemetry for Capabilities (errors, latency)

Offline & Sync (OFS)
OFS-01 Offline Scope (what works offline)
OFS-02 Sync Model (queues, conflict resolution)
OFS-03 Local Storage Schema (entities cached locally)
OFS-04 Reconciliation Rules (merge, LWW, prompts)
OFS-05 Sync Observability (stuck queues, retries, metrics)

Mobile Performance & Battery (MBAT)
MBAT-01 Background Work Rules (limits, schedules)
MBAT-02 Network Usage Policy (batching, retries, metered)
MBAT-03 Battery Budget & Constraints (targets)
MBAT-04 Perf Profiling Plan (mobile tooling)

Mobile Deep Links & Universal Links (MDL)
MDL-01 Link Scheme & Domains (app links, universal links)
MDL-02 Routing Rules (link → screen/action mapping)
MDL-03 Auth Gating & Safety (signed links, validation)
MDL-04 Fallback Behavior (not installed, unknown routes)

Push Notifications (MPUSH)
MPUSH-01 Notification Types Catalog (by notif_id)
MPUSH-02 Payload Contract (fields, routing, localization keys)
MPUSH-03 Permission & Opt-In Rules (platform specific)
MPUSH-04 Delivery/Retry Rules (quiet hours, collapse keys)
MPUSH-05 Deep Link Routing (notif tap behavior)
MPUSH-06 Abuse/Spam Controls for Notifications

App Store Release & Signing (SIGN)
SIGN-01 Build Artifact Types (debug/beta/prod)
SIGN-02 Signing Keys & Rotation Policy
SIGN-03 Store Submission Checklist (iOS/Android)
SIGN-04 Versioning Rules (build numbers, semver mapping)
SIGN-05 Release Channel Policy (internal/beta/GA)

Backend/API (API)

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API-01 Endpoint Catalog
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API-05 Rate Limit & Abuse Controls (per endpoint class)
API-06 Background Jobs Spec (API-triggered + system jobs)
API-07 WebSocket API Spec

API-01

API-01 — Endpoint Catalog

Header Block

- template_id: API-01
- title: Endpoint Catalog
- type: backend_api
- template_version: 1.0.0
- output_path: 10_app/api/API-01_Endpoint_Catalog.md
- compliance_gate_id: TMP-05.PRIMARY.API
- upstream_dependencies: ["PRD-04", "MAP-02", "APIG-01", "PMAD-02", "ERR-02", "ERR-03", "PFS-01"]
- inputs_required: ["PRD-04", "MAP-02", "APIG-01", "PMAD-02", "ERR-02", "ERR-03", "DATA-01", "PFS-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical inventory of backend endpoints (HTTP + internal-only surfaces if applicable): what exists, what feature it supports, what auth applies, what entities it touches, and what contract docs must exist for it. This is the source list used to generate per-endpoint specs, tests, and mapping.

Inputs Required

- PRD-04: {{xref:PRD-04}} | OPTIONAL
- MAP-02: {{xref:MAP-02}} | OPTIONAL
- APIG-01: {{xref:APIG-01}} | OPTIONAL
- PMAD-02: {{xref:PMAD-02}} | OPTIONAL
- ERR-02: {{xref:ERR-02}} | OPTIONAL
- ERR-03: {{xref:ERR-03}} | OPTIONAL
- DATA-01: {{xref:DATA-01}} | OPTIONAL
- PFS-01: {{xref:PFS-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Endpoint list (minimum 20 for non-trivial products; justify if smaller)
- For each endpoint row:

    - endpoint_id (stable)
    - method
    - path
    - title (1 line)
    - purpose (1–2 lines)
    - feature_id(s) served (PRD-04)
    - endpoint_kind (public/auth/admin/internal/webhook)
    - authn required (yes/no + scheme)
    - authz policy_id (PMAD) if authn required or if resource-gated
    - request_schema_ref (API-02 or DATA-06) | OPTIONAL
    - response_schema_ref (API-02) | OPTIONAL
    - entities_read (DATA entity_ids) | OPTIONAL
    - entities_written (DATA entity_ids) | OPTIONAL
    - list/query capable (yes/no)
    - pfs_profile_ref (PFS profile or “default contract”) | OPTIONAL
    - idempotency_required (yes/no + key rule pointer) | OPTIONAL
    - rate_limit_ref (RLIM limit_id/class) | OPTIONAL
    - error_contract (ERR-03 compliant: yes/no)
    - owner_boundary/service (SBDT boundary id) | OPTIONAL
    - versioning (APIG-02) | OPTIONAL
- Catalog coverage checks:
    - unique endpoint_id
    - unique method+path
    - each endpoint maps to ≥1 feature_id (unless internal/system endpoint with explicit exception)
    - authz policy present when required
    - list endpoints declare pfs_profile_ref (or explicit “no filters/sorts”)
    - mutation endpoints declare idempotency_required stance

Optional Fields

- Deprecation status (active/deprecated/sunset date) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must comply with APIG-01 naming/envelope and ERR-03 error payload.
- If endpoint_kind == admin, it must also appear in ADMIN-05.
- If list/query capable == true, PFS contract binding is required (PFS-01/02/03/04).
- Any state mutation endpoint must have an idempotency stance (required or explicitly not supported with rationale).
- Do not duplicate semantics via multiple endpoints with minor differences; use query contracts or versioning.

Output Format

1) Endpoint Catalog (canonical)

endpoint_id	kind	method	path	title	feature_ids	authn	policy_id	entities_read	entities_written	list_query	pfs_ref	idem_required	rate_ref	err_contract	owner	version	deprecation	notes
ep_001	{{endpoints[0].kind}}	{{endpoints[0].method}}	{{endpoints[0].path}}	{{endpoints[0].title}}	{{endpoints[0].features}}	{{endpoints[0].authn}}	{{endpoints[0].policy}}	{{endpoints[0].read}}	{{endpoints[0].write}}	{{endpoints[0].list}}	{{endpoints[0].pfs}}	{{endpoints[0].idem}}	{{endpoints[0].rate}}	{{endpoints[0].err}}	{{endpoints[0].owner}}	{{endpoints[0].version}}	{{endpoints[0].deprec}}	{{endpoints[0].notes}}
ep_002	{{endpoints[1].kind}}	{{endpoints[1].method}}	{{endpoints[1].path}}	{{endpoints[1].title}}	{{endpoints[1].features}}	{{endpoints[1].authn}}	{{endpoints[1].policy}}	{{endpoints[1].read}}	{{endpoints[1].write}}	{{endpoints[1].list}}	{{endpoints[1].pfs}}	{{endpoints[1].idem}}	{{endpoints[1].rate}}	{{endpoints[1].e	{{endpoints[1].owner}}	{{endpoints[1].version}}	{{endpoints[1].deprec}}	{{endpoints[1].notes}}

rr}
}

2) Coverage Checks (required)

- endpoint_ids unique: {{coverage.unique_endpoint_ids}}
- method+path unique: {{coverage.unique_method_path}}
- feature mapping complete: {{coverage.feature_mapping_complete}}
- authz refs complete when required: {{coverage.policy_complete}}
- pfs bound for list endpoints: {{coverage.pfs_bound}}
- idempotency stance for mutation endpoints: {{coverage.idem_stance_present}}

Cross-References

- Upstream: {{xref:APIG-01}} | OPTIONAL, {{xref:PMAD-02}} | OPTIONAL, {{xref:ERR-03}} | OPTIONAL
- Downstream: {{xref:API-02}} (per endpoint spec), {{xref:API-05}} (rate binding), {{xref:QA-02}} (contract tests) | OPTIONAL
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. kind/method/path/title/features/authn.
- intermediate: Required. Add policy_id + entities touched + list_query flags.
- advanced: Required. Add PFS bindings + idempotency stance + coverage checks.

Unknown Handling

- UNKNOWN_ALLOWED: `pfs_profile_ref` (only if list_query == false), `rate_limit_ref`, `owner_boundary`, `versioning`, `deprecation_status`, `notes`
- If authn required and policy_id is UNKNOWN → block Completeness Gate.
- If list_query == true and pfs_ref is UNKNOWN → block Completeness Gate.
- If endpoint is mutation and idem_required is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.API
- Pass conditions:

- required_fields_present == true
- endpoints_count >= 20 (or justified)
- unique_method_path == true
- feature_mapping_complete == true
- policy_complete == true
- pfs_bound == true
- idem_stance_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

API-02

API-02 — Endpoint Spec (per endpoint: request/response/auth/errors)

Header Block

- template_id: API-02
- title: Endpoint Spec (per endpoint: request/response/auth/errors)
- type: backend_api
- template_version: 1.0.0
- output_path: 10_app/api/API-02_Endpoint_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.API
- upstream_dependencies: ["API-01", "APIG-01", "APIG-02", "PMAD-04", "ERR-03", "PFS-01"]
- inputs_required: ["API-01", "APIG-01", "APIG-02", "PMAD-04", "ERR-03", "PFS-01", "DATA-01", "DATA-06", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Provide the authoritative per-endpoint contract: request/response shape, auth requirements, validation rules, error cases with reason codes, pagination/filtering rules (if list), and side effects. This is what the build and tests implement.

Inputs Required

- API-01: {{xref:API-01}} | OPTIONAL
- APIG-01: {{xref:APIG-01}} | OPTIONAL

- APIG-02: {{xref:APIG-02}} | OPTIONAL
- PMAD-04: {{xref:PMAD-04}} | OPTIONAL
- ERR-03: {{xref:ERR-03}} | OPTIONAL
- PFS-01: {{xref:PFS-01}} | OPTIONAL
- DATA-01: {{xref:DATA-01}} | OPTIONAL
- DATA-06: {{xref:DATA-06}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

For each endpoint spec block:

- endpoint_id (must exist in API-01)
- method + path
- purpose
- feature_id(s)
- auth:
    - authn required (yes/no + scheme)
    - authz policy ref (PMAD policy_id)
- request contract:
    - headers (required/optional)
    - path params (types)
    - query params (types) + PFS rules if list/search
    - body schema (fields/types) OR schema ref (DATA-06)

- response contract:
    - success status code(s)
    - response schema (fields/types)
    - pagination envelope (if list)
- side effects:
    - entities touched (reads/writes)
    - events emitted (EVT ids) | OPTIONAL
    - background jobs triggered (WFO/JBS ids) | OPTIONAL
- error cases:
    - list of failure conditions
    - http_status
    - reason_code (ERR-02)
    - retryability posture (ERR-05) | OPTIONAL
- rate limits class (RLIM) | OPTIONAL
- idempotency requirements (WFO-03/ERR-05) for writes | OPTIONAL
- observability requirements (required log fields) | OPTIONAL
- test requirements (contract tests) | OPTIONAL

Optional Fields

- Examples (request/response) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must comply with ERR-03 error payload shape.
- Must comply with APIG-01 naming + envelope standards.
- List endpoints must specify PFS rules: filters/sorts/pagination.
- Auth must be explicit; “protected” is not a spec.
- Every error case must map to a reason_code (or explicit fallback policy).

Output Format

Endpoint Spec Blocks (repeat; one per endpoint_id)

{{endpoint.endpoint_id}} — {{endpoint.title}}

- Method/Path: {{endpoint.method}} {{endpoint.path}}
- Purpose: {{endpoint.purpose}}
- Feature IDs: {{endpoint.feature_ids}}
- Owner boundary: {{endpoint.owner}} | OPTIONAL
- Version: {{endpoint.version}} | OPTIONAL

Auth

- Authn: {{endpoint.auth.authn}}
- Authz policy: {{endpoint.auth.authz_ref}}

Request

- Headers: {{endpoint.request.headers}} | OPTIONAL
- Path params: {{endpoint.request.path_params}} | OPTIONAL
- Query params: {{endpoint.request.query_params}} | OPTIONAL

- PFS rules (if list): {{endpoint.request.pfs_rules}} | OPTIONAL
- Body schema ref: {{endpoint.request.body_schema_ref}} | OPTIONAL

- Body fields (if inline):

field	type	required	validation	notes
{{endpoint.request.body_fields[0].name}}	{{endpoint.request.body_fields[0].type}}	{{endpoint.request.body_fields[0].required}}	{{endpoint.request.body_fields[0].validation}}	{{endpoint.request.body_fields[0].notes}}

-

Response

- Success status: {{endpoint.response.success_status}}
- Response schema ref: {{endpoint.response.schema_ref}} | OPTIONAL

- Response fields (if inline):

field	type	required	notes
{{endpoint.response.fields[0].name}}	{{endpoint.response.fields[0].type}}	{{endpoint.response.fields[0].required}}	{{endpoint.response.fields[0].notes}}

-
- Pagination envelope (if list): {{endpoint.response.pagination_envelope}} | OPTIONAL

Side Effects

- Entities read: {{endpoint.effects.entities_read}} | OPTIONAL
- Entities written: {{endpoint.effects.entities_written}} | OPTIONAL
- Events emitted: {{endpoint.effects.events_emitted}} | OPTIONAL
- Jobs triggered: {{endpoint.effects.jobs_triggered}} | OPTIONAL

Error Cases (required)

condition	http_status	reason_code	retryable	notes
{{endpoint.errors[0].condition}}	{{endpoint.errors[0].status}}	{{endpoint.errors[0].rc}}	{{endpoint.errors[0].retryable}}	{{endpoint.errors[0].notes}}

Operational Controls

- Rate limit class: {{endpoint.controls.rate_class}} | OPTIONAL
- Idempotency: {{endpoint.controls.idempotency}} | OPTIONAL

Observability

- Required log fields: {{endpoint.obs.required_fields}} | OPTIONAL
- Metrics: {{endpoint.obs.metrics}} | OPTIONAL

Tests

- Contract tests required: {{endpoint.tests.contract}} | OPTIONAL
- Negative cases required: {{endpoint.tests.negative}} | OPTIONAL

Cross-References

- Upstream: {{xref:API-01}} | OPTIONAL, {{xref:PMAD-04}} | OPTIONAL, {{xref:ERR-03}} | OPTIONAL, {{xref:PFS-01}} | OPTIONAL
- Downstream: {{xref:QA-02}} | OPTIONAL, {{xref:APIG-05}} | OPTIONAL, {{xref:RLIM-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Auth + request/response + error cases per endpoint.
- intermediate: Required. Add side effects + PFS rules where applicable.

- advanced: Required. Add idempotency + observability + test requirements.

Unknown Handling

- UNKNOWN_ALLOWED: `examples`, `notes`, `events_emitted`, `jobs_triggered`, `rate_class`, `observability`, `tests`
- If endpoint_id not found in API-01 → block Completeness Gate.
- If any error case lacks reason_code → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.API
- Pass conditions:
  - required_fields_present == true
  - all_endpoint_ids_exist_in_API_01 == true
  - auth_present == true
  - error_cases_have_reason_codes == true
  - list_endpoints_have_PFS_rules == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

API-03

API-03 — Error & Status Code Policy

Header Block

- template_id: API-03
- title: Error & Status Code Policy
- type: backend_api
- template_version: 1.0.0
- output_path: 10_app/api/API-03_Error_Status_Code_Policy.md
- compliance_gate_id: TMP-05.PRIMARY.API
- upstream_dependencies: ["ERR-01", "ERR-02", "ERR-03", "APIG-01"]
- inputs_required: ["ERR-01", "ERR-02", "ERR-03", "APIG-01", "APIG-06", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the enforceable policy for HTTP status codes and API error behavior: which statuses are allowed, how status maps to error_class and reason_code, what clients may rely on, and what must never be returned (leak prevention).

Inputs Required

- ERR-01: {{xref:ERR-01}} | OPTIONAL
- ERR-02: {{xref:ERR-02}} | OPTIONAL
- ERR-03: {{xref:ERR-03}} | OPTIONAL
- APIG-01: {{xref:APIG-01}} | OPTIONAL

- APIG-06: {{xref:APIG-06}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Allowed success status codes by method:
    - GET (200)
    - POST (201/200)
    - PUT/PATCH (200/204)
    - DELETE (200/204)
- Allowed error status codes and when:
    - 400/422 validation
    - 401 unauthenticated
    - 403 unauthorized
    - 404 not found
    - 409 conflict
    - 429 rate limited
    - 5xx dependency/system
- Status mapping table: error_class/category → default status
- Reason-code override policy (ERR-02 controls specific statuses)
- “Never return” list (e.g., 500 with stack traces, raw DB errors)
- Client contract statements:
    - what is stable

    - what is not guaranteed
- Verification checklist

Optional Fields

- Non-HTTP transports mapping (WS) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must conform to ERR-03 payload contract for all non-2xx.
- Avoid 404 vs 403 leakage: policy must specify concealment stance.
- 422 vs 400 must be consistent (pick one for validation; define both only with rules).
- 5xx classification must be consistent with error taxonomy.
- Status codes must not be “chosen per endpoint” outside defined override policy.

Output Format

1) Success Codes by Method (required)

method	allowed_success_codes	notes
GET	{{success.get}}	{{success.get_notes}}
POST	{{success.post}}	{{success.post_notes}}
PUT	{{success.put}}	{{success.put_notes}}
PATCH	{{success.patch}}	{{success.patch_notes}}
DELETE	{{success.delete}}	{{success.delete_notes}}

2) Error Status Codes Policy (required)

status	when_used	error_class	notes
400	{{errors.s400.when}}	validation	{{errors.s400.notes}}
401	{{errors.s401.when}}	authz	{{errors.s401.notes}}
403	{{errors.s403.when}}	authz	{{errors.s403.notes}}
404	{{errors.s404.when}}	domain_rule/dependency	{{errors.s404.notes}}
409	{{errors.s409.when}}	conflict	{{errors.s409.notes}}
429	{{errors.s429.when}}	dependency	{{errors.s429.notes}}
500	{{errors.s500.when}}	system_unknown	{{errors.s500.notes}}
503	{{errors.s503.when}}	dependency	{{errors.s503.notes}}

3) Status Mapping Table (required)

error_class	default_status	allowed_overrides	notes
validation	{{map.validation}}	{{map.validation_overrides}}	{{map.validation_notes}}
domain_rule	{{map.domain}}	{{map.domain_overrides}}	{{map.domain_notes}}
authz	{{map.authz}}	{{map.authz_overrides}}	{{map.authz_notes}}
conflict	{{map.conflict}}	{{map.conflict_overrides}}	{{map.conflict_notes}}
dependency	{{map.dependency}}	{{map.dependency_overrides}}	{{map.dependency_notes}}
system_unknown	{{map.system}}	{{map.system_overrides}}	{{map.system_notes}}

4) Reason-Code Override Policy (required)

- Overrides allowed when: {{overrides.allowed_when}}
- How overrides are documented: {{overrides.documentation}}
- Guardrails: {{overrides.guardrails}} | OPTIONAL

5) Never Return (required)

- {{never_return[0]}}
- {{never_return[1]}} | OPTIONAL
- {{never_return[2]}} | OPTIONAL

6) Client Contract Statements (required)

- Stable guarantees: {{client_contract.stable}}
- Non-guarantees: {{client_contract.not_guaranteed}} | OPTIONAL

7) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:ERR-03}} | OPTIONAL, {{xref:ERR-01}} | OPTIONAL
- Downstream: {{xref:API-02}} | OPTIONAL, {{xref:API-07}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Success codes + basic error statuses.
- intermediate: Required. Add mapping table + never-return list.
- advanced: Required. Add override policy + client contract statements.

Unknown Handling

- UNKNOWN_ALLOWED: `ws_mapping`, `notes`
- If status mapping table is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.API
- Pass conditions:
  - required_fields_present == true
  - success_codes_defined == true
  - error_codes_policy_defined == true
  - mapping_table_present == true
  - payload_contract_compliance == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

API-04

API-04 — AuthZ Rules (RBAC/ABAC checks)

Header Block

template_id: API-04
title: AuthZ Rules (RBAC/ABAC checks)
type: api_security_policy
template_version: 1.0.0
output_path: 10_app/api/API-04_AuthZ_Rules.md
compliance_gate_id: TMP-05.PRIMARY.API
upstream_dependencies: ["API-01", "API-02", "API-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-01", "API-02", "API-03", "SEC-01", "SEC-02", "ADMIN-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the single, canonical authorization model for the application API surface: **what roles/permissions exist (RBAC), what contextual constraints apply (ABAC), how endpoint specs reference these rules, and how denials are handled**. This document must not invent roles, permissions, or scopes that are not present in upstream inputs; any missing required elements must be marked UNKNOWN.

Inputs Required

SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-02 Endpoint Specs: {{api.endpoint_specs}}
API-03 Error & Status Code Policy: {{api.error_policy}}
SEC-01 Authentication: {{sec.authn}}
SEC-02 Security / Policy Controls: {{sec.authz_inputs}} | OPTIONAL
ADMIN-05 Privileged API Surface Catalog: {{admin.privileged_surface}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields

AuthZ model statement (RBAC + ABAC combination rule)
Principal model (what identity/claims fields exist for decisions)
Resource scope model (org/project/env/resource ownership boundaries)
Roles registry (role IDs and meanings)
Permissions registry (permission IDs and meanings)
Role → permission mapping (matrix)
ABAC conditions library (named conditions with deterministic evaluation rules)
Endpoint AuthZ binding rules (how API-02 must reference API-04)
Deny behavior policy (403 vs 404 concealment rule)
Admin override policy (if applicable)

Audit/evidence requirements for privileged actions
Verification requirements (tests that must exist per endpoint class)

Optional Fields
Service-to-service authz rules | OPTIONAL
Impersonation rules | OPTIONAL
Break-glass / emergency access rules | OPTIONAL
Two-person approval policy for high-risk actions | OPTIONAL
Row-level security notes (DB-level enforcement) | OPTIONAL
WebSocket/RPC authz mapping notes | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Must use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not introduce new role IDs or permission IDs. Use only what exists in: {{spec.roles_by_id}} and {{spec.permissions_by_id}} | OPTIONAL
Every endpoint spec (API-02) MUST reference AuthZ using the binding format defined here.
ABAC rules MUST be deterministic and evaluable from available principal/resource/context fields (no “human judgment” predicates).
Tenant isolation is a hard rule: cross-tenant access MUST be denied unless explicitly defined as allowed in inputs.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
If deny behavior uses concealment (404), the concealment rule MUST be stable per resource type (not ad-hoc per endpoint).
All privileged actions MUST emit audit evidence per this doc (or be marked UNKNOWN and flagged).

Output Format

1. AuthZ Model (Canonical)
   Model: {{authz.model}}
   Decision rule (normative):
   ALLOW iff (RBAC allows) AND (ABAC passes)
   Otherwise DENY.
   Notes: {{authz.model_notes}} | OPTIONAL

2. Principal (Identity/Claims) Model
   Principal fields available for authz decisions:
   user_id: {{principal.user_id}} | OPTIONAL
   service_id: {{principal.service_id}} | OPTIONAL
   roles: {{principal.roles}}
   permissions: {{principal.permissions}} | OPTIONAL
   org_ids: {{principal.org_ids}} | OPTIONAL
   project_ids: {{principal.project_ids}} | OPTIONAL
   env: {{principal.env}} | OPTIONAL

risk_class: {{principal.risk_class}} | OPTIONAL
Any missing required claim fields: {{principal.missing_fields}} | OPTIONAL

3. Resource Scope Model
   Define the scope boundaries used for ABAC:
   Org boundary: {{scope.org_rule}}
   Project boundary: {{scope.project_rule}}
   Environment boundary: {{scope.env_rule}}
   Ownership boundary: {{scope.ownership_rule}} | OPTIONAL
   Resource types covered (minimum):
   {{resources[0].type}} (scope: {{resources[0].scope_fields}})
   {{resources[1].type}} (scope: {{resources[1].scope_fields}})
   {{resources[2].type}} (scope: {{resources[2].scope_fields}})
   If any required resource scope is unknown: {{resources.unknown_scope_notes}} | OPTIONAL

4. Roles Registry
   List canonical roles (IDs must match upstream inputs):
   {{roles[0].role_id}} — {{roles[0].name}} — {{roles[0].summary}}
   {{roles[1].role_id}} — {{roles[1].name}} — {{roles[1].summary}}
   {{roles[2].role_id}} — {{roles[2].name}} — {{roles[2].summary}}
   Admin/privileged roles: {{roles.privileged}} | OPTIONAL

5. Permissions Registry
   List canonical permissions (IDs must match upstream inputs):
   {{perms[0].perm_id}} — {{perms[0].name}} — {{perms[0].summary}}
   {{perms[1].perm_id}} — {{perms[1].name}} — {{perms[1].summary}}
   {{perms[2].perm_id}} — {{perms[2].name}} — {{perms[2].summary}}

6. Role → Permission Matrix (RBAC)
   For each role, enumerate granted permissions:
   {{rbac[0].role_id}}:

- {{rbac[0].perm_ids[0]}}

- {{rbac[0].perm_ids[1]}}
  {{rbac[1].role_id}}:

- {{rbac[1].perm_ids[0]}}

- {{rbac[1].perm_ids[1]}}
  Unknown role mappings: {{rbac.unknowns}} | OPTIONAL

7. ABAC Conditions Library
   Define named conditions used by endpoint specs. Each condition MUST include: inputs, evaluation rule, deny reason code.
   Condition: {{abac[0].cond_id}}
   Name: {{abac[0].name}}
   Inputs: {{abac[0].inputs}}
   Rule: {{abac[0].rule}}
   Deny reason_code: {{abac[0].reason_code}}

Condition: {{abac[1].cond_id}}
Name: {{abac[1].name}}
Inputs: {{abac[1].inputs}}
Rule: {{abac[1].rule}}
Deny reason_code: {{abac[1].reason_code}}

Minimum required condition coverage:

- tenant/org match: {{abac.required.tenant_match}}
- project scope match: {{abac.required.project_match}}
- env gating (prod vs non-prod): {{abac.required.env_gate}}
- ownership (if applicable): {{abac.required.ownership}} | OPTIONAL
- state-based constraints (if applicable): {{abac.required.state_constraints}} | OPTIONAL

8. Endpoint AuthZ Binding (How API-02 Must Reference API-04)
   Each API-02 endpoint MUST include an authz block with: action_id, resource_type, required_permissions, required_abac_conditions, deny_behavior, audit_required.
   Canonical authz block format (for API-02):
   authz:
   action_id: {{api02.authz.action_id}}
   resource_type: {{api02.authz.resource_type}}
   required_permissions: {{api02.authz.required_permissions}}
   required_abac_conditions: {{api02.authz.required_abac_conditions}}
   deny_behavior: {{api02.authz.deny_behavior}}
   audit_required: {{api02.authz.audit_required}}

9. Deny Behavior Policy (403 vs 404)
   Default deny behavior: {{deny.default}} (expected: 403)
   Concealment policy (if used):
   Resource types using concealment: {{deny.conceal.resource_types}} | OPTIONAL
   Rule: {{deny.conceal.rule}} | OPTIONAL

If concealment is UNKNOWN, it must be flagged: {{deny.conceal.open_question}} | OPTIONAL

10. Admin Override Policy (Optional)
    Override allowed: {{admin.override.allowed}} | OPTIONAL
    Required permission(s): {{admin.override.perms}} | OPTIONAL
    Audit requirement: {{admin.override.audit_required}} | OPTIONAL
    Environments where override is allowed: {{admin.override.envs}} | OPTIONAL

11. Audit & Evidence Requirements
    For actions where audit_required == true, log fields MUST include:
    principal: {{audit.principal_fields}}
    action_id: {{audit.action_id}}
    resource: {{audit.resource_fields}}
    decision: {{audit.decision}}
    reason_code: {{audit.reason_code}}
    request_id: {{audit.request_id}}
    evidence_refs: {{audit.evidence_refs}} (e.g., {{xref:QA-03}}) | OPTIONAL

12. Verification Requirements (Minimum)
    Per protected endpoint class, tests MUST include:

- deny unauthenticated (per SEC-01)
- deny lacking permission
- deny failing tenant boundary ABAC
- allow with correct role + ABAC pass
- deny for at least one state-based constraint (if applicable)
- audit emission test when audit_required == true

13. References
    Endpoint catalog: {{xref:API-01}}
    Endpoint specs: {{xref:API-02}}
    Errors/status mapping: {{xref:API-03}}
    Rate limits/abuse: {{xref:API-05}} | OPTIONAL
    Privileged surface: {{xref:ADMIN-05}} | OPTIONAL
    Security/authn: {{xref:SEC-01}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:API-01}}, {{xref:API-02}}, {{xref:SEC-01}} | OPTIONAL, {{xref:STANDARDS_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-01}} | OPTIONAL, {{xref:ADMIN-05}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL, {{xref:QA-03}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Fill fields with UNKNOWN where needed; do not invent roles/permissions.
intermediate: Required. Populate roles/permissions from inputs; bind ABAC conditions to available principal/resource fields.
advanced: Required. Provide complete endpoint binding guidance and audit/verification rigor with stable reason codes.

Unknown Handling
UNKNOWN_ALLOWED: sec.authz_inputs, admin.override.*, impersonation_rules, break_glass_rules, ws_authz_notes, state_constraints (only if upstream state machine docs are missing)
If any Required Field is UNKNOWN, allow only if: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If tenant boundary rules are UNKNOWN → block Completeness Gate.
If roles or permissions registries are UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.API
Pass conditions:
required_fields_present == true
no new role_ids or perm_ids introduced (only references to existing IDs)
api02_binding_format_defined == true
tenant_isolation_rule_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

API-05

API-05 — Rate Limit & Abuse Controls (per endpoint class)

Header Block

- template_id: API-05
- title: Rate Limit & Abuse Controls (per endpoint class)
- type: backend_api
- template_version: 1.0.0
- output_path: 10_app/api/API-05_Rate_Limit_Abuse_Controls.md
- compliance_gate_id: TMP-05.PRIMARY.API
- upstream_dependencies: ["RLIM-01", "RLIM-02", "API-01", "ERR-02"]
- inputs_required: ["RLIM-01", "RLIM-02", "API-01", "ERR-02", "OBS-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define how rate limiting and abuse controls apply to API endpoint classes: classification, limits, enforcement points, error behaviors, exemptions, and observability. This is the API-specific binding of the global RLIM policy.

Inputs Required

- RLIM-01: {{xref:RLIM-01}} | OPTIONAL
- RLIM-02: {{xref:RLIM-02}} | OPTIONAL
- API-01: {{xref:API-01}} | OPTIONAL

- ERR-02: {{xref:ERR-02}} | OPTIONAL
- OBS-04: {{xref:OBS-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Endpoint classification scheme (classes)
    - public read
    - authenticated read
    - write/mutate
    - auth endpoints
    - admin endpoints
    - file/media endpoints
- Mapping: endpoint_id → class
- Limits per class:
    - scope (ip/user/tenant)
    - threshold (req/min)
    - burst policy
    - block duration/cooldown
- Enforcement point (edge/api gateway/app middleware)
- Error behavior (429 + reason_code + Retry-After)
- Abuse detection tie-ins (suspicious patterns) pointer
- Exemptions policy pointer

- Verification checklist

Optional Fields

- WAF/bot protection pointer | OPTIONAL
- Notes | OPTIONAL

Rules

- Rate limits must be enforceable server-side and consistent across endpoints of same class.
- 429 responses must conform to ERR-03 and include reason_code.
- Retry-After must be included where applicable.
- Admin endpoints require stricter default limits and auditing.

Output Format

1) Endpoint Classes (required)

class_id	description	default_scope	default_limit	burst_policy	notes
cls_public_read	{{classes.public.desc}}	{{classes.public.scope}}	{{classes.public.limit}}	{{classes.public.burst}}	{{classes.public.notes}}
cls_write	{{classes.write.desc}}	{{classes.write.scope}}	{{classes.write.limit}}	{{classes.write.burst}}	{{classes.write.notes}}

2) Endpoint → Class Mapping (required)

endpoint_id	class_id	notes
{{map[0].endpoint_id}}	{{map[0].class_id}}	{{map[0].notes}}
{{map[1].endpoint_id}}	{{map[1].class_id}}	{{map[1].notes}}

3) Limits by Class (required)

class_id	scope	threshold	burst	cooldown	enforcement_point	error_reason_code	retry_after_policy	notes
cls_public_read	{{limits.public.scope}}	{{limits.public.threshold}}	{{limits.public.burst}}	{{limits.public.cooldown}}	{{limits.public.enforce}}	{{limits.public.rc}}	{{limits.public.retry_after}}	{{limits.public.notes}}
cls_write	{{limits.write.scope}}	{{limits.write.threshold}}	{{limits.write.burst}}	{{limits.write.cooldown}}	{{limits.write.enforce}}	{{limits.write.rc}}	{{limits.write.retry_after}}	{{limits.write.notes}}

4) Error Behavior (required)

- 429 contract: {{error.contract}} (must follow ERR-03)
- Retry-After rule: {{error.retry_after}}
- Client guidance pointer: {{xref:ERR-04}} | OPTIONAL

5) Exemptions & Abuse Tie-ins (required)

- Exemptions policy pointer: {{xref:RLIM-05}} | OPTIONAL
- Abuse signals pointer: {{xref:RLIM-03}} | OPTIONAL

6) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RLIM-01}} | OPTIONAL, {{xref:API-01}} | OPTIONAL

- Downstream: {{xref:RLIM-06}} | OPTIONAL, {{xref:ALRT-*}} | OPTIONAL, {{xref:QA-04}} | OPTIONAL

- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Classes + mapping table.

- intermediate: Required. Add limits and enforcement points.

- advanced: Required. Add exemptions tie-ins and observability hooks.

Unknown Handling

- UNKNOWN_ALLOWED: `waf_pointer`, `notes`, `burst_policy`, `cooldown`

- If any class lacks enforcement_point or reason_code → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.API

- Pass conditions:

    - required_fields_present == true

    - classes_present == true

    - endpoint_mapping_present == true

    - limits_present == true

    - error_behavior_present == true

    - placeholder_resolution == true

- no_unapproved_unknowns == true

API-06

API-06 — Background Jobs Spec (API-triggered + system jobs)

Header Block

template_id: API-06
title: Background Jobs Spec (API-triggered + system jobs)
type: api_background_jobs
template_version: 1.0.0
output_path: 10_app/api/API-06_Background_Jobs_Spec.md
compliance_gate_id: TMP-05.PRIMARY.API
upstream_dependencies: ["API-01", "API-02", "API-03", "API-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-01", "API-02", "API-03", "API-05", "JBS-01", "RLIM-01", "FFCFG-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical specification format for **background jobs that are initiated by API actions or run as system jobs**, including triggers, inputs/outputs, idempotency, concurrency, retries, DLQ handling, status surfaces, and observability requirements. This template must be consistent with the Endpoint Catalog and per-endpoint specs and must not invent job IDs, triggers, or behavior not present in upstream inputs.

Inputs Required

SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-02 Endpoint Specs: {{api.endpoint_specs}}
API-03 Error & Status Code Policy: {{api.error_policy}}
API-05 Rate Limit & Abuse Controls: {{api.ratelimit_policy}} | OPTIONAL
JBS-01 Jobs Inventory: {{jbs.jobs_inventory}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.flags}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields

Job linkage model (how endpoints reference job specs)
Job identifiers (job_id) source rule (must come from JBS-01)
Trigger types (API-triggered vs system-triggered)
Trigger definition format (what fields define a trigger)
Job lifecycle model (states + transitions)
Inputs contract (schema + provenance)
Outputs contract (result schema + evidence refs)
Idempotency rules (job-level and trigger-level)
Concurrency rules (per job_id / per resource / global)

Retry policy binding rules (how to reference JBS-04 / defaults if absent)
DLQ / poison handling binding rules (how to reference JBS-04/JBS-07 equivalents)
Status surfacing rules (what API endpoints expose job status + polling model)
Cancellation policy (if supported)
Timeouts/SLO fields (target latencies per class)
Observability fields (logs/metrics/traces + correlation IDs)
Security constraints (PII redaction, access control for job artifacts)

Optional Fields
Job priority classes | OPTIONAL
Backfill/replay rules | OPTIONAL
Manual operator controls | OPTIONAL
Rate-limit coupling rules (job-trigger throttles) | OPTIONAL
Feature-flag enablement rules per env | OPTIONAL
Resource cost hints (CPU/mem/queue) | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new job_ids; use: {{jbs.jobs_inventory[job_*].job_id}} as given.
Every background job referenced by an API endpoint MUST have a job spec entry using this template OR be explicitly marked UNKNOWN (with pointer).
Every job MUST be traceable to its trigger source (endpoint, scheduler, event, operator action).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
All status surfaces MUST map errors to: {{xref:API-03}}
All access to job artifacts/status MUST obey: {{xref:API-04}} | OPTIONAL
Retries/DLQ MUST follow: {{xref:JBS-04}} | OPTIONAL
If idempotency is required, bind to: {{xref:JBS-05}} | OPTIONAL

Output Format

1. Job Linkage Model (API → Job)
   How endpoints declare job triggers: {{jobs.linkage_model}}
   Binding rule:

- API-02 endpoint spec MUST include: {{api02.job_binding_block}} when async work exists.
  Canonical job binding block (for API-02):
  job:
  job_id: {{api02.job.job_id}}
  trigger: {{api02.job.trigger}}
  response_mode: {{api02.job.response_mode}} (e.g., "202 + status_url" | "sync" | UNKNOWN)
  status_url: {{api02.job.status_url}} | OPTIONAL

2. Job Identifier Rules
   job_id source: {{jobs.job_id_source}} (expected: JBS-01)
   Allowed format: {{jobs.job_id_format}} | OPTIONAL
   Unknown handling note: {{jobs.job_id_unknown_note}} | OPTIONAL

3. Trigger Types
   API-triggered: {{jobs.trigger_types.api_triggered}}
   System-triggered: {{jobs.trigger_types.system_triggered}}
   Other triggers (events/schedules/operators): {{jobs.trigger_types.other}} | OPTIONAL

4. Trigger Definition Format
   Trigger fields:
   trigger_type: {{trigger.type}}
   source: {{trigger.source}} (endpoint_id/schedule_id/event_id/operator_action_id)
   conditions: {{trigger.conditions}} | OPTIONAL
   feature_flag: {{trigger.feature_flag}} | OPTIONAL
   throttle_scope: {{trigger.throttle_scope}} | OPTIONAL

5. Job Lifecycle Model
   States:
   {{job_lifecycle.states[0]}}
   {{job_lifecycle.states[1]}}
   {{job_lifecycle.states[2]}}
   Transitions: {{job_lifecycle.transitions}}
   Terminal states: {{job_lifecycle.terminal_states}}

6. Inputs Contract
   Inputs schema ref: {{job.inputs.schema_ref}} (e.g., {{xref:SCHEMA_*}}) | OPTIONAL
   Inputs fields:
   {{job.inputs.fields[0].name}} — {{job.inputs.fields[0].type}} — {{job.inputs.fields[0].source}}
   {{job.inputs.fields[1].name}} — {{job.inputs.fields[1].type}} — {{job.inputs.fields[1].source}}
   Provenance requirements: {{job.inputs.provenance_rules}}

7. Outputs Contract
   Outputs schema ref: {{job.outputs.schema_ref}} | OPTIONAL
   Outputs fields:
   {{job.outputs.fields[0].name}} — {{job.outputs.fields[0].type}}
   {{job.outputs.fields[1].name}} — {{job.outputs.fields[1].type}}
   Evidence refs (proof/log pointers): {{job.outputs.evidence_refs}} | OPTIONAL

8. Idempotency Rules
   Required? {{job.idempotency.required}}

Idempotency key source: {{job.idempotency.key_source}} (request_id/idempotency_key/resource_hash) | OPTIONAL
Dedup window: {{job.idempotency.window}} | OPTIONAL
Conflict behavior: {{job.idempotency.conflict_behavior}} (return existing job_id vs error) | OPTIONAL

9. Concurrency Rules
   Concurrency scope: {{job.concurrency.scope}} (per job_id / per resource / global)
   Max in-flight: {{job.concurrency.max_inflight}}
   Locking strategy: {{job.concurrency.locking}} | OPTIONAL
   Race resolution: {{job.concurrency.race_resolution}} | OPTIONAL

10. Retry & DLQ Binding
    Retry policy reference: {{xref:JBS-04}} | OPTIONAL
    Per-job overrides: {{job.retry.overrides}} | OPTIONAL
    DLQ policy reference: {{job.dlq.policy_ref}} | OPTIONAL
    Poison message handling: {{job.dlq.poison_handling}} | OPTIONAL

11. Status Surfacing (API Contracts)
    Status exposure model: {{job.status.model}} (polling endpoint / push / both)
    Status endpoint id: {{job.status.endpoint_id}} | OPTIONAL
    Status response fields:
    job_id: {{job.status.fields.job_id}}
    state: {{job.status.fields.state}}
    progress: {{job.status.fields.progress}} | OPTIONAL
    started_at: {{job.status.fields.started_at}} | OPTIONAL
    finished_at: {{job.status.fields.finished_at}} | OPTIONAL
    last_error: {{job.status.fields.last_error}} | OPTIONAL
    result_ref: {{job.status.fields.result_ref}} | OPTIONAL

12. Cancellation Policy
    Cancelable? {{job.cancel.cancelable}}
    Who can cancel: {{job.cancel.allowed_principals}} | OPTIONAL
    Cancellation behavior: {{job.cancel.behavior}} | OPTIONAL

13. Timeouts & SLOs
    Timeout: {{job.timeouts.hard_timeout_ms}}
    SLO target: {{job.slo.target_ms}} | OPTIONAL
    SLO class: {{job.slo.class}} | OPTIONAL

14. Observability Requirements
    Correlation IDs: {{obs.correlation_ids}} (request_id, job_id, run_id, trace_id)
    Logs (required fields): {{obs.logs.required_fields}}
    Metrics: {{obs.metrics}}

Traces: {{obs.traces}} | OPTIONAL
Alerts: {{obs.alerts}} | OPTIONAL

15. Security & Compliance
    PII in job payloads: {{security.pii_policy}} | OPTIONAL
    Redaction rules: {{security.redaction_rules}} | OPTIONAL
    Access control for status/results: {{security.access_control}} (bind to {{xref:API-04}}) | OPTIONAL
    Retention policy for job artifacts: {{security.retention}} | OPTIONAL

16. References
    Jobs inventory: {{xref:JBS-01}} | OPTIONAL
    Per-job spec: {{xref:JBS-02}} | OPTIONAL
    Scheduling rules: {{xref:JBS-03}} | OPTIONAL
    Retry/DLQ policy: {{xref:JBS-04}} | OPTIONAL
    Idempotency & concurrency: {{xref:JBS-05}} | OPTIONAL
    API endpoints: {{xref:API-01}}, {{xref:API-02}}
    Errors/status mapping: {{xref:API-03}}
    Rate limits: {{xref:API-05}} | OPTIONAL
    Feature flags: {{xref:FFCFG-01}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:API-01}}, {{xref:API-02}}, {{xref:JBS-01}} | OPTIONAL, {{xref:STANDARDS_INDEX}} | OPTIONAL
Downstream: {{xref:JBS-02}}, {{xref:JBS-03}}, {{xref:JBS-04}}, {{xref:JBS-05}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-OBSERVABILITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for missing trigger/job details; do not invent job IDs.
intermediate: Required. Bind jobs to endpoints using job_id from JBS-01; specify lifecycle + status surfaces.
advanced: Required. Provide strict idempotency/concurrency rules and full observability/audit linkages.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, rlim.policy, ffcfg.flags, job.priority_classes, replay_rules, manual_controls, feature_flag_enablement, resource_cost_hints, traces, alerts
If any Required Field is UNKNOWN, allow only if: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If job_id source rules are UNKNOWN → block Completeness Gate.
If linkage model is UNKNOWN → block Completeness Gate.

Completeness Gate

Gate ID: TMP-05.PRIMARY.API

Pass conditions:

required_fields_present == true

no new job_ids introduced (only references to existing IDs)

api02_job_binding_defined == true

placeholder_resolution == true

no_unapproved_unknowns == true

API-07

API-07 — WebSocket API Spec

Header Block

- template_id: API-07
- title: WebSocket API Spec
- type: backend_api
- template_version: 1.0.0
- output_path: 10_app/api/API-07_WebSocket_API_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.API
- upstream_dependencies: ["RTM-01", "RTM-02", "RTM-03", "RTM-04", "PMAD-02", "ERR-02", "ERR-03", "RLIM-01"]
- inputs_required: ["RTM-01", "RTM-02", "RTM-03", "RTM-04", "PMAD-02", "ERR-02", "ERR-03", "RLIM-01", "OBS-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the **build-authoritative** WebSocket interface: connection URL(s), authentication/authorization handshake, message envelope, message types, channel semantics, acknowledgments, delivery semantics, error/close behavior, rate limits, and required observability fields.

Inputs Required

- RTM-01: {{xref:RTM-01}} | OPTIONAL
- RTM-02: {{xref:RTM-02}} | OPTIONAL
- RTM-03: {{xref:RTM-03}} | OPTIONAL
- RTM-04: {{xref:RTM-04}} | OPTIONAL
- PMAD-02: {{xref:PMAD-02}} | OPTIONAL
- ERR-02: {{xref:ERR-02}} | OPTIONAL
- ERR-03: {{xref:ERR-03}} | OPTIONAL
- RLIM-01: {{xref:RLIM-01}} | OPTIONAL
- OBS-01: {{xref:OBS-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

A) Connection & Transport (required)

- WS URL(s) per environment
- Supported protocols/subprotocols
- Heartbeat/ping policy:
    - interval
    - timeout
    - client reconnection trigger
- Compression/binary policy (allowed/forbidden)
- Max message size (bytes)
- Origin/CORS-like policy (if browser clients)
- Verification checklist

B) Auth Handshake (required)

- Authn model:
    - token type (session/JWT)
    - where token is provided (query/header/first message)
    - refresh/rotation posture
- Authz model:
    - how role/permissions are derived
    - default deny policy
- Handshake sequence (ordered steps)
- Failure behaviors:
    - auth failure close code
    - reason_code mapping
    - retry guidance

C) Message Envelope (required)

- Required fields:
    - msg_id (client or server generated)
    - type
    - direction (c2s/s2c) | OPTIONAL
    - channel (string)
    - payload (object)
    - ts (ISO8601)
    - correlation_id
    - request_id | OPTIONAL
    - reply_to (msg_id) | OPTIONAL
- Versioning:
    - envelope_version
    - message schema version posture

D) Message Types Catalog (required; minimum 25)

For each type:

- type name
- direction
- payload schema ref OR inline schema
- auth required (yes/no)
- authz policy id (PMAD) if protected
- rate limit binding (RLIM limit_id/class) | OPTIONAL
- ack required (yes/no)
- idempotency/dedupe key stance (msg_id, payload key)
- error behavior on invalid payload (reason_code)

E) Channel Semantics (required)

- Channel naming rules pointer (RTM-03)
- Channel types supported (room, dm, system, stream, presence)
- Join/leave semantics:
    - who can join
    - who can publish
    - who can subscribe
- Subscription lifecycle:
    - auto-subscribe policy (if any)
    - resubscribe on reconnect rules
- Presence rules (if used) | OPTIONAL

F) Delivery Semantics & Acks (required)

- Delivery posture (at-least-once vs best-effort) by message type
- Ordering guarantees scope (per-channel/per-user/none)
- Ack model:
    - ack payload fields
    - ack timeout policy
    - retry policy
- Dedupe rules:
    - dedupe window
    - duplicate handling (drop/merge)
- Backpressure policy:
    - server → client throttling
    - client publish throttling

G) Errors & Close Codes (required)

- Error payload shape (must align with ERR-03):
    - error_id

    - reason_code
    - message_safe
    - retryable
    - details (redacted)
    - correlation_id
- Close codes catalog (minimum 10):
    - auth failed
    - policy violation
    - message too large
    - rate limited
    - server shutdown
- Mapping table:
    - condition → close_code → reason_code → client action

H) Observability (required)

- Required log fields:
    - connection_id
    - user_id/tenant_id (hashed where needed)
    - channel
    - msg_id
    - type
    - direction
    - reason_code (if error)
    - correlation_id
    - latency_ms (if measurable)
- Metrics catalog (minimum 15):
    - connections_opened
    - connections_active
    - reconnects
    - messages_in/out
    - publish_failures
    - auth_failures
    - ack_timeouts
- Alerts (minimum 8)

Optional Fields

- Binary frames spec | OPTIONAL
- Encryption/pinning notes (mobile) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must align with RTM delivery semantics (RTM-04). No “exactly once” promises unless proven.
- Protected message types must have explicit policy_id.
- All errors must provide a reason_code and use the standard error shape.
- Server must enforce authz on join and publish (not client-side).
- Rate limiting must be specified for publish-heavy message types.

Output Format

1) Connection & Transport

- WS URLs: {{ws.urls}}
- Subprotocols: {{ws.subprotocols}} | OPTIONAL
- Heartbeat: {{ws.heartbeat}}
- Max message size: {{ws.max_message_size}}
- Compression/binary: {{ws.compression}} | OPTIONAL
- Origin policy: {{ws.origin_policy}} | OPTIONAL

2) Handshake Sequence (ordered)

1. {{handshake.steps[0]}}
2. {{handshake.steps[1]}}
3. {{handshake.steps[2]}} | OPTIONAL
- Auth failure close code: {{handshake.auth_fail_close}}
- Auth failure reason_code: {{handshake.auth_fail_rc}}

3) Message Envelope (canonical)

```
{

  "envelope_version": "{{env.v}}",

  "msg_id": "{{msg.msg_id}}",

  "type": "{{msg.type}}",

  "channel": "{{msg.channel}}",

  "payload": {{msg.payload}},

  "ts": "{{msg.ts}}",

  "correlation_id": "{{msg.correlation_id}}",

  "reply_to": "{{msg.reply_to}}"
```

}

4) Message Types Catalog

type	direction	payload_ref	auth_required	policy_id	rate_limit	ack_required	dedupe_key	invalid_payload_rc	notes
join	c → s	{{types.join.payload_ref}}	true	{{types.join.policy}}	{{types.join.rlim}}	true	{{types.join.dedupe}}	{{types.join.invalid_rc}}	{{types.join.notes}}
publish	c → s	{{types.publish.payload_ref}}	true	{{types.publish.policy}}	{{types.publish.rlim}}	true	{{types.publish.dedupe}}	{{types.publish.invalid_rc}}	{{types.publish.notes}}
ack	s → c	{{types.ack.payload_ref}}	true	n/a	{{types.ack.rlim}}	false	{{types.ack.dedupe}}	{{types.ack.invalid_rc}}	{{types.ack.notes}}

5) Channel Semantics

- Naming rules pointer: {{xref:RTM-03}} | OPTIONAL
- Channel types: {{channels.types}}
- Join rules: {{channels.join}}
- Publish rules: {{channels.publish}}
- Subscribe rules: {{channels.subscribe}} | OPTIONAL
- Reconnect resubscribe: {{channels.resubscribe}}

6) Delivery & Ack Rules

- Delivery posture by type: {{delivery.by_type}}
- Ordering scope: {{delivery.ordering}}
- Ack timeout: {{acks.timeout}}

- Retry policy: {{acks.retry}} | OPTIONAL
- Dedupe window: {{dedupe.window}}
- Duplicate handling: {{dedupe.handling}}
- Backpressure: {{backpressure.policy}} | OPTIONAL

7) Errors & Close Codes

Error Payload (ERR-03 aligned)

```
{
  "error_id": "{{err.error_id}}",
  "reason_code": "{{err.reason_code}}",
  "message": "{{err.message_safe}}",
  "retryable": {{err.retryable}},
  "details": {{err.details}},
  "correlation_id": "{{err.correlation_id}}"
}
```

Close Codes Catalog

close_code	condition	reason_code	client_action	notes
{{close[0].code}}	{{close[0].condition}}	{{close[0].rc}}	{{close[0].action}}	{{close[0].notes}}
{{close[1].code}}	{{close[1].condition}}	{{close[1].rc}}	{{close[1].action}}	{{close[1].notes}}

8) Observability

- Required log fields: {{obs.log_fields}}

- Metrics: {{obs.metrics}}
- Alerts: {{obs.alerts}} | OPTIONAL

9) Verification Checklist

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RTM-04}} | OPTIONAL, {{xref:PMAD-02}} | OPTIONAL, {{xref:ERR-03}} | OPTIONAL
- Downstream: {{xref:SMD-04}} | OPTIONAL, {{xref:RLIM-02}} | OPTIONAL, {{xref:QA-04}} | OPTIONAL
- Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. URLs, handshake, envelope, message types list.
- intermediate: Required. Add channel rules + acks/dedupe + close codes.
- advanced: Required. Add rate limit bindings + observability rigor + mapping table.

Unknown Handling

- UNKNOWN_ALLOWED: `binary_frames_spec`, `encryption_pinning_notes`, `notes`, `presence_rules`, `origin_policy`, `compression_policy`
- If any protected message type lacks policy_id → block Completeness Gate.
- If any error condition lacks reason_code mapping → block Completeness Gate.
- If max message size is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.API
- Pass conditions:
  - required_fields_present == true
  - handshake_defined == true
  - message_types_count >= 25
  - channel_rules_defined == true

- error_contract_aligned == true
- close_codes_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

Background Jobs & Scheduling (JBS)

Background Jobs & Scheduling (JBS)
JBS-01 Jobs Inventory (by job_id)
JBS-02 Job Spec (per job: trigger, inputs, outputs, retries)
JBS-03 Scheduling Rules (cron/event-driven, env enablement)
JBS-04 Retry/DLQ Policy (backoff, poison messages)
JBS-05 Idempotency & Concurrency for Jobs
JBS-06 Job Observability (metrics/logs/traces, alerts)

JBS-01

API-07 — WebSocket API Spec

Header Block

- template_id: API-07
- title: WebSocket API Spec
- type: backend_api
- template_version: 1.0.0
- output_path: 10_app/api/API-07_WebSocket_API_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.API
- upstream_dependencies: ["RTM-02", "RTM-03", "RTM-04", "ERR-03", "PMAD-02"]
- inputs_required: ["RTM-02", "RTM-03", "RTM-04", "ERR-03", "PMAD-02", "RLIM-01", "OBS-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the authoritative WebSocket API contract: connection/auth handshake, message envelope, message types, channel join/leave semantics, acknowledgments, error behavior, rate limits, and observability. This binds RTM architecture decisions to a build-ready interface spec.

Inputs Required

- RTM-02: {{xref:RTM-02}} | OPTIONAL
- RTM-03: {{xref:RTM-03}} | OPTIONAL
- RTM-04: {{xref:RTM-04}} | OPTIONAL
- ERR-03: {{xref:ERR-03}} | OPTIONAL

- PMAD-02: {{xref:PMAD-02}} | OPTIONAL
- RLIM-01: {{xref:RLIM-01}} | OPTIONAL
- OBS-01: {{xref:OBS-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Connection details:
    - WS endpoint URL(s) + env variants
    - protocols/subprotocols
    - auth handshake steps
    - session/token validation point
- Message envelope schema:
    - msg_id
    - type
    - channel
    - payload
    - timestamp
    - correlation_id
- Message types catalog (minimum 20):
    - join/leave
    - publish message
    - ack/nack

    - presence updates
    - server notices
- Channel rules:
    - naming patterns (RTM-03)
    - join auth rules (policy_id)
    - publish auth rules
    - subscribe rules
- Delivery semantics:
    - ordering guarantee (per channel)
    - dedupe rule (msg_id)
    - ack rules + retry rules
- Error behavior:
    - error message shape (ERR-03 aligned)
    - close codes policy
    - reason_code mapping
- Rate limiting policy for WS (RLIM)
- Observability requirements (logs/metrics/traces)
- Verification checklist

Optional Fields

- Compression/binary frames rules | OPTIONAL
- Notes | OPTIONAL

Rules

- Must align to RTM-04 for delivery semantics (no surprise “exactly once”).
- Authz must be server enforced on join and publish.
- Errors must use the ERR-03 shape; include reason_code.
- All message types must be enumerated and versioned.

Output Format

1) Connection & Handshake (required)

- WS URL(s): {{ws.urls}}
- Subprotocol: {{ws.subprotocol}} | OPTIONAL
- Handshake steps: {{ws.handshake_steps}}
- Auth validation: {{ws.auth_validation}}

2) Message Envelope (required)

```
{
 "msg_id": "{{msg.msg_id}}",
 "type": "{{msg.type}}",
 "channel": "{{msg.channel}}",
 "payload": {{msg.payload}},
 "timestamp": "{{msg.timestamp}}",
 "correlation_id": "{{msg.correlation_id}}"
}
```

3) Message Types Catalog (canonical)

type	direction (c→s / s→c)	payload_schema_ref	requires_auth	authz_policy_ref	ack_required	notes

join	c→s	{{types.join.schema}}	true	{{types.join.policy}}	true	{{types.join.notes}}
publish	c→s	{{types.publish.schema}}	true	{{types.publish.policy}}	true	{{types.publish.notes}}
message	s→c	{{types.message.schema}}	true	n/a	false	{{types.message.notes}}

4) Channel Rules (required)

- Naming patterns pointer: {{xref:RTM-03}} | OPTIONAL
- Join auth rule: {{channels.join_auth}}
- Publish auth rule: {{channels.publish_auth}}
- Subscribe rule: {{channels.subscribe_rule}} | OPTIONAL

5) Delivery Semantics (required)

- Per-channel ordering: {{delivery.ordering}}
- Dedupe rule: {{delivery.dedupe}}
- Ack rules: {{delivery.ack}}
- Retry rules: {{delivery.retry}} | OPTIONAL

6) Errors & Close Codes (required)

- Error payload shape: {{errors.payload_shape}} (ERR-03 aligned)
- Reason code mapping: {{errors.reason_code_mapping}}
- Close codes policy: {{errors.close_codes}} | OPTIONAL

7) Rate Limits (required)

- WS rate limit pointer: {{xref:RLIM-02}} | OPTIONAL

- Per-connection limits: {{limits.per_connection}} | OPTIONAL

8) Observability (required)

- Required log fields: {{obs.log_fields}}
- Metrics: {{obs.metrics}} | OPTIONAL
- Tracing: {{obs.tracing}} | OPTIONAL

9) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RTM-02}} | OPTIONAL, {{xref:RTM-04}} | OPTIONAL, {{xref:ERR-03}} | OPTIONAL
- Downstream: {{xref:FE-07}} | OPTIONAL, {{xref:SMD-04}} | OPTIONAL, {{xref:QA-04}} | OPTIONAL
- Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Handshake + envelope + message types list.
- intermediate: Required. Add authz policies and delivery semantics.
- advanced: Required. Add error mapping, rate limits, and observability.

Unknown Handling

- UNKNOWN_ALLOWED: `compression_rules`, `notes`, `close_codes_policy`, `metrics`, `tracing`
- If any authenticated message type lacks authz_policy_ref → block Completeness Gate.
- If any error path lacks reason_code mapping → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.API
- Pass conditions:
  - required_fields_present == true
  - handshake_defined == true
  - message_types_count >= 20
  - authz_defined_for_protected_types == true
  - error_contract_aligned == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

JBS-02

JBS-01 — Jobs Inventory (by job_id)

Header Block

- template_id: JBS-01
- title: Jobs Inventory (by job_id)
- type: background_jobs_scheduling
- template_version: 1.0.0
- output_path: 10_app/jobs/JBS-01_Jobs_Inventory.md
- compliance_gate_id: TMP-05.PRIMARY.JBS
- upstream_dependencies: ["WFO-01", "WFO-04", "EVT-01", "ERR-05"]
- inputs_required: ["WFO-01", "WFO-04", "EVT-01", "ERR-05", "RELIA-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical registry of background jobs by stable `job_id`: what jobs exist, why they exist, how they're triggered, what they touch, and what reliability controls apply.

Inputs Required

- WFO-01: {{xref:WFO-01}} | OPTIONAL
- WFO-04: {{xref:WFO-04}} | OPTIONAL
- EVT-01: {{xref:EVT-01}} | OPTIONAL
- ERR-05: {{xref:ERR-05}} | OPTIONAL

- RELIA-01: {{xref:RELIA-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Jobs list (minimum 12 for non-trivial systems; justify if smaller)
- For each job:
    - job_id (stable)
    - name
    - purpose
    - type (scheduled/event-driven/api-triggered/manual)
    - trigger definition (cron/event/endpoint)
    - owner_service/boundary
    - inputs (high level)
    - outputs/side effects (events/writes)
    - data sensitivity level (PII)
    - idempotency requirement (yes/no + policy pointer)
    - concurrency posture (single/partitioned/parallel)
    - retry profile (ERR-05 profile id)
    - DLQ/quarantine behavior pointer (WFO-05)
    - observability requirements (metrics/log fields)
    - runbook pointer (OPS-06) | OPTIONAL

Optional Fields

- Priority class | OPTIONAL
- Notes | OPTIONAL

Rules

- job_id must be stable and never reused for different behavior.
- Jobs must declare retry profile and idempotency stance.
- High-PII jobs require explicit redaction rules.
- Scheduled jobs must declare env enablement and safety constraints.

Output Format

Jobs Inventory (canonical)

job_id	name	type	trigger	owner	inputs	outputs	pii	idempotent	concurrency	retry_profile	dlq_behavior	obs	notes
job_001	{{jobs[0].name}}	{{jobs[0].type}}	{{jobs[0].trigger}}	{{jobs[0].owner}}	{{jobs[0].inputs}}	{{jobs[0].outputs}}	{{jobs[0].pii}}	{{jobs[0].idem}}	{{jobs[0].conc}}	{{jobs[0].retry}}	{{jobs[0].dlq}}	{{jobs[0].obs}}	{{jobs[0].notes}}
job_002	{{jobs[1].name}}	{{jobs[1].type}}	{{jobs[1].trigger}}	{{jobs[1].owner}}	{{jobs[1].inputs}}	{{jobs[1].outputs}}	{{jobs[1].pii}}	{{jobs[1].idem}}	{{jobs[1].conc}}	{{jobs[1].retry}}	{{jobs[1].dlq}}	{{jobs[1].obs}}	{{jobs[1].notes}}

Coverage Checks (required)

- job_ids unique: {{coverage.unique_job_ids}}
- all jobs have retry_profile: {{coverage.retry_profiles_present}}
- all jobs have idempotency stance: {{coverage.idempotency_present}}

Cross-References

- Upstream: {{xref:WFO-01}} | OPTIONAL, {{xref:ERR-05}} | OPTIONAL
- Downstream: {{xref:JBS-02}}, {{xref:JBS-04}} | OPTIONAL, {{xref:JBS-06}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Inventory rows with trigger + owner + purpose.
- intermediate: Required. Add retry_profile + idempotency + concurrency.
- advanced: Required. Add DLQ behavior and observability fields.

Unknown Handling

- UNKNOWN_ALLOWED: `priority_class`, `notes`, `runbook_pointer`
- If any job lacks retry_profile or idempotency stance → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.JBS
- Pass conditions:

- required_fields_present == true
- jobs_count >= 12 (or justified)
- unique_job_ids == true
- retry_profiles_present == true
- idempotency_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

JBS-03

JBS-03 — Scheduling Rules (cron/event-driven, env enablement)

Header Block

- template_id: JBS-03
- title: Scheduling Rules (cron/event-driven, env enablement)
- type: background_jobs_scheduling
- template_version: 1.0.0
- output_path: 10_app/jobs/JBS-03_Scheduling_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.JBS
- upstream_dependencies: ["JBS-01", "WFO-04", "ENV-01"]
- inputs_required: ["JBS-01", "WFO-04", "ENV-01", "RELIA-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define deterministic scheduling rules for all jobs: cron policies, event-driven dispatch rules, environment enablement, timezone handling, and safety constraints to prevent runaway schedules.

Inputs Required

- JBS-01: {{xref:JBS-01}} | OPTIONAL
- WFO-04: {{xref:WFO-04}} | OPTIONAL
- ENV-01: {{xref:ENV-01}} | OPTIONAL

- RELIA-01: {{xref:RELIA-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Scheduling posture:
    - supported schedule types (cron, fixed-delay, event-driven)
    - timezone standard (UTC default or explicit)
- Environment enablement rules:
    - dev/stage/prod policy
    - default-on vs default-off policy
- Cron safety rules:
    - minimum interval allowed
    - approval requirements for high frequency
    - blackout windows (if any)
- Dispatch rules:
    - dedupe on dispatch policy
    - concurrency caps per job
- Per-job schedule bindings (minimum 10):
    - job_id
    - schedule type
    - schedule expression (cron or trigger)
    - timezone

    - env enablement
    - safety notes
- Verification checklist

Optional Fields

- Maintenance window policy | OPTIONAL
- Notes | OPTIONAL

Rules

- Schedules must be explicit and valid; no ambiguous natural language.
- All scheduled jobs must define timezone.
- High-frequency schedules require explicit approval and observability.
- Event-driven jobs must define dedupe keys for dispatch.

Output Format

1) Global Scheduling Rules (required)

- Supported schedule types: {{global.types}}
- Timezone standard: {{global.timezone}}
- Default enablement stance: {{global.default_enablement}}

2) Cron Safety Rules (required)

- Minimum interval: {{cron.min_interval}}
- High-frequency approval: {{cron.approval}}

- Blackout windows: {{cron.blackouts}} | OPTIONAL

3) Dispatch Rules (required)

- Dispatch dedupe policy: {{dispatch.dedupe}}
- Concurrency caps: {{dispatch.concurrency_caps}}
- Dispatch failure behavior: {{dispatch.failure_behavior}} | OPTIONAL

4) Job Schedule Bindings (canonical)

job_id	schedule_type	expression/trigger	timezone	envs	concurrency_cap	notes
{{bindings[0].job_id}}	{{bindings[0].type}}	{{bindings[0].expr}}	{{bindings[0].tz}}	{{bindings[0].envs}}	{{bindings[0].cap}}	{{bindings[0].notes}}
{{bindings[1].job_id}}	{{bindings[1].type}}	{{bindings[1].expr}}	{{bindings[1].tz}}	{{bindings[1].envs}}	{{bindings[1].cap}}	{{bindings[1].notes}}

5) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:ENV-01}} | OPTIONAL, {{xref:WFO-04}} | OPTIONAL
- Downstream: {{xref:JBS-04}}, {{xref:JBS-06}} | OPTIONAL, {{xref:ALRT-*}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Global scheduling rules + schedule bindings table.
- intermediate: Required. Add cron safety and dispatch dedupe policy.
- advanced: Required. Add approvals and maintenance/blackout policies.

Unknown Handling

- UNKNOWN_ALLOWED: `maintenance_window_policy`, `notes`, `dispatch_failure_behavior`, `blackout_windows`
- If any scheduled job lacks timezone or env enablement → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.JBS
- Pass conditions:
    - required_fields_present == true
    - bindings_present == true
    - timezone_defined_for_scheduled == true
    - env_enablement_defined == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

JBS-04

JBS-03 — Scheduling Rules (cron/event-driven, env enablement)

Header Block

- template_id: JBS-03
- title: Scheduling Rules (cron/event-driven, env enablement)
- type: background_jobs_scheduling
- template_version: 1.0.0
- output_path: 10_app/jobs/JBS-03_Scheduling_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.JBS
- upstream_dependencies: ["JBS-01", "WFO-04", "ENV-01"]
- inputs_required: ["JBS-01", "WFO-04", "ENV-01", "RELIA-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define deterministic scheduling rules for all jobs: cron policies, event-driven dispatch rules, environment enablement, timezone handling, and safety constraints to prevent runaway schedules.

Inputs Required

- JBS-01: {{xref:JBS-01}} | OPTIONAL
- WFO-04: {{xref:WFO-04}} | OPTIONAL
- ENV-01: {{xref:ENV-01}} | OPTIONAL

- RELIA-01: {{xref:RELIA-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Scheduling posture:
    - supported schedule types (cron, fixed-delay, event-driven)
    - timezone standard (UTC default or explicit)
- Environment enablement rules:
    - dev/stage/prod policy
    - default-on vs default-off policy
- Cron safety rules:
    - minimum interval allowed
    - approval requirements for high frequency
    - blackout windows (if any)
- Dispatch rules:
    - dedupe on dispatch policy
    - concurrency caps per job
- Per-job schedule bindings (minimum 10):
    - job_id
    - schedule type
    - schedule expression (cron or trigger)
    - timezone

    - env enablement
    - safety notes
- Verification checklist

Optional Fields

- Maintenance window policy | OPTIONAL
- Notes | OPTIONAL

Rules

- Schedules must be explicit and valid; no ambiguous natural language.
- All scheduled jobs must define timezone.
- High-frequency schedules require explicit approval and observability.
- Event-driven jobs must define dedupe keys for dispatch.

Output Format

1) Global Scheduling Rules (required)

- Supported schedule types: {{global.types}}
- Timezone standard: {{global.timezone}}
- Default enablement stance: {{global.default_enablement}}

2) Cron Safety Rules (required)

- Minimum interval: {{cron.min_interval}}
- High-frequency approval: {{cron.approval}}

- Blackout windows: {{cron.blackouts}} | OPTIONAL

3) Dispatch Rules (required)

- Dispatch dedupe policy: {{dispatch.dedupe}}
- Concurrency caps: {{dispatch.concurrency_caps}}
- Dispatch failure behavior: {{dispatch.failure_behavior}} | OPTIONAL

4) Job Schedule Bindings (canonical)

job_id	schedule_type	expression/trigger	timezone	envs	concurrency_cap	notes
{{bindings[0].job_id}}	{{bindings[0].type}}	{{bindings[0].expr}}	{{bindings[0].tz}}	{{bindings[0].envs}}	{{bindings[0].cap}}	{{bindings[0].notes}}
{{bindings[1].job_id}}	{{bindings[1].type}}	{{bindings[1].expr}}	{{bindings[1].tz}}	{{bindings[1].envs}}	{{bindings[1].cap}}	{{bindings[1].notes}}

5) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:ENV-01}} | OPTIONAL, {{xref:WFO-04}} | OPTIONAL
- Downstream: {{xref:JBS-04}}, {{xref:JBS-06}} | OPTIONAL, {{xref:ALRT-*}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Global scheduling rules + schedule bindings table.
- intermediate: Required. Add cron safety and dispatch dedupe policy.
- advanced: Required. Add approvals and maintenance/blackout policies.

Unknown Handling

- UNKNOWN_ALLOWED: `maintenance_window_policy`, `notes`, `dispatch_failure_behavior`, `blackout_windows`
- If any scheduled job lacks timezone or env enablement → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.JBS
- Pass conditions:
    - required_fields_present == true
    - bindings_present == true
    - timezone_defined_for_scheduled == true
    - env_enablement_defined == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

JBS-05

JBS-05 — Idempotency & Concurrency for Jobs

Header Block

- template_id: JBS-05
- title: Idempotency & Concurrency for Jobs
- type: background_jobs_scheduling
- template_version: 1.0.0
- output_path: 10_app/jobs/JBS-05_Idempotency_Concurrency_for_Jobs.md
- compliance_gate_id: TMP-05.PRIMARY.JBS
- upstream_dependencies: ["WFO-03", "JBS-01", "ERR-05"]
- inputs_required: ["WFO-03", "JBS-01", "ERR-05", "DATA-08", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define job-specific idempotency keys, concurrency limits, locking scopes, and dedupe rules so retries and parallel workers do not cause duplicate side effects or data corruption.

Inputs Required

- WFO-03: {{xref:WFO-03}} | OPTIONAL
- JBS-01: {{xref:JBS-01}} | OPTIONAL
- ERR-05: {{xref:ERR-05}} | OPTIONAL
- DATA-08: {{xref:DATA-08}} | OPTIONAL

- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Job idempotency key policy:
    - when required (all write jobs)
    - key derivation rules
    - TTL policy for keys
- Job concurrency posture:
    - default concurrency (workers)
    - per-job concurrency caps
    - per-entity/tenant partitioning rules
- Locking rules:
    - lock scope (entity_id, tenant_id, global)
    - lock timeout policy
- Dedupe rules:
    - duplicate detection window
    - handling on duplicate (skip/merge)
- Job mapping table (minimum 12 jobs):
    - job_id
    - idempotency key definition
    - partition key (if any)
    - concurrency cap

    - lock scope
    - dedupe window
    - duplicate behavior
- Verification checklist

Optional Fields

- Exactly-once disclaimer | OPTIONAL
- Notes | OPTIONAL

Rules

- No claims of exactly-once without proven semantics; default at-least-once with dedupe.
- Keys must be deterministic and stable for the job's logical unit of work.
- Partition keys must align with data isolation rules (tenant boundaries).
- Locking must be minimal and avoid global locks unless justified.

Output Format

1) Global Policy (required)

- Idempotency required for: {{policy.idem_required_for}}
- Key TTL policy: {{policy.key_ttl}}
- Default concurrency stance: {{policy.default_concurrency}}
- Lock timeout policy: {{policy.lock_timeout}}

2) Job Concurrency & Idempotency Map (canonical)

job_id	idem_key_def	key_ttl	partition_key	concurrency_cap	lock_scope	lock_timeout	dedupe_window	dup_behavior	notes
{{map[0].job_id}}	{{map[0].idem}}	{{map[0].ttl}}	{{map[0].partition}}	{{map[0].cap}}	{{map[0].lock}}	{{map[0].lock_timeout}}	{{map[0].dedupe}}	{{map[0].dup_behavior}}	{{map[0].notes}}
{{map[1].job_id}}	{{map[1].idem}}	{{map[1].ttl}}	{{map[1].partition}}	{{map[1].cap}}	{{map[1].lock}}	{{map[1].lock_timeout}}	{{map[1].dedupe}}	{{map[1].dup_behavior}}	{{map[1].notes}}

3) Dedupe Rules (required)

- Duplicate detection: {{dedupe.detect}}
- Duplicate handling: {{dedupe.handle}}
- Window defaults: {{dedupe.window_defaults}} | OPTIONAL

4) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:WFO-03}} | OPTIONAL, {{xref:ERR-05}} | OPTIONAL
- Downstream: {{xref:JBS-06}} | OPTIONAL, {{xref:QA-03}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Global policy + mapping table structure.
- intermediate: Required. Add partitioning and caps per job.
- advanced: Required. Add lock timeouts and dedupe behaviors.

Unknown Handling

- UNKNOWN_ALLOWED: `exactly_once_disclaimer`, `notes`
- If any write job lacks idem_key_def → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.JBS
- Pass conditions:
    - required_fields_present == true
    - map_present == true
    - idem_defined_for_write_jobs == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

JBS-06

JBS-05 — Idempotency & Concurrency for Jobs

Header Block

- template_id: JBS-05
- title: Idempotency & Concurrency for Jobs
- type: background_jobs_scheduling
- template_version: 1.0.0
- output_path: 10_app/jobs/JBS-05_Idempotency_Concurrency_for_Jobs.md
- compliance_gate_id: TMP-05.PRIMARY.JBS
- upstream_dependencies: ["WFO-03", "JBS-01", "ERR-05"]
- inputs_required: ["WFO-03", "JBS-01", "ERR-05", "DATA-08", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define job-specific idempotency keys, concurrency limits, locking scopes, and dedupe rules so retries and parallel workers do not cause duplicate side effects or data corruption.

Inputs Required

- WFO-03: {{xref:WFO-03}} | OPTIONAL
- JBS-01: {{xref:JBS-01}} | OPTIONAL
- ERR-05: {{xref:ERR-05}} | OPTIONAL
- DATA-08: {{xref:DATA-08}} | OPTIONAL

- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Job idempotency key policy:
    - when required (all write jobs)
    - key derivation rules
    - TTL policy for keys
- Job concurrency posture:
    - default concurrency (workers)
    - per-job concurrency caps
    - per-entity/tenant partitioning rules
- Locking rules:
    - lock scope (entity_id, tenant_id, global)
    - lock timeout policy
- Dedupe rules:
    - duplicate detection window
    - handling on duplicate (skip/merge)
- Job mapping table (minimum 12 jobs):
    - job_id
    - idempotency key definition
    - partition key (if any)
    - concurrency cap

    - lock scope
    - dedupe window
    - duplicate behavior
- Verification checklist

Optional Fields

- Exactly-once disclaimer | OPTIONAL
- Notes | OPTIONAL

Rules

- No claims of exactly-once without proven semantics; default at-least-once with dedupe.
- Keys must be deterministic and stable for the job's logical unit of work.
- Partition keys must align with data isolation rules (tenant boundaries).
- Locking must be minimal and avoid global locks unless justified.

Output Format

1) Global Policy (required)

- Idempotency required for: {{policy.idem_required_for}}
- Key TTL policy: {{policy.key_ttl}}
- Default concurrency stance: {{policy.default_concurrency}}
- Lock timeout policy: {{policy.lock_timeout}}

2) Job Concurrency & Idempotency Map (canonical)

job_id	idem_key_def	key_ttl	partition_key	concurrency_cap	lock_scope	lock_timeout	dedupe_window	dup_behavior	notes
{{map[0].job_id}}	{{map[0].idem}}	{{map[0].ttl}}	{{map[0].partition}}	{{map[0].cap}}	{{map[0].lock}}	{{map[0].lock_timeout}}	{{map[0].dedupe}}	{{map[0].dup_behavior}}	{{map[0].notes}}
{{map[1].job_id}}	{{map[1].idem}}	{{map[1].ttl}}	{{map[1].partition}}	{{map[1].cap}}	{{map[1].lock}}	{{map[1].lock_timeout}}	{{map[1].dedupe}}	{{map[1].dup_behavior}}	{{map[1].notes}}

3) Dedupe Rules (required)

- Duplicate detection: {{dedupe.detect}}
- Duplicate handling: {{dedupe.handle}}
- Window defaults: {{dedupe.window_defaults}} | OPTIONAL

4) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}} | OPTIONAL

Cross-References

- Upstream: {{xref:WFO-03}} | OPTIONAL, {{xref:ERR-05}} | OPTIONAL
- Downstream: {{xref:JBS-06}} | OPTIONAL, {{xref:QA-03}} | OPTIONAL
- Standards: {{standards.rules[STD-RELIABILITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Global policy + mapping table structure.
- intermediate: Required. Add partitioning and caps per job.
- advanced: Required. Add lock timeouts and dedupe behaviors.

Unknown Handling

- UNKNOWN_ALLOWED: `exactly_once_disclaimer`, `notes`
- If any write job lacks idem_key_def → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.JBS
- Pass conditions:
    - required_fields_present == true
    - map_present == true
    - idem_defined_for_write_jobs == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

Eventing & Webhooks (EVT)

Eventing & Webhooks (EVT)
EVT-01 Event Catalog (by event_id)
EVT-02 Event Schema Spec (payload, versioning, pii rules)
EVT-03 Producer/Consumer Map (who emits/consumes, contracts)
EVT-04 Delivery Semantics (ordering, retries, dedupe)
EVT-05 Webhook Producer Spec (outbound webhooks)
EVT-06 Webhook Consumer Spec (inbound webhooks)
EVT-07 Event Failure Handling (DLQ, replay, backfill)
EVT-08 Event Observability (lag, success rate, tracing)

EVT-01

EVT-01 — Event Catalog (by event_id)
Header Block
template_id: EVT-01
title: Event Catalog (by event_id)
type: eventing_catalog
template_version: 1.0.0
output_path: 10_app/eventing/EVT-01_Event_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.EVENTING
upstream_dependencies: []
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical catalog of all events emitted and/or consumed by the system, indexed by **event_id**. This document must be consistent with the Canonical Spec and must not invent events not present in upstream inputs. If events are missing, they must be marked UNKNOWN rather than invented.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Event catalog index (list of event entries)
event_id (stable identifier)
event_name (human-readable)
event_version (semver or integer policy)
event_description (one paragraph)
producer(s) (service/module/domain)
consumer(s) (service/module/domain)
delivery surface (internal bus / webhook / ws / other)
reliability class (best-effort / at-least-once / exactly-once | if supported)
ordering key / partition key (if applicable)
PII classification (none / low / medium / high)
schema_ref (pointer to EVT-02 schema spec)
triggering action(s) (what causes the event)
idempotency/dedupe guidance (consumer expectations)
retention (how long stored/replayable)
observability hooks (metrics names / trace attributes)

Optional Fields
Sample payload pointer | OPTIONAL
Backfill/replay support (yes/no + notes) | OPTIONAL
Dead-letter routing (dlq event_id) | OPTIONAL
Security sensitivity notes | OPTIONAL
Related feature IDs | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new event_ids. Use only: {{spec.events_by_id[event_*]}} as given.
Each event entry MUST be unique by event_id.
Each event entry MUST reference an EVT-02 schema via schema_ref (or be marked UNKNOWN).
PII classification MUST follow: {{standards.rules[STD-PII-CLASSIFICATION]}} | OPTIONAL
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not restate full payload details here; payload fields belong in EVT-02.
If UNKNOWN appears for producer/consumer, it must be flagged in “Open Questions.”

Output Format

1. Catalog Summary
   Total events: {{evt.catalog.total}}
   Domains covered: {{evt.catalog.domains}} | OPTIONAL
   Surfaces: {{evt.catalog.surfaces}} | OPTIONAL
   Versioning policy pointer: {{xref:EVT-02}} | OPTIONAL
2. Event Index (by event_id)
   For each event, include the following entry block:

Event
event_id: {{events[0].event_id}}
event_name: {{events[0].event_name}}
event_version: {{events[0].event_version}}
description: {{events[0].description}}
producer(s): {{events[0].producers}}
consumer(s): {{events[0].consumers}}
surface: {{events[0].surface}}
reliability: {{events[0].reliability}}
ordering_key: {{events[0].ordering_key}} | OPTIONAL
partition_key: {{events[0].partition_key}} | OPTIONAL
pii_class: {{events[0].pii_class}}
schema_ref: {{events[0].schema_ref}} (expected: {{xref:EVT-02}})
triggering_actions: {{events[0].triggering_actions}}
dedupe_guidance: {{events[0].dedupe_guidance}}

retention: {{events[0].retention}}
observability:
metrics: {{events[0].observability.metrics}}
trace_attrs: {{events[0].observability.trace_attrs}} | OPTIONAL
sample_payload_ref: {{events[0].sample_payload_ref}} | OPTIONAL
dlq_route: {{events[0].dlq_route}} | OPTIONAL
replay_support: {{events[0].replay_support}} | OPTIONAL
related_feature_ids: {{events[0].related_feature_ids}} | OPTIONAL
open_questions:
{{events[0].open_questions[0]}} | OPTIONAL

(Repeat the “Event” entry block for each event_id in the system.)

3. References
   Event schema specs: {{xref:EVT-02}}
   Producer/consumer map: {{xref:EVT-03}} | OPTIONAL
   Delivery semantics: {{xref:EVT-04}} | OPTIONAL
   Webhook specs: {{xref:EVT-05}}, {{xref:EVT-06}} | OPTIONAL
   Failure handling: {{xref:EVT-07}} | OPTIONAL
   Observability: {{xref:EVT-08}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:DOMAIN_MAP}} | OPTIONAL, {{xref:STANDARDS_INDEX}} | OPTIONAL
Downstream: {{xref:EVT-02}}, {{xref:EVT-03}}, {{xref:EVT-04}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-CLASSIFICATION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Fill required fields with UNKNOWN where needed; do not invent event_ids.
intermediate: Required. Populate producers/consumers/surfaces from inputs; ensure schema_ref points to EVT-02.
advanced: Required. Add ordering/partition keys, dedupe and retention guidance, and observability hooks.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, ordering_key, partition_key, sample_payload_ref, dlq_route, replay_support, related_feature_ids, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If event_id list is UNKNOWN → block Completeness Gate.
If schema_ref is UNKNOWN for any event → allow only if flagged in Open Questions.

Completeness Gate
Gate ID: TMP-05.PRIMARY.EVENTING
Pass conditions:
required_fields_present == true
all event_ids are unique (no duplicates)
no new event_ids introduced (only references to existing IDs)
placeholder_resolution == true
no_unapproved_unknowns == true

EVT-02

EVT-02 — Event Schema Spec (payload, versioning, pii rules)
Header Block
template_id: EVT-02
title: Event Schema Spec (payload, versioning, pii rules)
type: eventing_schema_spec
template_version: 1.0.0
output_path: 10_app/eventing/EVT-02_Event_Schema_Spec.md
compliance_gate_id: TMP-05.PRIMARY.EVENTING
upstream_dependencies: ["EVT-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "EVT-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical schema specification format for each event payload, including: required/optional fields, typing, versioning rules, backward/forward compatibility, and PII handling requirements. This template must be consistent with the Event Catalog and must not invent event_ids or payload fields not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Schema identifier (schema_id) and binding to event_id
Event versioning policy (what increments mean)
Envelope fields (required metadata around payload)
Payload schema (fields list with types + requiredness)
PII classification per field (or pointer to policy)
Redaction rules (logs/traces)
Compatibility rules (backward/forward)
Validation rules (required fields, enums, formats)
Serialization format (json/avro/proto/other)
Schema storage/registry location (where schema lives)
Change control rules (how schema changes are approved/published)
Test vectors (minimum: one valid + one invalid example pointer)

Optional Fields
Field-level constraints (min/max/pattern) | OPTIONAL
Deprecation policy | OPTIONAL

Localization fields policy | OPTIONAL
Compression/encryption policy | OPTIONAL
Sample payloads (inline) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new event_ids; use only: {{evt.catalog.events_by_id[event_*]}} as given.
Each EVT-02 spec MUST map to exactly one event_id and declare schema_id.
Envelope fields MUST be consistent across all events unless explicitly allowed otherwise.
PII classification MUST follow: {{standards.rules[STD-PII-CLASSIFICATION]}} | OPTIONAL
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not restate producer/consumer mapping here; that belongs in EVT-03.
If UNKNOWN appears in envelope fields or versioning policy, it must be flagged in Open Questions.

Output Format

1. Schema Identity
   event_id: {{schema.event_id}}
   schema_id: {{schema.schema_id}}
   event_version: {{schema.event_version}}
   serialization: {{schema.serialization}} (json/avro/proto/UNKNOWN)
   registry_location: {{schema.registry_location}}
   owner: {{schema.owner}} | OPTIONAL
2. Versioning Policy
   Versioning model: {{versioning.model}} (semver/integer/UNKNOWN)
   Compatible changes (allowed):
   {{versioning.compatible[0]}}
   {{versioning.compatible[1]}}
   Breaking changes (require major bump):
   {{versioning.breaking[0]}}
   {{versioning.breaking[1]}}
   Deprecation window: {{versioning.deprecation_window}} | OPTIONAL
3. Event Envelope (Required Metadata)
   The following envelope fields MUST exist for all events (unless overridden):
   event_id: {{envelope.event_id}}
   event_version: {{envelope.event_version}}
   event_ts: {{envelope.event_ts}}
   producer: {{envelope.producer}}
   trace_id: {{envelope.trace_id}} | OPTIONAL
   request_id: {{envelope.request_id}} | OPTIONAL
   idempotency_key: {{envelope.idempotency_key}} | OPTIONAL

tenant: {{envelope.tenant}} | OPTIONAL
schema_ref: {{envelope.schema_ref}} | OPTIONAL

4. Payload Schema
   List payload fields (do not include envelope fields here):
   {{payload.fields[0].name}} — {{payload.fields[0].type}} — required: {{payload.fields[0].required}} — pii: {{payload.fields[0].pii_class}}
   {{payload.fields[1].name}} — {{payload.fields[1].type}} — required: {{payload.fields[1].required}} — pii: {{payload.fields[1].pii_class}}
   {{payload.fields[2].name}} — {{payload.fields[2].type}} — required: {{payload.fields[2].required}} — pii: {{payload.fields[2].pii_class}}
5. Validation Rules
   Required fields: {{validation.required_fields}}
   Enums: {{validation.enums}} | OPTIONAL
   Formats: {{validation.formats}} | OPTIONAL
   Constraints: {{validation.constraints}} | OPTIONAL
6. PII & Redaction
   PII policy pointer: {{standards.rules[STD-PII-CLASSIFICATION]}} | OPTIONAL
   Field redaction rules (logs): {{pii.redaction.logs}}
   Field redaction rules (traces): {{pii.redaction.traces}} | OPTIONAL
   Storage encryption requirement: {{pii.encryption_required}} | OPTIONAL
7. Compatibility & Migration
   Backward compatibility rule: {{compat.backward}}
   Forward compatibility rule: {{compat.forward}} | OPTIONAL
   Consumer migration guidance: {{compat.migration_guidance}} | OPTIONAL
8. Change Control
   Who can change schemas: {{change_control.owners}}
   Approval workflow: {{change_control.approvals}} | OPTIONAL
   Publishing steps: {{change_control.publish_steps}} | OPTIONAL
   Rollout policy: {{change_control.rollout_policy}} | OPTIONAL
9. Test Vectors
   Valid example pointer: {{tests.valid_example_ref}}
   Invalid example pointer: {{tests.invalid_example_ref}}
   Contract tests required: {{tests.contract_tests_required}} | OPTIONAL
10. References
    Event catalog entry: {{xref:EVT-01}}
    Producer/consumer map: {{xref:EVT-03}} | OPTIONAL
    Delivery semantics: {{xref:EVT-04}} | OPTIONAL
    Failure handling: {{xref:EVT-07}} | OPTIONAL

Cross-References
Upstream: {{xref:EVT-01}}, {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:STANDARDS_INDEX}} | OPTIONAL
Downstream: {{xref:EVT-03}}, {{xref:EVT-04}}, {{xref:EVT-07}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-PII-CLASSIFICATION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where inputs do not define fields; do not invent event_ids or payload fields.
intermediate: Required. Populate envelope + payload field lists from inputs; define validation rules.
advanced: Required. Add compatibility detail, migration notes, and test vectors with clear pointers.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, owner, trace_id, request_id, idempotency_key, tenant, schema_ref, constraints, enums, formats, deprecation_window, encryption_required, migration_guidance, approvals, publish_steps, rollout_policy, contract_tests_required
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If payload schema is UNKNOWN → block Completeness Gate.
If PII classification policy is UNKNOWN → block Completeness Gate unless explicitly allowed by standards.

Completeness Gate
Gate ID: TMP-05.PRIMARY.EVENTING
Pass conditions:
required_fields_present == true
schema_id maps to an existing event_id from EVT-01 (no new event_ids introduced)
envelope_fields_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

EVT-03

EVT-03 — Producer/Consumer Map (who emits/consumes, contracts)
Header Block
template_id: EVT-03
title: Producer/Consumer Map (who emits/consumes, contracts)
type: eventing_producer_consumer_map
template_version: 1.0.0
output_path: 10_app/eventing/EVT-03_Producer_Consumer_Map.md
compliance_gate_id: TMP-05.PRIMARY.EVENTING
upstream_dependencies: ["EVT-01", "EVT-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "EVT-01", "EVT-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical mapping of **who emits** each event and **who consumes** it, including the binding to schema contracts, delivery surface, and consumer obligations (idempotency, ordering assumptions, error handling). This template must be consistent with EVT-01/EVT-02 and must not invent event_ids, producer IDs, or consumer IDs not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}}
EVT-02 Event Schema Specs: {{evt.schemas}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Producer/consumer map index (list of mappings)
event_id (must exist in EVT-01)
schema_ref (must exist in EVT-02)
producer_id(s) (stable identifier)
consumer_id(s) (stable identifier)
surface (bus/webhook/other)
producer contract obligations (when emitted, guarantees)
consumer contract obligations (idempotency, ordering, ack)
delivery semantics assumptions per mapping (at-least-once, etc.)
partition/ordering assumptions per mapping (if applicable)
failure handling binding (what happens on consumer failure)
ownership (who owns producer + who owns consumer)
version upgrade policy (how consumers migrate)

Optional Fields
Consumer priority tier | OPTIONAL
Backfill/replay consumer behavior | OPTIONAL
Consumer filtering rules | OPTIONAL
Webhook endpoints (if surface includes webhook) | OPTIONAL
SLO expectations | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new event_ids; use only those in {{xref:EVT-01}}.
Each mapping MUST reference a valid schema_ref from {{xref:EVT-02}}.
Every consumer MUST declare idempotency behavior (dedupe key rule or UNKNOWN).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not restate payload fields; payload details belong to EVT-02.
If UNKNOWN appears for producer_id or consumer_id, it must be flagged in Open Questions.

Output Format

1. Map Summary
   Total events mapped: {{map.total_events}}
   Total producer nodes: {{map.total_producers}} | OPTIONAL
   Total consumer nodes: {{map.total_consumers}} | OPTIONAL
   Surfaces in use: {{map.surfaces}} | OPTIONAL
   Contract versioning pointer: {{xref:EVT-02}} | OPTIONAL
2. Producer/Consumer Mapping (by event_id)
   For each event_id, include the following mapping block:

Mapping
event_id: {{mappings[0].event_id}}
schema_ref: {{mappings[0].schema_ref}}
surface: {{mappings[0].surface}}

producer(s):

- producer_id: {{mappings[0].producers[0].producer_id}}
  owner: {{mappings[0].producers[0].owner}} | OPTIONAL
  emits_when: {{mappings[0].producers[0].emits_when}}
  guarantees: {{mappings[0].producers[0].guarantees}}
  partition_key: {{mappings[0].producers[0].partition_key}} | OPTIONAL
  ordering_key: {{mappings[0].producers[0].ordering_key}} | OPTIONAL

consumer(s):

- consumer_id: {{mappings[0].consumers[0].consumer_id}}
  owner: {{mappings[0].consumers[0].owner}} | OPTIONAL
  consumes_for: {{mappings[0].consumers[0].consumes_for}}
  obligations:
  idempotency: {{mappings[0].consumers[0].idempotency}}
  ordering: {{mappings[0].consumers[0].ordering}} | OPTIONAL
  ack_mode: {{mappings[0].consumers[0].ack_mode}} | OPTIONAL
  retry_behavior: {{mappings[0].consumers[0].retry_behavior}} | OPTIONAL
  failure_handling_ref: {{mappings[0].consumers[0].failure_handling_ref}} (expected: {{xref:EVT-07}}) | OPTIONAL
  replay_behavior: {{mappings[0].consumers[0].replay_behavior}} | OPTIONAL
  upgrade_policy: {{mappings[0].consumers[0].upgrade_policy}}

delivery_semantics: {{mappings[0].delivery_semantics}}
partitioning_assumptions: {{mappings[0].partitioning_assumptions}} | OPTIONAL
slo: {{mappings[0].slo}} | OPTIONAL
open_questions:
{{mappings[0].open_questions[0]}} | OPTIONAL

(Repeat the “Mapping” block for each event_id.)

3. References
   Event catalog: {{xref:EVT-01}}
   Schema specs: {{xref:EVT-02}}
   Delivery semantics: {{xref:EVT-04}} | OPTIONAL
   Failure handling: {{xref:EVT-07}} | OPTIONAL
   Observability: {{xref:EVT-08}} | OPTIONAL

Cross-References
Upstream: {{xref:EVT-01}}, {{xref:EVT-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:EVT-04}}, {{xref:EVT-07}}, {{xref:EVT-08}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for producer/consumer IDs if missing; do not invent.
intermediate: Required. Bind each mapping to schema_ref; define consumer idempotency obligations.
advanced: Required. Add delivery semantics and upgrade policy guidance per consumer.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, owner, partition_key, ordering_key, ack_mode, retry_behavior, failure_handling_ref, replay_behavior, consumer_priority, filtering_rules, webhook_endpoints, slo, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If event_id is UNKNOWN → block Completeness Gate.
If schema_ref is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.EVENTING
Pass conditions:
required_fields_present == true
all event_ids exist in EVT-01 (no new event_ids introduced)
all schema_refs exist in EVT-02
placeholder_resolution == true
no_unapproved_unknowns == true

EVT-04

EVT-04 — Delivery Semantics (ordering, retries, dedupe)
Header Block
template_id: EVT-04
title: Delivery Semantics (ordering, retries, dedupe)
type: eventing_delivery_semantics
template_version: 1.0.0
output_path: 10_app/eventing/EVT-04_Delivery_Semantics.md
compliance_gate_id: TMP-05.PRIMARY.EVENTING
upstream_dependencies: ["EVT-01", "EVT-02", "EVT-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "EVT-01", "EVT-02", "EVT-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical delivery semantics for the eventing system, including ordering guarantees, retry behavior, deduplication expectations, acknowledgement modes, and consumer processing obligations. This template must be consistent with EVT-01/02/03 and must not invent semantics not supported by upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}}
EVT-02 Event Schema Specs: {{evt.schemas}}
EVT-03 Producer/Consumer Map: {{evt.map}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Delivery model statement (at-most-once / at-least-once / exactly-once / UNKNOWN)
Ack model (producer ack, broker ack, consumer ack)
Retry policy (producer side, broker side, consumer side)
Backoff strategy (fixed/exponential/jitter/UNKNOWN)
Max retry attempts / retry window
Deduplication model (where dedupe happens + key)
Idempotency requirements for consumers
Ordering guarantees (none / per partition / global / UNKNOWN)
Partitioning strategy (partition key rules)
Re-delivery behavior (when duplicates can occur)
Poison message policy binding (DLQ rules pointer)
Replay/backfill policy binding (if supported)
Failure classification (transient vs permanent) and handling
Observability requirements for delivery (lag, retry count, dupes)

Optional Fields
Transactional outbox pattern rules | OPTIONAL
Exactly-once constraints/requirements | OPTIONAL
Consumer concurrency limits | OPTIONAL
Cross-region delivery notes | OPTIONAL
Schema evolution interaction notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Semantics defined here MUST be referenced by EVT-03 mappings (or marked UNKNOWN).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Deduplication MUST be deterministic (key-based) if claimed.
If ordering is claimed, partition key rules MUST be specified (or UNKNOWN flagged).
If DLQ is supported, failure routing MUST reference EVT-07 (or be UNKNOWN flagged).

Output Format

1. Delivery Model
   delivery_mode: {{delivery.mode}} (at_most_once / at_least_once / exactly_once / UNKNOWN)
   notes: {{delivery.notes}} | OPTIONAL
2. Acknowledgement Model
   producer_ack: {{ack.producer_ack}}
   broker_ack: {{ack.broker_ack}} | OPTIONAL
   consumer_ack: {{ack.consumer_ack}}
   ack_timeout_ms: {{ack.timeout_ms}} | OPTIONAL
3. Retry Policy
   producer_retry: {{retry.producer}} | OPTIONAL
   broker_retry: {{retry.broker}} | OPTIONAL
   consumer_retry: {{retry.consumer}}
   backoff_strategy: {{retry.backoff_strategy}}
   max_attempts: {{retry.max_attempts}}
   retry_window: {{retry.window}} | OPTIONAL
   retryable_error_classes: {{retry.retryable_error_classes}} | OPTIONAL
   non_retryable_error_classes: {{retry.non_retryable_error_classes}} | OPTIONAL
4. Deduplication & Idempotency
   dedupe_location: {{dedupe.location}} (consumer / broker / producer / UNKNOWN)
   dedupe_key: {{dedupe.key}} (event_id + key fields / idempotency_key / UNKNOWN)
   dedupe_window: {{dedupe.window}} | OPTIONAL
   consumer_idempotency_required: {{dedupe.consumer_idempotency_required}}
   idempotency_guidance: {{dedupe.guidance}}
5. Ordering & Partitioning
   ordering_guarantee: {{ordering.guarantee}} (none / per_partition / global / UNKNOWN)

partition_key_rule: {{ordering.partition_key_rule}}
tie_break_rule: {{ordering.tie_break_rule}} | OPTIONAL
out_of_order_handling: {{ordering.out_of_order_handling}} | OPTIONAL

6. Redelivery & Duplicate Expectations
   when_duplicates_occur: {{redelivery.when_duplicates_occur}}
   consumer_duplicate_handling: {{redelivery.consumer_handling}}
   poison_message_definition: {{redelivery.poison_definition}} | OPTIONAL
7. Failure Handling Bindings
   dlq_supported: {{failure.dlq_supported}}
   dlq_policy_ref: {{failure.dlq_policy_ref}} (expected: {{xref:EVT-07}}) | OPTIONAL
   replay_supported: {{failure.replay_supported}} | OPTIONAL
   replay_policy_ref: {{failure.replay_policy_ref}} | OPTIONAL
   failure_classification:
   transient: {{failure.classification.transient}}
   permanent: {{failure.classification.permanent}}
8. Observability Requirements
   Lag metrics: {{obs.lag_metrics}}
   Retry metrics: {{obs.retry_metrics}}
   DLQ metrics: {{obs.dlq_metrics}} | OPTIONAL
   Duplicate metrics: {{obs.duplicate_metrics}} | OPTIONAL
   Tracing requirements: {{obs.tracing}} | OPTIONAL
   Alert thresholds: {{obs.alerts}} | OPTIONAL
9. References
   Event catalog: {{xref:EVT-01}}
   Schema specs: {{xref:EVT-02}}
   Producer/consumer map: {{xref:EVT-03}}
   Failure handling: {{xref:EVT-07}} | OPTIONAL
   Observability: {{xref:EVT-08}} | OPTIONAL

Cross-References
Upstream: {{xref:EVT-01}}, {{xref:EVT-02}}, {{xref:EVT-03}}
Downstream: {{xref:EVT-07}}, {{xref:EVT-08}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where semantics are not defined by inputs; do not invent guarantees.
intermediate: Required. Define retry/dedupe/order rules that match the platform's actual capabilities.
advanced: Required. Add crisp failure classification and observability/alerting requirements.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, producer_retry, broker_retry, retry_window, retryable_error_classes, non_retryable_error_classes, dedupe_window,

tie_break_rule, out_of_order_handling, poison_message_definition, dlq_policy_ref, replay_supported, replay_policy_ref, tracing, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If delivery_mode is UNKNOWN → allow only if flagged as Open Question.
If ordering_guarantee is claimed but partition_key_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.EVENTING
Pass conditions:
required_fields_present == true
semantics_consistent_with_evt03 == true
placeholder_resolution == true
no_unapproved_unknowns == true

EVT-05

EVT-05 — Webhook Producer Spec (outbound webhooks)
Header Block
template_id: EVT-05
title: Webhook Producer Spec (outbound webhooks)
type: eventing_webhook_producer_spec
template_version: 1.0.0
output_path: 10_app/eventing/EVT-05_Webhook_Producer_Spec.md
compliance_gate_id: TMP-05.PRIMARY.EVENTING
upstream_dependencies: ["EVT-01", "EVT-02", "EVT-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "EVT-01", "EVT-02", "EVT-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical specification format for outbound webhooks produced by the system, including subscription model, delivery/auth/signing, retries, dedupe, rate limiting, payload binding to EVT schemas, and observability. This template must be consistent with the Event Catalog/Schema specs and must not invent webhook event_ids or capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}}
EVT-02 Event Schema Specs: {{evt.schemas}}
EVT-04 Delivery Semantics: {{evt.delivery_semantics}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Webhook surface scope (which events can be delivered as webhooks)
Subscription model (static endpoints vs per-tenant subscriptions)
Webhook endpoint configuration fields (url, secrets, headers)
Auth/signing model (HMAC/signature headers/timestamps)
Delivery method (POST/PUT) and content type
Payload format (envelope + schema binding to EVT-02)
Delivery semantics (retries, backoff, max attempts, timeouts)
Deduplication/idempotency guidance for receivers
Ordering guarantees (if any)
Failure handling (disable rules, DLQ, dead endpoint behavior)
Rate limiting / abuse controls for outbound delivery
Security constraints (SSRF protections, allowlists, TLS requirements)
Observability requirements (delivery success rate, latency, retry counts)

Optional Fields
Webhook event filtering rules | OPTIONAL
Batching policy | OPTIONAL
Compression policy | OPTIONAL
Tenant-specific signing keys | OPTIONAL
Custom headers policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new event_ids; use only those in {{xref:EVT-01}}.
Every webhook payload MUST reference a schema_ref from {{xref:EVT-02}} (or be UNKNOWN).
Signing/auth model MUST be explicit; if unknown, mark UNKNOWN and flag.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Outbound delivery MUST follow delivery semantics in {{xref:EVT-04}} unless explicitly overridden.
If overrides exist, they must be documented and justified (or marked UNKNOWN).

Output Format

1. Webhook Scope
   enabled: {{webhooks.enabled}}
   supported_event_ids: {{webhooks.supported_event_ids}} (must come from {{xref:EVT-01}})
   subscription_model: {{webhooks.subscription_model}} (static/per_tenant/UNKNOWN)
   notes: {{webhooks.notes}} | OPTIONAL
2. Subscription & Endpoint Configuration
   Config fields:
   url: {{webhook.config.url}}
   secret_ref: {{webhook.config.secret_ref}} | OPTIONAL
   custom_headers: {{webhook.config.custom_headers}} | OPTIONAL
   event_filters: {{webhook.config.event_filters}} | OPTIONAL
   enabled_flag: {{webhook.config.enabled_flag}} | OPTIONAL
   validation_rules: {{webhook.config.validation_rules}}
3. Authentication / Signing
   signing_enabled: {{signing.enabled}}
   signing_type: {{signing.type}} (hmac/rs256/UNKNOWN)
   signature_header: {{signing.signature_header}} | OPTIONAL
   timestamp_header: {{signing.timestamp_header}} | OPTIONAL
   tolerance_seconds: {{signing.tolerance_seconds}} | OPTIONAL
   replay_protection: {{signing.replay_protection}} | OPTIONAL
4. Delivery Request Format
   http_method: {{delivery.http_method}} (POST/UNKNOWN)

content_type: {{delivery.content_type}} (application/json/UNKNOWN)
timeout_ms: {{delivery.timeout_ms}}
tls_required: {{delivery.tls_required}}
redirect_policy: {{delivery.redirect_policy}} | OPTIONAL

5. Payload Binding (to EVT-02)
   envelope_fields: {{payload.envelope_fields}}
   schema_ref: {{payload.schema_ref}} (expected: {{xref:EVT-02}})
   payload_ref: {{payload.payload_ref}} | OPTIONAL
   pii_policy: {{payload.pii_policy}} | OPTIONAL
6. Delivery Semantics (Retries/Dedupe/Ordering)
   retry_policy_ref: {{delivery.retry_policy_ref}} (expected: {{xref:EVT-04}})
   backoff_strategy: {{delivery.backoff_strategy}}
   max_attempts: {{delivery.max_attempts}}
   dedupe_key: {{delivery.dedupe_key}} | OPTIONAL
   ordering_guarantee: {{delivery.ordering_guarantee}} | OPTIONAL
7. Failure Handling
   disable_on_fail: {{failure.disable_on_fail}}
   disable_threshold: {{failure.disable_threshold}} | OPTIONAL
   dead_endpoint_behavior: {{failure.dead_endpoint_behavior}}
   dlq_supported: {{failure.dlq_supported}} | OPTIONAL
   dlq_ref: {{failure.dlq_ref}} | OPTIONAL
8. Rate Limits & Abuse Controls
   outbound_ratelimit_policy: {{abuse.ratelimit_policy}} | OPTIONAL
   per_destination_limits: {{abuse.per_destination_limits}} | OPTIONAL
   spike_protection: {{abuse.spike_protection}} | OPTIONAL
9. Security Constraints
   ssrf_protections: {{security.ssrf_protections}}
   allowlist_policy: {{security.allowlist_policy}} | OPTIONAL
   tls_min_version: {{security.tls_min_version}} | OPTIONAL
   secret_storage_policy: {{security.secret_storage_policy}} | OPTIONAL
10. Observability Requirements
    metrics:
    delivery_success_rate: {{obs.metrics.delivery_success_rate}}
    delivery_latency: {{obs.metrics.delivery_latency}}
    retry_count: {{obs.metrics.retry_count}}
    failure_count: {{obs.metrics.failure_count}}
    logs_required_fields: {{obs.logs_required_fields}}
    trace_attrs: {{obs.trace_attrs}} | OPTIONAL
    alerts: {{obs.alerts}} | OPTIONAL
11. References
    Event catalog: {{xref:EVT-01}}
    Schema specs: {{xref:EVT-02}}
    Delivery semantics: {{xref:EVT-04}}
    Failure handling: {{xref:EVT-07}} | OPTIONAL

Observability: {{xref:EVT-08}} | OPTIONAL
Rate limit policy: {{xref:RLIM-01}} | OPTIONAL

Cross-References
Upstream: {{xref:EVT-01}}, {{xref:EVT-02}}, {{xref:EVT-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:EVT-06}} | OPTIONAL, {{xref:EVT-07}} | OPTIONAL, {{xref:EVT-08}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where signing/delivery details are not specified by inputs; do not invent event_ids.
intermediate: Required. Bind payloads to EVT-02 and define retries/backoff/timeouts.
advanced: Required. Add SSRF protections, allowlist policy, and full observability/alerting.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, secret_ref, custom_headers, event_filters, enabled_flag, signature_header, timestamp_header, tolerance_seconds, replay_protection, redirect_policy, payload_ref, pii_policy, dedupe_key, ordering_guarantee, disable_threshold, dlq_supported, dlq_ref, ratelimit_policy, per_destination_limits, spike_protection, allowlist_policy, tls_min_version, secret_storage_policy, trace_attrs, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If signing_enabled is UNKNOWN → flag in Open Questions.
If tls_required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.EVENTING
Pass conditions:
required_fields_present == true
all webhook event_ids exist in EVT-01 (no new IDs introduced)
payload_schema_ref_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

EVT-06

EVT-06 — Webhook Consumer Spec (inbound webhooks)
Header Block
template_id: EVT-06
title: Webhook Consumer Spec (inbound webhooks)
type: eventing_webhook_consumer_spec
template_version: 1.0.0
output_path: 10_app/eventing/EVT-06_Webhook_Consumer_Spec.md
compliance_gate_id: TMP-05.PRIMARY.EVENTING
upstream_dependencies: ["EVT-01", "EVT-02", "EVT-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "EVT-01", "EVT-02", "EVT-04", "API-01", "API-02", "RLIM-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical specification format for inbound webhooks consumed by the system, including endpoint surface, authentication/verification, validation, idempotency/deduplication, replay protection, rate limiting/abuse controls, mapping into internal events or jobs, and observability. This template must be consistent with API endpoint specs and eventing delivery semantics and must not invent inbound sources or contracts not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}} | OPTIONAL
EVT-02 Event Schema Specs: {{evt.schemas}} | OPTIONAL
EVT-04 Delivery Semantics: {{evt.delivery_semantics}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-02 Endpoint Specs: {{api.endpoint_specs}}
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Inbound webhook sources catalog (by source_id)
Inbound endpoint mapping (source_id → endpoint_id)
Authentication/verification model (signature, secrets, IP allowlist, mTLS, etc.)
Replay protection model (timestamp tolerance, nonce, signature binding)
Payload validation rules (schema checks, required headers)
Normalization/mapping rules (payload → internal schema/event/job)
Idempotency/deduplication rules (dedupe key + window)
Rate limiting / abuse controls (per source, per tenant, global)
Failure response policy (what HTTP response codes are returned)
Retry semantics expectations (what the sender will do)

Security constraints (SSRF not applicable; but request size, header allowlist, parser hardening)
Observability requirements (accept rate, reject reasons, latency, dedupe hits)

Optional Fields
Batching support | OPTIONAL
Encryption requirements | OPTIONAL
Quarantine mode (store + manual review) | OPTIONAL
Per-source feature flag enablement | OPTIONAL
Dead-letter storage rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent inbound sources; source_id MUST come from inputs or be UNKNOWN.
Inbound webhook endpoints MUST be defined in API-01 and specified in API-02.
Verification MUST be explicit; if unknown, mark UNKNOWN and flag.
Idempotency MUST be deterministic if claimed (key-based).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Failure responses MUST follow: {{xref:API-03}} | OPTIONAL
AuthZ for any side-effects MUST follow: {{xref:API-04}} | OPTIONAL

Output Format

1. Inbound Sources Summary
   total_sources: {{inbound.total_sources}}
   sources: {{inbound.sources}} | OPTIONAL
   notes: {{inbound.notes}} | OPTIONAL
2. Source Registry (by source_id)
   Source
   source_id: {{sources[0].source_id}}
   name: {{sources[0].name}}
   owner: {{sources[0].owner}} | OPTIONAL
   expected_sender: {{sources[0].expected_sender}} | OPTIONAL
   enabled: {{sources[0].enabled}}
   feature_flag: {{sources[0].feature_flag}} | OPTIONAL
3. Endpoint Mapping
   Each inbound source MUST map to one or more API endpoints:
   {{mapping[0].source_id}} → endpoint_id: {{mapping[0].endpoint_id}}
   path: {{mapping[0].path}} | OPTIONAL
   method: {{mapping[0].method}} | OPTIONAL
4. Authentication / Verification
   verification_required: {{verify.required}}
   verification_type: {{verify.type}} (hmac/rs256/mTLS/ip_allowlist/UNKNOWN)
   secret_ref: {{verify.secret_ref}} | OPTIONAL

signature_header: {{verify.signature_header}} | OPTIONAL
timestamp_header: {{verify.timestamp_header}} | OPTIONAL
tolerance_seconds: {{verify.tolerance_seconds}} | OPTIONAL
ip_allowlist: {{verify.ip_allowlist}} | OPTIONAL
mtls_requirements: {{verify.mtls_requirements}} | OPTIONAL

5. Replay Protection
   replay_protection_enabled: {{replay.enabled}}
   nonce_required: {{replay.nonce_required}} | OPTIONAL
   nonce_store: {{replay.nonce_store}} | OPTIONAL
   binding_rule: {{replay.binding_rule}} (what inputs are signed)
   replay_window: {{replay.window}} | OPTIONAL
6. Payload Validation
   content_type_required: {{validation.content_type_required}}
   max_body_bytes: {{validation.max_body_bytes}}
   required_headers: {{validation.required_headers}} | OPTIONAL
   schema_validation: {{validation.schema_validation}} (schema_ref/UNKNOWN)
   invalid_payload_behavior: {{validation.invalid_payload_behavior}}
7. Normalization / Mapping
   internal_target: {{mapping_rules.internal_target}} (event/job/command/UNKNOWN)
   target_ref: {{mapping_rules.target_ref}} (e.g., {{xref:EVT-01}}/{{xref:JBS-01}}/{{xref:API-06}}) | OPTIONAL
   field_mapping_rules: {{mapping_rules.field_mapping_rules}} | OPTIONAL
   pii_handling: {{mapping_rules.pii_handling}} | OPTIONAL
8. Idempotency / Deduplication
   dedupe_enabled: {{dedupe.enabled}}
   dedupe_key_rule: {{dedupe.key_rule}}
   dedupe_window: {{dedupe.window}} | OPTIONAL
   on_duplicate: {{dedupe.on_duplicate}} (ignore/ack/overwrite/UNKNOWN)
9. Rate Limits & Abuse Controls
   ratelimit_policy_ref: {{abuse.ratelimit_policy_ref}} (expected: {{xref:RLIM-01}}) | OPTIONAL
   per_source_limits: {{abuse.per_source_limits}} | OPTIONAL
   global_limits: {{abuse.global_limits}} | OPTIONAL
   abuse_signals: {{abuse.signals}} | OPTIONAL
   enforcement_actions: {{abuse.enforcement_actions}} | OPTIONAL
10. Failure Response Policy
    success_response_code: {{responses.success_code}}
    invalid_signature_code: {{responses.invalid_signature_code}}
    invalid_payload_code: {{responses.invalid_payload_code}}
    rate_limited_code: {{responses.rate_limited_code}}
    server_error_code: {{responses.server_error_code}}
    error_body_policy: {{responses.error_body_policy}} (bind to {{xref:API-03}}) | OPTIONAL
11. Sender Retry Expectations
    sender_retries_on: {{sender.retry_on}}

sender_backoff: {{sender.backoff}} | OPTIONAL
max_retry_window: {{sender.max_retry_window}} | OPTIONAL

12. Security Constraints
    parser_hardening: {{security.parser_hardening}}
    header_allowlist: {{security.header_allowlist}} | OPTIONAL
    json_depth_limits: {{security.json_depth_limits}} | OPTIONAL
    logging_redaction: {{security.logging_redaction}} | OPTIONAL
13. Observability Requirements
    metrics:
    accept_count: {{obs.metrics.accept_count}}
    reject_count: {{obs.metrics.reject_count}}
    reject_reasons: {{obs.metrics.reject_reasons}} | OPTIONAL
    latency: {{obs.metrics.latency}}
    dedupe_hits: {{obs.metrics.dedupe_hits}} | OPTIONAL
    logs_required_fields: {{obs.logs_required_fields}}
    trace_attrs: {{obs.trace_attrs}} | OPTIONAL
    alerts: {{obs.alerts}} | OPTIONAL
14. References
    Endpoint catalog: {{xref:API-01}}
    Endpoint specs: {{xref:API-02}}
    Errors/status policy: {{xref:API-03}} | OPTIONAL
    Delivery semantics: {{xref:EVT-04}} | OPTIONAL
    Rate limits: {{xref:RLIM-01}} | OPTIONAL
    Background jobs: {{xref:API-06}} | OPTIONAL

Cross-References
Upstream: {{xref:API-01}}, {{xref:API-02}}, {{xref:EVT-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:EVT-07}} | OPTIONAL, {{xref:EVT-08}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where verification/mapping details are not provided; do not invent sources.
intermediate: Required. Define verification, validation, dedupe, and rate limits.
advanced: Required. Add replay protection rigor and full observability/alerting requirements.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, owner, expected_sender, feature_flag, secret_ref, signature_header, timestamp_header, tolerance_seconds, ip_allowlist, mtls_requirements, nonce_required, nonce_store, replay_window, required_headers, schema_validation, field_mapping_rules, pii_handling, dedupe_window, per_source_limits, global_limits, abuse_signals, enforcement_actions, error_body_policy, sender_backoff,

max_retry_window, header_allowlist, json_depth_limits, logging_redaction, trace_attrs, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If verification_required is UNKNOWN → flag in Open Questions.
If max_body_bytes is UNKNOWN → block Completeness Gate.
If dedupe_key_rule is UNKNOWN while dedupe_enabled == true → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.EVENTING
Pass conditions:
required_fields_present == true
all inbound endpoints exist in API-01/API-02
verification_model_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

EVT-07

EVT-07 — Event Failure Handling (DLQ, replay, backfill)
Header Block
template_id: EVT-07
title: Event Failure Handling (DLQ, replay, backfill)
type: eventing_failure_handling
template_version: 1.0.0
output_path: 10_app/eventing/EVT-07_Event_Failure_Handling.md
compliance_gate_id: TMP-05.PRIMARY.EVENTING
upstream_dependencies: ["EVT-01", "EVT-03", "EVT-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "EVT-01", "EVT-03", "EVT-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for handling event delivery and processing failures, including DLQ routing, retries vs dead-lettering, replay, backfill, and operator procedures. This template must be consistent with delivery semantics and producer/consumer mapping and must not invent failure modes or recovery capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}}
EVT-03 Producer/Consumer Map: {{evt.map}}
EVT-04 Delivery Semantics: {{evt.delivery_semantics}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Failure taxonomy (classes of failure)
Retry vs DLQ decision rules
DLQ model (where stored, how routed, identifiers)
DLQ schema (what fields captured)
Poison message definition and handling rules
Replay policy (who can replay, how, constraints)
Backfill policy (who can backfill, how, constraints)
Deduplication interaction (replay duplicates rules)
Retention rules (event retention, DLQ retention)
Operator controls (pause consumer, drain, replay, purge)
Safety controls (rate limits on replay/backfill)
Audit/evidence requirements for operator actions
Observability requirements (DLQ depth, replay success, failure rates)

Optional Fields
Per-event override rules | OPTIONAL
Partial replay rules (time range, key range) | OPTIONAL
Reprocessing idempotency guidelines | OPTIONAL
Cross-system compensation patterns | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Failure handling rules MUST be consistent with {{xref:EVT-04}} delivery semantics.
Do not invent DLQ capability; if DLQ is not defined in inputs, mark UNKNOWN and flag.
Replay/backfill MUST be access-controlled and audited (or UNKNOWN flagged).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Any per-event overrides MUST be declared explicitly and reference event_id from {{xref:EVT-01}}.

Output Format

1. Failure Taxonomy
   Failure classes:
   {{failure.classes[0]}}
   {{failure.classes[1]}}
   {{failure.classes[2]}}
   Transient definition: {{failure.transient_definition}}
   Permanent definition: {{failure.permanent_definition}}
   Poison definition: {{failure.poison_definition}}
2. Retry vs DLQ Decision Rules
   retry_allowed_when: {{decision.retry_allowed_when}}
   dlq_route_when: {{decision.dlq_route_when}}
   max_attempts_policy: {{decision.max_attempts_policy}}
   non_retryable_conditions: {{decision.non_retryable_conditions}} | OPTIONAL
3. DLQ Model
   dlq_supported: {{dlq.supported}}
   dlq_location: {{dlq.location}} (queue/topic/table/UNKNOWN)
   dlq_id_format: {{dlq.id_format}} | OPTIONAL
   routing_rule: {{dlq.routing_rule}}
   per_consumer_dlq: {{dlq.per_consumer}} | OPTIONAL
   per_event_dlq: {{dlq.per_event}} | OPTIONAL
4. DLQ Record Schema (Captured Fields)
   Required fields captured for each DLQ record:
   dlq_id: {{dlq_record.dlq_id}}
   event_id: {{dlq_record.event_id}}
   event_version: {{dlq_record.event_version}} | OPTIONAL
   producer: {{dlq_record.producer}} | OPTIONAL

consumer: {{dlq_record.consumer}} | OPTIONAL
failed_at: {{dlq_record.failed_at}}
failure_class: {{dlq_record.failure_class}}
failure_reason: {{dlq_record.failure_reason}}
attempt_count: {{dlq_record.attempt_count}}
payload_ref: {{dlq_record.payload_ref}} | OPTIONAL
trace_id: {{dlq_record.trace_id}} | OPTIONAL
request_id: {{dlq_record.request_id}} | OPTIONAL
dedupe_key: {{dlq_record.dedupe_key}} | OPTIONAL

5. Poison Message Handling
   poison_detection_rule: {{poison.detection_rule}}
   poison_action: {{poison.action}} (dlq/skip/quarantine/UNKNOWN)
   quarantine_location: {{poison.quarantine_location}} | OPTIONAL
   operator_notification: {{poison.operator_notification}} | OPTIONAL
6. Replay Policy
   replay_supported: {{replay.supported}}
   who_can_replay: {{replay.who_can_replay}} (bind to {{xref:API-04}}) | OPTIONAL
   replay_modes: {{replay.modes}} (single/dlq_batch/time_range/key_range/UNKNOWN)
   replay_constraints: {{replay.constraints}}
   replay_rate_limits: {{replay.rate_limits}} | OPTIONAL
   replay_dedupe_rule: {{replay.dedupe_rule}}
7. Backfill Policy
   backfill_supported: {{backfill.supported}}
   who_can_backfill: {{backfill.who_can_backfill}} | OPTIONAL
   backfill_sources: {{backfill.sources}} (events_store/snapshots/external/UNKNOWN)
   backfill_constraints: {{backfill.constraints}}
   backfill_rate_limits: {{backfill.rate_limits}} | OPTIONAL
   backfill_dedupe_rule: {{backfill.dedupe_rule}}
8. Retention Rules
   event_retention: {{retention.events}}
   dlq_retention: {{retention.dlq}}
   replay_window: {{retention.replay_window}} | OPTIONAL
9. Operator Controls & Procedures
   controls:
   pause_consumer: {{ops.pause_consumer}}
   resume_consumer: {{ops.resume_consumer}}
   drain_dlq: {{ops.drain_dlq}} | OPTIONAL
   replay_dlq: {{ops.replay_dlq}} | OPTIONAL
   purge_dlq: {{ops.purge_dlq}} | OPTIONAL
   required_audit: {{ops.required_audit}}
   safety_checks: {{ops.safety_checks}} | OPTIONAL
10. Audit & Evidence
    audit_required_for: {{audit.required_for}}

audit_fields: {{audit.fields}}
evidence_refs: {{audit.evidence_refs}} | OPTIONAL

11. Observability Requirements
    metrics:
    failure_rate: {{obs.metrics.failure_rate}}
    retry_rate: {{obs.metrics.retry_rate}} | OPTIONAL
    dlq_depth: {{obs.metrics.dlq_depth}} | OPTIONAL
    replay_success: {{obs.metrics.replay_success}} | OPTIONAL
    backfill_success: {{obs.metrics.backfill_success}} | OPTIONAL
    alerts: {{obs.alerts}} | OPTIONAL
    dashboards: {{obs.dashboards}} | OPTIONAL
12. Per-Event Overrides (Optional)
    override
    event_id: {{overrides[0].event_id}}
    override_fields: {{overrides[0].override_fields}}
    notes: {{overrides[0].notes}} | OPTIONAL
13. References
    Event catalog: {{xref:EVT-01}}
    Producer/consumer map: {{xref:EVT-03}}
    Delivery semantics: {{xref:EVT-04}}
    Webhook producer: {{xref:EVT-05}} | OPTIONAL
    Webhook consumer: {{xref:EVT-06}} | OPTIONAL
    Observability: {{xref:EVT-08}} | OPTIONAL

Cross-References
Upstream: {{xref:EVT-01}}, {{xref:EVT-03}}, {{xref:EVT-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:EVT-08}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-AUDIT]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for DLQ/replay/backfill if not defined; do not invent capabilities.
intermediate: Required. Define retry vs DLQ rules and captured DLQ schema fields.
advanced: Required. Add replay/backfill constraints, operator controls, audit + observability rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, dlq_id_format, per_consumer_dlq, per_event_dlq, payload_ref, trace_id, request_id, dedupe_key, quarantine_location, operator_notification, who_can_replay, replay_rate_limits, who_can_backfill, backfill_rate_limits, replay_window, safety_checks, evidence_refs, alerts, dashboards, overrides, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If dlq_supported is UNKNOWN → flag in Open Questions.
If replay_supported == true and replay_dedupe_rule is UNKNOWN → block Completeness Gate.
If backfill_supported == true and backfill_sources is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.EVENTING
Pass conditions:
required_fields_present == true
failure_handling_consistent_with_evt04 == true
placeholder_resolution == true
no_unapproved_unknowns == true

EVT-08

EVT-08 — Event Observability (lag, success rate, tracing)
Header Block
template_id: EVT-08
title: Event Observability (lag, success rate, tracing)
type: eventing_observability
template_version: 1.0.0
output_path: 10_app/eventing/EVT-08_Event_Observability.md
compliance_gate_id: TMP-05.PRIMARY.EVENTING
upstream_dependencies: ["EVT-01", "EVT-03", "EVT-04", "EVT-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "EVT-01", "EVT-03", "EVT-04", "EVT-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability requirements for the eventing system, including required metrics, logs, traces, dashboards, and alerting for event production, delivery, consumption, retries, dedupe, and DLQ/replay flows. This template must be consistent with delivery semantics and failure handling and must not invent instrumentation that contradicts upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
EVT-01 Event Catalog: {{evt.catalog}}
EVT-03 Producer/Consumer Map: {{evt.map}}
EVT-04 Delivery Semantics: {{evt.delivery_semantics}}
EVT-07 Failure Handling: {{evt.failure_handling}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Core metric set (produce/deliver/consume)
Lag measurement definition (where measured + units)
Success/failure rate definitions (per surface, per event_id, per consumer)
Retry metrics (attempts, backoff time, retry rate)
Dedupe metrics (duplicate count, dedupe hit rate)
DLQ metrics (depth, age, ingress rate, drain rate)
Replay/backfill metrics (requests, success, failure, throttling)
Log field requirements (correlation + redaction)
Trace propagation requirements (trace_id + span attributes)
Dashboard requirements (minimum panels)
Alerting requirements (threshold types + paging policy)
SLO/SLA hooks (how to bind to SLO docs if present)

Optional Fields
Per-event custom metrics | OPTIONAL
Per-consumer custom dashboards | OPTIONAL
Sampling rules for traces | OPTIONAL
Cost controls (metric cardinality limits) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Observability signals MUST be keyed by stable identifiers (event_id, consumer_id, producer_id) where applicable.
Cardinality MUST be controlled; do not use unbounded IDs as metric labels unless explicitly allowed.
PII MUST be redacted in logs/traces per: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Alerts MUST reference the failure handling procedures from {{xref:EVT-07}} where relevant.

Output Format

1. Observability Scope
   surfaces: {{obs.surfaces}} (bus/webhook/ws/UNKNOWN)
   coverage_model: {{obs.coverage_model}} (per event_id / per mapping / per surface)
   notes: {{obs.notes}} | OPTIONAL
2. Metrics — Core
   Required metric names (or naming rules):
   produce_count: {{metrics.produce_count}}
   deliver_count: {{metrics.deliver_count}} | OPTIONAL
   consume_count: {{metrics.consume_count}}
   failure_count: {{metrics.failure_count}}
   success_rate: {{metrics.success_rate}}
   latency_ms: {{metrics.latency_ms}}
   lag_ms: {{metrics.lag_ms}}
   Dimensions (labels):
   event_id: {{metrics.labels.event_id}}
   producer_id: {{metrics.labels.producer_id}} | OPTIONAL
   consumer_id: {{metrics.labels.consumer_id}} | OPTIONAL
   surface: {{metrics.labels.surface}} | OPTIONAL
   env: {{metrics.labels.env}} | OPTIONAL
3. Metrics — Retry/Dedupe
   retry_attempts: {{metrics.retry_attempts}}
   retry_rate: {{metrics.retry_rate}} | OPTIONAL
   backoff_time_ms: {{metrics.backoff_time_ms}} | OPTIONAL

dedupe_hits: {{metrics.dedupe_hits}}
duplicate_rate: {{metrics.duplicate_rate}} | OPTIONAL

4. Metrics — DLQ/Replay/Backfill
   dlq_depth: {{metrics.dlq_depth}} | OPTIONAL
   dlq_oldest_age_ms: {{metrics.dlq_oldest_age_ms}} | OPTIONAL
   dlq_ingress_rate: {{metrics.dlq_ingress_rate}} | OPTIONAL
   dlq_drain_rate: {{metrics.dlq_drain_rate}} | OPTIONAL
   replay_requests: {{metrics.replay_requests}} | OPTIONAL
   replay_success: {{metrics.replay_success}} | OPTIONAL
   replay_failure: {{metrics.replay_failure}} | OPTIONAL
   backfill_requests: {{metrics.backfill_requests}} | OPTIONAL
   backfill_success: {{metrics.backfill_success}} | OPTIONAL
   backfill_failure: {{metrics.backfill_failure}} | OPTIONAL
5. Logs — Required Fields
   All producer/consumer logs MUST include:
   timestamp: {{logs.fields.timestamp}}
   event_id: {{logs.fields.event_id}}
   event_version: {{logs.fields.event_version}} | OPTIONAL
   producer_id: {{logs.fields.producer_id}} | OPTIONAL
   consumer_id: {{logs.fields.consumer_id}} | OPTIONAL
   request_id: {{logs.fields.request_id}} | OPTIONAL
   trace_id: {{logs.fields.trace_id}} | OPTIONAL
   delivery_attempt: {{logs.fields.delivery_attempt}} | OPTIONAL
   dedupe_key: {{logs.fields.dedupe_key}} | OPTIONAL
   result: {{logs.fields.result}} (success/failure)
   failure_class: {{logs.fields.failure_class}} | OPTIONAL
   error_code: {{logs.fields.error_code}} | OPTIONAL
   redaction_applied: {{logs.fields.redaction_applied}} | OPTIONAL
6. Tracing — Propagation & Attributes
   trace_required: {{tracing.required}}
   propagation_headers: {{tracing.propagation_headers}} | OPTIONAL
   span_attributes_required:
   event_id: {{tracing.attrs.event_id}}
   consumer_id: {{tracing.attrs.consumer_id}} | OPTIONAL
   producer_id: {{tracing.attrs.producer_id}} | OPTIONAL
   surface: {{tracing.attrs.surface}} | OPTIONAL
   attempt: {{tracing.attrs.attempt}} | OPTIONAL
   sampling_policy: {{tracing.sampling_policy}} | OPTIONAL
7. Dashboards (Minimum)
   Dashboards to provide:
   overview_dashboard: {{dashboards.overview}}
   per_consumer_dashboard: {{dashboards.per_consumer}} | OPTIONAL
   dlq_dashboard: {{dashboards.dlq}} | OPTIONAL

webhook_dashboard: {{dashboards.webhook}} | OPTIONAL
Minimum panels per dashboard: {{dashboards.min_panels}}

8. Alerting Requirements
   Alert classes:
- lag_high: {{alerts.lag_high}}
- failure_rate_high: {{alerts.failure_rate_high}}
- dlq_depth_high: {{alerts.dlq_depth_high}} | OPTIONAL
- replay_failures: {{alerts.replay_failures}} | OPTIONAL

   Threshold definition model: {{alerts.threshold_model}} (static/baseline/UNKNOWN)
   Routing/paging policy: {{alerts.routing_policy}} | OPTIONAL
   Runbook/procedure ref: {{alerts.runbook_ref}} (expected: {{xref:EVT-07}}) | OPTIONAL
9. SLO Hooks (Optional)
   slo_doc_ref: {{obs.slo_doc_ref}} | OPTIONAL
   slo_targets: {{obs.slo_targets}} | OPTIONAL
10. Cardinality & Cost Controls (Optional)
    label_allowlist: {{cost.label_allowlist}} | OPTIONAL
    high_cardinality_labels_disallowed: {{cost.disallowed_labels}} | OPTIONAL
    sampling_controls: {{cost.sampling_controls}} | OPTIONAL
11. References
    Event catalog: {{xref:EVT-01}}
    Producer/consumer map: {{xref:EVT-03}}
    Delivery semantics: {{xref:EVT-04}}
    Failure handling: {{xref:EVT-07}}

Cross-References
Upstream: {{xref:EVT-01}}, {{xref:EVT-03}}, {{xref:EVT-04}}, {{xref:EVT-07}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OPS-OBS}} | OPTIONAL, {{xref:ALRT-01}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL, {{standards.rules[STD-OBSERVABILITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for tooling specifics; define required signal categories and keys.
intermediate: Required. Define metric/log/trace fields and minimum dashboards/alerts.
advanced: Required. Add cardinality controls, sampling rules, and SLO bindings.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, deliver_count, producer_id/consumer_id labels, retry_rate, backoff_time_ms, duplicate_rate, dlq_* metrics, replay/backfill metrics, propagation_headers, sampling_policy, dashboards.*, alerts.routing_policy, alerts.runbook_ref, obs.slo_doc_ref, obs.slo_targets, cost.* , open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If logs required fields are UNKNOWN → block Completeness Gate.
If tracing.required is UNKNOWN → flag in Open Questions.

Completeness Gate
Gate ID: TMP-05.PRIMARY.EVENTING
Pass conditions:
required_fields_present == true
metric_set_defined == true
log_fields_defined == true
dashboard_minimum_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Rate Limits & Abuse Controls (RLIM)

Rate Limits & Abuse Controls (RLIM)
RLIM-01 Rate Limit Policy (global rules, scopes)
RLIM-02 Rate Limit Catalog (limits by surface/endpoint)
RLIM-03 Abuse Signals & Detection Rules
RLIM-04 Enforcement Actions Matrix (throttle/ban/captcha)
RLIM-05 Exemptions & Allowlist Policy
RLIM-06 Rate Limit Observability (alerts, dashboards)

RLIM-01

RLIM-01 — Rate Limit Policy (global rules, scopes)
Header Block
template_id: RLIM-01
title: Rate Limit Policy (global rules, scopes)
type: rate_limit_policy
template_version: 1.0.0
output_path: 10_app/ratelimits/RLIM-01_Rate_Limit_Policy.md
compliance_gate_id: TMP-05.PRIMARY.RLIM
upstream_dependencies: []
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the single, canonical global rate limit policy: what surfaces are rate-limited, what scopes exist, how limits are computed and enforced, how bursts are handled, and how rate-limit decisions map to errors/status. This template must be consistent with standards and must not invent enforcement capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Rate-limited surfaces (API, WS, webhooks, jobs triggers, admin)
Scope model (per IP / per user / per token / per org / per project / per route / per endpoint class)
Identity keys used for enforcement (client identifiers)
Limit units (req/sec, req/min, tokens/min, bytes/min, concurrent)
Burst handling model (token bucket/leaky bucket/UNKNOWN)
Enforcement actions (throttle, reject, slow, ban)
Response behavior (headers + error mapping)
Exemptions model (allowlists, internal services, admin)
Override policy (who can change limits, audit requirements)
Observability requirements (metrics, logs, dashboards, alerts)
Abuse escalation linkage (bind to RLIM-03/RLIM-04 if present)

Optional Fields
Per-surface default limits | OPTIONAL
Geo-based policies | OPTIONAL
Cost-based throttling | OPTIONAL
Dynamic limit adjustments | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not enumerate per-endpoint limits here; those belong in RLIM-02.
Rate limit errors MUST map to: {{xref:API-03}} | OPTIONAL
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
If exemptions exist, they MUST be explicit and auditable.

Output Format

1. Policy Scope
   surfaces: {{policy.surfaces}}
   applies_in_envs: {{policy.envs}} | OPTIONAL
   notes: {{policy.notes}} | OPTIONAL
2. Scope Model (Keys)
   Supported scopes:
   per_ip: {{scope.per_ip}}
   per_user: {{scope.per_user}}
   per_token: {{scope.per_token}} | OPTIONAL
   per_org: {{scope.per_org}} | OPTIONAL
   per_project: {{scope.per_project}} | OPTIONAL
   per_route: {{scope.per_route}} | OPTIONAL
   per_endpoint_class: {{scope.per_endpoint_class}} | OPTIONAL
   Identity key rules: {{scope.identity_key_rules}}
3. Limit Units
   request_rate: {{units.request_rate}} (e.g., req/min)
   token_rate: {{units.token_rate}} | OPTIONAL
   byte_rate: {{units.byte_rate}} | OPTIONAL
   concurrency: {{units.concurrency}} | OPTIONAL
4. Burst Handling
   algorithm: {{burst.algorithm}} (token_bucket/leaky_bucket/UNKNOWN)
   burst_capacity: {{burst.capacity}} | OPTIONAL
   refill_rate: {{burst.refill_rate}} | OPTIONAL
   smoothing: {{burst.smoothing}} | OPTIONAL
5. Enforcement Actions
   actions_supported: {{enforcement.actions_supported}}
   throttle_behavior: {{enforcement.throttle_behavior}} | OPTIONAL
   reject_behavior: {{enforcement.reject_behavior}}
   ban_policy_ref: {{enforcement.ban_policy_ref}} | OPTIONAL
6. Response Behavior (Errors + Headers)
   error_code: {{responses.error_code}} (expected: rate-limit code from {{xref:API-03}})
   status_code: {{responses.status_code}} (expected: 429)
   headers:
   rate_limit_limit: {{responses.headers.limit}} | OPTIONAL
   rate_limit_remaining: {{responses.headers.remaining}} | OPTIONAL

rate_limit_reset: {{responses.headers.reset}} | OPTIONAL
retry_after: {{responses.headers.retry_after}} | OPTIONAL
body_policy: {{responses.body_policy}} | OPTIONAL

7. Exemptions / Allowlist Model
   exempt_principals: {{exempt.principals}} | OPTIONAL
   exempt_routes: {{exempt.routes}} | OPTIONAL
   allowlist_policy: {{exempt.allowlist_policy}} | OPTIONAL
   audit_required: {{exempt.audit_required}} | OPTIONAL
8. Overrides & Change Control
   who_can_change_limits: {{change_control.who}}
   audit_policy: {{change_control.audit_policy}} | OPTIONAL
   rollout_rules: {{change_control.rollout_rules}} | OPTIONAL
9. Observability Requirements
   metrics:
   limited_count: {{obs.metrics.limited_count}}
   throttle_count: {{obs.metrics.throttle_count}} | OPTIONAL
   ban_count: {{obs.metrics.ban_count}} | OPTIONAL
   top_keys: {{obs.metrics.top_keys}} | OPTIONAL
   logs_required_fields: {{obs.logs_required_fields}}
   dashboards: {{obs.dashboards}} | OPTIONAL
   alerts: {{obs.alerts}} | OPTIONAL
10. Abuse Escalation Linkage (Optional)
    signals_ref: {{xref:RLIM-03}} | OPTIONAL
    actions_matrix_ref: {{xref:RLIM-04}} | OPTIONAL
11. References
    Per-endpoint limits catalog: {{xref:RLIM-02}} | OPTIONAL
    Abuse signals: {{xref:RLIM-03}} | OPTIONAL
    Enforcement actions: {{xref:RLIM-04}} | OPTIONAL
    Exemptions: {{xref:RLIM-05}} | OPTIONAL
    Observability: {{xref:RLIM-06}} | OPTIONAL
    API error mapping: {{xref:API-03}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:STANDARDS_INDEX}} | OPTIONAL
Downstream: {{xref:RLIM-02}}, {{xref:RLIM-03}}, {{xref:RLIM-04}}, {{xref:RLIM-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where enforcement details are missing; define scope model and 429 mapping.
intermediate: Required. Define algorithm, headers, exemptions, and observability requirements.
advanced: Required. Add change control rigor and abuse escalation linkages.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, applies_in_envs, per_token, per_org, per_project, per_route, per_endpoint_class, token_rate, byte_rate, concurrency, burst_capacity, refill_rate, smoothing, throttle_behavior, ban_policy_ref, headers, body_policy, exempt_principals, exempt_routes, allowlist_policy, audit_required, audit_policy, rollout_rules, dashboards, alerts, signals_ref, actions_matrix_ref, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If scope.identity_key_rules is UNKNOWN → block Completeness Gate.
If responses.status_code is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.RLIM
Pass conditions:
required_fields_present == true
scope_model_defined == true
error_mapping_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

RLIM-02

RLIM-02 — Rate Limit Catalog (limits by surface/endpoint)
Header Block
template_id: RLIM-02
title: Rate Limit Catalog (limits by surface/endpoint)
type: rate_limit_catalog
template_version: 1.0.0
output_path: 10_app/ratelimits/RLIM-02_Rate_Limit_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.RLIM
upstream_dependencies: ["RLIM-01", "API-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "RLIM-01", "API-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of concrete rate limits applied across surfaces and endpoints, including scope keys, units, burst settings, and enforcement actions. This document must be consistent with the global rate limit policy and the endpoint catalog and must not invent endpoint IDs or limit classes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}}
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Catalog index (list of limit entries)
limit_id (stable identifier)
surface (api/ws/webhook/jobs/admin)
target binding (endpoint_id / route pattern / consumer_id / producer_id)
scope (per_ip/per_user/per_org/per_project/per_token/etc.)
unit (req/min, req/sec, bytes/min, concurrent, etc.)
limit value (number)
burst configuration (capacity/refill) or pointer to RLIM-01 defaults
enforcement action (reject/throttle/slow/ban)
error mapping (status + error_code pointer to API-03)
exemptions reference (allowlist) | OPTIONAL
observability tags (labels to attach)

Optional Fields
Time-window overrides | OPTIONAL

Environment overrides (dev/stage/prod) | OPTIONAL
Per-tenant overrides | OPTIONAL
Feature flag gating | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent endpoint_id; bindings MUST reference API-01 (or be UNKNOWN flagged).
Limit behavior MUST be consistent with RLIM-01 (scopes, units, headers, enforcement).
Error mapping MUST follow {{xref:API-03}} | OPTIONAL
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Each limit_id MUST be unique.

Output Format

1. Catalog Summary
   total_limits: {{catalog.total_limits}}
   surfaces: {{catalog.surfaces}} | OPTIONAL
   notes: {{catalog.notes}} | OPTIONAL
2. Limit Entries (by limit_id)
   Limit
   limit_id: {{limits[0].limit_id}}
   surface: {{limits[0].surface}}
   target_type: {{limits[0].target_type}}
   (endpoint_id/route_pattern/consumer_id/producer_id/UNKNOWN)
   target_ref: {{limits[0].target_ref}} (expected: {{xref:API-01}} when endpoint_id)
   scope: {{limits[0].scope}}
   unit: {{limits[0].unit}}
   value: {{limits[0].value}}
   burst_capacity: {{limits[0].burst_capacity}} | OPTIONAL
   refill_rate: {{limits[0].refill_rate}} | OPTIONAL
   enforcement_action: {{limits[0].enforcement_action}}
   error_code: {{limits[0].error_code}} | OPTIONAL
   status_code: {{limits[0].status_code}} | OPTIONAL
   headers_policy: {{limits[0].headers_policy}} | OPTIONAL
   env_overrides: {{limits[0].env_overrides}} | OPTIONAL
   tenant_overrides: {{limits[0].tenant_overrides}} | OPTIONAL
   exemptions_ref: {{limits[0].exemptions_ref}} | OPTIONAL
   feature_flag: {{limits[0].feature_flag}} | OPTIONAL
   observability_tags: {{limits[0].observability_tags}}
   open_questions:
   {{limits[0].open_questions[0]}} | OPTIONAL

(Repeat the “Limit” entry block for each limit_id.)

3. References
   Global policy: {{xref:RLIM-01}}
   Endpoint catalog: {{xref:API-01}}
   Error/status mapping: {{xref:API-03}} | OPTIONAL
   Exemptions policy: {{xref:RLIM-05}} | OPTIONAL
   Observability: {{xref:RLIM-06}} | OPTIONAL

Cross-References
Upstream: {{xref:RLIM-01}}, {{xref:API-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:RLIM-03}}, {{xref:RLIM-04}}, {{xref:RLIM-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for targets if missing; do not invent endpoint IDs.
intermediate: Required. Populate limits with concrete scopes/units/values consistent with RLIM-01.
advanced: Required. Add env/tenant overrides and observability tags with controlled cardinality.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, target_type, target_ref (only if binding is missing), burst_capacity, refill_rate, error_code, status_code, headers_policy, env_overrides, tenant_overrides, exemptions_ref, feature_flag, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If limit_id is UNKNOWN → block Completeness Gate.
If surface is UNKNOWN → block Completeness Gate.
If target_type == endpoint_id and target_ref is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.RLIM
Pass conditions:
required_fields_present == true
all limit_ids are unique
all endpoint_id targets exist in API-01 (no new IDs introduced)
placeholder_resolution == true
no_unapproved_unknowns == true

RLIM-03

RLIM-03 — Abuse Signals & Detection Rules
Header Block
template_id: RLIM-03
title: Abuse Signals & Detection Rules
type: abuse_signals_detection
template_version: 1.0.0
output_path: 10_app/ratelimits/RLIM-03_Abuse_Signals_Detection_Rules.md
compliance_gate_id: TMP-05.PRIMARY.RLIM
upstream_dependencies: ["RLIM-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "RLIM-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical abuse signal catalog and detection rule format used to identify abusive or anomalous behavior across rate-limited surfaces. This template must be consistent with the global rate limit policy and must not invent enforcement capabilities beyond what upstream inputs allow.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Signal registry (signal_id catalog)
Signal definitions (what is measured, units, where captured)
Detection rules (rule_id catalog)
Rule predicates/threshold model (static/baseline/UNKNOWN)
Evaluation window (time window)
Scope keys (per ip/user/org/project/token)
Trigger criteria (what constitutes a hit)
False-positive handling notes
Evidence capture requirements (what logs/metrics prove rule fired)
Severity levels (low/med/high/critical)
Action binding pointer (maps to RLIM-04 actions matrix)
Observability requirements (rule hit rate, top keys, alerting)

Optional Fields
Per-surface rule sets | OPTIONAL
Per-endpoint-class rules | OPTIONAL

ML/anomaly detection notes | OPTIONAL
Manual review workflow | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not define enforcement actions here; bind to {{xref:RLIM-04}}.
Rules MUST be deterministic and evaluable from captured signals (no “human judgment” predicates).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Cardinality controls MUST be respected for labels/keys (no unbounded identifiers unless allowed).

Output Format

1. Signal Registry (by signal_id)
   Signal
   signal_id: {{signals[0].signal_id}}
   name: {{signals[0].name}}
   surface: {{signals[0].surface}} (api/ws/webhook/jobs/admin/UNKNOWN)
   what_it_measures: {{signals[0].what_it_measures}}
   unit: {{signals[0].unit}}
   collection_point: {{signals[0].collection_point}}
   (edge/middleware/app/broker/UNKNOWN)
   dimensions: {{signals[0].dimensions}} | OPTIONAL
   pii_risk: {{signals[0].pii_risk}} | OPTIONAL
   notes: {{signals[0].notes}} | OPTIONAL

(Repeat for each signal_id.)

2. Detection Rules (by rule_id)
   Rule
   rule_id: {{rules[0].rule_id}}
   name: {{rules[0].name}}
   severity: {{rules[0].severity}}
   scope_key: {{rules[0].scope_key}}
   (per_ip/per_user/per_org/per_project/per_token/UNKNOWN)
   window: {{rules[0].window}} (e.g., 60s/5m/1h/UNKNOWN)
   predicate_model: {{rules[0].predicate_model}} (static/baseline/UNKNOWN)
   predicate: {{rules[0].predicate}}
   signals_used: {{rules[0].signals_used}}
   trigger_criteria: {{rules[0].trigger_criteria}}
   evidence_required: {{rules[0].evidence_required}}
   false_positive_handling: {{rules[0].false_positive_handling}} | OPTIONAL

action_ref: {{rules[0].action_ref}} (expected: {{xref:RLIM-04}})
alerting: {{rules[0].alerting}} | OPTIONAL
open_questions:
{{rules[0].open_questions[0]}} | OPTIONAL

(Repeat for each rule_id.)

3. Evidence Capture Requirements
   Minimum evidence fields when a rule fires:
   timestamp: {{evidence.timestamp}}
   rule_id: {{evidence.rule_id}}
   signal_snapshot: {{evidence.signal_snapshot}}
   scope_key: {{evidence.scope_key}}
   scope_value: {{evidence.scope_value}}
   decision: {{evidence.decision}}
   request_id: {{evidence.request_id}} | OPTIONAL
   trace_id: {{evidence.trace_id}} | OPTIONAL
4. Observability Requirements
   metrics:
   rule_hits: {{obs.metrics.rule_hits}}
   rule_hit_rate: {{obs.metrics.rule_hit_rate}} | OPTIONAL
   top_scope_values: {{obs.metrics.top_scope_values}} | OPTIONAL
   false_positive_rate: {{obs.metrics.false_positive_rate}} | OPTIONAL
   dashboards: {{obs.dashboards}} | OPTIONAL
   alerts: {{obs.alerts}} | OPTIONAL
5. References
   Global rate limit policy: {{xref:RLIM-01}}
   Actions matrix: {{xref:RLIM-04}} | OPTIONAL
   Exemptions policy: {{xref:RLIM-05}} | OPTIONAL
   Observability: {{xref:RLIM-06}} | OPTIONAL

Cross-References
Upstream: {{xref:RLIM-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:RLIM-04}}, {{xref:RLIM-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-OBSERVABILITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where signals/rules are not defined; do not invent action behavior.
intermediate: Required. Define rule windows, predicates, and evidence requirements.
advanced: Required. Add cardinality controls, alerting thresholds, and false-positive handling rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, dimensions, pii_risk, notes, predicate_model, false_positive_handling, alerting, dashboards, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If rule_id or signal_id is UNKNOWN → block Completeness Gate.
If action_ref is UNKNOWN → allow only if flagged in Open Questions.

Completeness Gate
Gate ID: TMP-05.PRIMARY.RLIM
Pass conditions:
required_fields_present == true
all signal_ids are unique
all rule_ids are unique
all action_refs (if present) point to RLIM-04
placeholder_resolution == true
no_unapproved_unknowns == true

-

RLIM-04

RLIM-04 — Enforcement Actions Matrix (throttle/ban/captcha)
Header Block
template_id: RLIM-04
title: Enforcement Actions Matrix (throttle/ban/captcha)
type: abuse_enforcement_actions_matrix
template_version: 1.0.0
output_path: 10_app/ratelimits/RLIM-04_Enforcement_Actions_Matrix.md
compliance_gate_id: TMP-05.PRIMARY.RLIM
upstream_dependencies: ["RLIM-01", "RLIM-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "RLIM-01", "RLIM-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical matrix of enforcement actions that can be applied when rate limiting or abuse rules trigger, including throttling, temporary blocks, bans, step-up challenges (captcha), and escalation procedures. This template must be consistent with the rate limit policy and detection rules and must not invent enforcement capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}}
RLIM-03 Abuse Signals & Detection Rules: {{abuse.rules}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Action catalog (action_id list)
Action definitions (what the action does)
Applicability (what surfaces/scopes can apply the action)
Trigger bindings (which rule severities map to which actions)
Duration model (fixed/variable/UNKNOWN)
Escalation ladder (repeat offender behavior)
Reversal/unban policy (who can reverse, audit required)
User-facing response behavior (status codes + messages mapping)
Audit/evidence requirements (record what action was applied)
Safety constraints (avoid lockouts for critical principals)
Observability requirements (action counts, top keys, outcomes)

Optional Fields
Captcha/challenge provider integration notes | OPTIONAL
Per-tenant enforcement overrides | OPTIONAL

Manual review workflow | OPTIONAL
Progressive trust scoring notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not define detection rules here; bind to {{xref:RLIM-03}}.
Actions MUST be deterministic in effect (clear scope + duration + enforcement point).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
User-facing responses MUST map to {{xref:API-03}} | OPTIONAL
All overrides/unbans MUST be audited (or UNKNOWN flagged).

Output Format

1. Action Catalog (by action_id)
   Action
   action_id: {{actions[0].action_id}}
   name: {{actions[0].name}}
   type: {{actions[0].type}} (throttle/reject/ban/captcha/UNKNOWN)
   applies_to_surfaces: {{actions[0].applies_to_surfaces}}
   applies_to_scopes: {{actions[0].applies_to_scopes}}
   enforcement_point: {{actions[0].enforcement_point}} (edge/middleware/app/UNKNOWN)
   effect: {{actions[0].effect}}
   duration_model: {{actions[0].duration_model}} (fixed/variable/UNKNOWN)
   duration: {{actions[0].duration}} | OPTIONAL
   reversal_allowed: {{actions[0].reversal_allowed}}
   reversal_who: {{actions[0].reversal_who}} | OPTIONAL
   audit_required: {{actions[0].audit_required}}
   response_mapping:
   status_code: {{actions[0].response.status_code}}
   error_code: {{actions[0].response.error_code}} | OPTIONAL
   message_policy: {{actions[0].response.message_policy}} | OPTIONAL
   notes: {{actions[0].notes}} | OPTIONAL

(Repeat for each action_id.)

2. Severity → Action Bindings (Matrix)
   Binding
   severity: {{bindings[0].severity}} (low/med/high/critical/UNKNOWN)
   rule_ids: {{bindings[0].rule_ids}} (expected: {{xref:RLIM-03}})
   action_id: {{bindings[0].action_id}} (expected: action_id from section 1)
   escalation_on_repeat: {{bindings[0].escalation_on_repeat}} | OPTIONAL
   repeat_window: {{bindings[0].repeat_window}} | OPTIONAL

(Repeat for each severity or mapping class.)

3. Escalation Ladder
   repeat_offender_model: {{escalation.model}} (count-based/score-based/UNKNOWN)
   stages:
   {{escalation.stages[0]}}
   {{escalation.stages[1]}}
   {{escalation.stages[2]}}
   reset_rules: {{escalation.reset_rules}} | OPTIONAL
4. Reversal / Unban Policy
   who_can_reverse: {{reversal.who_can_reverse}}
   audit_policy: {{reversal.audit_policy}} | OPTIONAL
   reversal_process: {{reversal.process}} | OPTIONAL
5. Safety Constraints
   protected_principals: {{safety.protected_principals}} | OPTIONAL
   max_lockout_duration: {{safety.max_lockout_duration}} | OPTIONAL
   bypass_conditions: {{safety.bypass_conditions}} | OPTIONAL
6. Audit & Evidence Requirements
   Required record fields when action applied:
   timestamp: {{audit.timestamp}}
   action_id: {{audit.action_id}}
   rule_id: {{audit.rule_id}} | OPTIONAL
   severity: {{audit.severity}} | OPTIONAL
   scope_key: {{audit.scope_key}}
   scope_value: {{audit.scope_value}}
   duration: {{audit.duration}} | OPTIONAL
   decision: {{audit.decision}}
   request_id: {{audit.request_id}} | OPTIONAL
   trace_id: {{audit.trace_id}} | OPTIONAL
7. Observability Requirements
   metrics:
   action_count: {{obs.metrics.action_count}}
   action_rate: {{obs.metrics.action_rate}} | OPTIONAL
   top_scope_values: {{obs.metrics.top_scope_values}} | OPTIONAL
   reversal_count: {{obs.metrics.reversal_count}} | OPTIONAL
   dashboards: {{obs.dashboards}} | OPTIONAL
   alerts: {{obs.alerts}} | OPTIONAL
8. References
   Global rate limit policy: {{xref:RLIM-01}}
   Abuse detection rules: {{xref:RLIM-03}}
   Exemptions policy: {{xref:RLIM-05}} | OPTIONAL
   Observability: {{xref:RLIM-06}} | OPTIONAL
   Error/status mapping: {{xref:API-03}} | OPTIONAL

Cross-References
Upstream: {{xref:RLIM-01}}, {{xref:RLIM-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:RLIM-05}}, {{xref:RLIM-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-AUDIT]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where action capabilities are not defined; do not invent integrations.
intermediate: Required. Define actions, durations, and severity bindings consistent with RLIM-03.
advanced: Required. Add escalation ladder, reversal process, and safety constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, duration, reversal_who, error_code, message_policy, notes, escalation_on_repeat, repeat_window, reset_rules, audit_policy, reversal_process, protected_principals, max_lockout_duration, bypass_conditions, dashboards, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If action_id is UNKNOWN → block Completeness Gate.
If severity→action bindings are UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.RLIM
Pass conditions:
required_fields_present == true
all action_ids are unique
all bindings reference existing rule_ids (RLIM-03) and action_ids
placeholder_resolution == true
no_unapproved_unknowns == true

RLIM-05

RLIM-05 — Exemptions & Allowlist Policy
Header Block
template_id: RLIM-05
title: Exemptions & Allowlist Policy
type: rate_limit_exemptions_policy
template_version: 1.0.0
output_path: 10_app/ratelimits/RLIM-05_Exemptions_Allowlist_Policy.md
compliance_gate_id: TMP-05.PRIMARY.RLIM
upstream_dependencies: ["RLIM-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "RLIM-01", "API-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical policy for rate limit exemptions and allowlists: what can be exempted, who can grant exemptions, how exemptions are scoped and time-bounded, how they are audited, and how exemptions interact with abuse enforcement. This template must be consistent with the global rate limit policy and must not invent exemption powers not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}}
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Exemption scope types (principal/route/surface/tenant/project/ip range)
Allowlist entry schema (fields required for an exemption record)
Grant policy (who can grant, required permissions)
Duration policy (permanent vs expiring; max duration)
Review/renewal policy (periodic review)
Audit requirements (who/what/when/why)
Revocation policy (who can revoke, immediate effect)
Safety constraints (no blanket exemptions for untrusted principals)
Interaction with abuse enforcement (RLIM-04)
Observability requirements (active exemptions count, usage impact)

Optional Fields
Emergency/break-glass exemptions | OPTIONAL
Per-environment differences | OPTIONAL
Approval workflow (two-person) | OPTIONAL

Per-tenant exemption limits | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Exemptions MUST be explicit, scoped, and auditable.
Exemptions SHOULD be time-bounded by default; permanent requires explicit policy.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Grant/revoke authority MUST align to {{xref:API-04}} (or be UNKNOWN flagged).
Exemptions MUST NOT disable abuse enforcement unless explicitly allowed.

Output Format

1. Exemption Scope Types
   supported_scopes: {{exempt.supported_scopes}}
   scope_definitions: {{exempt.scope_definitions}} | OPTIONAL
2. Allowlist Entry Schema
   Entry fields:
   exemption_id: {{entry.exemption_id}}
   scope_type: {{entry.scope_type}}
   scope_value: {{entry.scope_value}}
   surfaces: {{entry.surfaces}} | OPTIONAL
   routes_or_targets: {{entry.routes_or_targets}} | OPTIONAL
   limit_ids_exempted: {{entry.limit_ids_exempted}} | OPTIONAL
   reason: {{entry.reason}}
   granted_by: {{entry.granted_by}}
   granted_at: {{entry.granted_at}}
   expires_at: {{entry.expires_at}} | OPTIONAL
   review_at: {{entry.review_at}} | OPTIONAL
   status: {{entry.status}} (active/revoked/expired)
   audit_ref: {{entry.audit_ref}} | OPTIONAL
3. Grant Policy
   who_can_grant: {{grant.who_can_grant}}
   required_permission: {{grant.required_permission}} | OPTIONAL
   approval_workflow: {{grant.approval_workflow}} | OPTIONAL
   max_active_exemptions_per_principal: {{grant.max_active_per_principal}} | OPTIONAL
4. Duration & Review Policy
   default_duration: {{duration.default}}
   max_duration: {{duration.max}} | OPTIONAL
   permanent_allowed: {{duration.permanent_allowed}}
   review_frequency: {{duration.review_frequency}} | OPTIONAL
   renewal_rules: {{duration.renewal_rules}} | OPTIONAL
5. Revocation Policy
   who_can_revoke: {{revoke.who_can_revoke}}

revocation_effect: {{revoke.effect}} (immediate/next_window/UNKNOWN)
revocation_audit_required: {{revoke.audit_required}}

6. Safety Constraints
   disallowed_scopes: {{safety.disallowed_scopes}} | OPTIONAL
   protected_targets: {{safety.protected_targets}} | OPTIONAL
   require_reason_code: {{safety.require_reason_code}} | OPTIONAL
7. Interaction with Abuse Enforcement
   abuse_actions_still_apply: {{interaction.abuse_actions_still_apply}}
   exceptions_to_abuse: {{interaction.exceptions_to_abuse}} | OPTIONAL
   escalation_override_allowed: {{interaction.escalation_override_allowed}} | OPTIONAL
8. Observability Requirements
   metrics:
   active_exemptions: {{obs.metrics.active_exemptions}}
   exempted_requests: {{obs.metrics.exempted_requests}} | OPTIONAL
   top_exempt_scopes: {{obs.metrics.top_exempt_scopes}} | OPTIONAL
   dashboards: {{obs.dashboards}} | OPTIONAL
   alerts: {{obs.alerts}} | OPTIONAL
9. References
   Global policy: {{xref:RLIM-01}}
   Limits catalog: {{xref:RLIM-02}} | OPTIONAL
   Abuse detection: {{xref:RLIM-03}} | OPTIONAL
   Enforcement actions: {{xref:RLIM-04}} | OPTIONAL
   Observability: {{xref:RLIM-06}} | OPTIONAL
   AuthZ rules: {{xref:API-04}} | OPTIONAL

Cross-References
Upstream: {{xref:RLIM-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:RLIM-02}}, {{xref:RLIM-04}}, {{xref:RLIM-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-AUDIT]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where grant/revoke authority is missing; do not invent.
intermediate: Required. Define entry schema, duration/review, and revocation rules.
advanced: Required. Add approval workflow, per-tenant limits, and safety constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, scope_definitions, surfaces, routes_or_targets, limit_ids_exempted, audit_ref, required_permission, approval_workflow, max_active_per_principal, max_duration, review_frequency, renewal_rules, disallowed_scopes, protected_targets, require_reason_code, exceptions_to_abuse, escalation_override_allowed, dashboards, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If who_can_grant is UNKNOWN → block Completeness Gate.
If allowlist entry schema is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.RLIM
Pass conditions:
required_fields_present == true
entry_schema_defined == true
grant_policy_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

RLIM-06

RLIM-06 — Rate Limit Observability (alerts, dashboards)
Header Block
template_id: RLIM-06
title: Rate Limit Observability (alerts, dashboards)
type: rate_limit_observability
template_version: 1.0.0
output_path: 10_app/ratelimits/RLIM-06_Rate_Limit_Observability.md
compliance_gate_id: TMP-05.PRIMARY.RLIM
upstream_dependencies: ["RLIM-01", "RLIM-02", "RLIM-03", "RLIM-04", "RLIM-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "RLIM-01", "RLIM-02", "RLIM-03", "RLIM-04", "RLIM-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability requirements for rate limiting and abuse enforcement, including required metrics, logs, dashboards, and alerts for throttles/rejects/bans/exemptions and detection rule hits. This template must be consistent with RLIM policy, catalogs, and enforcement actions and must not invent instrumentation requirements that contradict upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}}
RLIM-02 Rate Limit Catalog: {{rlim.catalog}} | OPTIONAL
RLIM-03 Abuse Signals & Detection: {{abuse.signals_rules}} | OPTIONAL
RLIM-04 Enforcement Actions Matrix: {{abuse.actions}} | OPTIONAL
RLIM-05 Exemptions Policy: {{rlim.exemptions}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Core metrics (limited_count, throttled_count, banned_count, exempted_count)
Per-surface breakdown requirements
Top keys reporting requirements (controlled cardinality)
Rule hit metrics (RLIM-03)
Action apply metrics (RLIM-04)
Exemption usage metrics (RLIM-05)
Log field requirements (correlation, redaction)
Dashboard requirements (minimum panels)
Alert requirements (rate limit spikes, ban spikes, rule hit spikes)
Runbook references (where to look, what to do)
Cardinality and cost controls

Optional Fields
Per-limit-id dashboards | OPTIONAL
Anomaly detection alerts | OPTIONAL
Sampling rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Metrics MUST use stable identifiers as labels (surface, scope_type, action_id) and avoid unbounded labels.
PII MUST be redacted in logs per: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Alerts MUST be actionable and mapped to procedures (runbooks) when available.

Output Format

1. Observability Scope
   surfaces: {{obs.surfaces}}
   envs: {{obs.envs}} | OPTIONAL
   notes: {{obs.notes}} | OPTIONAL
2. Metrics — Core
   limited_count: {{metrics.limited_count}}
   throttled_count: {{metrics.throttled_count}} | OPTIONAL
   rejected_count: {{metrics.rejected_count}} | OPTIONAL
   banned_count: {{metrics.banned_count}} | OPTIONAL
   exempted_count: {{metrics.exempted_count}} | OPTIONAL
   Dimensions (labels):
   surface: {{metrics.labels.surface}}
   scope_type: {{metrics.labels.scope_type}} | OPTIONAL
   enforcement_action: {{metrics.labels.enforcement_action}} | OPTIONAL
   limit_id: {{metrics.labels.limit_id}} | OPTIONAL
   rule_id: {{metrics.labels.rule_id}} | OPTIONAL
   env: {{metrics.labels.env}} | OPTIONAL
3. Metrics — Rule Hits & Actions
   rule_hits: {{metrics.rule_hits}} | OPTIONAL
   action_applied: {{metrics.action_applied}} | OPTIONAL
   severity_breakdown: {{metrics.severity_breakdown}} | OPTIONAL
4. Metrics — Exemptions
   active_exemptions: {{metrics.active_exemptions}} | OPTIONAL
   exempted_requests: {{metrics.exempted_requests}} | OPTIONAL
5. Logs — Required Fields
   timestamp: {{logs.fields.timestamp}}
   surface: {{logs.fields.surface}}
   scope_type: {{logs.fields.scope_type}} | OPTIONAL

scope_value_hash: {{logs.fields.scope_value_hash}} | OPTIONAL
limit_id: {{logs.fields.limit_id}} | OPTIONAL
rule_id: {{logs.fields.rule_id}} | OPTIONAL
action_id: {{logs.fields.action_id}} | OPTIONAL
decision: {{logs.fields.decision}} (throttle/reject/ban/exempt/allow)
reason_code: {{logs.fields.reason_code}} | OPTIONAL
request_id: {{logs.fields.request_id}} | OPTIONAL
trace_id: {{logs.fields.trace_id}} | OPTIONAL
redaction_applied: {{logs.fields.redaction_applied}} | OPTIONAL

6. Dashboards (Minimum)
   overview_dashboard: {{dashboards.overview}}
   abuse_dashboard: {{dashboards.abuse}} | OPTIONAL
   exemptions_dashboard: {{dashboards.exemptions}} | OPTIONAL
   top_keys_dashboard: {{dashboards.top_keys}} | OPTIONAL
   minimum_panels: {{dashboards.minimum_panels}}
7. Alerts
   rate_limit_spike: {{alerts.rate_limit_spike}}
   ban_spike: {{alerts.ban_spike}} | OPTIONAL
   rule_hit_spike: {{alerts.rule_hit_spike}} | OPTIONAL
   exemption_change_alert: {{alerts.exemption_change_alert}} | OPTIONAL
   threshold_model: {{alerts.threshold_model}} (static/baseline/UNKNOWN)
   routing_policy: {{alerts.routing_policy}} | OPTIONAL
   runbook_ref: {{alerts.runbook_ref}} | OPTIONAL
8. Cardinality & Cost Controls
   disallowed_labels: {{cost.disallowed_labels}} | OPTIONAL
   label_allowlist: {{cost.label_allowlist}} | OPTIONAL
   top_k_limits: {{cost.top_k_limits}} | OPTIONAL
   sampling_policy: {{cost.sampling_policy}} | OPTIONAL
9. References
   Global policy: {{xref:RLIM-01}}
   Limits catalog: {{xref:RLIM-02}} | OPTIONAL
   Signals/rules: {{xref:RLIM-03}} | OPTIONAL
   Actions matrix: {{xref:RLIM-04}} | OPTIONAL
   Exemptions: {{xref:RLIM-05}} | OPTIONAL

Cross-References
Upstream: {{xref:RLIM-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OPS-OBS}} | OPTIONAL, {{xref:ALRT-01}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-PII-REDACTION]}} | OPTIONAL,
{{standards.rules[STD-OBSERVABILITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define core metric/log fields and minimum dashboards; use UNKNOWN for

tooling specifics.
intermediate: Required. Add alerts and rule/action/exemption instrumentation.
advanced: Required. Add cardinality controls and cost/sampling policy.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, envs, throttled_count, rejected_count, banned_count, exempted_count, labels.*, rule_hits/action_applied/severity_breakdown, scope_value_hash, reason_code, request_id, trace_id, dashboards.*, alerts.*, cost.* , open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If logs required fields are UNKNOWN → block Completeness Gate.
If alerts.threshold_model is UNKNOWN → flag in Open Questions.

Completeness Gate
Gate ID: TMP-05.PRIMARY.RLIM
Pass conditions:
required_fields_present == true
metric_set_defined == true
log_fields_defined == true
dashboard_minimum_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Feature Flags & Config (FFCFG)

Feature Flags & Config (FFCFG)
FFCFG-01 Feature Flag Registry (by flag_id)
FFCFG-02 Flag Behavior Spec (default, targeting, kill-switch)
FFCFG-03 Rollout Plan (phased, canary, % ramp)
FFCFG-04 Config Matrix (env vars/settings by env)
FFCFG-05 Safe Degradation Rules (flag off behavior)
FFCFG-06 Audit & Change Control (who can flip, logging)

FFCFG-01

FFCFG-01 — Feature Flag Registry (by flag_id)
Header Block
template_id: FFCFG-01
title: Feature Flag Registry (by flag_id)
type: feature_flag_registry
template_version: 1.0.0
output_path: 10_app/feature_flags/FFCFG-01_Feature_Flag_Registry.md
compliance_gate_id: TMP-05.PRIMARY.FFCFG
upstream_dependencies: []
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical registry of all feature flags in the system, indexed by **flag_id**. This document must be consistent with the Canonical Spec and must not invent flag_ids not present in upstream inputs. If flags are missing, they must be marked UNKNOWN rather than invented.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Feature flag index (list of flag entries)
flag_id (stable identifier)
flag_name (human-readable)
flag_description (one paragraph)
flag_type (release/ops/experiment/kill_switch/UNKNOWN)
owner/team
default_state (on/off) baseline
environments covered (dev/stage/prod/other)
status (active/deprecated/retired)
related feature IDs (if applicable)
references (pointers to behavior spec FFCFG-02 and rollout plan FFCFG-03)

Optional Fields
Created_at / updated_at | OPTIONAL
Expiration / sunset date | OPTIONAL
Notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new flag_ids. Use only: {{spec.flags_by_id[flag_*]}} as given.
Each flag entry MUST be unique by flag_id.
Each flag entry SHOULD have a behavior spec reference ({{xref:FFCFG-02}}) and rollout plan reference ({{xref:FFCFG-03}}) or be marked UNKNOWN and flagged.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not restate full flag behavior here; behavior belongs in FFCFG-02.

Output Format

1. Registry Summary
   total_flags: {{ffcfg.registry.total_flags}}
   environments: {{ffcfg.registry.envs}} | OPTIONAL
   flag_types_in_use: {{ffcfg.registry.flag_types}} | OPTIONAL
   notes: {{ffcfg.registry.notes}} | OPTIONAL
2. Flag Index (by flag_id)
   For each flag, include the following entry block:

Flag
flag_id: {{flags[0].flag_id}}
flag_name: {{flags[0].flag_name}}
description: {{flags[0].description}}
flag_type: {{flags[0].flag_type}}
owner: {{flags[0].owner}}
default_state: {{flags[0].default_state}}
envs:
dev: {{flags[0].envs.dev}}
stage: {{flags[0].envs.stage}} | OPTIONAL
prod: {{flags[0].envs.prod}}
status: {{flags[0].status}}
related_feature_ids: {{flags[0].related_feature_ids}} | OPTIONAL
behavior_spec_ref: {{flags[0].behavior_spec_ref}} (expected: {{xref:FFCFG-02}}) | OPTIONAL
rollout_plan_ref: {{flags[0].rollout_plan_ref}} (expected: {{xref:FFCFG-03}}) | OPTIONAL
created_at: {{flags[0].created_at}} | OPTIONAL
sunset_date: {{flags[0].sunset_date}} | OPTIONAL
notes: {{flags[0].notes}} | OPTIONAL
open_questions:
{{flags[0].open_questions[0]}} | OPTIONAL

(Repeat the “Flag” entry block for each flag_id.)

3. References
   Flag behavior specs: {{xref:FFCFG-02}}

Rollout plans: {{xref:FFCFG-03}} | OPTIONAL
Config matrix: {{xref:FFCFG-04}} | OPTIONAL
Safe degradation rules: {{xref:FFCFG-05}} | OPTIONAL
Audit & change control: {{xref:FFCFG-06}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:DOMAIN_MAP}} | OPTIONAL, {{xref:STANDARDS_INDEX}} | OPTIONAL
Downstream: {{xref:FFCFG-02}}, {{xref:FFCFG-03}}, {{xref:FFCFG-04}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Fill required fields with UNKNOWN where needed; do not invent flag_ids.
intermediate: Required. Populate owner/type/status and behavior/rollout references when inputs exist.
advanced: Required. Add env-specific defaults and lifecycle details (deprecated/retired) with traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, stage env, related_feature_ids, behavior_spec_ref, rollout_plan_ref, created_at, updated_at, sunset_date, notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If flag_id list is UNKNOWN → block Completeness Gate.
If defaults.prod is UNKNOWN for any flag → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FFCFG
Pass conditions:
required_fields_present == true
all flag_ids are unique (no duplicates)
no new flag_ids introduced (only references to existing IDs)
placeholder_resolution == true
no_unapproved_unknowns == true

FFCFG-02

FFCFG-02 — Flag Behavior Spec (default, targeting, kill-switch)
Header Block
template_id: FFCFG-02
title: Flag Behavior Spec (default, targeting, kill-switch)
type: feature_flag_behavior_spec
template_version: 1.0.0
output_path: 10_app/feature_flags/FFCFG-02_Flag_Behavior_Spec.md
compliance_gate_id: TMP-05.PRIMARY.FFCFG
upstream_dependencies: ["FFCFG-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FFCFG-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical specification format for a single feature flag's behavior: default state, targeting rules, rollout controls, kill-switch behavior, safe degradation expectations, and audit requirements. This template must be consistent with the feature flag registry and must not invent flag_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.registry}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
flag_id (must exist in FFCFG-01)
flag_name (human-readable)
flag_description (one paragraph)
default_state (on/off) per environment
targeting model (who gets it)
targeting predicates (rules list)
rollout controls (percent ramp, allowlist, segments)
kill_switch behavior (what happens when forced off)
safe degradation rules (what features disable / fallback paths)
dependencies (other flags/configs)
audit/change control fields (who can change, logging required)
observability hooks (metrics/logs when evaluated)

Optional Fields
Owner/team | OPTIONAL
User messaging copy | OPTIONAL

Experiment binding (A/B test id) | OPTIONAL
Expiration / sunset date | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new flag_ids; use only those in {{xref:FFCFG-01}}.
Each flag spec MUST define default_state for every defined environment (or UNKNOWN).
Kill-switch MUST specify safe degradation behavior (or UNKNOWN flagged).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not restate rollout plan details that belong in {{xref:FFCFG-03}}; reference it.

Output Format

1. Flag Identity
   flag_id: {{flag.flag_id}}
   flag_name: {{flag.name}}
   description: {{flag.description}}
   owner: {{flag.owner}} | OPTIONAL
2. Defaults (by Environment)
   default_state:
   dev: {{defaults.dev}}
   stage: {{defaults.stage}} | OPTIONAL
   prod: {{defaults.prod}}
   other_envs: {{defaults.other}} | OPTIONAL
3. Targeting Model
   targeting_enabled: {{targeting.enabled}}
   targeting_type: {{targeting.type}} (segment/allowlist/attribute_rules/UNKNOWN)
   target_population: {{targeting.population}} | OPTIONAL
4. Targeting Predicates (Rules List)
   rule
   rule_id: {{rules[0].rule_id}}
   priority: {{rules[0].priority}} | OPTIONAL
   when: {{rules[0].when}} (predicate expression / UNKNOWN)
   then_state: {{rules[0].then_state}} (on/off)
   notes: {{rules[0].notes}} | OPTIONAL

(Repeat for each targeting rule.)

5. Rollout Controls
   rollout_mode: {{rollout.mode}} (off/on/percent_ramp/allowlist/UNKNOWN)
   ramp_percent: {{rollout.ramp_percent}} | OPTIONAL
   segments: {{rollout.segments}} | OPTIONAL
   allowlist: {{rollout.allowlist}} | OPTIONAL

ramp_schedule_ref: {{rollout.ramp_schedule_ref}} (expected: {{xref:FFCFG-03}}) | OPTIONAL

6. Kill-Switch Behavior
   kill_switch_supported: {{kill.supported}}
   kill_switch_trigger: {{kill.trigger}} (manual/auto/UNKNOWN)
   forced_state: {{kill.forced_state}} (off/on/UNKNOWN)
   kill_switch_actions:
   disable_features: {{kill.actions.disable_features}}
   fallback_path: {{kill.actions.fallback_path}} | OPTIONAL
   data_write_behavior: {{kill.actions.data_write_behavior}} | OPTIONAL
   user_visible_behavior: {{kill.actions.user_visible_behavior}} | OPTIONAL
7. Safe Degradation Rules
   when_off_behavior: {{degrade.when_off_behavior}}
   fallback_components: {{degrade.fallback_components}} | OPTIONAL
   api_behavior: {{degrade.api_behavior}} | OPTIONAL
   data_consistency_notes: {{degrade.data_consistency_notes}} | OPTIONAL
8. Dependencies
   depends_on_flags: {{deps.flags}} | OPTIONAL
   depends_on_config: {{deps.config}} | OPTIONAL
   conflicts_with: {{deps.conflicts_with}} | OPTIONAL
9. Audit & Change Control
   who_can_change: {{audit.who_can_change}}
   logging_required: {{audit.logging_required}}
   audit_fields: {{audit.fields}} | OPTIONAL
   change_control_ref: {{audit.change_control_ref}} | OPTIONAL
10. Observability Hooks
    evaluation_metric: {{obs.evaluation_metric}}
    exposure_metric: {{obs.exposure_metric}} | OPTIONAL
    logs_required_fields: {{obs.logs_required_fields}}
    trace_attrs: {{obs.trace_attrs}} | OPTIONAL
11. References
    Flag registry: {{xref:FFCFG-01}}
    Rollout plan: {{xref:FFCFG-03}} | OPTIONAL
    Safe degradation rules: {{xref:FFCFG-05}} | OPTIONAL
    Audit & change control: {{xref:FFCFG-06}} | OPTIONAL

Cross-References
Upstream: {{xref:FFCFG-01}}, {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:STANDARDS_INDEX}} | OPTIONAL
Downstream: {{xref:FFCFG-03}}, {{xref:FFCFG-05}}, {{xref:FFCFG-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where targeting/rollout details are missing; do not invent

flag_ids.
intermediate: Required. Define defaults, targeting predicates, and kill-switch behavior.
advanced: Required. Add safe degradation rigor and observability/audit completeness.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, owner, stage defaults, other_envs, targeting.population, rule priority/notes, segments, allowlist, ramp_schedule_ref, kill switch details, fallback components, api behavior, data consistency notes, deps.*, audit_fields, change_control_ref, exposure_metric, trace_attrs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If flag_id is UNKNOWN → block Completeness Gate.
If defaults.prod is UNKNOWN → block Completeness Gate.
If kill_switch_supported is true and degrade.when_off_behavior is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FFCFG
Pass conditions:
required_fields_present == true
flag_id exists in FFCFG-01 (no new IDs introduced)
default_state_defined_for_envs == true
placeholder_resolution == true
no_unapproved_unknowns == true

FFCFG-03

FFCFG-03 — Rollout Plan (phased, canary, % ramp)
Header Block
template_id: FFCFG-03
title: Rollout Plan (phased, canary, % ramp)
type: feature_flag_rollout_plan
template_version: 1.0.0
output_path: 10_app/feature_flags/FFCFG-03_Rollout_Plan.md
compliance_gate_id: TMP-05.PRIMARY.FFCFG
upstream_dependencies: ["FFCFG-01", "FFCFG-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FFCFG-01", "FFCFG-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rollout plan format for a feature flag, including staged deployment (dev → stage → prod), canary cohorts, percent ramp schedules, monitoring requirements, rollback steps, and exit criteria. This template must be consistent with the flag registry and behavior specs and must not invent flag_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.registry}}
FFCFG-02 Flag Behavior Specs: {{ffcfg.behavior_specs}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
flag_id (must exist in FFCFG-01)
rollout objective (why/what success means)
environments sequence (dev/stage/prod)
phases (ordered list)
phase entry criteria
phase ramp plan (% or cohort)
monitoring plan (what to watch)
rollback plan (what to do if issues)
kill-switch instructions
exit criteria (when rollout is “done”)
audit/change control requirements (who can flip, logging)

Optional Fields
Stakeholders/approvers | OPTIONAL
Communication plan | OPTIONAL

Experiment design notes | OPTIONAL
Sunset plan (remove flag) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new flag_ids; use only those in {{xref:FFCFG-01}}.
Rollout phases MUST be strictly ordered and must define entry criteria.
Rollback MUST reference the kill-switch behavior from {{xref:FFCFG-02}} (or be UNKNOWN flagged).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not restate full flag behavior; reference FFCFG-02.

Output Format

1. Rollout Identity
   flag_id: {{rollout.flag_id}}
   objective: {{rollout.objective}}
   owner: {{rollout.owner}} | OPTIONAL
2. Environment Sequence
   env_sequence:
   {{rollout.env_sequence[0]}}
   {{rollout.env_sequence[1]}} | OPTIONAL
   {{rollout.env_sequence[2]}}
   notes: {{rollout.env_notes}} | OPTIONAL
3. Phases (Ordered)
   Phase
   phase_id: {{phases[0].phase_id}}
   name: {{phases[0].name}}
   env: {{phases[0].env}}
   targeting: {{phases[0].targeting}} (cohort/%/allowlist/UNKNOWN)
   ramp:
   percent: {{phases[0].ramp.percent}} | OPTIONAL
   cohort: {{phases[0].ramp.cohort}} | OPTIONAL
   duration: {{phases[0].duration}} | OPTIONAL
   entry_criteria: {{phases[0].entry_criteria}}
   monitoring: {{phases[0].monitoring}}
   exit_criteria: {{phases[0].exit_criteria}}
   rollback_steps: {{phases[0].rollback_steps}}
   notes: {{phases[0].notes}} | OPTIONAL

(Repeat the “Phase” block for each rollout phase.)

4. Monitoring Plan (Global)
   metrics_to_watch:
   {{monitoring.metrics[0]}}
   {{monitoring.metrics[1]}}
   dashboards: {{monitoring.dashboards}} | OPTIONAL
   alert_refs: {{monitoring.alert_refs}} | OPTIONAL
5. Rollback Plan (Global)
   rollback_trigger_conditions: {{rollback.triggers}}
   rollback_actions: {{rollback.actions}}
   kill_switch_steps: {{rollback.kill_switch_steps}} (bind to {{xref:FFCFG-02}}) | OPTIONAL
   recovery_validation: {{rollback.recovery_validation}} | OPTIONAL
6. Exit Criteria
   rollout_done_when: {{exit.done_when}}
   post_rollout_steps: {{exit.post_steps}} | OPTIONAL
7. Audit & Change Control
   who_can_flip: {{audit.who_can_flip}}
   logging_required: {{audit.logging_required}}
   approval_required: {{audit.approval_required}} | OPTIONAL
   audit_fields: {{audit.fields}} | OPTIONAL
8. References
   Flag registry: {{xref:FFCFG-01}}
   Behavior spec: {{xref:FFCFG-02}}
   Config matrix: {{xref:FFCFG-04}} | OPTIONAL
   Safe degradation rules: {{xref:FFCFG-05}} | OPTIONAL
   Audit/change control: {{xref:FFCFG-06}} | OPTIONAL

Cross-References
Upstream: {{xref:FFCFG-01}}, {{xref:FFCFG-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FFCFG-04}}, {{xref:FFCFG-05}}, {{xref:FFCFG-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for metrics/criteria if not defined; do not invent phases.
intermediate: Required. Define phase ordering, monitoring, rollback, and kill-switch steps.
advanced: Required. Add crisp exit criteria and change control rigor with approvers and audit fields.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, owner, env_notes, ramp percent/cohort, duration, notes, dashboards, alert_refs, kill_switch_steps, recovery_validation, post_rollout_steps, approval_required, audit_fields, stakeholders, comms, experiment notes, sunset plan, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If phases list is UNKNOWN → block Completeness Gate.
If rollback plan is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FFCFG
Pass conditions:
required_fields_present == true
flag_id exists in FFCFG-01 (no new IDs introduced)
phase_order_defined == true
rollback_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FFCFG-04

FFCFG-04 — Config Matrix (env vars/settings by env)
Header Block
template_id: FFCFG-04
title: Config Matrix (env vars/settings by env)
type: config_matrix
template_version: 1.0.0
output_path: 10_app/feature_flags/FFCFG-04_Config_Matrix.md
compliance_gate_id: TMP-05.PRIMARY.FFCFG
upstream_dependencies: ["FFCFG-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FFCFG-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical matrix of configuration values by environment (env vars, runtime settings, service configs), including defaults, requiredness, secrets handling, and change control. This template must be consistent with the feature/config registry and must not invent config keys not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.registry}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Config key registry (config_id/config_key list)
Description/purpose for each config key
Type (string/int/bool/json/secret/UNKNOWN)
Requiredness (required/optional)
Default value policy (global default + env overrides)
Environment columns (dev/stage/prod/other)
Secret handling (where stored, not in repo)
Validation rules (format/range)
Change control policy (who can change, audit/logging)
Runtime reload behavior (hot reload vs restart required)
Dependencies (what relies on the config)

Optional Fields
Owner/team | OPTIONAL
Rotation policy (for secrets) | OPTIONAL

Deprecation/sunset plan | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new config keys; use only: {{spec.config_by_id[cfg_*]}} as given.
Secrets MUST NOT be stored as plaintext in this document (store references only).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
If a config is required in prod, prod value MUST be set (not UNKNOWN) unless explicitly allowed by standards.

Output Format

1. Matrix Summary
   environments: {{matrix.envs}}
   total_keys: {{matrix.total_keys}}
   notes: {{matrix.notes}} | OPTIONAL
2. Config Matrix (by config_id)
   Config
   config_id: {{configs[0].config_id}}
   config_key: {{configs[0].config_key}}
   description: {{configs[0].description}}
   type: {{configs[0].type}}
   required: {{configs[0].required}}
   default_policy: {{configs[0].default_policy}} | OPTIONAL
   validation: {{configs[0].validation}} | OPTIONAL
   reload_behavior: {{configs[0].reload_behavior}}
   secret_ref: {{configs[0].secret_ref}} | OPTIONAL
   dependencies: {{configs[0].dependencies}} | OPTIONAL

values_by_env:
dev: {{configs[0].values.dev}}
stage: {{configs[0].values.stage}} | OPTIONAL
prod: {{configs[0].values.prod}}
other_envs: {{configs[0].values.other}} | OPTIONAL

change_control:
who_can_change: {{configs[0].change_control.who_can_change}}
logging_required: {{configs[0].change_control.logging_required}}
approval_required: {{configs[0].change_control.approval_required}} | OPTIONAL

owner: {{configs[0].owner}} | OPTIONAL
rotation_policy: {{configs[0].rotation_policy}} | OPTIONAL
deprecation_plan: {{configs[0].deprecation_plan}} | OPTIONAL

open_questions:
{{configs[0].open_questions[0]}} | OPTIONAL

(Repeat the “Config” block for each config_id.)

3. References
   Feature flags: {{xref:FFCFG-01}} | OPTIONAL
   Audit/change control: {{xref:FFCFG-06}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:STANDARDS_INDEX}} | OPTIONAL
Downstream: {{xref:FFCFG-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-SECRETS]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where values are missing; do not invent config keys.
intermediate: Required. Populate env values and validation rules from inputs.
advanced: Required. Add reload behavior, dependencies, and change control rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, default_policy, validation, secret_ref (as reference only), dependencies, stage/other envs, approval_required, owner, rotation_policy, deprecation_plan, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If configs list is UNKNOWN → block Completeness Gate.
If required == true and values.prod is UNKNOWN → block Completeness Gate unless explicitly allowed by standards.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FFCFG
Pass conditions:
required_fields_present == true
no new config keys introduced (only references to existing IDs)
secret_values_not_in_plaintext == true
placeholder_resolution == true
no_unapproved_unknowns == true

FFCFG-05

FFCFG-05 — Safe Degradation Rules (flag off behavior)
Header Block
template_id: FFCFG-05
title: Safe Degradation Rules (flag off behavior)
type: feature_flag_safe_degradation
template_version: 1.0.0
output_path: 10_app/feature_flags/FFCFG-05_Safe_Degradation_Rules.md
compliance_gate_id: TMP-05.PRIMARY.FFCFG
upstream_dependencies: ["FFCFG-01", "FFCFG-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FFCFG-01", "FFCFG-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical safe degradation requirements when a feature flag is OFF (or kill-switched), including UI/API fallbacks, data write behavior, compatibility guarantees, and user experience rules. This template must be consistent with flag behavior specs and must not invent flag_ids or fallback behaviors not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.registry}}
FFCFG-02 Flag Behavior Specs: {{ffcfg.behavior_specs}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Degradation policy statement (system-wide expectations)
Per-flag degradation entries (flag_id binding)
When-off behavior (what is disabled)
Fallback path (UI routing/component fallback)
API behavior when off (error vs no-op vs alternate impl)
Data write behavior when off (block writes, queue, alternate storage)
Backward compatibility rules (old clients/new servers)
User messaging policy (what user sees)
Security implications (ensure disabled means inaccessible)
Observability hooks (metric/log for flag-off path usage)

Optional Fields
Per-environment degradation differences | OPTIONAL
Manual override behaviors | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new flag_ids; use only those in {{xref:FFCFG-01}}.
Safe degradation MUST prevent data corruption and avoid breaking core flows.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Security rule: if feature is off, protected operations MUST remain denied (no “hidden but callable” endpoints).
Do not restate full targeting/rollout details; those belong in FFCFG-02/03.

Output Format

1. Global Safe Degradation Policy
   principles:
   {{policy.principles[0]}}
   {{policy.principles[1]}}
   {{policy.principles[2]}}
   default_api_off_behavior: {{policy.default_api_off_behavior}}
   default_ui_off_behavior: {{policy.default_ui_off_behavior}}
   default_data_off_behavior: {{policy.default_data_off_behavior}}
2. Per-Flag Degradation Entries
   Entry
   flag_id: {{entries[0].flag_id}}
   disabled_features: {{entries[0].disabled_features}}
   ui_fallback: {{entries[0].ui_fallback}}
   api_off_behavior: {{entries[0].api_off_behavior}}
   data_write_behavior: {{entries[0].data_write_behavior}}
   compatibility_notes: {{entries[0].compatibility_notes}} | OPTIONAL
   user_message_policy: {{entries[0].user_message_policy}} | OPTIONAL
   security_notes: {{entries[0].security_notes}} | OPTIONAL
   observability:
   metric_off_path_hits: {{entries[0].observability.metric_off_path_hits}}
   logs_required_fields: {{entries[0].observability.logs_required_fields}} | OPTIONAL
   open_questions:
   {{entries[0].open_questions[0]}} | OPTIONAL

(Repeat the “Entry” block for each flag_id.)

3. Compatibility Rules
   client_server_matrix: {{compat.matrix}} | OPTIONAL
   old_client_behavior: {{compat.old_client_behavior}}
   new_server_behavior: {{compat.new_server_behavior}}
   version_gating_policy: {{compat.version_gating_policy}} | OPTIONAL
4. Observability Requirements (Global)
   metrics:

off_path_hits_total: {{obs.metrics.off_path_hits_total}}
off_path_hits_by_flag: {{obs.metrics.off_path_hits_by_flag}} | OPTIONAL
error_rate_when_off: {{obs.metrics.error_rate_when_off}} | OPTIONAL
dashboards: {{obs.dashboards}} | OPTIONAL
alerts: {{obs.alerts}} | OPTIONAL

5. References
   Flag registry: {{xref:FFCFG-01}}
   Behavior spec: {{xref:FFCFG-02}}
   Rollout plan: {{xref:FFCFG-03}} | OPTIONAL
   Config matrix: {{xref:FFCFG-04}} | OPTIONAL

Cross-References
Upstream: {{xref:FFCFG-01}}, {{xref:FFCFG-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FFCFG-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where fallback behavior is not defined; do not invent per-flag entries.
intermediate: Required. Define UI/API/data behaviors and ensure security-deny when off.
advanced: Required. Add compatibility matrix and robust observability requirements.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, compatibility_notes, user_message_policy, security_notes, logs_required_fields, compat.matrix, version_gating_policy, dashboards, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If entries list is UNKNOWN → block Completeness Gate.
If api_off_behavior is UNKNOWN for any entry → flag in Open Questions.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FFCFG
Pass conditions:
required_fields_present == true
all flag_ids exist in FFCFG-01 (no new IDs introduced)
off_behavior_defined_per_flag == true
placeholder_resolution == true
no_unapproved_unknowns == true

FFCFG-06

FFCFG-06 — Audit & Change Control (who can flip, logging)
Header Block
template_id: FFCFG-06
title: Audit & Change Control (who can flip, logging)
type: feature_flag_audit_change_control
template_version: 1.0.0
output_path: 10_app/feature_flags/FFCFG-06_Audit_Change_Control.md
compliance_gate_id: TMP-05.PRIMARY.FFCFG
upstream_dependencies: ["FFCFG-01", "FFCFG-02", "FFCFG-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FFCFG-01", "FFCFG-02", "FFCFG-03", "API-04", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical audit and change control policy for feature flags: who can change flags, what approvals are required, what must be logged, how changes are reviewed, and how emergency changes are handled. This template must be consistent with AuthZ rules and must not invent permissions or actors not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.registry}}
FFCFG-02 Flag Behavior Specs: {{ffcfg.behavior_specs}} | OPTIONAL
FFCFG-03 Rollout Plans: {{ffcfg.rollout_plans}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Authority model (who can change flags; required permissions)
Change types (create/update/delete/retire, targeting changes, ramp changes, kill-switch)
Approval policy (none/one-person/two-person/UNKNOWN)
Logging requirements (fields captured for each change)
Audit storage location / retention policy
Emergency change policy (break-glass)
Rollback policy for mistaken changes
Review cadence (periodic audit review)
Access to audit logs (who can view)
Observability requirements (flag change metrics/alerts)

Optional Fields
Ticket/change request linkage | OPTIONAL
Notifications policy (who gets notified) | OPTIONAL
Per-environment stricter controls | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new roles/permissions; bind to {{xref:API-04}}.
All flag changes MUST be logged with actor + before/after + reason (or UNKNOWN flagged).
Emergency changes MUST be audited and time-bounded if possible.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Audit trail format MUST bind to {{xref:ADMIN-03}} if present (or UNKNOWN).

Output Format

1. Authority Model
   who_can_change_flags: {{auth.who_can_change_flags}}
   required_permissions: {{auth.required_permissions}} | OPTIONAL
   env_restrictions: {{auth.env_restrictions}} | OPTIONAL
2. Change Types
   supported_change_types:
   {{changes.types[0]}}
   {{changes.types[1]}}
   {{changes.types[2]}}
3. Approval Policy
   approval_model: {{approval.model}} (none/one_person/two_person/UNKNOWN)
   approval_required_for: {{approval.required_for}} | OPTIONAL
   approvers: {{approval.approvers}} | OPTIONAL
4. Logging Requirements (Audit Record)
   audit_record_fields:
   timestamp: {{audit.fields.timestamp}}
   flag_id: {{audit.fields.flag_id}}
   change_type: {{audit.fields.change_type}}
   actor: {{audit.fields.actor}}
   env: {{audit.fields.env}} | OPTIONAL
   before: {{audit.fields.before}}
   after: {{audit.fields.after}}
   reason: {{audit.fields.reason}}
   request_id: {{audit.fields.request_id}} | OPTIONAL
   trace_id: {{audit.fields.trace_id}} | OPTIONAL
   ticket_ref: {{audit.fields.ticket_ref}} | OPTIONAL

audit_storage: {{audit.storage}}
retention: {{audit.retention}} | OPTIONAL

5. Emergency Change Policy (Break-Glass)
   break_glass_supported: {{emergency.supported}}
   who_can_break_glass: {{emergency.who}} | OPTIONAL
   allowed_changes: {{emergency.allowed_changes}} | OPTIONAL
   time_bounds: {{emergency.time_bounds}} | OPTIONAL
   postmortem_required: {{emergency.postmortem_required}} | OPTIONAL
6. Rollback Policy
   rollback_supported: {{rollback.supported}}
   rollback_method: {{rollback.method}} (revert/audit_replay/UNKNOWN)
   rollback_requirements: {{rollback.requirements}} | OPTIONAL
7. Review Cadence
   review_frequency: {{review.frequency}}
   review_owners: {{review.owners}} | OPTIONAL
   review_checklist: {{review.checklist}} | OPTIONAL
8. Audit Log Access
   who_can_view_audit: {{access.who_can_view_audit}}
   redaction_policy: {{access.redaction_policy}} | OPTIONAL
9. Observability Requirements
   metrics:
   flag_change_count: {{obs.metrics.flag_change_count}}
   kill_switch_activations: {{obs.metrics.kill_switch_activations}} | OPTIONAL
   unauthorized_change_attempts: {{obs.metrics.unauthorized_change_attempts}} | OPTIONAL
   alerts: {{obs.alerts}} | OPTIONAL
   notifications: {{obs.notifications}} | OPTIONAL
10. References
    Flag registry: {{xref:FFCFG-01}}
    Behavior specs: {{xref:FFCFG-02}} | OPTIONAL
    Rollout plans: {{xref:FFCFG-03}} | OPTIONAL
    AuthZ rules: {{xref:API-04}} | OPTIONAL
    Audit trail spec: {{xref:ADMIN-03}} | OPTIONAL

Cross-References
Upstream: {{xref:FFCFG-01}}, {{xref:API-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-03}} | OPTIONAL, {{xref:OPS-OBS}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-AUDIT]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for approvals/retention if missing; do not invent permissions.

intermediate: Required. Define logging fields and rollback/emergency policy.
advanced: Required. Add env-specific controls, review cadence, and alerting/notifications rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, required_permissions, env_restrictions, approval_required_for, approvers, retention, request_id/trace_id, ticket_ref, who_can_break_glass, allowed_changes, time_bounds, postmortem_required, rollback_requirements, review_owners, review_checklist, redaction_policy, alerts, notifications, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If audit_storage is UNKNOWN → block Completeness Gate.
If who_can_change_flags is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FFCFG
Pass conditions:
required_fields_present == true
authority_model_defined == true
audit_record_fields_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

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API Pagination/Filtering/Sorting (PFS)

API Pagination/Filtering/Sorting (PFS)
PFS-01 Query Contract (filter/sort operators, syntax)
PFS-02 Pagination Rules (cursor/offset, stability guarantees)
PFS-03 Default Ordering & Tie-Break Rules
PFS-04 Validation & Error Mapping (bad query → reason codes)
PFS-05 Performance Constraints (limits, indexed fields policy)

PFS-01

PFS-01 — Query Contract (filter/sort operators, syntax)
Header Block
template_id: PFS-01
title: Query Contract (filter/sort operators, syntax)
type: api_query_contract
template_version: 1.0.0
output_path: 10_app/api_query/PFS-01_Query_Contract.md
compliance_gate_id: TMP-05.PRIMARY.PFS
upstream_dependencies: ["API-01", "API-02", "API-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-01", "API-02", "API-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical query contract for API list/search endpoints: filter syntax, sort syntax, operator set, validation rules, reserved keywords, and how query errors map to API error policy. This template must be consistent with endpoint specs and must not invent query capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-02 Endpoint Specs: {{api.endpoint_specs}}
API-03 Error & Status Code Policy: {{api.error_policy}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Supported query parameters (filter, sort, page, cursor, limit, include, fields, etc.)
Filter grammar (syntax definition)
Supported filter operators (eq, ne, lt, lte, gt, gte, in, contains, prefix, exists, etc.)
Operator typing rules (numeric/date/string/boolean)
Sort grammar (single/multi-field, direction)
Field allowlist model (what fields are filterable/sortable)
Reserved keywords and escaping rules
Default behaviors (default sort, default limit)
Validation rules (bad operator, bad field, bad value)
Error mapping for invalid queries (bind to API-03)
Security constraints (prevent injection, prevent unbounded queries)

Optional Fields
Full-text search parameter contract | OPTIONAL

Nested field query support | OPTIONAL
Join/include expansion rules | OPTIONAL
Query complexity limits | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Field allowlist MUST be explicit; do not allow arbitrary field names unless explicitly supported.
Validation failures MUST map to {{xref:API-03}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Query contract MUST be applied consistently across all list endpoints (or deviations must be explicitly documented).

Output Format

1. Parameters (Top-Level)
   filter_param: {{query.params.filter}}
   sort_param: {{query.params.sort}}
   page_param: {{query.params.page}} | OPTIONAL
   cursor_param: {{query.params.cursor}} | OPTIONAL
   limit_param: {{query.params.limit}}
   include_param: {{query.params.include}} | OPTIONAL
   fields_param: {{query.params.fields}} | OPTIONAL
   search_param: {{query.params.search}} | OPTIONAL
2. Filter Grammar
   grammar: {{filter.grammar}}
   examples:
   {{filter.examples[0]}}
   {{filter.examples[1]}} | OPTIONAL
3. Filter Operators
   operators:
   eq: {{ops.eq}}
   ne: {{ops.ne}} | OPTIONAL
   lt: {{ops.lt}} | OPTIONAL
   lte: {{ops.lte}} | OPTIONAL
   gt: {{ops.gt}} | OPTIONAL
   gte: {{ops.gte}} | OPTIONAL
   in: {{ops.in}} | OPTIONAL
   contains: {{ops.contains}} | OPTIONAL
   prefix: {{ops.prefix}} | OPTIONAL
   exists: {{ops.exists}} | OPTIONAL

Operator typing rules:
string_ops: {{ops.typing.string_ops}}

number_ops: {{ops.typing.number_ops}} | OPTIONAL
date_ops: {{ops.typing.date_ops}} | OPTIONAL
bool_ops: {{ops.typing.bool_ops}} | OPTIONAL

4. Sort Grammar
   sort_syntax: {{sort.syntax}}
   multi_sort_supported: {{sort.multi_supported}}
   direction_tokens: {{sort.direction_tokens}} (asc/desc/UNKNOWN)
   examples:
   {{sort.examples[0]}}
   {{sort.examples[1]}} | OPTIONAL
5. Field Allowlist Model
   filterable_fields_source: {{fields.filterable_source}} (per-endpoint allowlist / global allowlist / UNKNOWN)
   sortable_fields_source: {{fields.sortable_source}}
   fields_format: {{fields.format}} (flat/nested_paths/UNKNOWN)
   allowlist_ref: {{fields.allowlist_ref}} | OPTIONAL
6. Reserved Keywords & Escaping
   reserved_tokens: {{reserved.tokens}}
   escape_rules: {{reserved.escape_rules}} | OPTIONAL
7. Defaults & Limits
   default_limit: {{limits.default_limit}}
   max_limit: {{limits.max_limit}} | OPTIONAL
   default_sort: {{limits.default_sort}}
   max_filter_clauses: {{limits.max_filter_clauses}} | OPTIONAL
   max_sort_fields: {{limits.max_sort_fields}} | OPTIONAL
8. Validation & Error Mapping
   invalid_field_behavior: {{validation.invalid_field_behavior}}
   invalid_operator_behavior: {{validation.invalid_operator_behavior}}
   invalid_value_behavior: {{validation.invalid_value_behavior}}
   error_code: {{validation.error_code}} (bind to {{xref:API-03}})
   status_code: {{validation.status_code}} (expected: 400)
   error_details_format: {{validation.error_details_format}} | OPTIONAL
9. Security Constraints
   injection_prevention: {{security.injection_prevention}}
   unbounded_query_prevention: {{security.unbounded_query_prevention}}
   pii_filtering_restrictions: {{security.pii_filtering_restrictions}} | OPTIONAL
10. References
    Endpoint specs using this contract: {{xref:API-02}}
    Pagination rules: {{xref:PFS-02}} | OPTIONAL
    Default ordering: {{xref:PFS-03}} | OPTIONAL
    Error mapping: {{xref:PFS-04}} | OPTIONAL
    Performance constraints: {{xref:PFS-05}} | OPTIONAL
    API error policy: {{xref:API-03}} | OPTIONAL

Cross-References
Upstream: {{xref:API-01}}, {{xref:API-02}}, {{xref:API-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PFS-02}}, {{xref:PFS-03}}, {{xref:PFS-04}}, {{xref:PFS-05}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for grammar specifics if not provided; do not invent capabilities.
intermediate: Required. Define operators, sort syntax, defaults, and validation mapping.
advanced: Required. Add explicit allowlist model and security/complexity limits.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, include/fields/search params, nested field support, query complexity limits, operator optional set, sort multi support, escape rules, max_* limits, error_details_format, pii filtering restrictions, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If filter.grammar is UNKNOWN → block Completeness Gate.
If validation.error_code is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PFS
Pass conditions:
required_fields_present == true
filter_grammar_defined == true
operator_set_defined == true
validation_mapping_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PFS-02

PFS-02 — Pagination Rules (cursor/offset, stability guarantees)
Header Block
template_id: PFS-02
title: Pagination Rules (cursor/offset, stability guarantees)
type: api_pagination_rules
template_version: 1.0.0
output_path: 10_app/api_query/PFS-02_Pagination_Rules.md
compliance_gate_id: TMP-05.PRIMARY.PFS
upstream_dependencies: ["PFS-01", "API-01", "API-02", "API-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PFS-01", "API-01", "API-02", "API-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical pagination rules for list endpoints, including cursor vs offset, parameter names, stability guarantees, tie-break rules, max limits, and error mapping. This template must be consistent with the query contract and endpoint specs and must not invent pagination modes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PFS-01 Query Contract: {{pfs.query_contract}}
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-02 Endpoint Specs: {{api.endpoint_specs}}
API-03 Error & Status Code Policy: {{api.error_policy}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Pagination mode (cursor/offset/UNKNOWN)
Parameter names (cursor/page/limit)
Cursor format rules (opaque vs structured)
Stability guarantees (consistent ordering across pages)
Default ordering requirement (bind to PFS-03)
Tie-break rule (required for stable ordering)
Max limit policy (max page size)
Backward/forward navigation behavior (next/prev)
Empty page behavior
Validation rules (bad cursor, bad page, bad limit)
Error mapping (bind to API-03 and PFS-04)

Optional Fields
Total count policy (whether total is returned) | OPTIONAL
Snapshot pagination (consistent snapshot) | OPTIONAL
Time-based pagination notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Pagination MUST be stable given the ordering defined in PFS-03 (or explicit per-endpoint override).
Cursor MUST be treated as opaque by clients unless explicitly defined otherwise.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Validation failures MUST map to {{xref:API-03}}.

Output Format

1. Pagination Mode
   mode: {{pagination.mode}} (cursor/offset/UNKNOWN)
   notes: {{pagination.notes}} | OPTIONAL
2. Parameters
   limit_param: {{pagination.params.limit}}
   cursor_param: {{pagination.params.cursor}} | OPTIONAL
   page_param: {{pagination.params.page}} | OPTIONAL
   direction_param: {{pagination.params.direction}} | OPTIONAL
3. Cursor Rules (if cursor mode)
   cursor_opaque: {{cursor.opaque}}
   cursor_encoding: {{cursor.encoding}} | OPTIONAL
   cursor_fields_included: {{cursor.fields_included}} | OPTIONAL
   cursor_expiry: {{cursor.expiry}} | OPTIONAL
4. Offset Rules (if offset mode)
   page_base: {{offset.page_base}} | OPTIONAL
   offset_param: {{offset.offset_param}} | OPTIONAL
   page_param: {{offset.page_param}} | OPTIONAL
5. Stability Guarantees
   stability_statement: {{stability.statement}}
   ordering_ref: {{stability.ordering_ref}} (expected: {{xref:PFS-03}})
   tie_break_rule: {{stability.tie_break_rule}}
   mutation_handling: {{stability.mutation_handling}} | OPTIONAL
6. Navigation Behavior
   next_page_rule: {{nav.next_page_rule}}
   prev_page_supported: {{nav.prev_supported}} | OPTIONAL
   prev_page_rule: {{nav.prev_page_rule}} | OPTIONAL
7. Limits & Defaults
   default_limit: {{limits.default_limit}}

max_limit: {{limits.max_limit}}
default_sort: {{limits.default_sort}} | OPTIONAL

8. Response Fields
   items_field: {{response.items_field}}
   next_cursor_field: {{response.next_cursor_field}} | OPTIONAL
   prev_cursor_field: {{response.prev_cursor_field}} | OPTIONAL
   has_more_field: {{response.has_more_field}} | OPTIONAL
   total_count_field: {{response.total_count_field}} | OPTIONAL
9. Validation & Error Mapping
   invalid_cursor_behavior: {{validation.invalid_cursor_behavior}}
   invalid_page_behavior: {{validation.invalid_page_behavior}} | OPTIONAL
   invalid_limit_behavior: {{validation.invalid_limit_behavior}}
   error_code: {{validation.error_code}} (bind to {{xref:PFS-04}} / {{xref:API-03}})
   status_code: {{validation.status_code}} (expected: 400)
10. References
    Query contract: {{xref:PFS-01}}
    Default ordering: {{xref:PFS-03}} | OPTIONAL
    Validation/error mapping: {{xref:PFS-04}} | OPTIONAL
    Performance constraints: {{xref:PFS-05}} | OPTIONAL
    API error policy: {{xref:API-03}} | OPTIONAL

Cross-References
Upstream: {{xref:PFS-01}}, {{xref:API-01}}, {{xref:API-02}}, {{xref:API-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PFS-03}}, {{xref:PFS-04}}, {{xref:PFS-05}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for cursor specifics if not defined; do not invent modes.
intermediate: Required. Define stability guarantees and tie-break rule.
advanced: Required. Add response field contract and total-count/snapshot policy if supported.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, pagination.notes, direction_param, cursor_encoding, cursor_fields_included, cursor_expiry, offset rules, mutation_handling, prev page rules, total_count_field, snapshot pagination, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If pagination.mode is UNKNOWN → block Completeness Gate.
If stability.tie_break_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PFS
Pass conditions:

required_fields_present == true
pagination_mode_defined == true
stability_guarantees_defined == true
tie_break_rule_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

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PFS-03

PFS-03 — Default Ordering & Tie-Break Rules
Header Block
template_id: PFS-03
title: Default Ordering & Tie-Break Rules
type: api_default_ordering
template_version: 1.0.0
output_path: 10_app/api_query/PFS-03_Default_Ordering_Tie_Break_Rules.md
compliance_gate_id: TMP-05.PRIMARY.PFS
upstream_dependencies: ["PFS-01", "PFS-02", "API-01", "API-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PFS-01", "PFS-02", "API-01", "API-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical default ordering rules for list endpoints, including the required tie-break rule to guarantee stable pagination and deterministic results when multiple records share the same primary sort value. This template must be consistent with query and pagination contracts and must not invent ordering guarantees not supported by upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PFS-01 Query Contract: {{pfs.query_contract}}
PFS-02 Pagination Rules: {{pfs.pagination_rules}}
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-02 Endpoint Specs: {{api.endpoint_specs}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Default ordering policy statement (system-wide)
Primary ordering fields (per resource/list type)
Direction defaults (asc/desc)
Tie-break field (stable unique field)
Null ordering policy (nulls first/last)
Ordering allowlist (what can be sorted) binding to PFS-01
Override rules (when endpoints can override defaults)
Consistency rules with cursor pagination (cursor includes ordering fields)

Optional Fields
Per-endpoint ordering overrides | OPTIONAL
Locale/collation rules | OPTIONAL

Case sensitivity rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Tie-break field MUST be unique and stable (e.g., id).
Ordering MUST be stable across pages as required by PFS-02.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
If ordering overrides exist, they MUST be explicit in API-02 per endpoint.

Output Format

1. Global Default Ordering Policy
   policy_statement: {{ordering.policy_statement}}
   null_ordering: {{ordering.null_ordering}}
   case_sensitivity: {{ordering.case_sensitivity}} | OPTIONAL
   collation_rules: {{ordering.collation_rules}} | OPTIONAL
2. Default Ordering By Resource/List Type
   Ordering Rule
   list_type: {{rules[0].list_type}} (resource/entity name or endpoint class)
   primary_sort_field: {{rules[0].primary_sort_field}}
   direction: {{rules[0].direction}}
   tie_break_field: {{rules[0].tie_break_field}}
   secondary_sort_fields: {{rules[0].secondary_sort_fields}} | OPTIONAL
   notes: {{rules[0].notes}} | OPTIONAL

(Repeat the “Ordering Rule” block for each list type.)

3. Sorting Allowlist Binding
   sortable_fields_source: {{allowlist.sortable_fields_source}} (PFS-01 per-endpoint allowlist / global / UNKNOWN)
   allowlist_ref: {{allowlist.ref}} | OPTIONAL
4. Override Rules
   when_overrides_allowed: {{override.when_allowed}}
   how_to_specify_override: {{override.how_to_specify}} (bind to API-02)
   override_validation: {{override.validation}} | OPTIONAL
5. Cursor Consistency Rules
   cursor_must_include: {{cursor.must_include}} (ordering fields + tie-break)
   cursor_change_behavior: {{cursor.change_behavior}} | OPTIONAL
6. References
   Query contract: {{xref:PFS-01}}
   Pagination rules: {{xref:PFS-02}}
   Validation/error mapping: {{xref:PFS-04}} | OPTIONAL

Performance constraints: {{xref:PFS-05}} | OPTIONAL
Endpoint specs: {{xref:API-02}}

Cross-References
Upstream: {{xref:PFS-01}}, {{xref:PFS-02}}, {{xref:API-01}}, {{xref:API-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PFS-04}}, {{xref:PFS-05}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for list_type ordering if not defined; do not invent unique keys.
intermediate: Required. Define primary + tie-break ordering rules for each list type.
advanced: Required. Add override validation rules and cursor consistency details.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, secondary_sort_fields, notes, allowlist_ref, override.validation, cursor.change_behavior, collation rules, case sensitivity, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If any rules[*].tie_break_field is UNKNOWN → block Completeness Gate.
If ordering.policy_statement is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PFS
Pass conditions:
required_fields_present == true
tie_break_defined_for_each_list_type == true
ordering_consistent_with_pfs02 == true
placeholder_resolution == true
no_unapproved_unknowns == true

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PFS-04

PFS-04 — Validation & Error Mapping (bad query → reason codes)
Header Block
template_id: PFS-04
title: Validation & Error Mapping (bad query → reason codes)
type: api_query_validation_error_mapping
template_version: 1.0.0
output_path: 10_app/api_query/PFS-04_Validation_Error_Mapping.md
compliance_gate_id: TMP-05.PRIMARY.PFS
upstream_dependencies: ["PFS-01", "PFS-02", "API-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PFS-01", "PFS-02", "API-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical validation rules and error mapping for query, filtering, sorting, and pagination parameters, including stable reason codes and how errors are represented under the API error policy. This template must be consistent with PFS query/pagination contracts and must not invent error codes beyond what upstream inputs allow.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PFS-01 Query Contract: {{pfs.query_contract}}
PFS-02 Pagination Rules: {{pfs.pagination_rules}}
API-03 Error & Status Code Policy: {{api.error_policy}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Validation scope (which params are validated)
Reason code registry (reason_code list)
Validation failure categories (invalid_field, invalid_operator, invalid_value, invalid_cursor, etc.)
Error status mapping (expected 400)
Error code mapping (bind to API-03)
Error details format for query errors (field/operator/value pointers)
Examples of common failures (optional but recommended for clarity)
Security validation rules (injection prevention, excessive complexity)

Optional Fields
Per-endpoint overrides | OPTIONAL
Localization of messages | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Reason codes MUST be stable and MUST NOT be renamed once published.
All query validation failures MUST map to API-03 error envelope/policy.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not invent new API error codes; reason codes are internal classification under the mapped API-03 error_code.

Output Format

1. Validation Scope
   validated_params: {{validation.params}}
   applies_to_endpoints: {{validation.applies_to}} | OPTIONAL
2. Reason Code Registry
   Reason
   reason_code: {{reasons[0].reason_code}}
   name: {{reasons[0].name}}
   category: {{reasons[0].category}}
   trigger: {{reasons[0].trigger}}
   details_fields: {{reasons[0].details_fields}} | OPTIONAL
   message_template: {{reasons[0].message_template}} | OPTIONAL

(Repeat for each reason_code.)

3. Failure Categories
   invalid_field: {{categories.invalid_field}}
   invalid_operator: {{categories.invalid_operator}}
   invalid_value: {{categories.invalid_value}}
   invalid_sort: {{categories.invalid_sort}} | OPTIONAL
   invalid_cursor: {{categories.invalid_cursor}} | OPTIONAL
   invalid_limit: {{categories.invalid_limit}}
   invalid_page: {{categories.invalid_page}} | OPTIONAL
   query_too_complex: {{categories.query_too_complex}} | OPTIONAL
4. Error Mapping (API-03 Binding)
   status_code: {{mapping.status_code}} (expected: 400)
   error_code: {{mapping.error_code}} (bind to {{xref:API-03}})
   error_category: {{mapping.error_category}} (expected: validation) | OPTIONAL
   retryable: {{mapping.retryable}} (expected: false) | OPTIONAL
5. Error Details Format
   details:
   param: {{details.param}}
   field: {{details.field}} | OPTIONAL
   operator: {{details.operator}} | OPTIONAL
   value: {{details.value}} | OPTIONAL

reason_code: {{details.reason_code}}
hint: {{details.hint}} | OPTIONAL

6. Security Validation Rules
   injection_checks: {{security.injection_checks}}
   complexity_limits: {{security.complexity_limits}} | OPTIONAL
   allowlist_enforcement: {{security.allowlist_enforcement}}
   reserved_token_handling: {{security.reserved_token_handling}} | OPTIONAL
7. Examples (Optional)
   example_invalid_field: {{examples.invalid_field}} | OPTIONAL
   example_invalid_operator: {{examples.invalid_operator}} | OPTIONAL
   example_invalid_cursor: {{examples.invalid_cursor}} | OPTIONAL
8. References
   Query contract: {{xref:PFS-01}}
   Pagination rules: {{xref:PFS-02}}
   Default ordering: {{xref:PFS-03}} | OPTIONAL
   Performance constraints: {{xref:PFS-05}} | OPTIONAL
   API error policy: {{xref:API-03}}

Cross-References
Upstream: {{xref:PFS-01}}, {{xref:PFS-02}}, {{xref:API-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PFS-05}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for reason codes if not defined; do not invent new API error codes.
intermediate: Required. Define reason registry and error details format.
advanced: Required. Add security/complexity validation and examples.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, applies_to_endpoints, details_fields, message_template, invalid_sort, invalid_cursor, invalid_page, query_too_complex, complexity_limits, reserved_token_handling, examples, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If mapping.error_code is UNKNOWN → block Completeness Gate.
If reason_code registry is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PFS
Pass conditions:
required_fields_present == true
reason_code_registry_defined == true

error_mapping_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PFS-05

PFS-05 — Performance Constraints (limits, indexed fields policy)
Header Block
template_id: PFS-05
title: Performance Constraints (limits, indexed fields policy)
type: api_query_performance_constraints
template_version: 1.0.0
output_path: 10_app/api_query/PFS-05_Performance_Constraints.md
compliance_gate_id: TMP-05.PRIMARY.PFS
upstream_dependencies: ["PFS-01", "PFS-02", "PFS-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PFS-01", "PFS-02", "PFS-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical performance constraints for queryable endpoints, including max page sizes, query complexity limits, indexed fields policy for filters/sorts, and safeguards against expensive scans. This template must be consistent with query/pagination contracts and must not invent datastore/index capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PFS-01 Query Contract: {{pfs.query_contract}}
PFS-02 Pagination Rules: {{pfs.pagination_rules}}
PFS-03 Default Ordering Rules: {{pfs.ordering_rules}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Performance budget statement (API query performance expectations)
Max limit policy (hard cap)
Complexity constraints (max filters, max sorts, max joins/includes)
Indexed fields policy (what fields must be indexed to be filterable/sortable)
Full-scan prevention rules (deny or degrade)
Timeout policy (query timeout thresholds)
Safe defaults (default limit, default ordering)
Caching policy (if any)
Per-endpoint performance overrides model
Observability requirements (slow query logs, p95 latency, query plan sampling)

Optional Fields
Search engine integration notes | OPTIONAL
Adaptive query throttling | OPTIONAL

Warm cache vs cold cache notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Fields MUST NOT be marked filterable/sortable unless they comply with indexed fields policy (or explicit override).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Performance failures SHOULD map to API-03 as dependency/internal per policy (or validation if rejected up-front).

Output Format

1. Performance Budget
   latency_targets: {{budget.latency_targets}}
   throughput_targets: {{budget.throughput_targets}} | OPTIONAL
   notes: {{budget.notes}} | OPTIONAL
2. Hard Limits
   max_limit: {{limits.max_limit}}
   default_limit: {{limits.default_limit}}
   max_filter_clauses: {{limits.max_filter_clauses}}
   max_sort_fields: {{limits.max_sort_fields}}
   max_include_depth: {{limits.max_include_depth}} | OPTIONAL
3. Complexity Constraints
   allowed_operators: {{complexity.allowed_operators}} | OPTIONAL
   disallowed_patterns: {{complexity.disallowed_patterns}} | OPTIONAL
   query_cost_model: {{complexity.cost_model}} (static/estimated/UNKNOWN)
   reject_when: {{complexity.reject_when}}
   degrade_when: {{complexity.degrade_when}} | OPTIONAL
4. Indexed Fields Policy
   policy_statement: {{index.policy_statement}}
   filter_index_required: {{index.filter_index_required}}
   sort_index_required: {{index.sort_index_required}}
   allowed_filter_fields: {{index.allowed_filter_fields}} | OPTIONAL
   allowed_sort_fields: {{index.allowed_sort_fields}} | OPTIONAL
   override_process: {{index.override_process}} | OPTIONAL
5. Full-Scan Prevention
   full_scan_detection: {{scan.detection}}
   behavior_on_full_scan: {{scan.behavior}} (reject/degrade/UNKNOWN)
   error_mapping_ref: {{scan.error_mapping_ref}} | OPTIONAL
6. Timeout Policy
   query_timeout_ms: {{timeouts.query_timeout_ms}}
   db_timeout_ms: {{timeouts.db_timeout_ms}} | OPTIONAL
   client_timeout_ms: {{timeouts.client_timeout_ms}} | OPTIONAL

7. Caching Policy
   cache_supported: {{cache.supported}}
   cache_keys: {{cache.keys}} | OPTIONAL
   ttl_seconds: {{cache.ttl_seconds}} | OPTIONAL
   invalidation_rules: {{cache.invalidation_rules}} | OPTIONAL
8. Per-Endpoint Overrides Model
   override_allowed: {{override.allowed}}
   how_to_specify: {{override.how_to_specify}} (bind to API-02)
   required_justification: {{override.required_justification}} | OPTIONAL
9. Observability Requirements
   metrics:
   p50_latency: {{obs.metrics.p50_latency}}
   p95_latency: {{obs.metrics.p95_latency}}
   slow_query_count: {{obs.metrics.slow_query_count}} | OPTIONAL
   rejected_query_count: {{obs.metrics.rejected_query_count}} | OPTIONAL
   logs:
   slow_query_log_fields: {{obs.logs.slow_query_log_fields}}
   query_plan_sampling: {{obs.logs.query_plan_sampling}} | OPTIONAL
   dashboards: {{obs.dashboards}} | OPTIONAL
   alerts: {{obs.alerts}} | OPTIONAL
10. References
    Query contract: {{xref:PFS-01}}
    Pagination rules: {{xref:PFS-02}}
    Default ordering: {{xref:PFS-03}}
    Validation/error mapping: {{xref:PFS-04}} | OPTIONAL
    API error policy: {{xref:API-03}} | OPTIONAL

Cross-References
Upstream: {{xref:PFS-01}}, {{xref:PFS-02}}, {{xref:PFS-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:API-02}} | OPTIONAL, {{xref:OPS-OBS}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-OBSERVABILITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define hard limits and timeouts; use UNKNOWN for datastore specifics.
intermediate: Required. Define indexed fields policy and full-scan prevention behavior.
advanced: Required. Add cost model, override process, and observability/alerting rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, throughput_targets, max_include_depth, allowed_operators, disallowed_patterns, cost_model, degrade_when, allowed_filter_fields, allowed_sort_fields, override_process, error_mapping_ref, db/client timeouts, cache keys/ttl/invalidation, required_justification, query_plan_sampling, dashboards, alerts, open_questions

If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If limits.max_limit is UNKNOWN → block Completeness Gate.
If index.policy_statement is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PFS
Pass conditions:
required_fields_present == true
hard_limits_defined == true
indexed_fields_policy_defined == true
timeout_policy_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

File Processing & Media Pipelines (FPMP)

File Processing & Media Pipelines (FPMP)
FPMP-01 Upload Contract (types, limits, auth, scanning)
FPMP-02 Storage Strategy (buckets, paths, access modes)
FPMP-03 Processing Pipeline Stages (transcode/resize/parse)
FPMP-04 Async Processing & Status Model (jobs, retries, DLQ)
FPMP-05 CDN/Delivery Rules (cache headers, variants)
FPMP-06 Security & Compliance for Files (PII, retention, malware)
FPMP-07 Media Observability (latency, failure rates, QoS)

FPMP-01

FPMP-01 — Upload Contract (types, limits, auth, scanning)
Header Block
template_id: FPMP-01
title: Upload Contract (types, limits, auth, scanning)
type: file_processing_upload_contract
template_version: 1.0.0
output_path: 10_app/file_processing/FPMP-01_Upload_Contract.md
compliance_gate_id: TMP-05.PRIMARY.FPMP
upstream_dependencies: ["API-01", "API-02", "API-03", "API-04", "RLIM-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-01", "API-02", "API-03", "API-04", "RLIM-01", "FFCFG-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical upload contract for files/media, including accepted types, size limits, authentication/authorization, scanning/validation, upload flow (direct vs signed URL), error mapping, and observability. This template must be consistent with API security policies and must not invent upload capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-02 Endpoint Specs: {{api.endpoint_specs}}
API-03 Error & Status Code Policy: {{api.error_policy}}
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.registry}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Upload surface(s) (endpoints and/or presigned URL flow)
Authentication requirement (who can upload)
Authorization requirement (what perms/roles)
Accepted file types (mime/extension allowlist)
Size limits (max bytes per file, per request)
Count limits (max files per request)
Upload flow (multipart/form-data, chunked, direct-to-storage)
Integrity checks (checksum, content-length validation)
Scanning policy (malware/virus scanning)
PII/compliance classification rules (what’s allowed)
Error mapping (validation vs forbidden vs too-large)

Rate limiting policy for uploads
Observability requirements (upload success/fail, scan outcomes)

Optional Fields
Chunking/resumable upload rules | OPTIONAL
Client-side prevalidation rules | OPTIONAL
Encryption requirements | OPTIONAL
Quarantine behavior | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Accepted types MUST be allowlisted (no “any file” unless explicitly allowed).
If scanning is required, files MUST NOT be made available until scan passes (or explicit policy).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
AuthZ MUST bind to {{xref:API-04}}; rate limits bind to {{xref:RLIM-01}}.
Errors MUST map to {{xref:API-03}}.

Output Format

1. Upload Surfaces
   upload_endpoints: {{upload.surfaces.endpoints}}
   presigned_flow_supported: {{upload.surfaces.presigned_supported}} | OPTIONAL
   chunked_upload_supported: {{upload.surfaces.chunked_supported}} | OPTIONAL
   notes: {{upload.surfaces.notes}} | OPTIONAL
2. AuthN/AuthZ
   auth_required: {{auth.required}}
   required_permissions: {{auth.required_permissions}} | OPTIONAL
   resource_scope_rules: {{auth.resource_scope_rules}} | OPTIONAL
   deny_behavior_ref: {{auth.deny_behavior_ref}} | OPTIONAL
3. Accepted Types
   mime_allowlist:
   {{types.mime[0]}}
   {{types.mime[1]}}
   extension_allowlist:
   {{types.extensions[0]}}
   {{types.extensions[1]}}
   content_sniffing_policy: {{types.sniffing_policy}} | OPTIONAL
4. Limits
   max_file_bytes: {{limits.max_file_bytes}}
   max_request_bytes: {{limits.max_request_bytes}} | OPTIONAL
   max_files_per_request: {{limits.max_files_per_request}}
   max_concurrent_uploads: {{limits.max_concurrent_uploads}} | OPTIONAL

5. Upload Flow
   request_format: {{flow.request_format}} (multipart/chunked/json+base64/UNKNOWN)
   direct_to_storage: {{flow.direct_to_storage}} | OPTIONAL
   presign_endpoint: {{flow.presign_endpoint}} | OPTIONAL
   required_headers: {{flow.required_headers}} | OPTIONAL
   checksum_required: {{flow.checksum_required}} | OPTIONAL
   resumable_rules: {{flow.resumable_rules}} | OPTIONAL
6. Integrity & Validation
   checksum_algo: {{integrity.checksum_algo}} | OPTIONAL
   content_length_required: {{integrity.content_length_required}}
   file_name_rules: {{integrity.file_name_rules}} | OPTIONAL
   metadata_validation: {{integrity.metadata_validation}} | OPTIONAL
7. Scanning & Quarantine
   scan_required: {{scan.required}}
   scan_type: {{scan.type}} (malware/virus/content_policy/UNKNOWN)
   scan_timing: {{scan.timing}} (pre_store/post_store_pre_release/UNKNOWN)
   quarantine_location: {{scan.quarantine_location}} | OPTIONAL
   release_condition: {{scan.release_condition}}
   failed_scan_behavior: {{scan.failed_behavior}}
8. Compliance & PII
   pii_policy_ref: {{compliance.pii_policy_ref}} | OPTIONAL
   allowed_content_rules: {{compliance.allowed_content_rules}}
   retention_policy: {{compliance.retention_policy}} | OPTIONAL
9. Rate Limits
   ratelimit_ref: {{ratelimit.ref}} (expected: {{xref:RLIM-01}}) | OPTIONAL
   upload_specific_limits: {{ratelimit.upload_specific_limits}} | OPTIONAL
10. Error Mapping
    too_large_error: {{errors.too_large_error}}
    invalid_type_error: {{errors.invalid_type_error}}
    scan_failed_error: {{errors.scan_failed_error}} | OPTIONAL
    unauthorized_error: {{errors.unauthorized_error}} | OPTIONAL
    forbidden_error: {{errors.forbidden_error}} | OPTIONAL
    status_codes: {{errors.status_codes}} | OPTIONAL
11. Observability Requirements
    metrics:
    upload_attempts: {{obs.metrics.upload_attempts}}
    upload_success: {{obs.metrics.upload_success}}
    upload_failures: {{obs.metrics.upload_failures}}
    scan_pass: {{obs.metrics.scan_pass}} | OPTIONAL
    scan_fail: {{obs.metrics.scan_fail}} | OPTIONAL
    latency: {{obs.metrics.latency}} | OPTIONAL
    logs_required_fields: {{obs.logs_required_fields}}
    alerts: {{obs.alerts}} | OPTIONAL

12. References
    Storage strategy: {{xref:FPMP-02}} | OPTIONAL
    Processing stages: {{xref:FPMP-03}} | OPTIONAL
    Async status model: {{xref:FPMP-04}} | OPTIONAL
    Security/compliance: {{xref:FPMP-06}} | OPTIONAL
    Observability: {{xref:FPMP-07}} | OPTIONAL
    API error policy: {{xref:API-03}} | OPTIONAL
    AuthZ: {{xref:API-04}} | OPTIONAL
    Rate limits: {{xref:RLIM-01}} | OPTIONAL

Cross-References
Upstream: {{xref:API-01}}, {{xref:API-02}}, {{xref:API-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FPMP-02}}, {{xref:FPMP-03}}, {{xref:FPMP-04}}, {{xref:FPMP-06}}, {{xref:FPMP-07}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-PII-CLASSIFICATION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where scanning/flow details are missing; do not invent allowed types.
intermediate: Required. Define allowlists, limits, scan policy, and error mapping.
advanced: Required. Add presigned/chunked support rules and full observability/alerting.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, presigned/chunked support, sniffing_policy, max_request_bytes, max_concurrent_uploads, direct_to_storage, presign_endpoint, required_headers, checksum_required, resumable_rules, checksum_algo, file_name_rules, metadata_validation, scan_type, quarantine_location, pii_policy_ref, retention_policy, ratelimit.ref, upload_specific_limits, scan_failed_error, status_codes, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If mime_allowlist is UNKNOWN → block Completeness Gate.
If limits.max_file_bytes is UNKNOWN → block Completeness Gate.
If scan_required is UNKNOWN → flag in Open Questions.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FPMP
Pass conditions:
required_fields_present == true
type_allowlist_defined == true
limits_defined == true
error_mapping_defined == true

placeholder_resolution == true
no_unapproved_unknowns == true

FPMP-02

FPMP-02 — Storage Strategy (buckets, paths, access modes)
Header Block
template_id: FPMP-02
title: Storage Strategy (buckets, paths, access modes)
type: file_processing_storage_strategy
template_version: 1.0.0
output_path: 10_app/file_processing/FPMP-02_Storage_Strategy.md
compliance_gate_id: TMP-05.PRIMARY.FPMP
upstream_dependencies: ["FPMP-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FPMP-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical storage strategy for uploaded files/media, including bucket/container layout, object key/path conventions, access modes, signed URL policies, encryption, retention, and lifecycle rules. This template must be consistent with the upload contract and must not invent storage capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FPMP-01 Upload Contract: {{fpmp.upload_contract}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Storage provider/surface (bucket/blob store/UNKNOWN)
Bucket/container inventory (by bucket_id)
Path/key naming convention (prefix rules)
Tenant scoping in paths (org/project/user boundaries)
Access modes (private/public/signed)
Signed URL policy (who can mint, TTL, scope)
Encryption policy (at rest/in transit)
Retention and lifecycle rules (expire/archive/delete)
Metadata model (what metadata stored alongside objects)
Versioning policy (overwrite vs versioned objects)
Audit requirements for access (downloads)
Observability requirements (storage errors, access logs)

Optional Fields
Multi-region replication | OPTIONAL
CDN integration pointer | OPTIONAL

Cold storage tiers | OPTIONAL
Legal hold policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Paths MUST include tenant scope identifiers if multi-tenant.
Access MUST default to private unless explicitly allowed.
Signed URL minting MUST obey AuthZ policy ({{xref:API-04}}) | OPTIONAL
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not store secrets; store references only.

Output Format

1. Storage Provider
   provider: {{storage.provider}} (s3/gcs/r2/azure_blob/UNKNOWN)
   notes: {{storage.notes}} | OPTIONAL
2. Buckets/Containers Inventory
   Bucket
   bucket_id: {{buckets[0].bucket_id}}
   name: {{buckets[0].name}}
   purpose: {{buckets[0].purpose}}
   region: {{buckets[0].region}} | OPTIONAL
   encryption: {{buckets[0].encryption}} | OPTIONAL
   versioning: {{buckets[0].versioning}} | OPTIONAL
   lifecycle_rules_ref: {{buckets[0].lifecycle_rules_ref}} | OPTIONAL
   notes: {{buckets[0].notes}} | OPTIONAL

(Repeat for each bucket_id.)

3. Path / Key Naming Convention
   base_prefix: {{paths.base_prefix}}
   tenant_prefix_rule: {{paths.tenant_prefix_rule}}
   object_key_template: {{paths.object_key_template}}
   allowed_chars: {{paths.allowed_chars}} | OPTIONAL
   collision_policy: {{paths.collision_policy}} (uuid/nonce/content_hash/UNKNOWN)
   examples:
   {{paths.examples[0]}}
   {{paths.examples[1]}} | OPTIONAL
4. Tenant Scoping
   org_scoping: {{scoping.org_scoping}}
   project_scoping: {{scoping.project_scoping}} | OPTIONAL
   user_scoping: {{scoping.user_scoping}} | OPTIONAL

5. Access Modes
   default_access: {{access.default}} (private/UNKNOWN)
   public_objects_allowed: {{access.public_allowed}}
   signed_urls_supported: {{access.signed_urls_supported}}
   access_control_model: {{access.control_model}}
   (bucket_policy/app_enforced/UNKNOWN)
6. Signed URL Policy
   who_can_mint: {{signed.who_can_mint}}
   mint_endpoint_ref: {{signed.mint_endpoint_ref}} | OPTIONAL
   ttl_seconds: {{signed.ttl_seconds}}
   scope_constraints: {{signed.scope_constraints}}
   single_use: {{signed.single_use}} | OPTIONAL
   revocation_supported: {{signed.revocation_supported}} | OPTIONAL
7. Encryption Policy
   in_transit_required: {{encryption.in_transit_required}}
   at_rest_required: {{encryption.at_rest_required}}
   kms_key_ref: {{encryption.kms_key_ref}} | OPTIONAL
8. Retention & Lifecycle
   retention_default: {{lifecycle.retention_default}}
   retention_by_class: {{lifecycle.retention_by_class}} | OPTIONAL
   delete_policy: {{lifecycle.delete_policy}}
   archive_policy: {{lifecycle.archive_policy}} | OPTIONAL
   legal_hold_supported: {{lifecycle.legal_hold_supported}} | OPTIONAL
9. Metadata Model
   stored_metadata_fields: {{metadata.fields}}
   pii_redaction_policy: {{metadata.pii_redaction_policy}} | OPTIONAL
10. Versioning Policy
    object_versioning: {{versioning.object_versioning}} (overwrite/versioned/UNKNOWN)
    overwrite_rules: {{versioning.overwrite_rules}} | OPTIONAL
    dedupe_policy: {{versioning.dedupe_policy}} | OPTIONAL
11. Audit & Observability
    audit_access_logs: {{audit.access_logs}}
    download_audit_required: {{audit.download_audit_required}} | OPTIONAL
    metrics: {{obs.metrics}}
    logs_required_fields: {{obs.logs_required_fields}}
    alerts: {{obs.alerts}} | OPTIONAL
12. References
    Upload contract: {{xref:FPMP-01}}
    Processing pipeline: {{xref:FPMP-03}} | OPTIONAL
    CDN/delivery rules: {{xref:FPMP-05}} | OPTIONAL
    Security/compliance: {{xref:FPMP-06}} | OPTIONAL
    Observability: {{xref:FPMP-07}} | OPTIONAL
    AuthZ rules: {{xref:API-04}} | OPTIONAL

Cross-References
Upstream: {{xref:FPMP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:STANDARDS_INDEX}} | OPTIONAL
Downstream: {{xref:FPMP-03}}, {{xref:FPMP-05}}, {{xref:FPMP-06}}, {{xref:FPMP-07}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECRETS]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for provider specifics if missing; define path + access defaults.
intermediate: Required. Define buckets, signed URL TTL/scope, retention/lifecycle.
advanced: Required. Add encryption/KMS details, audit + observability rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, region, encryption, versioning, lifecycle_rules_ref, allowed_chars, mint_endpoint_ref, single_use, revocation_supported, kms_key_ref, retention_by_class, archive_policy, legal_hold_supported, pii_redaction_policy, overwrite_rules, dedupe_policy, download_audit_required, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If paths.object_key_template is UNKNOWN → block Completeness Gate.
If access.default is UNKNOWN → block Completeness Gate.
If signed_urls_supported == true and signed.ttl_seconds is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FPMP
Pass conditions:
required_fields_present == true
bucket_inventory_defined == true
path_convention_defined == true
access_policy_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FPMP-03

FPMP-03 — Processing Pipeline Stages (transcode/resize/parse)
Header Block
template_id: FPMP-03
title: Processing Pipeline Stages (transcode/resize/parse)
type: file_processing_pipeline_stages
template_version: 1.0.0
output_path: 10_app/file_processing/FPMP-03_Processing_Pipeline_Stages.md
compliance_gate_id: TMP-05.PRIMARY.FPMP
upstream_dependencies: ["FPMP-01", "FPMP-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FPMP-01", "FPMP-02", "API-06", "JBS-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical specification format for file/media processing pipelines, including stage inventory, stage inputs/outputs, transforms (transcode/resize/parse), error handling, and how stages map to background jobs. This template must be consistent with upload/storage strategy and must not invent pipeline stages or job_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FPMP-01 Upload Contract: {{fpmp.upload_contract}}
FPMP-02 Storage Strategy: {{fpmp.storage_strategy}}
API-06 Background Jobs Spec: {{api.background_jobs}} | OPTIONAL
JBS-01 Jobs Inventory: {{jbs.jobs_inventory}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Pipeline inventory (pipeline_id list)
Stage inventory (stage_id list per pipeline)
Stage ordering rules (deterministic order)
Stage inputs (object refs + metadata)
Stage outputs (derived artifacts + refs)
Stage type (scan/parse/transcode/resize/thumbnail/extract/UNKNOWN)
Stage execution model (sync/async/job-backed)
Job binding rules (stage → job_id when async)
Failure handling per stage (retry vs fail vs quarantine)
Resource limits (cpu/mem/timeouts)
Observability requirements (per stage metrics/logs)
Security/compliance constraints (PII redaction, sandboxing)

Optional Fields
Conditional stages (type-based routing) | OPTIONAL
Parallel stages policy | OPTIONAL
Caching/dedupe of processing | OPTIONAL
Manual reprocess controls | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new job_ids; bind to {{xref:JBS-01}} when used.
Stage order MUST be deterministic; if conditional, conditions must be explicit.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Failure handling MUST bind to FPMP-04/JBS retry policy if present (or UNKNOWN).

Output Format

1. Pipeline Inventory
   Total pipelines: {{pipelines.total}}
   pipeline_ids: {{pipelines.pipeline_ids}} | OPTIONAL
2. Pipeline Spec (by pipeline_id)
   Pipeline
   pipeline_id: {{pipeline[0].pipeline_id}}
   name: {{pipeline[0].name}} | OPTIONAL
   trigger: {{pipeline[0].trigger}} (upload/event/api/UNKNOWN)
   applies_to_types: {{pipeline[0].applies_to_types}} | OPTIONAL
   entry_conditions: {{pipeline[0].entry_conditions}} | OPTIONAL

Stages (ordered):
Stage
stage_id: {{pipeline[0].stages[0].stage_id}}
stage_type: {{pipeline[0].stages[0].stage_type}}
order: {{pipeline[0].stages[0].order}}
execution_model: {{pipeline[0].stages[0].execution_model}} (sync/async/UNKNOWN)
job_id: {{pipeline[0].stages[0].job_id}} | OPTIONAL
inputs: {{pipeline[0].stages[0].inputs}}
outputs: {{pipeline[0].stages[0].outputs}}
failure_handling: {{pipeline[0].stages[0].failure_handling}}
resource_limits: {{pipeline[0].stages[0].resource_limits}} | OPTIONAL
observability:
metrics: {{pipeline[0].stages[0].observability.metrics}}
logs_required_fields: {{pipeline[0].stages[0].observability.logs_required_fields}} | OPTIONAL
security: {{pipeline[0].stages[0].security}} | OPTIONAL
notes: {{pipeline[0].stages[0].notes}} | OPTIONAL

(Repeat the “Stage” block for each stage in the pipeline. Repeat the “Pipeline” block for each pipeline_id.)

3. Stage Ordering Rules
   ordering_guarantee: {{ordering.guarantee}}
   tie_break_rule: {{ordering.tie_break_rule}} | OPTIONAL
   parallelism_policy: {{ordering.parallelism_policy}} | OPTIONAL
4. Job Binding Rules
   when_job_required: {{jobs.when_required}}
   job_id_source: {{jobs.job_id_source}} (expected: {{xref:JBS-01}})
   job_trigger_ref: {{jobs.trigger_ref}} | OPTIONAL
5. Failure Handling Rules
   retry_policy_ref: {{failure.retry_policy_ref}} | OPTIONAL
   quarantine_policy: {{failure.quarantine_policy}} | OPTIONAL
   reprocess_policy: {{failure.reprocess_policy}} | OPTIONAL
6. Security & Compliance
   sandboxing_required: {{security.sandboxing_required}}
   pii_redaction_policy: {{security.pii_redaction_policy}} | OPTIONAL
   artifact_access_control: {{security.artifact_access_control}} | OPTIONAL
7. Observability (Global)
   pipeline_metrics: {{obs.pipeline_metrics}}
   stage_metrics_minimum: {{obs.stage_metrics_minimum}}
   dashboards: {{obs.dashboards}} | OPTIONAL
   alerts: {{obs.alerts}} | OPTIONAL
8. References
   Upload contract: {{xref:FPMP-01}}
   Storage strategy: {{xref:FPMP-02}}
   Async processing/status model: {{xref:FPMP-04}} | OPTIONAL
   CDN/delivery: {{xref:FPMP-05}} | OPTIONAL
   Security/compliance: {{xref:FPMP-06}} | OPTIONAL
   Observability: {{xref:FPMP-07}} | OPTIONAL
   Background jobs: {{xref:API-06}} | OPTIONAL
   Jobs inventory: {{xref:JBS-01}} | OPTIONAL

Cross-References
Upstream: {{xref:FPMP-01}}, {{xref:FPMP-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FPMP-04}}, {{xref:FPMP-06}}, {{xref:FPMP-07}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-OBSERVABILITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for pipeline specifics if missing; do not invent job_ids/stages.

intermediate: Required. Define stage order, IO, execution model, and failure handling.
advanced: Required. Add security sandboxing, resource limits, and observability rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, pipeline name, applies_to_types, entry_conditions, job_id, resource_limits, logs_required_fields, security, notes, tie_break_rule, parallelism_policy, trigger_ref, retry_policy_ref, quarantine_policy, reprocess_policy, pii_redaction_policy, artifact_access_control, dashboards, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If stage ordering is UNKNOWN → block Completeness Gate.
If execution_model == async and job_id is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FPMP
Pass conditions:
required_fields_present == true
pipeline_inventory_defined == true
stage_order_defined == true
async_stages_have_job_ids == true
placeholder_resolution == true
no_unapproved_unknowns == true

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FPMP-04

FPMP-04 — Async Processing & Status Model (jobs, retries, DLQ)
Header Block
template_id: FPMP-04
title: Async Processing & Status Model (jobs, retries, DLQ)
type: file_processing_async_status_model
template_version: 1.0.0
output_path: 10_app/file_processing/FPMP-04_Async_Processing_Status_Model.md
compliance_gate_id: TMP-05.PRIMARY.FPMP
upstream_dependencies: ["FPMP-01", "FPMP-03", "API-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FPMP-01", "FPMP-03", "API-06", "JBS-04", "JBS-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical async processing model and status contract for file/media workflows, including job lifecycle states, status endpoints/fields, retry/DLQ bindings, idempotency/concurrency rules, and user-visible progress semantics. This template must be consistent with background job specs and processing pipeline stages and must not invent job capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FPMP-01 Upload Contract: {{fpmp.upload_contract}}
FPMP-03 Processing Pipeline Stages: {{fpmp.pipeline_stages}}
API-06 Background Jobs Spec: {{api.background_jobs}} | OPTIONAL
JBS-04 Retry/DLQ Policy: {{jbs.retry_dlq}} | OPTIONAL
JBS-05 Idempotency & Concurrency for Jobs: {{jbs.idempotency_concurrency}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Status model statement (how async processing is represented)
Status object schema (status_id/job_id/file_id mapping)
Lifecycle states (queued/running/succeeded/failed/etc.)
State transition rules (what causes transitions)
Progress fields (percent/stage/current step)
Status access control (who can view status/results)
Retry policy binding (to JBS-04)
DLQ/quarantine binding rules (when used)
Idempotency rules (dedupe job triggers per file)
Concurrency rules (per file/per user/per pipeline)
Status endpoints contract (polling fields, response shape)

Error mapping (to API-03)
Observability requirements (status latency, failures, stuck jobs)

Optional Fields
Push updates (WS/events) | OPTIONAL
Partial success semantics | OPTIONAL
Cancellation semantics | OPTIONAL
Backfill/reprocess rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Status schema MUST be stable and MUST be used by all async file pipelines.
Retries/DLQ MUST follow {{xref:JBS-04}} when present.
Idempotency/concurrency MUST follow {{xref:JBS-05}} when present.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Status endpoints MUST map errors to {{xref:API-03}} and enforce access via {{xref:API-04}} | OPTIONAL.

Output Format

1. Async Status Model
   model: {{status.model}} (job-backed/status-record/UNKNOWN)
   primary_identifier: {{status.primary_identifier}} (status_id/job_id/UNKNOWN)
   notes: {{status.notes}} | OPTIONAL
2. Status Object Schema
   status_id: {{schema.status_id}}
   file_id: {{schema.file_id}}
   pipeline_id: {{schema.pipeline_id}} | OPTIONAL
   job_id: {{schema.job_id}} | OPTIONAL
   state: {{schema.state}}
   stage_id: {{schema.stage_id}} | OPTIONAL
   progress: {{schema.progress}} | OPTIONAL
   created_at: {{schema.created_at}}
   updated_at: {{schema.updated_at}}
   started_at: {{schema.started_at}} | OPTIONAL
   finished_at: {{schema.finished_at}} | OPTIONAL
   result_ref: {{schema.result_ref}} | OPTIONAL
   last_error: {{schema.last_error}} | OPTIONAL
3. Lifecycle States
   states:
   {{lifecycle.states[0]}}
   {{lifecycle.states[1]}}
   {{lifecycle.states[2]}}

terminal_states: {{lifecycle.terminal_states}}
transitions: {{lifecycle.transitions}} | OPTIONAL

4. State Transition Rules
   queued_to_running: {{transitions.queued_to_running}}
   running_to_succeeded: {{transitions.running_to_succeeded}}
   running_to_failed: {{transitions.running_to_failed}}
   failed_to_retrying: {{transitions.failed_to_retrying}} | OPTIONAL
   retrying_to_running: {{transitions.retrying_to_running}} | OPTIONAL
5. Progress Semantics
   progress_model: {{progress.model}} (percent/stage_based/UNKNOWN)
   progress_field_rules: {{progress.field_rules}} | OPTIONAL
   stage_progress_mapping: {{progress.stage_mapping}} | OPTIONAL
6. Access Control
   who_can_view_status: {{access.who_can_view_status}}
   who_can_view_results: {{access.who_can_view_results}} | OPTIONAL
   authz_ref: {{access.authz_ref}} (expected: {{xref:API-04}}) | OPTIONAL
7. Retry & DLQ Binding
   retry_policy_ref: {{retry.policy_ref}} (expected: {{xref:JBS-04}}) | OPTIONAL
   dlq_supported: {{dlq.supported}}
   dlq_policy_ref: {{dlq.policy_ref}} | OPTIONAL
   quarantine_supported: {{dlq.quarantine_supported}} | OPTIONAL
8. Idempotency & Concurrency
   idempotency_key_rule: {{idem.key_rule}}
   dedupe_window: {{idem.window}} | OPTIONAL
   concurrency_scope: {{conc.scope}} (per_file/per_user/per_pipeline/UNKNOWN)
   max_inflight: {{conc.max_inflight}}
   locking_strategy: {{conc.locking_strategy}} | OPTIONAL
9. Status Endpoints Contract
   status_endpoint_id: {{endpoints.status_endpoint_id}} | OPTIONAL
   get_status_route: {{endpoints.get_status_route}} | OPTIONAL
   list_status_route: {{endpoints.list_status_route}} | OPTIONAL
   response_fields: {{endpoints.response_fields}}
   polling_guidance: {{endpoints.polling_guidance}} | OPTIONAL
10. Error Mapping
    error_policy_ref: {{errors.policy_ref}} (expected: {{xref:API-03}})
    status_codes: {{errors.status_codes}} | OPTIONAL
    last_error_format: {{errors.last_error_format}} | OPTIONAL
11. Observability Requirements
    metrics:
    jobs_started: {{obs.metrics.jobs_started}}
    jobs_succeeded: {{obs.metrics.jobs_succeeded}}
    jobs_failed: {{obs.metrics.jobs_failed}}
    jobs_stuck: {{obs.metrics.jobs_stuck}} | OPTIONAL
    avg_duration: {{obs.metrics.avg_duration}} | OPTIONAL

p95_duration: {{obs.metrics.p95_duration}} | OPTIONAL
dlq_count: {{obs.metrics.dlq_count}} | OPTIONAL
logs_required_fields: {{obs.logs_required_fields}}
alerts: {{obs.alerts}} | OPTIONAL

12. References
    Upload contract: {{xref:FPMP-01}}
    Pipeline stages: {{xref:FPMP-03}}
    Background jobs: {{xref:API-06}} | OPTIONAL
    Retry/DLQ: {{xref:JBS-04}} | OPTIONAL
    Idempotency/concurrency: {{xref:JBS-05}} | OPTIONAL
    CDN/delivery: {{xref:FPMP-05}} | OPTIONAL
    Security/compliance: {{xref:FPMP-06}} | OPTIONAL
    Observability: {{xref:FPMP-07}} | OPTIONAL

Cross-References
Upstream: {{xref:FPMP-01}}, {{xref:FPMP-03}}, {{xref:API-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FPMP-05}}, {{xref:FPMP-06}}, {{xref:FPMP-07}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-OBSERVABILITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for optional endpoints/push updates; define states + schema skeleton.
intermediate: Required. Bind retry/idempotency to JBS docs and define access control + polling contract.
advanced: Required. Add stuck-job detection, DLQ/quarantine semantics, and observability/alerts rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, pipeline_id, job_id, stage_id, progress, started_at/finished_at, result_ref, lifecycle.transitions, transition retrying rules, progress field rules, stage mapping, who_can_view_results, authz_ref, retry/dlq policy refs, quarantine_supported, dedupe_window, locking_strategy, endpoint IDs/routes, polling guidance, status_codes, last_error_format, jobs_stuck metrics, duration metrics, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If lifecycle.states is UNKNOWN → block Completeness Gate.
If idempotency_key_rule is UNKNOWN → block Completeness Gate.
If max_inflight is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FPMP
Pass conditions:

required_fields_present == true
status_schema_defined == true
lifecycle_defined == true
idempotency_concurrency_defined == true
error_mapping_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FPMP-05

FPMP-05 — CDN/Delivery Rules (cache headers, variants)
Header Block
template_id: FPMP-05
title: CDN/Delivery Rules (cache headers, variants)
type: file_processing_cdn_delivery_rules
template_version: 1.0.0
output_path: 10_app/file_processing/FPMP-05_CDN_Delivery_Rules.md
compliance_gate_id: TMP-05.PRIMARY.FPMP
upstream_dependencies: ["FPMP-02", "FPMP-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FPMP-02", "FPMP-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for delivering stored files/media to clients, including CDN integration, cache headers, access patterns (signed URLs vs proxy), content variants (sizes/transcodes), and invalidation rules. This template must be consistent with storage strategy and processing pipelines and must not invent delivery capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FPMP-02 Storage Strategy: {{fpmp.storage_strategy}}
FPMP-03 Processing Pipeline Stages: {{fpmp.pipeline_stages}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Delivery surface (direct bucket, CDN, app proxy, UNKNOWN)
Public vs private delivery policy
Signed URL delivery rules (TTL, scope, headers)
Cache header policy (cache-control, etag, vary)
Variant model (original vs processed variants)
Variant selection rules (by type/size)
URL/path rules for variants
Invalidation/purge policy (when to purge)
Range requests support (media streaming)
Rate limiting/abuse controls for delivery
Observability requirements (cache hit rate, delivery latency, 4xx/5xx)

Optional Fields
Watermarking rules | OPTIONAL
Hotlink protection | OPTIONAL

Compression rules | OPTIONAL
Cross-region delivery notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Private content MUST not be cacheable in shared caches unless explicitly allowed.
Variant naming MUST be deterministic and must not collide.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Delivery access control MUST align to {{xref:FPMP-02}} and {{xref:API-04}} | OPTIONAL.

Output Format

1. Delivery Surface
   mode: {{delivery.mode}} (cdn/direct/proxy/UNKNOWN)
   cdn_provider: {{delivery.cdn_provider}} | OPTIONAL
   base_url: {{delivery.base_url}} | OPTIONAL
   notes: {{delivery.notes}} | OPTIONAL
2. Public vs Private Policy
   public_delivery_allowed: {{policy.public_allowed}}
   private_delivery_model: {{policy.private_model}} (signed_url/proxy/UNKNOWN)
   auth_required_for_private: {{policy.auth_required}} | OPTIONAL
3. Signed URL Rules (if used)
   ttl_seconds: {{signed.ttl_seconds}}
   scope_constraints: {{signed.scope_constraints}}
   headers_bound: {{signed.headers_bound}} | OPTIONAL
   single_use: {{signed.single_use}} | OPTIONAL
4. Cache Header Policy
   cache_control_public: {{cache.public}}
   cache_control_private: {{cache.private}}
   etag_enabled: {{cache.etag_enabled}}
   vary_headers: {{cache.vary_headers}} | OPTIONAL
   stale_while_revalidate: {{cache.swr}} | OPTIONAL
5. Variant Model
   variant_types: {{variants.types}} (original/thumbnail/resize/transcode/UNKNOWN)
   variant_id_format: {{variants.id_format}} | OPTIONAL
   storage_binding_rule: {{variants.storage_binding_rule}}
   default_variant: {{variants.default}} | OPTIONAL
6. Variant Selection Rules
   selection_inputs: {{variants.selection_inputs}} (accept header, query param, device class, UNKNOWN)
   selection_rules: {{variants.selection_rules}}
   fallback_rules: {{variants.fallback_rules}} | OPTIONAL

7. URL/Path Rules
   url_template: {{urls.template}}
   query_params_allowed: {{urls.query_params_allowed}} | OPTIONAL
   path_encoding_rules: {{urls.encoding_rules}} | OPTIONAL
8. Invalidation / Purge Policy
   purge_supported: {{purge.supported}}
   purge_triggers: {{purge.triggers}}
   purge_scope: {{purge.scope}} (object/variant/prefix/UNKNOWN)
   purge_api_ref: {{purge.api_ref}} | OPTIONAL
9. Range Requests / Streaming
   range_requests_supported: {{streaming.range_supported}}
   range_header_rules: {{streaming.range_header_rules}} | OPTIONAL
   partial_content_cache_policy: {{streaming.partial_cache_policy}} | OPTIONAL
10. Rate Limits & Abuse Controls
    ratelimit_policy_ref: {{abuse.ratelimit_policy_ref}} | OPTIONAL
    hotlink_protection: {{abuse.hotlink_protection}} | OPTIONAL
11. Observability Requirements
    metrics:
    cache_hit_rate: {{obs.metrics.cache_hit_rate}} | OPTIONAL
    delivery_latency: {{obs.metrics.delivery_latency}}
    status_4xx: {{obs.metrics.status_4xx}} | OPTIONAL
    status_5xx: {{obs.metrics.status_5xx}} | OPTIONAL
    bytes_served: {{obs.metrics.bytes_served}} | OPTIONAL
    logs_required_fields: {{obs.logs_required_fields}}
    alerts: {{obs.alerts}} | OPTIONAL
12. References
    Upload contract: {{xref:FPMP-01}} | OPTIONAL
    Storage strategy: {{xref:FPMP-02}}
    Processing stages: {{xref:FPMP-03}}
    Async status model: {{xref:FPMP-04}} | OPTIONAL
    Security/compliance: {{xref:FPMP-06}} | OPTIONAL
    Observability: {{xref:FPMP-07}} | OPTIONAL
    Rate limit policy: {{xref:RLIM-01}} | OPTIONAL

Cross-References
Upstream: {{xref:FPMP-02}}, {{xref:FPMP-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FPMP-06}}, {{xref:FPMP-07}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for CDN provider specifics; define cache vs private policy skeleton.
intermediate: Required. Define variant selection rules, invalidation policy, and signed URL/cache

headers.
advanced: Required. Add streaming/range rules, hotlink protection, and observability/alerts rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, cdn_provider, base_url, auth_required_for_private, headers_bound, single_use, vary_headers, swr, variant_id_format, default_variant, selection_inputs, fallback_rules, query_params_allowed, encoding_rules, purge_supported/triggers/scope/api_ref, range header rules, partial cache policy, ratelimit_policy_ref, hotlink_protection, cache hit rate, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If delivery.mode is UNKNOWN → block Completeness Gate.
If cache_control_private is UNKNOWN → block Completeness Gate.
If variants.selection_rules is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FPMP
Pass conditions:
required_fields_present == true
delivery_mode_defined == true
cache_policy_defined == true
variant_model_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FPMP-06

FPMP-06 — Security & Compliance for Files (PII, retention, malware)
Header Block
template_id: FPMP-06
title: Security & Compliance for Files (PII, retention, malware)
type: file_processing_security_compliance
template_version: 1.0.0
output_path: 10_app/file_processing/FPMP-06_Security_Compliance_For_Files.md
compliance_gate_id: TMP-05.PRIMARY.FPMP
upstream_dependencies: ["FPMP-01", "FPMP-02", "FPMP-03", "FPMP-04", "FPMP-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FPMP-01", "FPMP-02", "FPMP-03", "FPMP-04", "FPMP-05", "API-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical security and compliance rules for file/media handling end-to-end: allowed content, malware scanning, quarantine, PII classification/redaction, retention/deletion, access controls, and audit requirements. This template must be consistent with upload/storage/delivery rules and must not invent compliance obligations not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FPMP-01 Upload Contract: {{fpmp.upload_contract}}
FPMP-02 Storage Strategy: {{fpmp.storage_strategy}}
FPMP-03 Processing Pipeline Stages: {{fpmp.pipeline_stages}}
FPMP-04 Async Status Model: {{fpmp.async_status}} | OPTIONAL
FPMP-05 CDN/Delivery Rules: {{fpmp.delivery_rules}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Content policy (allowed/disallowed categories)
Malware scanning requirements (required/optional + timing)
Quarantine rules (where stored, who can access)
PII classification rules for files + derived metadata
Redaction rules (logs, previews, extracted text)
Encryption requirements (at rest/in transit)
Access control model (who can upload/view/download/delete)
Retention and deletion policy (default + overrides)
Legal hold / preservation policy (if applicable)
Audit requirements (uploads, downloads, deletes, overrides)
Incident response hooks (what happens on malware/PII violation)

Optional Fields
DLP integration notes | OPTIONAL
Customer data export policy | OPTIONAL
Regional residency rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
If malware scanning is required, content MUST NOT be served until scan passes (unless explicitly allowed).
PII handling MUST follow {{standards.rules[STD-PII-CLASSIFICATION]}} | OPTIONAL
Access control MUST align to {{xref:API-04}} and storage access modes in {{xref:FPMP-02}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Retention/deletion MUST be enforceable (not “best effort”) unless explicitly allowed.

Output Format

1. Content Policy
   allowed_content: {{policy.allowed_content}}
   disallowed_content: {{policy.disallowed_content}}
   file_type_exceptions: {{policy.file_type_exceptions}} | OPTIONAL
2. Malware Scanning
   scan_required: {{malware.scan_required}}
   scan_timing: {{malware.scan_timing}} (pre_release/post_upload/UNKNOWN)
   scanner_type: {{malware.scanner_type}} | OPTIONAL
   on_detected_behavior: {{malware.on_detected_behavior}} (quarantine/delete/block/UNKNOWN)
   notification_policy: {{malware.notification_policy}} | OPTIONAL
3. Quarantine Rules
   quarantine_supported: {{quarantine.supported}}
   quarantine_location: {{quarantine.location}}
   who_can_access_quarantine: {{quarantine.who_can_access}} | OPTIONAL
   release_conditions: {{quarantine.release_conditions}} | OPTIONAL
4. PII Classification
   pii_policy_ref: {{pii.policy_ref}} | OPTIONAL
   file_pii_classes: {{pii.file_classes}}
   metadata_pii_classes: {{pii.metadata_classes}} | OPTIONAL
   derived_data_rules: {{pii.derived_data_rules}} | OPTIONAL
5. Redaction Rules
   logs_redaction: {{redaction.logs}}
   preview_redaction: {{redaction.preview}} | OPTIONAL
   extracted_text_redaction: {{redaction.extracted_text}} | OPTIONAL
6. Encryption Requirements
   in_transit_required: {{encryption.in_transit_required}}

at_rest_required: {{encryption.at_rest_required}}
kms_key_ref: {{encryption.kms_key_ref}} | OPTIONAL

7. Access Control Model
   upload_authz: {{access.upload_authz}}
   download_authz: {{access.download_authz}}
   delete_authz: {{access.delete_authz}}
   sharing_policy: {{access.sharing_policy}} | OPTIONAL
8. Retention & Deletion
   default_retention: {{retention.default}}
   retention_overrides: {{retention.overrides}} | OPTIONAL
   deletion_policy: {{retention.deletion_policy}}
   secure_delete_required: {{retention.secure_delete_required}} | OPTIONAL
   legal_hold_supported: {{retention.legal_hold_supported}} | OPTIONAL
9. Audit Requirements
   audit_events:
   upload_event: {{audit.events.upload}}
   download_event: {{audit.events.download}} | OPTIONAL
   delete_event: {{audit.events.delete}}
   override_event: {{audit.events.override}} | OPTIONAL
   audit_fields: {{audit.fields}}
   retention: {{audit.retention}} | OPTIONAL
10. Incident Response Hooks
    malware_incident_flow: {{ir.malware_flow}}
    pii_incident_flow: {{ir.pii_flow}} | OPTIONAL
    escalation_contacts: {{ir.escalation_contacts}} | OPTIONAL
11. References
    Upload contract: {{xref:FPMP-01}}
    Storage strategy: {{xref:FPMP-02}}
    Processing stages: {{xref:FPMP-03}}
    Async status model: {{xref:FPMP-04}} | OPTIONAL
    CDN/delivery rules: {{xref:FPMP-05}} | OPTIONAL
    Observability: {{xref:FPMP-07}} | OPTIONAL
    AuthZ rules: {{xref:API-04}} | OPTIONAL

Cross-References
Upstream: {{xref:FPMP-01}}, {{xref:FPMP-02}}, {{xref:FPMP-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FPMP-07}} | OPTIONAL, {{xref:ADMIN-03}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-CLASSIFICATION]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-RETENTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where scanner/quarantine specifics are missing; do not invent compliance obligations.
intermediate: Required. Define scanning, PII classes, retention/deletion, and access control rules.
advanced: Required. Add incident response flows, audit rigor, and derived-data redaction policy.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, file type exceptions, scanner_type, notification_policy, who_can_access_quarantine, release_conditions, policy_ref, metadata_classes, derived_data_rules, preview/extracted text redaction, kms_key_ref, sharing_policy, retention_overrides, secure_delete_required, legal_hold_supported, audit retention, pii incident flow, escalation contacts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If malware.scan_required is UNKNOWN → flag in Open Questions.
If access control rules are UNKNOWN → block Completeness Gate.
If deletion_policy is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FPMP
Pass conditions:
required_fields_present == true
content_policy_defined == true
access_control_defined == true
retention_deletion_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FPMP-07

FPMP-07 — Media Observability (latency, failure rates, QoS)
Header Block
template_id: FPMP-07
title: Media Observability (latency, failure rates, QoS)
type: file_processing_observability
template_version: 1.0.0
output_path: 10_app/file_processing/FPMP-07_Media_Observability.md
compliance_gate_id: TMP-05.PRIMARY.FPMP
upstream_dependencies: ["FPMP-01", "FPMP-03", "FPMP-04", "FPMP-05", "FPMP-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FPMP-01", "FPMP-03", "FPMP-04", "FPMP-05", "FPMP-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability requirements for file/media upload, processing, storage, and delivery, including required metrics, logs, traces, dashboards, alerts, and QoS/SLO hooks. This template must be consistent with pipeline and delivery semantics and must not invent instrumentation that contradicts upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FPMP-01 Upload Contract: {{fpmp.upload_contract}}
FPMP-03 Processing Pipeline Stages: {{fpmp.pipeline_stages}}
FPMP-04 Async Status Model: {{fpmp.async_status}} | OPTIONAL
FPMP-05 CDN/Delivery Rules: {{fpmp.delivery_rules}} | OPTIONAL
FPMP-06 Security/Compliance: {{fpmp.security_compliance}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Core metric set (upload/process/delivery)
Latency definitions (upload latency, processing duration, delivery latency)
Success/failure rate definitions (by stage, by pipeline, by delivery surface)
Scan/quarantine metrics (if applicable)
Variant generation metrics (if applicable)
Cache metrics (if CDN)
Storage error metrics
Log field requirements (correlation + redaction)
Trace propagation requirements (request_id/trace_id/file_id)
Dashboard minimum panels
Alert requirements (failure spikes, latency regressions, stuck jobs, scan failures)
QoS/SLO hooks (targets by class if defined)

Optional Fields
Per-type metrics (image/video/pdf/etc.) | OPTIONAL
Sampling policy | OPTIONAL
Cost controls/cardinality limits | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Metrics MUST be keyed by stable identifiers (pipeline_id, stage_id, variant_type) and avoid unbounded labels.
PII MUST be redacted in logs/traces per: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Alerts MUST be actionable and map to operational procedures where available.

Output Format

1. Observability Scope
   surfaces: {{obs.surfaces}} (upload/process/storage/delivery/UNKNOWN)
   coverage_model: {{obs.coverage_model}} (per pipeline/stage/type/UNKNOWN)
   notes: {{obs.notes}} | OPTIONAL
2. Metrics — Upload
   upload_attempts: {{metrics.upload.attempts}}
   upload_success: {{metrics.upload.success}}
   upload_failures: {{metrics.upload.failures}}
   upload_latency_ms: {{metrics.upload.latency_ms}} | OPTIONAL
   bytes_uploaded: {{metrics.upload.bytes}} | OPTIONAL
3. Metrics — Processing
   jobs_started: {{metrics.processing.jobs_started}}
   jobs_succeeded: {{metrics.processing.jobs_succeeded}}
   jobs_failed: {{metrics.processing.jobs_failed}}
   stage_duration_ms: {{metrics.processing.stage_duration_ms}} | OPTIONAL
   pipeline_duration_ms: {{metrics.processing.pipeline_duration_ms}} | OPTIONAL
   stuck_jobs: {{metrics.processing.stuck_jobs}} | OPTIONAL
4. Metrics — Security/Scanning (if applicable)
   scan_pass: {{metrics.scan.pass}} | OPTIONAL
   scan_fail: {{metrics.scan.fail}} | OPTIONAL
   quarantine_count: {{metrics.scan.quarantine_count}} | OPTIONAL
5. Metrics — Variants/CDN/Delivery
   variant_generated: {{metrics.delivery.variant_generated}} | OPTIONAL
   delivery_latency_ms: {{metrics.delivery.latency_ms}}
   cache_hit_rate: {{metrics.delivery.cache_hit_rate}} | OPTIONAL
   status_4xx: {{metrics.delivery.status_4xx}} | OPTIONAL

status_5xx: {{metrics.delivery.status_5xx}} | OPTIONAL
bytes_served: {{metrics.delivery.bytes_served}} | OPTIONAL

6. Metrics — Storage
   storage_put_failures: {{metrics.storage.put_failures}} | OPTIONAL
   storage_get_failures: {{metrics.storage.get_failures}} | OPTIONAL
   storage_latency_ms: {{metrics.storage.latency_ms}} | OPTIONAL
7. Logs — Required Fields
   timestamp: {{logs.fields.timestamp}}
   file_id: {{logs.fields.file_id}} | OPTIONAL
   pipeline_id: {{logs.fields.pipeline_id}} | OPTIONAL
   stage_id: {{logs.fields.stage_id}} | OPTIONAL
   variant_type: {{logs.fields.variant_type}} | OPTIONAL
   request_id: {{logs.fields.request_id}} | OPTIONAL
   trace_id: {{logs.fields.trace_id}} | OPTIONAL
   result: {{logs.fields.result}} (success/failure)
   error_code: {{logs.fields.error_code}} | OPTIONAL
   redaction_applied: {{logs.fields.redaction_applied}} | OPTIONAL
8. Tracing — Propagation & Attributes
   tracing_required: {{tracing.required}}
   propagation_headers: {{tracing.propagation_headers}} | OPTIONAL
   span_attributes_required:
   file_id: {{tracing.attrs.file_id}} | OPTIONAL
   pipeline_id: {{tracing.attrs.pipeline_id}} | OPTIONAL
   stage_id: {{tracing.attrs.stage_id}} | OPTIONAL
   variant_type: {{tracing.attrs.variant_type}} | OPTIONAL
   sampling_policy: {{tracing.sampling_policy}} | OPTIONAL
9. Dashboards (Minimum)
   overview_dashboard: {{dashboards.overview}}
   processing_dashboard: {{dashboards.processing}} | OPTIONAL
   delivery_dashboard: {{dashboards.delivery}} | OPTIONAL
   security_dashboard: {{dashboards.security}} | OPTIONAL
   minimum_panels: {{dashboards.minimum_panels}}
10. Alerts
    upload_failure_spike: {{alerts.upload_failure_spike}}
    processing_failure_spike: {{alerts.processing_failure_spike}}
    delivery_latency_regression: {{alerts.delivery_latency_regression}}
    stuck_jobs_alert: {{alerts.stuck_jobs_alert}} | OPTIONAL
    scan_fail_spike: {{alerts.scan_fail_spike}} | OPTIONAL
    threshold_model: {{alerts.threshold_model}} (static/baseline/UNKNOWN)
    routing_policy: {{alerts.routing_policy}} | OPTIONAL
    runbook_ref: {{alerts.runbook_ref}} | OPTIONAL
11. QoS/SLO Hooks (Optional)
    slo_doc_ref: {{obs.slo_doc_ref}} | OPTIONAL
    slo_targets: {{obs.slo_targets}} | OPTIONAL

12. Cardinality & Cost Controls (Optional)
    label_allowlist: {{cost.label_allowlist}} | OPTIONAL
    disallowed_labels: {{cost.disallowed_labels}} | OPTIONAL
    sampling_controls: {{cost.sampling_controls}} | OPTIONAL
13. References
    Upload contract: {{xref:FPMP-01}}
    Pipeline stages: {{xref:FPMP-03}}
    Async status model: {{xref:FPMP-04}} | OPTIONAL
    CDN/delivery rules: {{xref:FPMP-05}} | OPTIONAL
    Security/compliance: {{xref:FPMP-06}} | OPTIONAL

Cross-References
Upstream: {{xref:FPMP-01}}, {{xref:FPMP-03}}, {{xref:FPMP-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OPS-OBS}} | OPTIONAL, {{xref:ALRT-01}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL, {{standards.rules[STD-OBSERVABILITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define core metric/log fields and minimum dashboards; use UNKNOWN for tooling specifics.
intermediate: Required. Add stage/pipeline metrics and alert triggers.
advanced: Required. Add cardinality controls, sampling, and QoS/SLO bindings.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, upload latency, stage/pipeline duration, stuck jobs, scan metrics, cache metrics, storage metrics, file_id/pipeline_id/stage_id/variant_type, tracing propagation/sampling, dashboards.*, alerts routing/runbook, slo hooks, cost controls, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If logs required fields are UNKNOWN → block Completeness Gate.
If alerts.threshold_model is UNKNOWN → flag in Open Questions.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FPMP
Pass conditions:
required_fields_present == true
metric_set_defined == true
log_fields_defined == true
dashboard_minimum_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Admin & Internal Tools APIs (ADMIN)

Admin & Internal Tools APIs (ADMIN)
ADMIN-01 Admin Capabilities Matrix
ADMIN-02 Moderation/Support Tools Spec (queues, actions)
ADMIN-03 Audit Trail Spec (actions + retention)
ADMIN-04 Data Repair Procedures (safe backfills)
ADMIN-05 Privileged API Surface Catalog (endpoints + authz)
ADMIN-06 Admin Observability & Safeguards (rate limits, alerts)

ADMIN-01

ADMIN-01 — Admin Capabilities Matrix
Header Block
template_id: ADMIN-01
title: Admin Capabilities Matrix
type: admin_capabilities_matrix
template_version: 1.0.0
output_path: 10_app/admin/ADMIN-01_Admin_Capabilities_Matrix.md
compliance_gate_id: TMP-05.PRIMARY.ADMIN
upstream_dependencies: ["API-01", "API-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-01", "API-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical matrix of admin/internal tool capabilities: what actions exist, who can perform them, what scope they operate on, what audit requirements apply, and what safeguards/rate limits must be enforced. This template must be consistent with AuthZ rules and must not invent admin actions not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Capability registry (capability_id list)
Capability description (what it does)
Surface (admin UI, admin API, CLI)
Action binding (permission/action_id)
Scope (global/org/project/resource)
Risk class (low/med/high/critical)
Safeguards required (confirmations, approvals, limits)
Audit required (yes/no) and audit ref
Rate limits / abuse controls for admin surfaces
Implementation references (what endpoints/tools provide it)

Optional Fields
Runbook/procedure pointers | OPTIONAL
UI location notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new permissions/roles; bind to {{xref:API-04}}.
Every capability MUST bind to a concrete implementation surface (endpoint/tool) or be marked UNKNOWN and flagged.
High-risk capabilities SHOULD require extra safeguards (two-step confirm, approvals) unless explicitly disallowed.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Audit requirements SHOULD bind to {{xref:ADMIN-03}} when available.

Output Format

1. Matrix Summary
   total_capabilities: {{admin.matrix.total}}
   surfaces: {{admin.matrix.surfaces}} | OPTIONAL
   risk_classes_in_use: {{admin.matrix.risk_classes}} | OPTIONAL
   notes: {{admin.matrix.notes}} | OPTIONAL
2. Capability Entries (by capability_id)
   Capability
   capability_id: {{capabilities[0].capability_id}}
   name: {{capabilities[0].name}}
   description: {{capabilities[0].description}}
   surface: {{capabilities[0].surface}}
   action_binding: {{capabilities[0].action_binding}} (perm/action_id)
   scope: {{capabilities[0].scope}}
   risk_class: {{capabilities[0].risk_class}}
   safeguards: {{capabilities[0].safeguards}}
   audit_required: {{capabilities[0].audit_required}}
   audit_ref: {{capabilities[0].audit_ref}} | OPTIONAL
   rate_limit_ref: {{capabilities[0].rate_limit_ref}} | OPTIONAL
   implementation_refs: {{capabilities[0].implementation_refs}} (e.g., endpoint_id(s) from {{xref:API-01}})
   runbook_ref: {{capabilities[0].runbook_ref}} | OPTIONAL
   ui_location: {{capabilities[0].ui_location}} | OPTIONAL
   open_questions:
   {{capabilities[0].open_questions[0]}} | OPTIONAL

(Repeat the “Capability” block for each capability_id.)

3. References
   Privileged surface catalog: {{xref:ADMIN-05}} | OPTIONAL
   Audit trail spec: {{xref:ADMIN-03}} | OPTIONAL
   Moderation/support tools: {{xref:ADMIN-02}} | OPTIONAL
   Admin observability/safeguards: {{xref:ADMIN-06}} | OPTIONAL

AuthZ rules: {{xref:API-04}} | OPTIONAL
Rate limit policy: {{xref:RLIM-01}} | OPTIONAL

Cross-References
Upstream: {{xref:API-01}}, {{xref:API-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-02}}, {{xref:ADMIN-03}}, {{xref:ADMIN-05}}, {{xref:ADMIN-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-AUDIT]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where implementations are missing; do not invent permissions.
intermediate: Required. Populate safeguards/audit/risk classes and bind to endpoints.
advanced: Required. Add runbook pointers and strong controls for critical capabilities.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, surfaces/risk class summaries, audit_ref, rate_limit_ref, runbook_ref, ui_location, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If capability_id list is UNKNOWN → block Completeness Gate.
If action_binding is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ADMIN
Pass conditions:
required_fields_present == true
all capability_ids are unique
no new permissions/roles introduced (binds to API-04)
placeholder_resolution == true
no_unapproved_unknowns == true

ADMIN-02

ADMIN-02 — Moderation/Support Tools Spec (queues, actions)
Header Block
template_id: ADMIN-02
title: Moderation/Support Tools Spec (queues, actions)
type: admin_moderation_support_tools_spec
template_version: 1.0.0
output_path: 10_app/admin/ADMIN-02_Moderation_Support_Tools_Spec.md
compliance_gate_id: TMP-05.PRIMARY.ADMIN
upstream_dependencies: ["ADMIN-01", "API-01", "API-02", "API-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ADMIN-01", "API-01", "API-02", "API-04", "RLIM-01", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical specification format for admin moderation/support tools: work queues, case/ticket model, actions available, safeguards, audit logging, and observability. This template must be consistent with the admin capabilities matrix and AuthZ rules and must not invent queues/actions not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ADMIN-01 Capabilities Matrix: {{admin.capabilities}}
API-01 Endpoint Catalog: {{api.endpoint_catalog}} | OPTIONAL
API-02 Endpoint Specs: {{api.endpoint_specs}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Queue registry (queue_id list)
Queue definition (what items enter, priority)
Case/work item schema (case_id, subject, status, assignee)
Action registry (action_id list)
Action definitions (what they do)
Action safeguards (confirmations, approvals)
Action AuthZ binding (permissions/roles)
Action audit requirements (fields, retention)
Rate limits/abuse controls for admin actions
Observability requirements (queue depth, handling time, action outcomes)

Optional Fields
SLA targets | OPTIONAL
Escalation routing | OPTIONAL
Templates/macros | OPTIONAL
User communication policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new permissions/roles; bind to {{xref:API-04}}.
All actions MUST be auditable (or UNKNOWN flagged).
High-risk actions SHOULD require extra safeguards unless explicitly disallowed.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Queue Registry (by queue_id)
   Queue
   queue_id: {{queues[0].queue_id}}
   name: {{queues[0].name}}
   description: {{queues[0].description}}
   entry_criteria: {{queues[0].entry_criteria}}
   priority_model: {{queues[0].priority_model}} | OPTIONAL
   status_model: {{queues[0].status_model}}
   assignment_model: {{queues[0].assignment_model}} | OPTIONAL
   sla_target: {{queues[0].sla_target}} | OPTIONAL
   observability_tags: {{queues[0].observability_tags}} | OPTIONAL

(Repeat for each queue_id.)

2. Work Item / Case Schema
   case_id: {{case.case_id}}
   subject_type: {{case.subject_type}}
   subject_id: {{case.subject_id}} | OPTIONAL
   status: {{case.status}}
   priority: {{case.priority}} | OPTIONAL
   assignee: {{case.assignee}} | OPTIONAL
   created_at: {{case.created_at}}
   updated_at: {{case.updated_at}}
   notes_field: {{case.notes_field}} | OPTIONAL
   evidence_refs: {{case.evidence_refs}} | OPTIONAL
3. Action Registry (by action_id)
   Action
   action_id: {{actions[0].action_id}}

name: {{actions[0].name}}
description: {{actions[0].description}}
applies_to_queue_ids: {{actions[0].applies_to_queue_ids}} | OPTIONAL
preconditions: {{actions[0].preconditions}} | OPTIONAL
effect: {{actions[0].effect}}
safeguards: {{actions[0].safeguards}}
authz_binding: {{actions[0].authz_binding}} (bind to {{xref:API-04}})
audit_required: {{actions[0].audit_required}}
audit_fields: {{actions[0].audit_fields}} | OPTIONAL
rate_limit_ref: {{actions[0].rate_limit_ref}} | OPTIONAL
implementation_refs: {{actions[0].implementation_refs}} | OPTIONAL
rollback_supported: {{actions[0].rollback_supported}} | OPTIONAL
rollback_steps: {{actions[0].rollback_steps}} | OPTIONAL
open_questions:
{{actions[0].open_questions[0]}} | OPTIONAL

(Repeat for each action_id.)

4. Audit Requirements
   audit_ref: {{audit.ref}} (expected: {{xref:ADMIN-03}}) | OPTIONAL
   retention: {{audit.retention}} | OPTIONAL
   redaction_policy: {{audit.redaction_policy}} | OPTIONAL
5. Observability Requirements
   metrics:
   queue_depth: {{obs.metrics.queue_depth}}
   time_to_first_action: {{obs.metrics.time_to_first_action}} | OPTIONAL
   time_to_resolve: {{obs.metrics.time_to_resolve}} | OPTIONAL
   action_success: {{obs.metrics.action_success}} | OPTIONAL
   action_failure: {{obs.metrics.action_failure}} | OPTIONAL
   dashboards: {{obs.dashboards}} | OPTIONAL
   alerts: {{obs.alerts}} | OPTIONAL
6. References
   Capabilities matrix: {{xref:ADMIN-01}}
   Audit trail: {{xref:ADMIN-03}} | OPTIONAL
   Privileged API catalog: {{xref:ADMIN-05}} | OPTIONAL
   Admin safeguards: {{xref:ADMIN-06}} | OPTIONAL
   AuthZ rules: {{xref:API-04}} | OPTIONAL
   Rate limits: {{xref:RLIM-01}} | OPTIONAL

Cross-References
Upstream: {{xref:ADMIN-01}}, {{xref:API-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-03}}, {{xref:ADMIN-05}}, {{xref:ADMIN-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-AUDIT]}}
| OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for queue/action specifics if missing; do not invent permissions.
intermediate: Required. Define safeguards, authz bindings, and audit fields.
advanced: Required. Add SLA/escalation policies and rollback procedures where applicable.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, priority_model, assignment_model, sla_target, observability_tags, subject_id, priority, assignee, notes_field, evidence_refs, applies_to_queue_ids, preconditions, audit_fields, rate_limit_ref, implementation_refs, rollback_supported, rollback_steps, audit.ref, retention, redaction_policy, dashboards, alerts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If queue_id list is UNKNOWN → block Completeness Gate.
If action_id list is UNKNOWN → block Completeness Gate.
If authz_binding is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ADMIN
Pass conditions:
required_fields_present == true
queue_registry_defined == true
action_registry_defined == true
audit_requirements_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ADMIN-03

ADMIN-03 — Audit Trail Spec (actions + retention)
Header Block
template_id: ADMIN-03
title: Audit Trail Spec (actions + retention)
type: admin_audit_trail_spec
template_version: 1.0.0
output_path: 10_app/admin/ADMIN-03_Audit_Trail_Spec.md
compliance_gate_id: TMP-05.PRIMARY.ADMIN
upstream_dependencies: ["API-04", "ADMIN-01", "ADMIN-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-04", "ADMIN-01", "ADMIN-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical audit trail model for privileged/admin actions and other security-relevant events: what must be logged, record schema, retention rules, immutability/tamper-evidence expectations, access control to audit logs, and how audit records support investigations. This template must be consistent with AuthZ rules and must not invent audit capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
ADMIN-01 Admin Capabilities Matrix: {{admin.capabilities}} | OPTIONAL
ADMIN-02 Moderation/Support Tools Spec: {{admin.tools_spec}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Audit scope statement (what categories of actions are audited)
Audit event types registry (event_type list)
Audit record schema (fields required)
Actor model (user/service, roles, ip/device if applicable)
Target/resource model (what was acted on)
Before/after capture policy (diff vs snapshot)
Reason requirement (why field required)
Retention policy (duration by event type/class)
Immutability/tamper-evidence policy (append-only, hashing)
Access control policy (who can view/export)
Redaction policy (no secrets, PII handling)
Observability requirements (audit write failures, volume)

Optional Fields
Export format/policy | OPTIONAL
Legal hold policy | OPTIONAL
Correlation to tickets/approvals | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Audit logs MUST be append-only (or UNKNOWN flagged) and resistant to tampering.
Audit records MUST not store secrets; store references/hashes when needed.
All privileged/admin actions MUST emit an audit record (or be UNKNOWN flagged).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Access to audit logs MUST be least-privilege and bind to {{xref:API-04}}.

Output Format

1. Audit Scope
   audited_categories: {{audit.scope.categories}}
   audited_surfaces: {{audit.scope.surfaces}} | OPTIONAL
   notes: {{audit.scope.notes}} | OPTIONAL
2. Audit Event Types Registry
   Event Type
   event_type: {{types[0].event_type}}
   name: {{types[0].name}}
   description: {{types[0].description}}
   risk_class: {{types[0].risk_class}} | OPTIONAL
   retention_class: {{types[0].retention_class}} | OPTIONAL

(Repeat for each event_type.)

3. Audit Record Schema
   audit_id: {{schema.audit_id}}
   event_type: {{schema.event_type}}
   timestamp: {{schema.timestamp}}
   actor:
   actor_type: {{schema.actor.actor_type}} (user/service/UNKNOWN)
   actor_id: {{schema.actor.actor_id}}
   roles: {{schema.actor.roles}} | OPTIONAL
   ip: {{schema.actor.ip}} | OPTIONAL
   user_agent: {{schema.actor.user_agent}} | OPTIONAL
   session_id: {{schema.actor.session_id}} | OPTIONAL

target:
resource_type: {{schema.target.resource_type}}

resource_id: {{schema.target.resource_id}} | OPTIONAL
org_id: {{schema.target.org_id}} | OPTIONAL
project_id: {{schema.target.project_id}} | OPTIONAL

action:
action_id: {{schema.action.action_id}}
decision: {{schema.action.decision}} (allow/deny/UNKNOWN) | OPTIONAL
reason: {{schema.action.reason}}

before: {{schema.before}} | OPTIONAL
after: {{schema.after}} | OPTIONAL
diff: {{schema.diff}} | OPTIONAL

correlation:
request_id: {{schema.correlation.request_id}} | OPTIONAL
trace_id: {{schema.correlation.trace_id}} | OPTIONAL
ticket_ref: {{schema.correlation.ticket_ref}} | OPTIONAL
approval_ref: {{schema.correlation.approval_ref}} | OPTIONAL

redaction_applied: {{schema.redaction_applied}} | OPTIONAL
evidence_refs: {{schema.evidence_refs}} | OPTIONAL

4. Before/After Capture Policy
   capture_mode: {{capture.mode}} (diff/snapshot/both/UNKNOWN)
   capture_required_for: {{capture.required_for}} | OPTIONAL
   sensitive_field_handling: {{capture.sensitive_field_handling}}
5. Reason Policy
   reason_required: {{reason.required}}
   reason_format: {{reason.format}} | OPTIONAL
   reason_codes: {{reason.codes}} | OPTIONAL
6. Retention Policy
   default_retention: {{retention.default}}
   retention_by_class: {{retention.by_class}} | OPTIONAL
   legal_hold_supported: {{retention.legal_hold_supported}} | OPTIONAL
7. Immutability / Tamper-Evidence
   append_only: {{immutability.append_only}}
   tamper_evidence_model: {{immutability.model}} (hash_chain/worm/UNKNOWN)
   integrity_verification: {{immutability.integrity_verification}} | OPTIONAL
8. Access Control
   who_can_view: {{access.who_can_view}}
   who_can_export: {{access.who_can_export}} | OPTIONAL
   export_approval_required: {{access.export_approval_required}} | OPTIONAL
9. Redaction Policy
   pii_redaction: {{redaction.pii_redaction}}

secrets_redaction: {{redaction.secrets_redaction}}
field_allowlist: {{redaction.field_allowlist}} | OPTIONAL

10. Observability Requirements
    metrics:
    audit_write_success: {{obs.metrics.audit_write_success}}
    audit_write_failure: {{obs.metrics.audit_write_failure}}
    audit_volume: {{obs.metrics.audit_volume}} | OPTIONAL
    alerts: {{obs.alerts}} | OPTIONAL
    dashboards: {{obs.dashboards}} | OPTIONAL
11. References
    Admin capabilities: {{xref:ADMIN-01}} | OPTIONAL
    Moderation/support tools: {{xref:ADMIN-02}} | OPTIONAL
    Privileged surface catalog: {{xref:ADMIN-05}} | OPTIONAL
    Admin safeguards: {{xref:ADMIN-06}} | OPTIONAL
    AuthZ rules: {{xref:API-04}} | OPTIONAL

Cross-References
Upstream: {{xref:API-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-06}} | OPTIONAL, {{xref:GOV-04}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-AUDIT]}} | OPTIONAL, {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for immutability specifics if missing; define record schema skeleton.
intermediate: Required. Define retention, access control, and redaction policies.
advanced: Required. Add tamper-evidence model and integrity verification procedures.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, risk/retention class, actor metadata (ip/user_agent/session), resource_id/org/project, before/after/diff fields, correlation fields, evidence_refs, capture.mode, capture.required_for, reason.format/codes, retention_by_class, legal_hold_supported, tamper_evidence_model, integrity_verification, who_can_export, export approvals, field_allowlist, alerts, dashboards, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If schema.audit_id or schema.event_type is UNKNOWN → block Completeness Gate.
If access.who_can_view is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ADMIN
Pass conditions:
required_fields_present == true
audit_record_schema_defined == true

retention_policy_defined == true
access_control_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ADMIN-04

ADMIN-04 — Data Repair Procedures (safe backfills)
Header Block
template_id: ADMIN-04
title: Data Repair Procedures (safe backfills)
type: admin_data_repair_procedures
template_version: 1.0.0
output_path: 10_app/admin/ADMIN-04_Data_Repair_Procedures.md
compliance_gate_id: TMP-05.PRIMARY.ADMIN
upstream_dependencies: ["ADMIN-01", "ADMIN-03", "API-04", "EVT-07", "JBS-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ADMIN-01", "ADMIN-03", "API-04", "EVT-07", "JBS-04", "JBS-05", "FFCFG-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical procedures and constraints for performing safe data repairs (manual fixes, backfills, replays, reconciliations), including approvals, dry runs, idempotency, rate limits, audit logging, and rollback/verification. This template must be consistent with audit/AuthZ rules and must not invent repair capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ADMIN-01 Capabilities Matrix: {{admin.capabilities}}
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}}
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
EVT-07 Failure Handling (replay/backfill): {{evt.failure_handling}} | OPTIONAL
JBS-04 Retry/DLQ Policy: {{jbs.retry_dlq}} | OPTIONAL
JBS-05 Idempotency & Concurrency: {{jbs.idempotency_concurrency}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.registry}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Repair action types (backfill, replay, patch, delete, reconcile)
Eligibility rules (what can be repaired; what cannot)
Approval policy (required roles, two-person if needed)
Safety checklist (pre-flight checks)
Dry-run policy (required/optional + output expectations)
Execution method (script/job/admin endpoint/UNKNOWN)
Idempotency requirements for repair operations
Concurrency/rate limit constraints for repairs
Audit logging requirements (what fields, retention)
Rollback/compensation policy

Verification steps (how to confirm correctness)
Communication policy (who is notified)

Optional Fields
Feature-flag gated repairs | OPTIONAL
Quarantine mode | OPTIONAL
Incident/ticket linkage | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
All repair actions MUST be auditable per {{xref:ADMIN-03}}.
AuthZ for repairs MUST bind to {{xref:API-04}}.
Repairs that replay events MUST respect {{xref:EVT-07}} semantics.
Job-based repairs MUST respect {{xref:JBS-04}} and {{xref:JBS-05}} when applicable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Repair Action Types
   supported_actions: {{repair.actions}}
   notes: {{repair.notes}} | OPTIONAL
2. Eligibility Rules
   eligible_data_domains: {{eligibility.domains}}
   disallowed_operations: {{eligibility.disallowed_ops}}
   pii_constraints: {{eligibility.pii_constraints}} | OPTIONAL
3. Approval Policy
   approval_required: {{approval.required}}
   who_can_approve: {{approval.who_can_approve}}
   two_person_required_for: {{approval.two_person_required_for}} | OPTIONAL
   ticket_required: {{approval.ticket_required}} | OPTIONAL
4. Safety Checklist (Pre-Flight)
   checklist:
   {{safety.checklist[0]}}
   {{safety.checklist[1]}}
   {{safety.checklist[2]}}
5. Dry-Run Policy
   dry_run_supported: {{dryrun.supported}}
   dry_run_required_for: {{dryrun.required_for}} | OPTIONAL
   dry_run_output: {{dryrun.output}} (diff/summary/sample/UNKNOWN)
   dry_run_validation: {{dryrun.validation}} | OPTIONAL
6. Execution Method
   execution_modes: {{exec.modes}} (script/job/admin_endpoint/UNKNOWN)

entry_points: {{exec.entry_points}} | OPTIONAL
feature_flag_gate: {{exec.feature_flag_gate}} | OPTIONAL

7. Idempotency & Concurrency
   idempotency_key_rule: {{idem.key_rule}}
   dedupe_window: {{idem.window}} | OPTIONAL
   concurrency_limits: {{conc.limits}}
   locking_strategy: {{conc.locking_strategy}} | OPTIONAL
   rate_limits: {{conc.rate_limits}} | OPTIONAL
8. Audit Logging
   audit_ref: {{audit.ref}} (expected: {{xref:ADMIN-03}})
   audit_fields: {{audit.fields}}
   retention: {{audit.retention}} | OPTIONAL
9. Rollback / Compensation
   rollback_supported: {{rollback.supported}}
   rollback_methods: {{rollback.methods}} | OPTIONAL
   compensation_events: {{rollback.compensation_events}} | OPTIONAL
10. Verification Steps
    verification_steps:
    {{verify.steps[0]}}
    {{verify.steps[1]}}
    success_criteria: {{verify.success_criteria}} | OPTIONAL
11. Communication Policy
    notify_roles: {{comms.notify_roles}} | OPTIONAL
    notify_channels: {{comms.notify_channels}} | OPTIONAL
    post_repair_report_required: {{comms.post_repair_report_required}} | OPTIONAL
12. References
    Admin capabilities: {{xref:ADMIN-01}}
    Audit trail: {{xref:ADMIN-03}}
    AuthZ rules: {{xref:API-04}} | OPTIONAL
    Event replay/backfill: {{xref:EVT-07}} | OPTIONAL
    Retry/DLQ: {{xref:JBS-04}} | OPTIONAL
    Idempotency/concurrency: {{xref:JBS-05}} | OPTIONAL
    Admin safeguards: {{xref:ADMIN-06}} | OPTIONAL

Cross-References
Upstream: {{xref:ADMIN-01}}, {{xref:ADMIN-03}}, {{xref:API-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-06}} | OPTIONAL, {{xref:RUNBOOK-REPAIR}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-AUDIT]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for execution entry points if missing; do not invent repair powers.

intermediate: Required. Define dry-run, idempotency, audit, and verification steps.
advanced: Required. Add two-person approvals for high-risk repairs and robust rollback/compensation.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, repair notes, pii constraints, two_person_required_for, ticket_required, dryrun.required_for, dryrun.validation, entry_points, feature_flag_gate, dedupe_window, locking_strategy, rate_limits, retention, rollback methods, compensation_events, success_criteria, comms.*, post_repair report, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If idempotency_key_rule is UNKNOWN → block Completeness Gate.
If audit.fields is UNKNOWN → block Completeness Gate.
If verification_steps is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ADMIN
Pass conditions:
required_fields_present == true
approval_policy_defined == true
safety_checklist_defined == true
idempotency_concurrency_defined == true
audit_logging_defined == true
verification_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ADMIN-05

ADMIN-05 — Privileged API Surface Catalog (endpoints + authz)
Header Block
template_id: ADMIN-05
title: Privileged API Surface Catalog (endpoints + authz)
type: admin_privileged_api_catalog
template_version: 1.0.0
output_path: 10_app/admin/ADMIN-05_Privileged_API_Surface_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.ADMIN
upstream_dependencies: ["API-01", "API-02", "API-04", "ADMIN-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-01", "API-02", "API-04", "ADMIN-01", "RLIM-01", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of privileged/admin API endpoints, including endpoint identifiers, operations, required permissions, ABAC constraints, risk class, audit requirements, and safeguards. This template must be consistent with API endpoint specs and AuthZ rules and must not invent endpoint IDs not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-02 Endpoint Specs: {{api.endpoint_specs}}
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
ADMIN-01 Capabilities Matrix: {{admin.capabilities}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Privileged endpoints list (endpoint_id list)
Endpoint path/method (or route_id)
Purpose/description
Required permissions/roles
ABAC constraints (tenant/env)
Risk class (low/med/high/critical)
Audit required (yes/no) and audit ref
Safeguards (confirmations, approvals)
Rate limit policy for privileged surface
Error mapping reference (API-03)

Optional Fields
Concealment policy (403 vs 404) | OPTIONAL
Two-person approval requirement | OPTIONAL
Runbook pointer | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent endpoint_ids; use only those in {{xref:API-01}}.
AuthZ MUST bind to {{xref:API-04}}; audit MUST bind to {{xref:ADMIN-03}} when present.
High-risk endpoints SHOULD have safeguards unless explicitly disallowed.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Catalog Summary
   total_privileged_endpoints: {{catalog.total}}
   risk_classes_in_use: {{catalog.risk_classes}} | OPTIONAL
   surfaces: {{catalog.surfaces}} | OPTIONAL
   notes: {{catalog.notes}} | OPTIONAL
2. Privileged Endpoint Entries (by endpoint_id)
   Endpoint
   endpoint_id: {{endpoints[0].endpoint_id}}
   method: {{endpoints[0].method}}
   path: {{endpoints[0].path}}
   name: {{endpoints[0].name}} | OPTIONAL
   description: {{endpoints[0].description}}
   risk_class: {{endpoints[0].risk_class}}
   required_permissions: {{endpoints[0].required_permissions}}
   abac_constraints: {{endpoints[0].abac_constraints}}
   deny_behavior: {{endpoints[0].deny_behavior}} | OPTIONAL
   audit_required: {{endpoints[0].audit_required}}
   audit_ref: {{endpoints[0].audit_ref}} | OPTIONAL
   safeguards: {{endpoints[0].safeguards}}
   rate_limit_ref: {{endpoints[0].rate_limit_ref}} | OPTIONAL
   error_policy_ref: {{endpoints[0].error_policy_ref}} | OPTIONAL
   implementation_notes: {{endpoints[0].implementation_notes}} | OPTIONAL
   runbook_ref: {{endpoints[0].runbook_ref}} | OPTIONAL
   open_questions:
   {{endpoints[0].open_questions[0]}} | OPTIONAL

(Repeat the “Endpoint” block for each privileged endpoint_id.)

3. References
   Endpoint catalog: {{xref:API-01}}
   Endpoint specs: {{xref:API-02}}
   AuthZ rules: {{xref:API-04}} | OPTIONAL
   Audit trail: {{xref:ADMIN-03}} | OPTIONAL
   Admin capabilities: {{xref:ADMIN-01}} | OPTIONAL
   Admin safeguards: {{xref:ADMIN-06}} | OPTIONAL
   Rate limits: {{xref:RLIM-01}} | OPTIONAL

Cross-References
Upstream: {{xref:API-01}}, {{xref:API-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-06}} | OPTIONAL, {{xref:RUNBOOK-ADMIN}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where risk/audit/safeguards are missing; do not invent endpoint IDs.
intermediate: Required. Bind authz/audit and specify ABAC constraints.
advanced: Required. Add runbooks and two-person approval for critical endpoints if applicable.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, name, deny_behavior, audit_ref, rate_limit_ref, error_policy_ref, implementation_notes, runbook_ref, two-person approvals, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If endpoints list is UNKNOWN → block Completeness Gate.
If required_permissions is UNKNOWN → block Completeness Gate.
If audit_required is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ADMIN
Pass conditions:
required_fields_present == true
all endpoint_ids exist in API-01 (no new IDs introduced)
authz_constraints_defined == true
audit_requirements_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ADMIN-06

ADMIN-06 — Admin Observability & Safeguards (rate limits, alerts)
Header Block
template_id: ADMIN-06
title: Admin Observability & Safeguards (rate limits, alerts)
type: admin_observability_safeguards
template_version: 1.0.0
output_path: 10_app/admin/ADMIN-06_Admin_Observability_Safeguards.md
compliance_gate_id: TMP-05.PRIMARY.ADMIN
upstream_dependencies: ["ADMIN-01", "ADMIN-03", "ADMIN-05", "RLIM-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ADMIN-01", "ADMIN-03", "ADMIN-05", "RLIM-01", "RLIM-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical safeguards and observability requirements for admin/internal tools and privileged API usage, including confirmations/approvals, rate limiting, anomaly detection, auditing integrity, dashboards, and alerts. This template must be consistent with admin capabilities and audit trail specs and must not invent safeguards not supported by upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ADMIN-01 Admin Capabilities Matrix: {{admin.capabilities}}
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}}
ADMIN-05 Privileged API Surface Catalog: {{admin.privileged_api_catalog}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
RLIM-02 Rate Limit Catalog: {{rlim.catalog}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Safeguard policy statement (system-wide)
Confirmation patterns (two-step confirm, typed confirm)
Approval patterns (two-person, ticket required)
Break-glass controls (time-bounded, logged)
Rate limit policy for admin surfaces (bind to RLIM)
Anomaly detection signals (abnormal volume, unusual actions)
Audit integrity checks (missing audit record detection)
Dashboard requirements (admin actions, failures, audit health)
Alert requirements (privileged action spikes, audit write failures)
Runbook/procedure references (what to do on alert)

Optional Fields
Per-capability safeguard overrides | OPTIONAL
Geo/IP restrictions | OPTIONAL
Device posture requirements | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
High-risk/critical capabilities MUST have stronger safeguards (or UNKNOWN flagged).
Admin surfaces MUST be rate-limited unless explicitly exempted.
Audit write failures MUST alert.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Safeguards MUST be consistent with AuthZ ({{xref:API-04}}) and audit policy ({{xref:ADMIN-03}}).

Output Format

1. Safeguard Policy
   policy_statement: {{policy.statement}}
   risk_classes: {{policy.risk_classes}} | OPTIONAL
   default_safeguards_by_risk: {{policy.defaults_by_risk}} | OPTIONAL
2. Confirmation Patterns
   two_step_confirm: {{confirm.two_step}}
   typed_confirm: {{confirm.typed}} | OPTIONAL
   cooldown_seconds: {{confirm.cooldown_seconds}} | OPTIONAL
3. Approval Patterns
   two_person_supported: {{approval.two_person_supported}}
   ticket_required_supported: {{approval.ticket_required_supported}} | OPTIONAL
   approval_required_for: {{approval.required_for}} | OPTIONAL
4. Break-Glass Controls
   break_glass_supported: {{breakglass.supported}}
   time_bounded: {{breakglass.time_bounded}} | OPTIONAL
   max_duration: {{breakglass.max_duration}} | OPTIONAL
   audit_required: {{breakglass.audit_required}}
   postmortem_required: {{breakglass.postmortem_required}} | OPTIONAL
5. Rate Limits for Admin Surfaces
   ratelimit_policy_ref: {{ratelimit.policy_ref}} (expected: {{xref:RLIM-01}}) | OPTIONAL
   admin_limit_entries_ref: {{ratelimit.catalog_ref}} (expected: {{xref:RLIM-02}}) | OPTIONAL
   exemptions_ref: {{ratelimit.exemptions_ref}} (expected: {{xref:RLIM-05}}) | OPTIONAL
6. Anomaly Detection Signals
   signals:
   {{anomaly.signals[0]}}

{{anomaly.signals[1]}}
detection_rules_ref: {{anomaly.rules_ref}} (expected: {{xref:RLIM-03}}) | OPTIONAL

7. Audit Integrity Checks
   audit_write_failure_alert: {{audit_checks.write_failure_alert}}
   missing_audit_record_detection: {{audit_checks.missing_record_detection}}
   integrity_verification: {{audit_checks.integrity_verification}} | OPTIONAL
8. Dashboards
   admin_actions_dashboard: {{dashboards.actions}}
   audit_health_dashboard: {{dashboards.audit_health}} | OPTIONAL
   privileged_api_dashboard: {{dashboards.privileged_api}} | OPTIONAL
   minimum_panels: {{dashboards.minimum_panels}}
9. Alerts
   privileged_action_spike: {{alerts.privileged_action_spike}}
   audit_write_failures: {{alerts.audit_write_failures}}
   high_risk_action_detected: {{alerts.high_risk_action_detected}} | OPTIONAL
   break_glass_used: {{alerts.break_glass_used}} | OPTIONAL
   threshold_model: {{alerts.threshold_model}} (static/baseline/UNKNOWN)
   routing_policy: {{alerts.routing_policy}} | OPTIONAL
   runbook_ref: {{alerts.runbook_ref}} | OPTIONAL
10. Per-Capability Overrides (Optional)
    override
    capability_id: {{overrides[0].capability_id}}
    safeguards: {{overrides[0].safeguards}}
    notes: {{overrides[0].notes}} | OPTIONAL
11. References
    Capabilities matrix: {{xref:ADMIN-01}}
    Audit trail: {{xref:ADMIN-03}}
    Privileged API catalog: {{xref:ADMIN-05}} | OPTIONAL
    Rate limit policy: {{xref:RLIM-01}} | OPTIONAL
    Rate limit catalog: {{xref:RLIM-02}} | OPTIONAL
    Abuse signals/rules: {{xref:RLIM-03}} | OPTIONAL
    Enforcement actions: {{xref:RLIM-04}} | OPTIONAL
    Exemptions: {{xref:RLIM-05}} | OPTIONAL

Cross-References
Upstream: {{xref:ADMIN-01}}, {{xref:ADMIN-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:RUNBOOK-ADMIN}} | OPTIONAL, {{xref:OPS-OBS}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-AUDIT]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define baseline safeguards and core alerts; use UNKNOWN for anomaly tooling.

intermediate: Required. Bind safeguards to risk classes and define admin rate-limit linkage.
advanced: Required. Add audit integrity verification and per-capability overrides.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, policy defaults by risk, typed confirm, cooldown, ticket requirements, approval_required_for, breakglass time bounds/max duration, postmortem_required, ratelimit refs, anomaly signals, anomaly rules ref, integrity verification, dashboards optional, alerts routing/runbook, overrides, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If policy.statement is UNKNOWN → block Completeness Gate.
If alerts.audit_write_failures is UNKNOWN → block Completeness Gate.
If minimum_panels is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ADMIN
Pass conditions:
required_fields_present == true
safeguards_defined == true
admin_rate_limit_linkage_defined == true
dashboard_minimum_defined == true
alert_set_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Frontend/UI Implementation (FE)

Frontend/UI Implementation (FE)
FE-01 Route Map + Layout Spec
FE-02 Component Implementation Spec
FE-03 UI State Model (loading/empty/error/success)
FE-04 Data Binding Rules (what calls what, when)
FE-05 Accessibility Implementation Notes (a11y enforcement)
FE-06 Theming/Tokens Integration (DSYS binding)
FE-07 Error Handling UX Implementation (surface rules)

FE-01

FE-01 — Route Map + Layout Spec
Header Block
template_id: FE-01
title: Route Map + Layout Spec
type: frontend_route_layout_spec
template_version: 1.0.0
output_path: 10_app/frontend/FE-01_Route_Map_Layout_Spec.md
compliance_gate_id: TMP-05.PRIMARY.FE
upstream_dependencies: ["ROUTE-01", "ROUTE-02", "PRD-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-01", "ROUTE-02", "ROUTE-03", "UICP-01", "FE-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical route map and layout structure for the frontend: all routes/screens, their layout shells, navigation placement, required guards, and layout composition rules. This template must be consistent with route contracts and must not invent route IDs or screen IDs not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-01 Route Contract: {{route.contract}}
ROUTE-02 Navigation Map: {{route.nav_map}}
ROUTE-03 Deep Link Map: {{route.deep_link_map}} | OPTIONAL
UICP-01 Page Composition Patterns: {{ui.page_patterns}} | OPTIONAL
FE-06 Theming/Tokens Integration: {{ui.theming}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Route registry (route_id list)
Route path(s) and param schema refs
Screen binding (route_id → screen_id)
Layout shells (app shell, auth shell, admin shell, modal shell)
Navigation placement per shell (tabs/sidebar/topbar)
Guard bindings (auth gating, role gating)
Default route and not-found route
Route grouping (by domain/feature)
Layout composition rules (what regions exist)
Cross-route transition rules (modal vs push vs replace)

Optional Fields
Breadcrumb model | OPTIONAL
Route metadata (title, analytics id) | OPTIONAL
A/B routing via feature flags | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new route_ids or screen_ids; use only those in {{spec.routes_by_id}} and {{spec.screens_by_id}}.
All paths/params MUST conform to {{xref:ROUTE-01}}.
All guard rules MUST conform to {{xref:ROUTE-04}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
If unknown routes exist, fallback MUST conform to {{xref:ROUTE-06}}.

Output Format

1. Shells (Layout Containers)
   Shell
   shell_id: {{shells[0].shell_id}}
   name: {{shells[0].name}}
   description: {{shells[0].description}}
   regions: {{shells[0].regions}} (e.g., header/sidebar/content/footer)
   navigation_type: {{shells[0].navigation_type}} (tabs/sidebar/topbar/none/UNKNOWN)
   default_route_id: {{shells[0].default_route_id}}
   not_found_route_id: {{shells[0].not_found_route_id}} | OPTIONAL

(Repeat for each shell.)

2. Route Registry (by route_id)
   Route
   route_id: {{routes[0].route_id}}
   path: {{routes[0].path}}
   path_aliases: {{routes[0].path_aliases}} | OPTIONAL
   screen_id: {{routes[0].screen_id}}
   shell_id: {{routes[0].shell_id}}
   params_schema_ref: {{routes[0].params_schema_ref}} | OPTIONAL
   query_params_allowed: {{routes[0].query_params_allowed}} | OPTIONAL
   guards:
   auth_required: {{routes[0].guards.auth_required}}
   role_gates: {{routes[0].guards.role_gates}} | OPTIONAL
   feature_flag_gate: {{routes[0].guards.feature_flag_gate}} | OPTIONAL
   transition: {{routes[0].transition}} (push/replace/modal/UNKNOWN)
   nav_visibility: {{routes[0].nav_visibility}} (visible/hidden/UNKNOWN)

nav_label: {{routes[0].nav_label}} | OPTIONAL
icon_ref: {{routes[0].icon_ref}} | OPTIONAL
breadcrumbs: {{routes[0].breadcrumbs}} | OPTIONAL
analytics_id: {{routes[0].analytics_id}} | OPTIONAL
open_questions:
{{routes[0].open_questions[0]}} | OPTIONAL

(Repeat the “Route” block for each route_id.)

3. Route Grouping
   grouping_model: {{groups.model}} (by domain/feature/shell/UNKNOWN)
   groups:
   {{groups.items[0]}}
   {{groups.items[1]}} | OPTIONAL
4. Layout Composition Rules
   global_regions: {{layout.global_regions}}
   region_rules: {{layout.region_rules}}
   page_padding_rules: {{layout.page_padding_rules}} | OPTIONAL
   safe_area_rules: {{layout.safe_area_rules}} | OPTIONAL
5. Transition Rules
   modal_rules: {{transitions.modal_rules}}
   replace_rules: {{transitions.replace_rules}} | OPTIONAL
   back_behavior: {{transitions.back_behavior}} | OPTIONAL
6. References
   Route contract: {{xref:ROUTE-01}}
   Navigation map: {{xref:ROUTE-02}}
   Deep links: {{xref:ROUTE-03}} | OPTIONAL
   Guard rules: {{xref:ROUTE-04}} | OPTIONAL
   Unknown route handling: {{xref:ROUTE-06}} | OPTIONAL
   UI composition patterns: {{xref:UICP-01}} | OPTIONAL

Cross-References
Upstream: {{xref:ROUTE-01}}, {{xref:ROUTE-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FE-02}}, {{xref:FE-03}}, {{xref:FE-04}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for optional metadata; do not invent route_ids/screen_ids.
intermediate: Required. Bind every route to shell + guards + transition type.
advanced: Required. Add grouping and composition rules with deterministic transitions.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, path_aliases, params_schema_ref, query_params_allowed, role_gates, feature_flag_gate, nav_label, icon_ref, breadcrumbs,

analytics_id, grouping model, padding/safe area rules, transitions details, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If route registry is UNKNOWN → block Completeness Gate.
If any route has unknown path or unknown screen_id → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FE
Pass conditions:
required_fields_present == true
all route_ids/screen_ids reference existing IDs (no new IDs introduced)
shells_defined == true
guards_bound == true
placeholder_resolution == true
no_unapproved_unknowns == true

FE-02

FE-02 — Component Implementation Spec
Header Block
template_id: FE-02
title: Component Implementation Spec
type: frontend_component_implementation_spec
template_version: 1.0.0
output_path: 10_app/frontend/FE-02_Component_Implementation_Spec.md
compliance_gate_id: TMP-05.PRIMARY.FE
upstream_dependencies: ["FE-01", "UICP-02", "DSYS-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-01", "UICP-02", "FE-06", "FORM-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical specification format for implementing frontend components, including props/slots/variants, state handling, accessibility requirements, data binding hooks, theming/token usage, and test/acceptance references. This template must be consistent with composition patterns and must not invent component IDs not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-01 Route Map + Layout Spec: {{fe.route_layout}} | OPTIONAL
UICP-02 Component Composition Rules: {{ui.component_composition}} | OPTIONAL
FE-06 Theming/Tokens Integration: {{ui.theming}} | OPTIONAL
FORM-02 Field Spec: {{forms.field_specs}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
component_id (must exist in SPEC_INDEX)
component_name (human-readable)
purpose/usage
variants (variant_id list)
props contract (name/type/required/default)
slots/children contract (if applicable)
state model (loading/empty/error/success)
interaction behavior (events/callbacks)
a11y requirements (roles, labels, focus)
styling/token binding rules (theme tokens)
data dependencies (queries/mutations) if any
error handling behavior (bind to FE-07/CER)
test hooks / acceptance refs

Optional Fields
Performance notes | OPTIONAL
Analytics hooks | OPTIONAL
i18n notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new component_ids; use only {{spec.components_by_id}}.
Component behavior MUST align with UI composition rules ({{xref:UICP-02}}) and theming rules ({{xref:FE-06}}).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
If component touches forms, it MUST align with {{xref:FORM-02}} and {{xref:FORM-03}}.

Output Format

1. Component Identity
   component_id: {{component.component_id}}
   name: {{component.name}}
   purpose: {{component.purpose}}
   where_used: {{component.where_used}} | OPTIONAL
2. Variants
   variant
   variant_id: {{variants[0].variant_id}}
   name: {{variants[0].name}}
   when_to_use: {{variants[0].when_to_use}}
   visual_notes: {{variants[0].visual_notes}} | OPTIONAL

(Repeat for each variant.)

3. Props Contract
   prop
   name: {{props[0].name}}
   type: {{props[0].type}}
   required: {{props[0].required}}
   default: {{props[0].default}} | OPTIONAL
   description: {{props[0].description}} | OPTIONAL

(Repeat for each prop.)

4. Slots / Children (Optional)
   slots: {{slots}} | OPTIONAL
   children_rules: {{children.rules}} | OPTIONAL

5. State Model
   states_supported: {{state.states_supported}} (loading/empty/error/success/UNKNOWN)
   loading_behavior: {{state.loading_behavior}} | OPTIONAL
   empty_behavior: {{state.empty_behavior}} | OPTIONAL
   error_behavior: {{state.error_behavior}}
   success_behavior: {{state.success_behavior}} | OPTIONAL
6. Interaction Behavior
   events:
   {{interaction.events[0]}}
   {{interaction.events[1]}} | OPTIONAL
   callbacks: {{interaction.callbacks}} | OPTIONAL
   keyboard_interactions: {{interaction.keyboard}} | OPTIONAL
7. Accessibility (A11y)
   roles: {{a11y.roles}}
   labels: {{a11y.labels}}
   focus_order: {{a11y.focus_order}} | OPTIONAL
   aria_props: {{a11y.aria_props}} | OPTIONAL
8. Styling / Tokens
   token_usage: {{style.token_usage}}
   variants_token_rules: {{style.variants_token_rules}} | OPTIONAL
   responsive_rules: {{style.responsive_rules}} | OPTIONAL
9. Data Dependencies (Optional)
   queries: {{data.queries}} | OPTIONAL
   mutations: {{data.mutations}} | OPTIONAL
   cache_invalidation: {{data.cache_invalidation}} | OPTIONAL
10. Error Handling
    error_surface_rules: {{errors.surface_rules}} (bind to {{xref:FE-07}}/{{xref:CER-01}}) | OPTIONAL
    retry_behavior: {{errors.retry_behavior}} | OPTIONAL
11. Tests / Acceptance
    test_ids: {{tests.test_ids}} | OPTIONAL
    acceptance_refs: {{tests.acceptance_refs}} | OPTIONAL
12. References
    Route/layout context: {{xref:FE-01}} | OPTIONAL
    Composition rules: {{xref:UICP-02}} | OPTIONAL
    Theming/tokens: {{xref:FE-06}} | OPTIONAL
    Forms: {{xref:FORM-02}} | OPTIONAL
    Client error UX: {{xref:FE-07}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:FE-01}} | OPTIONAL
Downstream: {{xref:FE-03}}, {{xref:FE-04}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL,

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where prop/variant details are missing; do not invent component_ids.
intermediate: Required. Define state model, a11y, styling tokens, and interaction events.
advanced: Required. Add data dependency hooks, performance notes, and test/acceptance traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, where_used, visual_notes, prop defaults/descriptions, slots/children rules, loading/empty/success behaviors, interaction callbacks/keyboard, focus order/aria props, variants token rules, responsive rules, data deps, error surface rules, tests, analytics/i18n notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If component_id is UNKNOWN → block Completeness Gate.
If state.states_supported is UNKNOWN → block Completeness Gate.
If a11y.roles is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FE
Pass conditions:
required_fields_present == true
component_id exists in SPEC_INDEX (no new IDs introduced)
state_model_defined == true
a11y_requirements_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FE-03

FE-03 — UI State Model (loading/empty/error/success)
Header Block
template_id: FE-03
title: UI State Model (loading/empty/error/success)
type: frontend_ui_state_model
template_version: 1.0.0
output_path: 10_app/frontend/FE-03_UI_State_Model.md
compliance_gate_id: TMP-05.PRIMARY.FE
upstream_dependencies: ["FE-01", "CER-01", "CER-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-01", "FE-07", "CER-01", "CER-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical UI state model used across screens and components: the required state set (loading/empty/error/success), how transitions happen, what content appears in each state, and consistency rules for retries and recoveries. This template must be consistent with client error handling strategy and must not invent state types not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-01 Route Map + Layout Spec: {{fe.route_layout}} | OPTIONAL
FE-07 Error Handling UX Implementation: {{fe.error_ux}} | OPTIONAL
CER-01 Error Boundary Strategy: {{cer.error_boundaries}} | OPTIONAL
CER-02 Retry & Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Canonical state set (required states)
State definitions (what each means)
State transition rules (what events cause transitions)
Loading UX rules (spinners, skeletons, blocking vs non-blocking)
Empty state UX rules (copy, actions)
Error state UX rules (message, retry CTA, escalation)
Success state rules (rendering + partial load)
Retry rules (what is retryable, backoff)
Consistency rules across screens/components
Observability hooks (optional)

Optional Fields
Partial/streaming state handling | OPTIONAL

Optimistic UI notes | OPTIONAL
Accessibility notes for state changes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
All screens/components MUST implement the canonical state set unless explicitly exempted.
Error display MUST follow {{xref:FE-07}} and recovery MUST follow {{xref:CER-02}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Canonical State Set
   states:
   {{states[0]}}
   {{states[1]}}
   {{states[2]}}
   {{states[3]}}
2. State Definitions
   loading: {{defs.loading}}
   empty: {{defs.empty}}
   error: {{defs.error}}
   success: {{defs.success}}
3. Transition Rules
   loading_to_success: {{transitions.loading_to_success}}
   loading_to_empty: {{transitions.loading_to_empty}}
   loading_to_error: {{transitions.loading_to_error}}
   error_to_loading_retry: {{transitions.error_to_loading_retry}}
   success_to_loading_refresh: {{transitions.success_to_loading_refresh}} | OPTIONAL
4. Loading UX Rules
   loading_types_supported: {{loading.types_supported}}
   (spinner/skeleton/progress/UNKNOWN)
   blocking_loading_rules: {{loading.blocking_rules}} | OPTIONAL
   non_blocking_loading_rules: {{loading.non_blocking_rules}} | OPTIONAL
   timeout_behavior: {{loading.timeout_behavior}} | OPTIONAL
5. Empty State UX Rules
   empty_copy_rules: {{empty.copy_rules}}
   empty_actions: {{empty.actions}} | OPTIONAL
   empty_illustration_policy: {{empty.illustration_policy}} | OPTIONAL
6. Error State UX Rules
   error_message_policy: {{error.message_policy}}
   retry_cta_policy: {{error.retry_cta_policy}}
   escalation_policy: {{error.escalation_policy}} | OPTIONAL
   error_details_visibility: {{error.details_visibility}} | OPTIONAL

7. Success State Rules
   render_policy: {{success.render_policy}}
   partial_success_policy: {{success.partial_success_policy}} | OPTIONAL
   refresh_policy: {{success.refresh_policy}} | OPTIONAL
8. Retry Rules
   retryable_error_classes: {{retry.retryable_error_classes}}
   retry_backoff_policy: {{retry.backoff_policy}} | OPTIONAL
   max_retry_attempts: {{retry.max_attempts}} | OPTIONAL
9. Consistency Rules
   component_contract: {{consistency.component_contract}}
   screen_contract: {{consistency.screen_contract}}
   copy_tone_policy: {{consistency.copy_tone_policy}} | OPTIONAL
10. Accessibility Notes (Optional)
    announce_state_changes: {{a11y.announce_state_changes}} | OPTIONAL
    focus_management_rules: {{a11y.focus_management_rules}} | OPTIONAL
11. References
    Route/layout: {{xref:FE-01}} | OPTIONAL
    Error UX: {{xref:FE-07}} | OPTIONAL
    Error boundaries: {{xref:CER-01}} | OPTIONAL
    Retry patterns: {{xref:CER-02}} | OPTIONAL

Cross-References
Upstream: {{xref:FE-01}} | OPTIONAL, {{xref:CER-01}} | OPTIONAL, {{xref:CER-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FE-02}}, {{xref:FE-04}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for optional rules; implement the canonical state set.
intermediate: Required. Define transitions and retry behavior that matches API error policy.
advanced: Required. Add accessibility state announcements and consistent copy/pattern enforcement.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, success refresh policy, partial/streaming, optimistic UI, a11y notes, illustration policy, escalation policy, error details visibility, retry backoff/max attempts, copy tone policy, observability hooks, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If canonical state set is UNKNOWN → block Completeness Gate.
If error_message_policy is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FE

Pass conditions:
required_fields_present == true
canonical_states_defined == true
transition_rules_defined == true
error_state_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FE-04

FE-04 — Data Binding Rules (what calls what, when)
Header Block
template_id: FE-04
title: Data Binding Rules (what calls what, when)
type: frontend_data_binding_rules
template_version: 1.0.0
output_path: 10_app/frontend/FE-04_Data_Binding_Rules.md
compliance_gate_id: TMP-05.PRIMARY.FE
upstream_dependencies: ["FE-01", "API-01", "API-02", "SMD-02", "SMD-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-01", "API-01", "API-02", "SMD-01", "SMD-02", "SMD-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for frontend data binding: which screens/components call which endpoints, when calls happen (on mount, on focus, on user action), cache usage, invalidation rules, and error handling bindings. This template must be consistent with state management strategy and API endpoint specs and must not invent endpoint IDs or call patterns not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-01 Route Map + Layout Spec: {{fe.route_layout}}
API-01 Endpoint Catalog: {{api.endpoint_catalog}}
API-02 Endpoint Specs: {{api.endpoint_specs}}
SMD-01 State Architecture: {{smd.state_arch}} | OPTIONAL
SMD-02 Query/Cache Strategy: {{smd.cache_strategy}} | OPTIONAL
SMD-03 Mutation Patterns: {{smd.mutation_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Binding registry (screen_id/component_id → endpoint_id list)
Call timing rules (on mount/focus/interval/user action)
Parameter sourcing rules (route params, state, form values)
Caching rules (what is cached, TTL/stale time)
Invalidation rules (what mutations invalidate what queries)
Optimistic update policy (if allowed)
Error handling binding (to FE-07/CER patterns)
Auth/session requirements (re-auth on 401, etc.)
Rate limit/backoff behavior for repeated calls
Observability hooks (optional)

Optional Fields
Prefetch rules | OPTIONAL
Polling rules | OPTIONAL
Realtime subscription hooks | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new endpoint_ids; bind only to {{xref:API-01}} / {{xref:API-02}}.
Bindings MUST be explicit; no “implicit calls.”
Caching/invalidation MUST conform to {{xref:SMD-02}}/{{xref:SMD-03}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Error handling MUST follow {{xref:FE-07}} and {{xref:CER-02}} when applicable.

Output Format

1. Binding Registry
   Binding
   binding_id: {{bindings[0].binding_id}}
   screen_id: {{bindings[0].screen_id}} | OPTIONAL
   component_id: {{bindings[0].component_id}} | OPTIONAL
   endpoint_id: {{bindings[0].endpoint_id}}
   call_timing: {{bindings[0].call_timing}} (mount/focus/user_action/interval/UNKNOWN)
   trigger: {{bindings[0].trigger}} | OPTIONAL
   params_source: {{bindings[0].params_source}}
   cache_policy: {{bindings[0].cache_policy}} | OPTIONAL
   invalidation_rules: {{bindings[0].invalidation_rules}} | OPTIONAL
   optimistic_update: {{bindings[0].optimistic_update}} | OPTIONAL
   error_handling_ref: {{bindings[0].error_handling_ref}} | OPTIONAL
   auth_behavior: {{bindings[0].auth_behavior}} | OPTIONAL
   ratelimit_backoff: {{bindings[0].ratelimit_backoff}} | OPTIONAL
   notes: {{bindings[0].notes}} | OPTIONAL
   open_questions:
   {{bindings[0].open_questions[0]}} | OPTIONAL

(Repeat for each binding entry.)

2. Call Timing Rules (Global)
   mount_rules: {{timing.mount_rules}}
   focus_rules: {{timing.focus_rules}} | OPTIONAL
   interval_rules: {{timing.interval_rules}} | OPTIONAL
   user_action_rules: {{timing.user_action_rules}}
3. Parameter Sourcing Rules
   route_params: {{params.route_params}}

state_params: {{params.state_params}} | OPTIONAL
form_params: {{params.form_params}} | OPTIONAL
default_values: {{params.default_values}} | OPTIONAL

4. Caching Rules
   cache_enabled: {{cache.enabled}}
   stale_time: {{cache.stale_time}} | OPTIONAL
   ttl: {{cache.ttl}} | OPTIONAL
   cache_key_rules: {{cache.key_rules}}
   prefetch_policy: {{cache.prefetch_policy}} | OPTIONAL
5. Invalidation Rules
   invalidation_model: {{invalidate.model}} (key-based/tag-based/UNKNOWN)
   mutation_to_query_map: {{invalidate.mutation_to_query_map}}
   invalidate_on_success: {{invalidate.on_success}} | OPTIONAL
   invalidate_on_error: {{invalidate.on_error}} | OPTIONAL
6. Optimistic Update Policy
   optimistic_allowed: {{optimistic.allowed}}
   rollback_rules: {{optimistic.rollback_rules}} | OPTIONAL
7. Error Handling Binding
   error_ux_ref: {{errors.ux_ref}} (expected: {{xref:FE-07}}) | OPTIONAL
   retry_policy_ref: {{errors.retry_policy_ref}} (expected: {{xref:CER-02}}) | OPTIONAL
   session_expiry_ref: {{errors.session_expiry_ref}} | OPTIONAL
8. Auth & Session Behavior
   on_401_behavior: {{auth.on_401}}
   on_403_behavior: {{auth.on_403}} | OPTIONAL
   token_refresh_rules: {{auth.token_refresh_rules}} | OPTIONAL
9. Rate Limit / Backoff
   backoff_policy: {{ratelimit.backoff_policy}}
   max_retries: {{ratelimit.max_retries}} | OPTIONAL
10. Observability Hooks (Optional)
    client_logs: {{obs.client_logs}} | OPTIONAL
    metrics: {{obs.metrics}} | OPTIONAL
11. References
    Route map/layout: {{xref:FE-01}}
    Endpoint catalog/specs: {{xref:API-01}}, {{xref:API-02}}
    State architecture: {{xref:SMD-01}} | OPTIONAL
    Cache strategy: {{xref:SMD-02}} | OPTIONAL
    Mutation patterns: {{xref:SMD-03}} | OPTIONAL
    Client error handling: {{xref:FE-07}} | OPTIONAL

Cross-References
Upstream: {{xref:FE-01}}, {{xref:API-01}}, {{xref:API-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FE-07}}, {{xref:SMD-02}}, {{xref:SMD-03}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for cache/invalidation specifics; do not invent endpoint IDs.
intermediate: Required. Define binding registry entries and caching/invalidation rules.
advanced: Required. Add optimistic update + backoff rigor and session-expiry handling bindings.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, trigger, cache_policy, invalidation_rules, optimistic_update, error refs, auth behavior, ratelimit backoff, focus/interval rules, state/form params, stale_time/ttl, prefetch policy, invalidate on error, rollback rules, token refresh rules, max retries, observability hooks, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If binding registry is UNKNOWN → block Completeness Gate.
If any binding has UNKNOWN endpoint_id → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FE
Pass conditions:
required_fields_present == true
bindings_reference_existing_endpoint_ids == true
cache_invalidation_model_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FE-05

FE-05 — Accessibility Implementation Notes (a11y enforcement)
Header Block
template_id: FE-05
title: Accessibility Implementation Notes (a11y enforcement)
type: frontend_accessibility_notes
template_version: 1.0.0
output_path: 10_app/frontend/FE-05_Accessibility_Implementation_Notes.md
compliance_gate_id: TMP-05.PRIMARY.FE
upstream_dependencies: ["FE-01", "FE-02", "UICP-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-01", "FE-02", "UICP-05", "FORM-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical accessibility (a11y) implementation requirements across the frontend, including semantic roles, labels, focus management, keyboard navigation, screen reader announcements, color contrast considerations, and enforcement/testing practices. This template must be consistent with UI composition patterns and must not invent platform requirements not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-01 Route Map + Layout Spec: {{fe.route_layout}} | OPTIONAL
FE-02 Component Implementation Spec: {{fe.component_specs}} | OPTIONAL
UICP-05 Interaction Patterns: {{ui.interaction_patterns}} | OPTIONAL
FORM-03 Validation UX Rules: {{forms.validation_ux}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
A11y baseline standard target (WCAG level or internal standard)
Semantic structure rules (headings, landmarks)
Labeling rules (inputs, buttons, icons)
Focus management rules (modals, drawers, route changes)
Keyboard navigation rules (tab order, shortcuts)
Screen reader announcement rules (toasts, errors, loading)
Color/contrast rules (tokens)
Form accessibility rules (errors, required fields)
Media accessibility rules (alt text, captions)
Testing/enforcement plan (linting, CI checks, manual QA)

Optional Fields
Platform-specific notes (web vs mobile) | OPTIONAL
Reduced motion rules | OPTIONAL
Localization/i18n a11y notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
A11y rules MUST be applied consistently across FE-02 component implementations.
Forms MUST follow {{xref:FORM-03}} validation a11y rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. A11y Standard Target
   standard: {{a11y.standard}} (WCAG_2.1_AA/UNKNOWN)
   notes: {{a11y.notes}} | OPTIONAL
2. Semantic Structure
   landmarks_required: {{semantic.landmarks_required}}
   heading_hierarchy_rules: {{semantic.heading_rules}}
   list_table_rules: {{semantic.list_table_rules}} | OPTIONAL
3. Labeling Rules
   inputs_label_rule: {{labels.inputs}}
   buttons_label_rule: {{labels.buttons}}
   icon_button_rule: {{labels.icon_buttons}}
   aria_label_policy: {{labels.aria_label_policy}} | OPTIONAL
4. Focus Management
   route_change_focus: {{focus.route_change_focus}}
   modal_focus_trap: {{focus.modal_focus_trap}}
   drawer_focus_rules: {{focus.drawer_focus_rules}} | OPTIONAL
   return_focus_rule: {{focus.return_focus_rule}}
5. Keyboard Navigation
   tab_order_rules: {{keyboard.tab_order_rules}}
   skip_links_supported: {{keyboard.skip_links_supported}} | OPTIONAL
   shortcut_policy: {{keyboard.shortcut_policy}} | OPTIONAL
6. Screen Reader Announcements
   toast_announce_policy: {{sr.toasts}}
   error_announce_policy: {{sr.errors}}
   loading_announce_policy: {{sr.loading}} | OPTIONAL
   live_region_rules: {{sr.live_region_rules}} | OPTIONAL
7. Color / Contrast
   contrast_target: {{color.contrast_target}}
   token_rules: {{color.token_rules}}
   focus_ring_rules: {{color.focus_ring_rules}} | OPTIONAL

8. Forms Accessibility
   required_field_indication: {{forms.required_field_indication}}
   error_association_rules: {{forms.error_association_rules}}
   inline_error_focus_rules: {{forms.inline_error_focus_rules}} | OPTIONAL
9. Media Accessibility
   alt_text_policy: {{media.alt_text_policy}}
   caption_policy: {{media.caption_policy}} | OPTIONAL
   transcript_policy: {{media.transcript_policy}} | OPTIONAL
10. Testing / Enforcement
    linting_rules: {{tests.linting_rules}}
    ci_checks: {{tests.ci_checks}} | OPTIONAL
    manual_qa_checklist: {{tests.manual_qa_checklist}}
    tools: {{tests.tools}} | OPTIONAL
11. References
    Components: {{xref:FE-02}} | OPTIONAL
    Interaction patterns: {{xref:UICP-05}} | OPTIONAL
    Form validation UX: {{xref:FORM-03}} | OPTIONAL

Cross-References
Upstream: {{xref:FE-01}} | OPTIONAL, {{xref:FE-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FE-07}}, {{xref:CER-01}} | OPTIONAL
Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for platform-specific tooling; implement baseline a11y rules.
intermediate: Required. Define focus/keyboard/screen reader policies and enforcement.
advanced: Required. Add CI gating and audit-style checklists with measurable targets.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, list/table rules, aria label policy, drawer rules, skip links, shortcut policy, loading/live region rules, focus ring rules, inline error focus, captions/transcripts, ci checks, tools, platform-specific notes, reduced motion, i18n notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If a11y.standard is UNKNOWN → block Completeness Gate.
If focus.modal_focus_trap is UNKNOWN → block Completeness Gate.
If tests.manual_qa_checklist is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FE
Pass conditions:

required_fields_present == true
a11y_standard_defined == true
focus_and_keyboard_rules_defined == true
enforcement_plan_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FE-06

FE-06 — Theming/Tokens Integration (DSYS binding)
Header Block
template_id: FE-06
title: Theming/Tokens Integration (DSYS binding)
type: frontend_theming_tokens_integration
template_version: 1.0.0
output_path: 10_app/frontend/FE-06_Theming_Tokens_Integration.md
compliance_gate_id: TMP-05.PRIMARY.FE
upstream_dependencies: ["DSYS-01", "UICP-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "DSYS-01", "DSYS-02", "UICP-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical integration rules between the frontend implementation and the design system tokens: token sources, naming, usage rules, theme switching, component token binding, and enforcement checks. This template must be consistent with design system docs and must not invent token names or theme mechanics not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
DSYS-01 Tokens Catalog: {{dsys.tokens_catalog}} | OPTIONAL
DSYS-02 Theme Definitions: {{dsys.themes}} | OPTIONAL
UICP-04 Content Hierarchy Rules: {{ui.typography_spacing}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Token source of truth (where tokens are defined)
Token naming conventions
Theme list (light/dark/brand variants)
Theme switching model (system/user/UNKNOWN)
Token usage rules (no raw colors, no magic numbers)
Component binding rules (how components consume tokens)
Typography scale binding (heading/body tokens)
Spacing/radius/shadow token binding
Contrast requirements / accessibility alignment
Enforcement plan (lint rules, CI checks)

Optional Fields
Runtime theming (dynamic tokens) | OPTIONAL
Platform differences (web vs mobile) | OPTIONAL

Custom theme support | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new token names; use only those in {{xref:DSYS-01}}.
UI implementations MUST reference tokens instead of raw values unless explicitly allowed.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
A11y contrast requirements MUST align with {{xref:FE-05}} when present.

Output Format

1. Token Source of Truth
   source_location: {{tokens.source_location}}
   format: {{tokens.format}} (json/css_vars/tailwind/UNKNOWN)
   versioning: {{tokens.versioning}} | OPTIONAL
2. Naming Conventions
   token_prefixes: {{naming.prefixes}}
   token_categories: {{naming.categories}}
   (color/typography/spacing/radius/shadow/UNKNOWN)
   example_tokens:
   {{naming.examples[0]}}
   {{naming.examples[1]}} | OPTIONAL
3. Themes
   themes:
   {{themes.list[0]}}
   {{themes.list[1]}} | OPTIONAL
   default_theme: {{themes.default}}
   theme_switching_model: {{themes.switching_model}} (system/user/UNKNOWN)
   theme_storage: {{themes.storage}} | OPTIONAL
4. Token Usage Rules
   no_raw_colors_rule: {{usage.no_raw_colors_rule}}
   no_magic_numbers_rule: {{usage.no_magic_numbers_rule}}
   fallback_rules: {{usage.fallback_rules}} | OPTIONAL
5. Component Binding Rules
   binding_pattern: {{binding.pattern}} (css_vars/context/provider/UNKNOWN)
   component_token_map_ref: {{binding.component_token_map_ref}} | OPTIONAL
   variant_token_binding: {{binding.variant_binding}} | OPTIONAL
6. Typography Binding
   typography_scale_ref: {{typography.scale_ref}} | OPTIONAL
   heading_tokens: {{typography.heading_tokens}}
   body_tokens: {{typography.body_tokens}}
   line_height_rules: {{typography.line_height_rules}} | OPTIONAL

7. Spacing / Radius / Shadow Binding
   spacing_tokens: {{layout.spacing_tokens}}
   radius_tokens: {{layout.radius_tokens}}
   shadow_tokens: {{layout.shadow_tokens}} | OPTIONAL
8. Contrast / A11y Alignment
   contrast_target: {{a11y.contrast_target}}
   token_contrast_rules: {{a11y.token_contrast_rules}} | OPTIONAL
9. Enforcement Plan
   lint_rules: {{enforce.lint_rules}}
   ci_checks: {{enforce.ci_checks}} | OPTIONAL
   manual_review_checklist: {{enforce.manual_review_checklist}} | OPTIONAL
10. References
   Tokens catalog: {{xref:DSYS-01}} | OPTIONAL
   Theme definitions: {{xref:DSYS-02}} | OPTIONAL
   A11y notes: {{xref:FE-05}} | OPTIONAL
   Typography/spacing hierarchy: {{xref:UICP-04}} | OPTIONAL

Cross-References
Upstream: {{xref:DSYS-01}} | OPTIONAL, {{xref:DSYS-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FE-02}}, {{xref:FE-05}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for platform specifics; enforce “no raw values” rules.
intermediate: Required. Define theme switching and component binding patterns.
advanced: Required. Add enforcement automation (lint/CI) and contrast validation rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, tokens versioning, naming examples, theme storage, fallback rules, component token map ref, variant token binding, typography scale ref, line height rules, shadow tokens, token contrast rules, ci checks, manual review checklist, runtime theming, platform differences, custom themes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If tokens.source_location is UNKNOWN → block Completeness Gate.
If usage.no_raw_colors_rule is UNKNOWN → block Completeness Gate.
If themes.default is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FE
Pass conditions:
required_fields_present == true
token_source_defined == true

theme_model_defined == true
usage_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FE-07

FE-07 — Error Handling UX Implementation (surface rules)
Header Block
template_id: FE-07
title: Error Handling UX Implementation (surface rules)
type: frontend_error_handling_ux
template_version: 1.0.0
output_path: 10_app/frontend/FE-07_Error_Handling_UX_Implementation.md
compliance_gate_id: TMP-05.PRIMARY.FE
upstream_dependencies: ["API-03", "CER-01", "CER-02", "CER-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-03", "CER-01", "CER-02", "CER-03", "CER-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical UX rules for surfacing and recovering from errors in the client: how different error classes map to UI surfaces (inline, toast, modal, full-screen), what copy to show, what actions to offer (retry, re-auth, report), and how to keep behavior consistent across screens and components. This template must be consistent with API error policy and client recovery strategies and must not invent error classes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-03 Error & Status Code Policy: {{api.error_policy}}
CER-01 Error Boundary Strategy: {{cer.error_boundaries}} | OPTIONAL
CER-02 Retry & Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
CER-03 Offline/Error Mode UX: {{cer.offline_mode}} | OPTIONAL
CER-04 Session Expiry Handling: {{cer.session_expiry}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Error class taxonomy (by API status/error_code categories)
Error surface mapping (class → UI surface)
Copy policy (tone, what not to expose)
Retry CTA rules (when to show retry; backoff guidance)
Inline field error rules (validation errors)
Global error boundary behavior (crash/fatal)
Offline mode behavior
Session expiry behavior (401 handling)
Escalation/reporting pathway (optional)
Logging/crash reporting hooks (fields, redaction)

Optional Fields
Per-route overrides | OPTIONAL
Localization/i18n rules | OPTIONAL
User support links | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not leak sensitive details from server errors; copy must be safe.
Validation errors MUST surface at field-level when available.
Session expiry MUST follow {{xref:CER-04}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Error Taxonomy
   classes:
   validation: {{taxonomy.validation}}
   auth: {{taxonomy.auth}}
   forbidden: {{taxonomy.forbidden}}
   not_found: {{taxonomy.not_found}}
   conflict: {{taxonomy.conflict}}
   rate_limit: {{taxonomy.rate_limit}}
   dependency: {{taxonomy.dependency}}
   internal: {{taxonomy.internal}}
   offline: {{taxonomy.offline}} | OPTIONAL
   fatal: {{taxonomy.fatal}} | OPTIONAL
2. Surface Mapping (Class → UX)
   validation_surface: {{surfaces.validation_surface}} (inline/summary/UNKNOWN)
   auth_surface: {{surfaces.auth_surface}}
   forbidden_surface: {{surfaces.forbidden_surface}}
   not_found_surface: {{surfaces.not_found_surface}}
   conflict_surface: {{surfaces.conflict_surface}} | OPTIONAL
   rate_limit_surface: {{surfaces.rate_limit_surface}}
   dependency_surface: {{surfaces.dependency_surface}}
   internal_surface: {{surfaces.internal_surface}}
   offline_surface: {{surfaces.offline_surface}} | OPTIONAL
   fatal_surface: {{surfaces.fatal_surface}} | OPTIONAL
3. Copy Policy
   tone: {{copy.tone}}
   safe_message_rules: {{copy.safe_message_rules}}
   redaction_rules: {{copy.redaction_rules}} | OPTIONAL
4. Retry CTA Rules
   retryable_classes: {{retry.retryable_classes}}

retry_button_label: {{retry.button_label}} | OPTIONAL
backoff_policy: {{retry.backoff_policy}} | OPTIONAL
max_attempts: {{retry.max_attempts}} | OPTIONAL

5. Field Validation Errors
   field_error_binding: {{validation.field_error_binding}}
   focus_first_error: {{validation.focus_first_error}} | OPTIONAL
   announce_errors: {{validation.announce_errors}} | OPTIONAL
6. Global Error Boundary Behavior
   boundary_scope: {{boundary.scope}} (global/per_route/UNKNOWN)
   fallback_ui: {{boundary.fallback_ui}}
   reset_rules: {{boundary.reset_rules}} | OPTIONAL
7. Offline Mode Behavior
   offline_detect_rule: {{offline.detect_rule}} | OPTIONAL
   offline_ui: {{offline.ui}} | OPTIONAL
   offline_recovery: {{offline.recovery}} | OPTIONAL
8. Session Expiry / Auth Failures
   on_401_behavior: {{auth.on_401_behavior}}
   re_auth_flow_ref: {{auth.re_auth_flow_ref}} | OPTIONAL
   state_restore_policy: {{auth.state_restore_policy}} | OPTIONAL
9. Reporting / Escalation (Optional)
   report_button_supported: {{report.supported}} | OPTIONAL
   report_payload_fields: {{report.payload_fields}} | OPTIONAL
10. Logging / Crash Reporting
    log_required_fields: {{logs.required_fields}}
    crash_reporting_enabled: {{logs.crash_reporting_enabled}} | OPTIONAL
    redaction_policy: {{logs.redaction_policy}} | OPTIONAL
11. References
    API error policy: {{xref:API-03}}
    Error boundaries: {{xref:CER-01}} | OPTIONAL
    Retry patterns: {{xref:CER-02}} | OPTIONAL
    Offline UX: {{xref:CER-03}} | OPTIONAL
    Session expiry: {{xref:CER-04}} | OPTIONAL
    Forms validation UX: {{xref:FORM-03}} | OPTIONAL

Cross-References
Upstream: {{xref:API-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FE-03}}, {{xref:CER-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Map classes to surfaces; use UNKNOWN for optional reporting/offline specifics.
intermediate: Required. Define retry/backoff and safe copy policy; bind validation errors to fields.
advanced: Required. Add crash logging/redaction rigor and boundary reset patterns.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, conflict surface, offline/fatal classes, copy redaction rules, retry labels/backoff/max attempts, validation focus/announce, boundary reset rules, offline section, reporting section, crash reporting enabled, redaction policy, per-route overrides, i18n rules, support links, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If surfaces.internal_surface is UNKNOWN → block Completeness Gate.
If on_401_behavior is UNKNOWN → block Completeness Gate.
If log_required_fields is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FE
Pass conditions:
required_fields_present == true
error_taxonomy_defined == true
surface_mapping_defined == true
session_expiry_behavior_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

State Management & Data Fetching (SMD)

State Management & Data Fetching (SMD)
SMD-01 State Architecture (global vs local, stores)
SMD-02 Query/Cache Strategy (client cache rules, invalidation)
SMD-03 Mutation Patterns (optimistic updates, rollback)
SMD-04 Realtime Subscription Patterns (WS hooks, reconciliation)
SMD-05 Offline State Handling (queueing, retry)
SMD-06 Observability for Client Data Layer (logs/metrics)

SMD-01

SMD-01 — State Architecture (global vs local, stores)
Header Block
template_id: SMD-01
title: State Architecture (global vs local, stores)
type: client_state_architecture
template_version: 1.0.0
output_path: 10_app/state/SMD-01_State_Architecture.md
compliance_gate_id: TMP-05.PRIMARY.SMD
upstream_dependencies: ["FE-01", "FE-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-01", "FE-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical client state architecture: what state is global vs local, what stores exist, ownership boundaries, state shape conventions, and how state interacts with navigation, caching, and offline behavior. This template must be consistent with frontend route/data binding rules and must not invent stores/state domains not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-01 Route Map + Layout: {{fe.route_layout}} | OPTIONAL
FE-04 Data Binding Rules: {{fe.data_binding}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
State philosophy statement (global vs local criteria)
Global store registry (store_id list)
Store ownership model (which domain owns what)
State shape conventions (normalized entities vs denormalized)
Local UI state guidance (component/screen state)
Derived state rules (selectors/memoization)
Persistence policy (what persists across sessions)
Hydration rules (startup state)
Reset rules (logout, tenant switch)
Interaction with cache layer (SMD-02)

Optional Fields
Debug tooling notes | OPTIONAL
Dev-only state instrumentation | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent new store_ids; use only those present in upstream inputs (or mark UNKNOWN).
Global stores MUST have clear ownership and reset behavior.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Architecture Overview
   philosophy: {{arch.philosophy}}
   global_vs_local_rule: {{arch.global_vs_local_rule}}
   notes: {{arch.notes}} | OPTIONAL
2. Global Store Registry
   Store
   store_id: {{stores[0].store_id}}
   name: {{stores[0].name}}
   owner_domain: {{stores[0].owner_domain}}
   purpose: {{stores[0].purpose}}
   state_shape: {{stores[0].state_shape}} | OPTIONAL
   persistence: {{stores[0].persistence}} (none/session/disk/UNKNOWN)
   reset_on: {{stores[0].reset_on}}
   selectors: {{stores[0].selectors}} | OPTIONAL
   actions: {{stores[0].actions}} | OPTIONAL
   open_questions:
   {{stores[0].open_questions[0]}} | OPTIONAL

(Repeat for each store_id.)

3. State Shape Conventions
   entity_normalization: {{shape.normalization}}
   ids_vs_objects_rule: {{shape.ids_vs_objects_rule}} | OPTIONAL
   derived_state_rule: {{shape.derived_state_rule}} | OPTIONAL
4. Local UI State Guidance
   when_local: {{local.when}}
   when_global: {{local.when_global}}
   screen_state_pattern: {{local.screen_state_pattern}} | OPTIONAL
5. Persistence & Hydration
   persisted_stores: {{persist.stores}} | OPTIONAL
   hydration_order: {{persist.hydration_order}}
   migration_rules: {{persist.migration_rules}} | OPTIONAL
6. Reset Rules
   logout_reset: {{reset.logout}}
   tenant_switch_reset: {{reset.tenant_switch}} | OPTIONAL
   soft_reset_rules: {{reset.soft_reset_rules}} | OPTIONAL

7. Cache Layer Interaction
   cache_source_of_truth: {{cache.source_of_truth}} (server_cache/store/UNKNOWN)
   cache_binding_ref: {{cache.binding_ref}} (expected: {{xref:SMD-02}}) | OPTIONAL
8. References
   Data binding rules: {{xref:FE-04}} | OPTIONAL
   Cache strategy: {{xref:SMD-02}} | OPTIONAL
   Mutation patterns: {{xref:SMD-03}} | OPTIONAL
   Offline handling: {{xref:SMD-05}} | OPTIONAL

Cross-References
Upstream: {{xref:FE-01}} | OPTIONAL, {{xref:FE-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SMD-02}}, {{xref:SMD-03}}, {{xref:SMD-05}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for store details if missing; define philosophy and reset rules.
intermediate: Required. Populate global store registry and persistence/hydration rules.
advanced: Required. Add normalized state conventions and selector/memoization guidance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, arch notes, state_shape, selectors/actions, ids_vs_objects rule, derived state rule, screen patterns, persisted stores, migration rules, tenant switch reset, soft reset, cache binding ref, debug tooling, instrumentation, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If global store registry is UNKNOWN → block Completeness Gate.
If reset.logout is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SMD
Pass conditions:
required_fields_present == true
global_store_registry_defined == true
reset_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SMD-02

SMD-02 — Query/Cache Strategy (client cache rules, invalidation)
Header Block
template_id: SMD-02
title: Query/Cache Strategy (client cache rules, invalidation)
type: client_query_cache_strategy
template_version: 1.0.0
output_path: 10_app/state/SMD-02_Query_Cache_Strategy.md
compliance_gate_id: TMP-05.PRIMARY.SMD
upstream_dependencies: ["SMD-01", "FE-04", "API-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SMD-01", "FE-04", "API-03", "PFS-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical client query and caching strategy: cache ownership, key conventions, stale/TTL rules, refetch triggers, invalidation model, pagination caching, and how errors map into client state. This template must be consistent with data binding rules and API pagination/error contracts and must not invent caching mechanisms not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SMD-01 State Architecture: {{smd.state_arch}}
FE-04 Data Binding Rules: {{fe.data_binding}}
API-03 Error & Status Code Policy: {{api.error_policy}} | OPTIONAL
PFS-02 Pagination Rules: {{pfs.pagination_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Cache philosophy (source of truth: server cache vs store)
Query key conventions (stable, deterministic)
Cache time rules (stale time, TTL)
Refetch triggers (focus, reconnect, interval, mutation success)
Pagination caching rules (cursor/offset)
Invalidation model (key-based/tag-based)
Error caching policy (cache error? when)
Data normalization rules (if any)
Security constraints (don’t cache secrets/PII)
Observability hooks (optional)

Optional Fields
Prefetch rules | OPTIONAL
Offline cache behavior binding (SMD-05) | OPTIONAL
Cache persistence rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Cache keys MUST be deterministic and based on endpoint_id + params.
Pagination caching MUST align with {{xref:PFS-02}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Do not cache sensitive fields unless explicitly allowed by security standards.

Output Format

1. Cache Philosophy
   source_of_truth: {{cache.source_of_truth}} (server_cache/store/UNKNOWN)
   ownership: {{cache.ownership}}
   notes: {{cache.notes}} | OPTIONAL
2. Query Key Conventions
   key_format: {{keys.format}}
   include_params_rule: {{keys.include_params_rule}}
   namespace_rules: {{keys.namespace_rules}} | OPTIONAL
   examples:
   {{keys.examples[0]}}
   {{keys.examples[1]}} | OPTIONAL
3. Cache Time Rules
   stale_time_ms: {{times.stale_time_ms}}
   ttl_ms: {{times.ttl_ms}} | OPTIONAL
   gc_time_ms: {{times.gc_time_ms}} | OPTIONAL
4. Refetch Triggers
   on_focus: {{refetch.on_focus}}
   on_reconnect: {{refetch.on_reconnect}} | OPTIONAL
   on_interval: {{refetch.on_interval}} | OPTIONAL
   on_mutation_success: {{refetch.on_mutation_success}}
5. Pagination Caching
   mode: {{pagination.mode}} (cursor/offset/UNKNOWN)
   page_key_rule: {{pagination.page_key_rule}}
   merge_rule: {{pagination.merge_rule}} | OPTIONAL
   reset_on_sort_filter_change: {{pagination.reset_on_sort_filter_change}} | OPTIONAL
6. Invalidation Model
   model: {{invalidate.model}} (key-based/tag-based/UNKNOWN)
   tags_schema: {{invalidate.tags_schema}} | OPTIONAL

invalidation_on_mutation: {{invalidate.on_mutation}}
scoped_invalidation: {{invalidate.scoped_invalidation}} | OPTIONAL

7. Error Caching Policy
   cache_errors: {{errors.cache_errors}}
   retry_policy_ref: {{errors.retry_policy_ref}} | OPTIONAL
   error_normalization: {{errors.normalization}} | OPTIONAL
8. Data Normalization Rules
   normalize_enabled: {{normalize.enabled}}
   entity_key_rule: {{normalize.entity_key_rule}} | OPTIONAL
   merge_policy: {{normalize.merge_policy}} | OPTIONAL
9. Security Constraints
   pii_cache_policy: {{security.pii_cache_policy}}
   secret_cache_policy: {{security.secret_cache_policy}}
   logout_cache_clear_rule: {{security.logout_cache_clear_rule}}
10. Observability Hooks (Optional)
    cache_hit_metric: {{obs.cache_hit_metric}} | OPTIONAL
    cache_miss_metric: {{obs.cache_miss_metric}} | OPTIONAL
    stale_served_metric: {{obs.stale_served_metric}} | OPTIONAL
11. References
    State architecture: {{xref:SMD-01}}
    Data binding: {{xref:FE-04}}
    Pagination rules: {{xref:PFS-02}} | OPTIONAL
    Mutation patterns: {{xref:SMD-03}} | OPTIONAL
    Offline handling: {{xref:SMD-05}} | OPTIONAL
    API error policy: {{xref:API-03}} | OPTIONAL

Cross-References
Upstream: {{xref:SMD-01}}, {{xref:FE-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SMD-03}}, {{xref:SMD-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define key format and stale time; use UNKNOWN for optional TTL/persistence.
intermediate: Required. Define invalidation model and pagination caching rules.
advanced: Required. Add normalization, security cache policies, and observability hooks.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, cache notes, namespace rules, example keys, ttl/gc times, reconnect/interval refetch, merge/reset rules, tags schema, scoped invalidation, retry policy ref, error normalization, normalization rules, observability hooks, prefetch/offline/persistence notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If keys.format is UNKNOWN → block Completeness Gate.
If invalidate.model is UNKNOWN → block Completeness Gate.
If security.logout_cache_clear_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SMD
Pass conditions:
required_fields_present == true
query_key_conventions_defined == true
invalidation_model_defined == true
security_cache_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SMD-03

SMD-03 — Mutation Patterns (optimistic updates, rollback)
Header Block
template_id: SMD-03
title: Mutation Patterns (optimistic updates, rollback)
type: client_mutation_patterns
template_version: 1.0.0
output_path: 10_app/state/SMD-03_Mutation_Patterns.md
compliance_gate_id: TMP-05.PRIMARY.SMD
upstream_dependencies: ["SMD-02", "FE-04", "API-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SMD-02", "FE-04", "API-02", "API-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical client mutation patterns: how writes are performed, how optimistic updates work, rollback rules, invalidation on success/failure, conflict handling, and how errors are surfaced. This template must be consistent with cache strategy and API error policy and must not invent mutation behaviors not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SMD-02 Query/Cache Strategy: {{smd.cache_strategy}}
FE-04 Data Binding Rules: {{fe.data_binding}}
API-02 Endpoint Specs: {{api.endpoint_specs}} | OPTIONAL
API-03 Error & Status Code Policy: {{api.error_policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Mutation initiation rules (user action vs auto)
Mutation request pattern (endpoint binding, payload construction)
Optimistic update policy (allowed/where)
Optimistic update mechanics (update cache/store)
Rollback rules (on failure)
Invalidation rules (what queries to invalidate)
Conflict handling pattern (409, version mismatch)
Retry policy (when to retry writes)
Error surfacing (bind to FE-07/CER)
Idempotency guidance for writes (client idempotency keys if used)

Optional Fields
Batch mutations | OPTIONAL

Offline queued writes (bind to SMD-05) | OPTIONAL
Background refresh after mutation | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Optimistic updates MUST be rollback-safe and must not violate server invariants.
Invalidation MUST conform to {{xref:SMD-02}}.
Errors MUST map to {{xref:API-03}} and surface per {{xref:FE-07}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Mutation Initiation
   when_mutations_run: {{mutations.when}}
   guard_rules: {{mutations.guard_rules}} | OPTIONAL
2. Request Pattern
   endpoint_binding_rule: {{request.endpoint_binding_rule}}
   payload_construction_rule: {{request.payload_construction_rule}}
   headers_required: {{request.headers_required}} | OPTIONAL
   idempotency_key_supported: {{request.idempotency_key_supported}} | OPTIONAL
3. Optimistic Update Policy
   optimistic_allowed: {{optimistic.allowed}}
   allowed_for_actions: {{optimistic.allowed_for_actions}} | OPTIONAL
   disallowed_for_actions: {{optimistic.disallowed_for_actions}} | OPTIONAL
4. Optimistic Update Mechanics
   update_target: {{optimistic.update_target}} (cache/store/both/UNKNOWN)
   apply_update_rule: {{optimistic.apply_rule}}
   temporary_id_policy: {{optimistic.temp_id_policy}} | OPTIONAL
5. Rollback Rules
   rollback_on_error: {{rollback.on_error}}
   rollback_mechanism: {{rollback.mechanism}}
   rollback_scope: {{rollback.scope}} | OPTIONAL
6. Invalidation Rules
   invalidation_model_ref: {{invalidate.model_ref}} (expected: {{xref:SMD-02}})
   invalidate_keys_on_success: {{invalidate.on_success}}
   invalidate_keys_on_error: {{invalidate.on_error}} | OPTIONAL
   refetch_after_success: {{invalidate.refetch_after_success}} | OPTIONAL
7. Conflict Handling
   on_409_behavior: {{conflict.on_409}}
   version_field_rule: {{conflict.version_field_rule}} | OPTIONAL
   merge_resolution_policy: {{conflict.merge_policy}} | OPTIONAL
8. Retry Policy
   retry_writes: {{retry.retry_writes}}

retryable_errors: {{retry.retryable_errors}} | OPTIONAL
backoff_policy: {{retry.backoff_policy}} | OPTIONAL
max_attempts: {{retry.max_attempts}} | OPTIONAL

9. Error Surfacing
   error_ux_ref: {{errors.ux_ref}} (expected: {{xref:FE-07}}) | OPTIONAL
   field_error_binding: {{errors.field_error_binding}} | OPTIONAL
10. Idempotency Guidance
    client_idempotency_key_rule: {{idem.client_key_rule}} | OPTIONAL
    dedupe_behavior: {{idem.dedupe_behavior}} | OPTIONAL
11. References
    Cache strategy: {{xref:SMD-02}}
    Data binding: {{xref:FE-04}}
    API endpoint specs: {{xref:API-02}} | OPTIONAL
    API error policy: {{xref:API-03}} | OPTIONAL
    Error UX: {{xref:FE-07}} | OPTIONAL
    Offline handling: {{xref:SMD-05}} | OPTIONAL

Cross-References
Upstream: {{xref:SMD-02}}, {{xref:FE-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SMD-05}} | OPTIONAL, {{xref:CER-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for optimistic/rollback details if not provided; define invalidation skeleton.
intermediate: Required. Define optimistic update mechanics, rollback, and conflict handling.
advanced: Required. Add idempotency keys, retry/backoff rigor, and offline queue binding if used.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, guard rules, headers required, idempotency key supported, allowed/disallowed action lists, update_target, temp id policy, rollback scope, invalidate on error, refetch after success, version field rule, merge policy, retryable errors/backoff/max attempts, field error binding, idempotency guidance, batch/offline notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If optimistic.allowed is UNKNOWN → block Completeness Gate.
If rollback.mechanism is UNKNOWN → block Completeness Gate.
If invalidate.on_success is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SMD
Pass conditions:
required_fields_present == true

optimistic_policy_defined == true
rollback_rules_defined == true
invalidation_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SMD-04

SMD-04 — Realtime Subscription Patterns (WS hooks, reconciliation)
Header Block
template_id: SMD-04
title: Realtime Subscription Patterns (WS hooks, reconciliation)
type: client_realtime_subscription_patterns
template_version: 1.0.0
output_path: 10_app/state/SMD-04_Realtime_Subscription_Patterns.md
compliance_gate_id: TMP-05.PRIMARY.SMD
upstream_dependencies: ["SMD-02", "EVT-01", "API-07", "CER-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SMD-02", "EVT-01", "EVT-04", "API-07", "CER-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical client realtime subscription patterns: how the client subscribes to realtime streams (typically WebSocket), how messages are normalized into cache/store, reconciliation rules with server truth, ordering/dedupe handling, and recovery patterns on disconnect. This template must be consistent with event delivery semantics and WS API specs and must not invent realtime capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SMD-02 Query/Cache Strategy: {{smd.cache_strategy}}
EVT-01 Event Catalog: {{evt.catalog}} | OPTIONAL
EVT-04 Delivery Semantics: {{evt.delivery_semantics}} | OPTIONAL
API-07 WebSocket API Spec: {{api.websocket_spec}} | OPTIONAL
CER-02 Retry & Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Subscription model (WS topics/channels/rooms/UNKNOWN)
Subscription lifecycle (connect, auth, subscribe, unsubscribe)
Hook/pattern naming (client API surface)
Message normalization rules (to cache/store)
Reconciliation strategy (server snapshot vs stream truth)
Ordering rules (sequence numbers, timestamps)
Deduplication rules (idempotency keys)
Backfill/replay strategy on reconnect
Error handling (disconnects, auth failures)
Observability hooks (connection state, message lag)

Optional Fields
Presence/typing indicators | OPTIONAL
Client-side rate limits for subscriptions | OPTIONAL
Compression/binary frames | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Realtime patterns MUST reconcile with server truth; the stream alone must not be assumed complete unless explicitly guaranteed.
Ordering/dedupe MUST align with {{xref:EVT-04}} semantics when applicable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL
Auth failures MUST bind to session expiry handling ({{xref:CER-04}}) when present.

Output Format

1. Subscription Model
   model: {{rt.model}} (topics/rooms/channels/UNKNOWN)
   transport: {{rt.transport}} (ws/sse/UNKNOWN)
   notes: {{rt.notes}} | OPTIONAL
2. Lifecycle
   connect_rule: {{lifecycle.connect_rule}}
   auth_rule: {{lifecycle.auth_rule}} | OPTIONAL
   subscribe_rule: {{lifecycle.subscribe_rule}}
   unsubscribe_rule: {{lifecycle.unsubscribe_rule}} | OPTIONAL
   reconnect_rule: {{lifecycle.reconnect_rule}} | OPTIONAL
3. Client Hook/Surface
   hook_names: {{client.hooks}}
   usage_rules: {{client.usage_rules}} | OPTIONAL
4. Message Normalization
   message_types: {{messages.types}} | OPTIONAL
   normalization_target: {{messages.normalization_target}} (cache/store/both/UNKNOWN)
   apply_message_rule: {{messages.apply_rule}}
   entity_key_rule: {{messages.entity_key_rule}} | OPTIONAL
5. Reconciliation Strategy
   baseline_source: {{reconcile.baseline_source}} (snapshot/api_query/UNKNOWN)
   when_to_snapshot: {{reconcile.when_to_snapshot}}
   conflict_resolution: {{reconcile.conflict_resolution}} (LWW/version/UNKNOWN)
   stale_data_policy: {{reconcile.stale_data_policy}} | OPTIONAL
6. Ordering Rules
   ordering_key: {{ordering.key}} (seq/timestamp/UNKNOWN)
   out_of_order_handling: {{ordering.out_of_order_handling}}
   buffer_window_ms: {{ordering.buffer_window_ms}} | OPTIONAL

7. Deduplication Rules
   dedupe_key_rule: {{dedupe.key_rule}}
   dedupe_window: {{dedupe.window}} | OPTIONAL
8. Reconnect Backfill / Replay
   backfill_supported: {{backfill.supported}}
   backfill_method: {{backfill.method}} (cursor/from_seq/time_range/UNKNOWN)
   backfill_request_rule: {{backfill.request_rule}} | OPTIONAL
   gap_detection: {{backfill.gap_detection}} | OPTIONAL
9. Error Handling
   disconnect_behavior: {{errors.disconnect_behavior}}
   auth_failure_behavior: {{errors.auth_failure_behavior}} | OPTIONAL
   retry_policy_ref: {{errors.retry_policy_ref}} (expected: {{xref:CER-02}}) | OPTIONAL
10. Observability Hooks
    connection_state_metric: {{obs.connection_state_metric}}
    message_lag_metric: {{obs.message_lag_metric}} | OPTIONAL
    disconnect_count_metric: {{obs.disconnect_count_metric}} | OPTIONAL
    logs_required_fields: {{obs.logs_required_fields}} | OPTIONAL
11. References
    Cache strategy: {{xref:SMD-02}}
    WebSocket API: {{xref:API-07}} | OPTIONAL
    Event catalog: {{xref:EVT-01}} | OPTIONAL
    Delivery semantics: {{xref:EVT-04}} | OPTIONAL
    Retry patterns: {{xref:CER-02}} | OPTIONAL

Cross-References
Upstream: {{xref:SMD-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SMD-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for transport specifics; define lifecycle + reconciliation skeleton.
intermediate: Required. Define ordering/dedupe + reconnect backfill strategy.
advanced: Required. Add buffering/gap detection and observability rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, auth rule, reconnect rule, hook names, message types, entity key rule, stale data policy, buffer window, dedupe window, backfill details, auth failure behavior, retry policy ref, message lag/disconnect metrics, logs fields, presence/compression notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If rt.model is UNKNOWN → block Completeness Gate.
If messages.apply_rule is UNKNOWN → block Completeness Gate.
If dedupe.key_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SMD
Pass conditions:
required_fields_present == true
subscription_lifecycle_defined == true
reconciliation_defined == true
ordering_and_dedupe_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SMD-05

SMD-05 — Offline State Handling (queueing, retry)
Header Block
template_id: SMD-05
title: Offline State Handling (queueing, retry)
type: client_offline_state_handling
template_version: 1.0.0
output_path: 10_app/state/SMD-05_Offline_State_Handling.md
compliance_gate_id: TMP-05.PRIMARY.SMD
upstream_dependencies: ["SMD-02", "SMD-03", "CER-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SMD-02", "SMD-03", "CER-03", "CER-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical offline handling model for the client: what works offline, how reads are served from cache, how writes are queued, retry/backoff rules on reconnect, conflict handling, and user-visible status. This template must be consistent with cache and mutation patterns and must not invent offline capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SMD-02 Query/Cache Strategy: {{smd.cache_strategy}}
SMD-03 Mutation Patterns: {{smd.mutation_patterns}}
CER-03 Offline/Error Mode UX: {{cer.offline_mode}} | OPTIONAL
CER-02 Retry & Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Offline scope (what functions offline)
Network state detection rule
Read behavior when offline (cache/stale)
Write queue model (what queued, schema)
Queue persistence model (disk/memory/UNKNOWN)
Retry policy on reconnect (backoff, max attempts)
Conflict handling when reconnecting (merge/ask user)
User-visible offline indicators (bind to CER-03)
Data safety rules (do not lose queued ops)
Observability hooks (queued ops count, failures)

Optional Fields
Per-route offline exceptions | OPTIONAL

Background sync rules | OPTIONAL
Battery/network constraints | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Queued writes MUST be idempotent or have dedupe rules.
Offline UX MUST align with {{xref:CER-03}} when present.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Offline Scope
   works_offline: {{offline.works_offline}}
   does_not_work_offline: {{offline.does_not_work}} | OPTIONAL
   notes: {{offline.notes}} | OPTIONAL
2. Network Detection
   network_state_source: {{net.source}} (os/api/ping/UNKNOWN)
   offline_threshold_rule: {{net.threshold_rule}} | OPTIONAL
3. Reads When Offline
   read_source: {{reads.source}} (cache_only/cache_stale/UNKNOWN)
   stale_allowed: {{reads.stale_allowed}}
   stale_max_age_ms: {{reads.stale_max_age_ms}} | OPTIONAL
4. Write Queue Model
   queue_enabled: {{queue.enabled}}
   queue_item_schema:
   op_id: {{queue.schema.op_id}}
   endpoint_id: {{queue.schema.endpoint_id}} | OPTIONAL
   method: {{queue.schema.method}} | OPTIONAL
   payload_ref: {{queue.schema.payload_ref}} | OPTIONAL
   created_at: {{queue.schema.created_at}}
   dedupe_key: {{queue.schema.dedupe_key}} | OPTIONAL
   retry_count: {{queue.schema.retry_count}} | OPTIONAL
   status: {{queue.schema.status}} | OPTIONAL
5. Queue Persistence
   persistence: {{queue.persistence}} (disk/memory/UNKNOWN)
   encryption: {{queue.encryption}} | OPTIONAL
   max_queue_size: {{queue.max_size}} | OPTIONAL
6. Retry on Reconnect
   retry_policy_ref: {{retry.policy_ref}} (expected: {{xref:CER-02}}) | OPTIONAL
   backoff_policy: {{retry.backoff_policy}}
   max_attempts: {{retry.max_attempts}} | OPTIONAL
   drain_order: {{retry.drain_order}} | OPTIONAL

7. Conflict Handling
   conflict_detection: {{conflict.detection}}
   on_conflict: {{conflict.on_conflict}} (merge/prompt/drop/UNKNOWN)
   version_field_rule: {{conflict.version_field_rule}} | OPTIONAL
8. User Experience
   offline_indicator: {{ux.offline_indicator}}
   queued_write_indicator: {{ux.queued_write_indicator}} | OPTIONAL
   error_copy_policy: {{ux.error_copy_policy}} | OPTIONAL
9. Data Safety Rules
   no_drop_rule: {{safety.no_drop_rule}}
   recovery_on_crash: {{safety.recovery_on_crash}} | OPTIONAL
10. Observability Hooks
    queued_ops_metric: {{obs.queued_ops_metric}}
    queue_failures_metric: {{obs.queue_failures_metric}} | OPTIONAL
    logs_required_fields: {{obs.logs_required_fields}} | OPTIONAL
11. References
    Cache strategy: {{xref:SMD-02}}
    Mutation patterns: {{xref:SMD-03}}
    Offline UX: {{xref:CER-03}} | OPTIONAL
    Retry patterns: {{xref:CER-02}} | OPTIONAL

Cross-References
Upstream: {{xref:SMD-02}}, {{xref:SMD-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OFS-01}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for persistence details; define offline scope + queue schema skeleton.
intermediate: Required. Define retry/backoff and conflict handling.
advanced: Required. Add crash recovery, encryption, and background sync rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, offline threshold, stale max age, endpoint_id/method/payload_ref, dedupe key, retry count/status, encryption, max queue size, retry policy ref, drain order, version field rule, queued indicators, error copy policy, crash recovery, queue failure metrics/log fields, route exceptions, background sync, battery/network constraints, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If offline.works_offline is UNKNOWN → block Completeness Gate.
If queue.schema.op_id is UNKNOWN → block Completeness Gate.
If retry.backoff_policy is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SMD
Pass conditions:
required_fields_present == true
offline_scope_defined == true
queue_schema_defined == true
retry_policy_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SMD-06

SMD-06 — Observability for Client Data Layer (logs/metrics)
Header Block
template_id: SMD-06
title: Observability for Client Data Layer (logs/metrics)
type: client_data_layer_observability
template_version: 1.0.0
output_path: 10_app/state/SMD-06_Client_Data_Layer_Observability.md
compliance_gate_id: TMP-05.PRIMARY.SMD
upstream_dependencies: ["SMD-02", "SMD-03", "SMD-04", "SMD-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SMD-02", "SMD-03", "SMD-04", "SMD-05", "FE-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability requirements for the client data layer: query/mutation metrics, cache hit/miss, retry/backoff behavior, offline queue telemetry, realtime connection metrics, and required log fields with redaction. This template must be consistent with state/caching/mutation patterns and must not invent telemetry fields not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SMD-02 Query/Cache Strategy: {{smd.cache_strategy}}
SMD-03 Mutation Patterns: {{smd.mutation_patterns}}
SMD-04 Realtime Subscription Patterns: {{smd.realtime_patterns}} | OPTIONAL
SMD-05 Offline Handling: {{smd.offline_handling}} | OPTIONAL
FE-07 Error Handling UX: {{fe.error_ux}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Core metrics set (query/mutation)
Cache metrics (hit/miss/stale served)
Retry metrics (attempts, backoff time)
Error metrics (by class)
Offline queue metrics (queue depth, drain success/fail)
Realtime metrics (connect/disconnect, lag) if applicable
Log field requirements (request_id, endpoint_id, error_code)
Redaction policy (no PII/secrets)
Dashboards minimum panels
Alerts (data layer failures/spikes)

Optional Fields
Sampling rules | OPTIONAL
User opt-out policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Telemetry MUST avoid high-cardinality labels (no raw IDs) unless explicitly allowed.
PII MUST be redacted in logs per: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Metrics — Queries
   query_count: {{metrics.queries.query_count}}
   query_success: {{metrics.queries.query_success}} | OPTIONAL
   query_failure: {{metrics.queries.query_failure}} | OPTIONAL
   query_latency_ms: {{metrics.queries.query_latency_ms}} | OPTIONAL
2. Metrics — Cache
   cache_hit: {{metrics.cache.hit}}
   cache_miss: {{metrics.cache.miss}}
   stale_served: {{metrics.cache.stale_served}} | OPTIONAL
3. Metrics — Mutations
   mutation_count: {{metrics.mutations.count}}
   mutation_success: {{metrics.mutations.success}} | OPTIONAL
   mutation_failure: {{metrics.mutations.failure}} | OPTIONAL
   optimistic_rollback_count: {{metrics.mutations.optimistic_rollback}} | OPTIONAL
4. Metrics — Retry/Backoff
   retry_attempts: {{metrics.retry.attempts}}
   backoff_time_ms: {{metrics.retry.backoff_time_ms}} | OPTIONAL
5. Metrics — Errors
   errors_by_class: {{metrics.errors.by_class}}
   rate_limit_hits: {{metrics.errors.rate_limit_hits}} | OPTIONAL
   session_expiry_events: {{metrics.errors.session_expiry_events}} | OPTIONAL
6. Metrics — Offline Queue (if applicable)
   queue_depth: {{metrics.offline.queue_depth}} | OPTIONAL
   queue_drain_success: {{metrics.offline.drain_success}} | OPTIONAL
   queue_drain_failure: {{metrics.offline.drain_failure}} | OPTIONAL
7. Metrics — Realtime (if applicable)
   ws_connected: {{metrics.realtime.connected}} | OPTIONAL
   ws_disconnects: {{metrics.realtime.disconnects}} | OPTIONAL
   ws_message_lag: {{metrics.realtime.message_lag}} | OPTIONAL
8. Logs — Required Fields
   timestamp: {{logs.fields.timestamp}}

endpoint_id: {{logs.fields.endpoint_id}} | OPTIONAL
request_id: {{logs.fields.request_id}} | OPTIONAL
trace_id: {{logs.fields.trace_id}} | OPTIONAL
operation_type: {{logs.fields.operation_type}} (query/mutation/subscription/UNKNOWN)
result: {{logs.fields.result}}
error_code: {{logs.fields.error_code}} | OPTIONAL
status_code: {{logs.fields.status_code}} | OPTIONAL
retry_count: {{logs.fields.retry_count}} | OPTIONAL
redaction_applied: {{logs.fields.redaction_applied}} | OPTIONAL

9. Redaction Policy
   pii_redaction: {{redaction.pii_redaction}}
   secrets_redaction: {{redaction.secrets_redaction}}
   field_allowlist: {{redaction.field_allowlist}} | OPTIONAL
10. Dashboards
    data_layer_dashboard: {{dashboards.data_layer}}
    offline_dashboard: {{dashboards.offline}} | OPTIONAL
    realtime_dashboard: {{dashboards.realtime}} | OPTIONAL
    minimum_panels: {{dashboards.minimum_panels}}
11. Alerts
    query_failure_spike: {{alerts.query_failure_spike}}
    mutation_failure_spike: {{alerts.mutation_failure_spike}} | OPTIONAL
    offline_queue_stuck: {{alerts.offline_queue_stuck}} | OPTIONAL
    realtime_disconnect_spike: {{alerts.realtime_disconnect_spike}} | OPTIONAL
    threshold_model: {{alerts.threshold_model}} (static/baseline/UNKNOWN)
    routing_policy: {{alerts.routing_policy}} | OPTIONAL
12. References
    Cache strategy: {{xref:SMD-02}}
    Mutation patterns: {{xref:SMD-03}}
    Realtime patterns: {{xref:SMD-04}} | OPTIONAL
    Offline handling: {{xref:SMD-05}} | OPTIONAL
    Error UX: {{xref:FE-07}} | OPTIONAL

Cross-References
Upstream: {{xref:SMD-02}}, {{xref:SMD-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OPS-OBS}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-OBSERVABILITY]}} | OPTIONAL,
{{standards.rules[STD-PII-REDACTION]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define core metrics + log fields; use UNKNOWN for dashboards/alerts tooling.
intermediate: Required. Add cache/retry/error/queue metrics and dashboard minimums.
advanced: Required. Add cardinality controls, alert thresholds, and opt-out/sampling policy if needed.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional metrics fields, request/trace IDs, status code, retry count, field allowlist, offline/realtime dashboards, alert routing, sampling/opt-out, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If metrics.cache.hit/miss are UNKNOWN → block Completeness Gate.
If logs.fields.operation_type is UNKNOWN → block Completeness Gate.
If redaction policy is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SMD
Pass conditions:
required_fields_present == true
core_metrics_defined == true
log_fields_defined == true
redaction_policy_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Client Performance & Rendering (CPR)

Client Performance & Rendering (CPR)
CPR-01 Performance Budget (web + mobile targets)
CPR-02 Rendering Strategy (SSR/CSR, list virtualization)
CPR-03 Asset Loading Strategy (code split, lazy load)
CPR-04 Profiling Plan (what to measure, tools)
CPR-05 Perf Regression Gates (thresholds, CI checks)

CPR-01

CPR-01 — Performance Budget (web + mobile targets)
Header Block
template_id: CPR-01
title: Performance Budget (web + mobile targets)
type: client_performance_budget
template_version: 1.0.0
output_path: 10_app/client_perf/CPR-01_Performance_Budget.md
compliance_gate_id: TMP-05.PRIMARY.CPR
upstream_dependencies: ["PRD-06", "FE-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PRD-06", "FE-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical performance budget targets for web and mobile clients, including latency targets, frame/render targets, bundle size targets, and network usage targets. This template must be consistent with product NFRs and must not invent performance targets not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PRD-06 NFRs: {{prd.nfrs}} | OPTIONAL
FE-01 Route Map + Layout: {{fe.route_layout}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Platforms covered (web/mobile)
Core user journeys covered (startup, navigation, list scroll, form submit)
Latency targets (p50/p95) by journey
Render/frame targets (fps, frame time)
Bundle/asset size targets (initial, per route)
Network targets (requests per screen, payload sizes)
Memory/battery targets (mobile)
Regression thresholds (what triggers fail gate)
Measurement plan (tools, where measured)

Optional Fields
Device tiers (low/med/high) | OPTIONAL
Regional network assumptions | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Targets MUST be measurable and expressed as numbers with units.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Platforms
   platforms: {{perf.platforms}}
   device_tiers: {{perf.device_tiers}} | OPTIONAL
2. Journeys Covered
   journeys:
   {{journeys[0]}}
   {{journeys[1]}}
   {{journeys[2]}}
3. Latency Targets
   startup_p50_ms: {{latency.startup.p50}}
   startup_p95_ms: {{latency.startup.p95}} | OPTIONAL
   nav_p50_ms: {{latency.navigation.p50}}
   nav_p95_ms: {{latency.navigation.p95}} | OPTIONAL
   submit_p50_ms: {{latency.submit.p50}} | OPTIONAL
   submit_p95_ms: {{latency.submit.p95}} | OPTIONAL
4. Render / Frame Targets
   target_fps: {{render.target_fps}}
   max_frame_time_ms: {{render.max_frame_time_ms}} | OPTIONAL
   scroll_jank_budget: {{render.scroll_jank_budget}} | OPTIONAL
5. Bundle / Asset Budgets
   initial_bundle_kb: {{bundle.initial_kb}}
   route_bundle_kb: {{bundle.route_kb}} | OPTIONAL
   image_budget_kb: {{bundle.image_budget_kb}} | OPTIONAL
6. Network Budgets
   max_requests_per_screen: {{network.max_requests_per_screen}}
   max_payload_kb_per_screen: {{network.max_payload_kb_per_screen}} | OPTIONAL
   cache_hit_target: {{network.cache_hit_target}} | OPTIONAL
7. Mobile Memory/Battery
   max_memory_mb: {{mobile.max_memory_mb}} | OPTIONAL
   battery_constraints: {{mobile.battery_constraints}} | OPTIONAL
8. Regression Thresholds
   fail_when: {{regression.fail_when}}
   warn_when: {{regression.warn_when}} | OPTIONAL
9. Measurement Plan
   tools: {{measure.tools}}
   where_measured: {{measure.where_measured}}
   cadence: {{measure.cadence}} | OPTIONAL

10. References
    Profiling plan: {{xref:CPR-04}} | OPTIONAL
    Perf regression gates: {{xref:CPR-05}} | OPTIONAL
    NFRs: {{xref:PRD-06}} | OPTIONAL

Cross-References
Upstream: {{xref:PRD-06}} | OPTIONAL, {{xref:FE-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CPR-04}}, {{xref:CPR-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where numeric targets are missing; do not invent numbers.
intermediate: Required. Populate measurable targets and measurement tooling from inputs.
advanced: Required. Add device tiers and regression gate rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, device tiers, p95 targets, submit targets, frame time/jank budget, route/image budgets, payload/caching targets, memory/battery, warn thresholds, cadence, regional assumptions, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If perf.platforms is UNKNOWN → block Completeness Gate.
If latency.startup.p50 is UNKNOWN → block Completeness Gate.
If bundle.initial_kb is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CPR
Pass conditions:
required_fields_present == true
platforms_defined == true
latency_targets_defined == true
bundle_and_network_budgets_defined == true
measurement_plan_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CPR-02

CPR-02 — Rendering Strategy (SSR/CSR, list virtualization)
Header Block
template_id: CPR-02
title: Rendering Strategy (SSR/CSR, list virtualization)
type: client_rendering_strategy
template_version: 1.0.0
output_path: 10_app/client_perf/CPR-02_Rendering_Strategy.md
compliance_gate_id: TMP-05.PRIMARY.CPR
upstream_dependencies: ["CPR-01", "FE-01", "UICP-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CPR-01", "FE-01", "UICP-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rendering strategy for clients, including SSR vs CSR decisions (web), navigation/render pipelines (mobile), list virtualization rules, and rendering performance safeguards. This template must be consistent with the performance budget and must not invent platform capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CPR-01 Performance Budget: {{cpr.budget}}
FE-01 Route Map + Layout: {{fe.route_layout}} | OPTIONAL
UICP-01 Page Composition Patterns: {{ui.page_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Web rendering model (SSR/CSR/ISR/UNKNOWN)
Mobile rendering model (native nav/render constraints)
Hydration strategy (if SSR)
Route-level rendering rules (which routes SSR)
List virtualization rules (when required)
Image rendering strategy (lazy load, decoding)
Skeleton/loading rendering rules (bind to FE-03)
Animation/motion performance rules
Accessibility implications (reduced motion)
Measurement hooks (profiling plan binding)

Optional Fields
Streaming rendering notes | OPTIONAL

Edge caching notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Rendering choices MUST fit within budgets from {{xref:CPR-01}}.
Long lists MUST be virtualized if above threshold (or UNKNOWN).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Web Rendering Model
   model: {{web.model}} (ssr/csr/isr/UNKNOWN)
   hydration: {{web.hydration}} | OPTIONAL
   route_rules: {{web.route_rules}} | OPTIONAL
2. Mobile Rendering Model
   model: {{mobile.model}} (rn/unknown)
   navigation_render_constraints: {{mobile.constraints}} | OPTIONAL
3. List Virtualization
   virtualization_required_when: {{lists.required_when}}
   threshold_items: {{lists.threshold_items}} | OPTIONAL
   library_or_pattern: {{lists.pattern}} | OPTIONAL
4. Image & Asset Rendering
   lazy_load_policy: {{assets.lazy_load_policy}}
   decode_policy: {{assets.decode_policy}} | OPTIONAL
   placeholder_policy: {{assets.placeholder_policy}} | OPTIONAL
5. Loading/Skeleton Rules
   skeleton_policy: {{loading.skeleton_policy}}
   bind_state_model_ref: {{loading.state_model_ref}} (expected: {{xref:FE-03}}) | OPTIONAL
6. Animations / Motion
   motion_allowed: {{motion.allowed}}
   reduced_motion_rule: {{motion.reduced_motion_rule}} | OPTIONAL
   fps_targets_ref: {{motion.fps_targets_ref}} (expected: {{xref:CPR-01}}) | OPTIONAL
7. Profiling Hooks
   profiling_plan_ref: {{profile.plan_ref}} (expected: {{xref:CPR-04}}) | OPTIONAL
   perf_markers: {{profile.perf_markers}} | OPTIONAL
8. References
   Performance budget: {{xref:CPR-01}}
   Asset loading: {{xref:CPR-03}} | OPTIONAL
   Profiling plan: {{xref:CPR-04}} | OPTIONAL
   Perf gates: {{xref:CPR-05}} | OPTIONAL
   UI state model: {{xref:FE-03}} | OPTIONAL

Cross-References
Upstream: {{xref:CPR-01}} , {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CPR-03}}, {{xref:CPR-04}}, {{xref:CPR-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for SSR/hydration specifics if not defined; define list virtualization threshold placeholder.
intermediate: Required. Define web/mobile model and skeleton/asset policies.
advanced: Required. Add profiling hooks and reduced-motion/perf marker rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, hydration, route rules, list thresholds/pattern, decode/placeholder policy, state model ref, reduced motion rules, fps refs, profiling hooks/markers, streaming/edge notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If web.model is UNKNOWN → block Completeness Gate (for web clients).
If lists.required_when is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CPR
Pass conditions:
required_fields_present == true
rendering_model_defined == true
virtualization_rules_defined == true
loading_asset_policies_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

- ○

CPR-03

CPR-03 — Asset Loading Strategy (code split, lazy load)
Header Block
template_id: CPR-03
title: Asset Loading Strategy (code split, lazy load)
type: client_asset_loading_strategy
template_version: 1.0.0
output_path: 10_app/client_perf/CPR-03_Asset_Loading_Strategy.md
compliance_gate_id: TMP-05.PRIMARY.CPR
upstream_dependencies: ["CPR-01", "CPR-02", "FE-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CPR-01", "CPR-02", "FE-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical asset loading strategy for clients, including code splitting, route-level lazy loading, image/font loading policies, and caching/preload rules. This template must be consistent with the performance budget and rendering strategy and must not invent bundling/caching mechanisms not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CPR-01 Performance Budget: {{cpr.budget}}
CPR-02 Rendering Strategy: {{cpr.rendering_strategy}}
FE-01 Route Map + Layout: {{fe.route_layout}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Code splitting model (route-based/component-based)
Lazy loading rules (what is deferred)
Critical path assets (what must load first)
Prefetch/preload policy
Image loading policy (responsive sizes, lazy)
Font loading policy (preload, fallback)
Caching policy (service worker/http cache/UNKNOWN)
Third-party scripts policy
Error handling for failed loads
Measurement hooks (bundle analysis)

Optional Fields
CDN strategy notes | OPTIONAL

Offline asset caching notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Asset loading MUST meet budgets in {{xref:CPR-01}}.
Lazy loading MUST not break navigation UX.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Code Splitting
   model: {{split.model}} (route/component/UNKNOWN)
   route_chunk_rules: {{split.route_chunk_rules}} | OPTIONAL
   shared_chunk_rules: {{split.shared_chunk_rules}} | OPTIONAL
2. Lazy Loading Rules
   defer_components: {{lazy.defer_components}}
   defer_routes: {{lazy.defer_routes}} | OPTIONAL
   defer_assets: {{lazy.defer_assets}} | OPTIONAL
   fallback_ui_policy: {{lazy.fallback_ui_policy}} | OPTIONAL
3. Critical Path Assets
   critical_js: {{critical.js}}
   critical_css: {{critical.css}} | OPTIONAL
   critical_fonts: {{critical.fonts}} | OPTIONAL
   critical_images: {{critical.images}} | OPTIONAL
4. Prefetch / Preload
   prefetch_policy: {{prefetch.policy}}
   preload_policy: {{prefetch.preload_policy}} | OPTIONAL
   conditions: {{prefetch.conditions}} | OPTIONAL
5. Image Loading
   responsive_images: {{images.responsive_images}}
   lazy_load: {{images.lazy_load}}
   placeholder_policy: {{images.placeholder_policy}} | OPTIONAL
6. Font Loading
   font_strategy: {{fonts.strategy}} (preload/swap/block/UNKNOWN)
   fallback_fonts: {{fonts.fallback_fonts}} | OPTIONAL
7. Caching Policy
   cache_model: {{cache.model}} (http_cache/sw/service_worker/UNKNOWN)
   cache_headers_policy: {{cache.headers_policy}} | OPTIONAL
   offline_cache_supported: {{cache.offline_supported}} | OPTIONAL
8. Third-Party Scripts
   allowed_third_party: {{third_party.allowed}}
   load_timing_rules: {{third_party.load_timing_rules}} | OPTIONAL
   security_policy: {{third_party.security_policy}} | OPTIONAL

9. Error Handling for Loads
   on_chunk_fail: {{errors.on_chunk_fail}}
   on_image_fail: {{errors.on_image_fail}} | OPTIONAL
   retry_policy: {{errors.retry_policy}} | OPTIONAL
10. Measurement Hooks
    bundle_analysis_tools: {{measure.bundle_tools}}
    bundle_reports: {{measure.bundle_reports}} | OPTIONAL
    cadence: {{measure.cadence}} | OPTIONAL
11. References
    Performance budget: {{xref:CPR-01}}
    Rendering strategy: {{xref:CPR-02}}
    Profiling plan: {{xref:CPR-04}} | OPTIONAL
    Perf gates: {{xref:CPR-05}} | OPTIONAL

Cross-References
Upstream: {{xref:CPR-01}}, {{xref:CPR-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CPR-04}}, {{xref:CPR-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for cache tooling; define split/lazy/critical assets skeleton.
intermediate: Required. Define prefetch/preload and image/font policies.
advanced: Required. Add third-party security policy and bundle measurement cadence.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, route/shared chunk rules, defer routes/assets, fallback UI, critical css/fonts/images, preload policy/conditions, placeholder policy, fallback fonts, cache headers/offline support, third-party timing/security, image fail policy, retry policy, bundle reports/cadence, CDN/offline notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If split.model is UNKNOWN → block Completeness Gate.
If critical.js is UNKNOWN → block Completeness Gate.
If errors.on_chunk_fail is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CPR
Pass conditions:
required_fields_present == true
splitting_and_lazy_rules_defined == true
critical_path_defined == true
error_handling_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CPR-04

CPR-04 — Profiling Plan (what to measure, tools)
Header Block
template_id: CPR-04
title: Profiling Plan (what to measure, tools)
type: client_profiling_plan
template_version: 1.0.0
output_path: 10_app/client_perf/CPR-04_Profiling_Plan.md
compliance_gate_id: TMP-05.PRIMARY.CPR
upstream_dependencies: ["CPR-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CPR-01", "CPR-02", "CPR-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical profiling plan for client performance: what to measure, which tools to use, test scenarios, and how results map to performance budgets and regression gates. This template must be consistent with the performance budget and must not invent measurement tooling not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CPR-01 Performance Budget: {{cpr.budget}}
CPR-02 Rendering Strategy: {{cpr.rendering_strategy}} | OPTIONAL
CPR-03 Asset Loading Strategy: {{cpr.asset_loading}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Profiling scope (web/mobile)
Scenarios to measure (startup, navigation, list scroll, uploads, etc.)
Metrics collected (p50/p95 latency, fps, memory, bundle)
Tooling list (profilers, lighthouse, flamegraphs, RN perf tools)
Test device/network profiles
Baseline capture rules (what is “good”)
Reporting format (where results stored)
Frequency/cadence (when to run profiling)
Ownership (who runs and reviews)

Optional Fields
Automation in CI | OPTIONAL
Sampling strategy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Collected metrics MUST align with CPR-01 targets.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Profiling Scope
   platforms: {{profile.platforms}}
   notes: {{profile.notes}} | OPTIONAL
2. Scenarios
   scenarios:
   {{scenarios[0]}}
   {{scenarios[1]}}
   {{scenarios[2]}}
3. Metrics Collected
   latency_metrics: {{metrics.latency}}
   render_metrics: {{metrics.render}} | OPTIONAL
   memory_metrics: {{metrics.memory}} | OPTIONAL
   bundle_metrics: {{metrics.bundle}} | OPTIONAL
   network_metrics: {{metrics.network}} | OPTIONAL
4. Tools
   web_tools: {{tools.web}} | OPTIONAL
   mobile_tools: {{tools.mobile}} | OPTIONAL
   bundle_tools: {{tools.bundle}} | OPTIONAL
5. Test Profiles
   devices: {{tests.devices}}
   network_profiles: {{tests.network_profiles}} | OPTIONAL
6. Baseline & Comparison
   baseline_source: {{baseline.source}}
   comparison_method: {{baseline.comparison_method}} | OPTIONAL
   acceptance_thresholds_ref: {{baseline.thresholds_ref}} (expected: {{xref:CPR-05}}) | OPTIONAL
7. Reporting
   report_location: {{report.location}}
   report_format: {{report.format}} (md/json/dashboard/UNKNOWN)
   required_fields: {{report.required_fields}} | OPTIONAL
8. Cadence & Ownership
   cadence: {{ops.cadence}}
   owner: {{ops.owner}}
   review_process: {{ops.review_process}} | OPTIONAL
9. References
   Performance budget: {{xref:CPR-01}}
   Perf regression gates: {{xref:CPR-05}} | OPTIONAL

Cross-References
Upstream: {{xref:CPR-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CPR-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for tooling specifics; define scenarios + metrics collected.
intermediate: Required. Define test profiles and reporting format/location.
advanced: Required. Add CI automation and baseline comparison thresholds.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, optional metrics, tool lists, network profiles, comparison method, thresholds ref, report required fields, review process, CI automation, sampling strategy, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If scenarios list is UNKNOWN → block Completeness Gate.
If report.location is UNKNOWN → block Completeness Gate.
If ops.owner is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CPR
Pass conditions:
required_fields_present == true
scenarios_defined == true
metrics_defined == true
reporting_defined == true
cadence_and_owner_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CPR-05

CPR-05 — Perf Regression Gates (thresholds, CI checks)
Header Block
template_id: CPR-05
title: Perf Regression Gates (thresholds, CI checks)
type: client_perf_regression_gates
template_version: 1.0.0
output_path: 10_app/client_perf/CPR-05_Perf_Regression_Gates.md
compliance_gate_id: TMP-05.PRIMARY.CPR
upstream_dependencies: ["CPR-01", "CPR-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CPR-01", "CPR-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical performance regression gate rules: what metrics are gated, threshold definitions, how checks run (CI/local), what constitutes fail vs warn, and the remediation workflow. This template must be consistent with the performance budget and profiling plan and must not invent gating systems not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CPR-01 Performance Budget: {{cpr.budget}}
CPR-04 Profiling Plan: {{cpr.profiling_plan}}
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Gated metrics list (startup, nav, fps, bundle size, etc.)
Threshold definitions (numeric with units)
Fail vs warn rule (what blocks merge)
Measurement method per metric (tool/scenario)
CI execution rule (when checks run)
Baseline comparison method (absolute vs delta)
Result reporting (artifacts, logs)
Remediation workflow (what to do on fail)

Optional Fields
Per-branch differences | OPTIONAL
Manual override policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Thresholds MUST be measurable and bind to CPR-01 targets.
CI checks MUST produce artifacts usable for debugging.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Gated Metrics
   metrics:
   {{gates.metrics[0]}}
   {{gates.metrics[1]}}
   {{gates.metrics[2]}}
2. Thresholds
   Threshold
   metric: {{thresholds[0].metric}}
   unit: {{thresholds[0].unit}}
   fail_when: {{thresholds[0].fail_when}}
   warn_when: {{thresholds[0].warn_when}} | OPTIONAL
   method: {{thresholds[0].method}}
   scenario: {{thresholds[0].scenario}} | OPTIONAL

(Repeat for each gated metric.)

3. Fail vs Warn Rules
   fail_blocks_merge: {{rules.fail_blocks_merge}}
   warn_allows_merge: {{rules.warn_allows_merge}} | OPTIONAL
   override_policy: {{rules.override_policy}} | OPTIONAL
4. CI Execution
   when_to_run: {{ci.when_to_run}}
   required_branches: {{ci.required_branches}} | OPTIONAL
   runtime_budget_minutes: {{ci.runtime_budget_minutes}} | OPTIONAL
5. Baseline Comparison
   comparison_model: {{baseline.model}} (absolute/delta/UNKNOWN)
   baseline_source: {{baseline.source}} | OPTIONAL
   artifact_storage: {{baseline.artifact_storage}} | OPTIONAL
6. Reporting
   report_artifacts: {{report.artifacts}}
   log_format: {{report.log_format}} | OPTIONAL
7. Remediation Workflow
   on_fail_steps:
   {{remediate.on_fail_steps[0]}}
   {{remediate.on_fail_steps[1]}}
   owner: {{remediate.owner}} | OPTIONAL

8. References
   Performance budget: {{xref:CPR-01}}
   Profiling plan: {{xref:CPR-04}}

Cross-References
Upstream: {{xref:CPR-01}}, {{xref:CPR-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:RELEASE-GATE}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for CI details; do not invent thresholds.
intermediate: Required. Define fail/warn and measurement methods/scenarios.
advanced: Required. Add baseline artifact storage and override workflow policies.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, warn_when, scenario, warn rules, override policy, required branches, runtime budget, baseline source, artifact storage, log format, remediation owner, branch differences, manual override, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If gates.metrics is UNKNOWN → block Completeness Gate.
If thresholds list is UNKNOWN → block Completeness Gate.
If report.artifacts is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CPR
Pass conditions:
required_fields_present == true
thresholds_defined == true
ci_execution_defined == true
reporting_defined == true
remediation_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Forms & Validation (FORM)

Forms & Validation (FORM)
FORM-01 Forms Inventory (by form_id/screen_id)
FORM-02 Field Spec (per field: type, rules, copy)
FORM-03 Validation UX Rules (inline, focus, announce)
FORM-04 Schema Mapping (forms ↔ DATA-06/DQV-03)
FORM-05 Submission & Error Recovery Patterns
FORM-06 Anti-Abuse for Forms (spam, throttles, bot defense)

FORM-01

FORM-01 — Forms Inventory (by form_id/screen_id)
Header Block
template_id: FORM-01
title: Forms Inventory (by form_id/screen_id)
type: forms_inventory
template_version: 1.0.0
output_path: 10_app/forms/FORM-01_Forms_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.FORM
upstream_dependencies: ["FE-01", "SPEC_INDEX"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical inventory of all forms in the application, indexed by **form_id**, and mapped to screens/routes. This document must be consistent with the Canonical Spec and must not invent form_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-01 Route Map + Layout: {{fe.route_layout}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Forms index (form_id list)
form_id (stable identifier)
form_name (human-readable)
screen_id (where the form lives)
route_id (optional)
purpose/intent (what the form does)
submit action binding (endpoint_id or action_id)
validation schema ref (FORM-04 binding)
success behavior (what happens after submit)
error behavior (bind to FORM-05/FE-07)

Optional Fields
Multi-step indicator | OPTIONAL
Draft/autosave support | OPTIONAL
Related feature IDs | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new form_ids; use only {{spec.forms_by_id}} as given.
Submit action bindings MUST reference existing endpoints/actions ({{xref:API-01}}/{{xref:API-02}}) if used.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Inventory Summary
   total_forms: {{forms.total}}
   screens_covered: {{forms.screens}} | OPTIONAL
   notes: {{forms.notes}} | OPTIONAL
2. Form Index (by form_id)
   Form
   form_id: {{items[0].form_id}}
   form_name: {{items[0].form_name}}
   screen_id: {{items[0].screen_id}}
   route_id: {{items[0].route_id}} | OPTIONAL
   purpose: {{items[0].purpose}}
   submit_binding: {{items[0].submit_binding}}
   validation_schema_ref: {{items[0].validation_schema_ref}}
   success_behavior: {{items[0].success_behavior}}
   error_behavior: {{items[0].error_behavior}}
   multi_step: {{items[0].multi_step}} | OPTIONAL
   draft_support: {{items[0].draft_support}} | OPTIONAL
   related_feature_ids: {{items[0].related_feature_ids}} | OPTIONAL
   open_questions:
   {{items[0].open_questions[0]}} | OPTIONAL

(Repeat the “Form” block for each form_id.)

3. References
   Field specs: {{xref:FORM-02}} | OPTIONAL
   Validation UX rules: {{xref:FORM-03}} | OPTIONAL
   Schema mapping: {{xref:FORM-04}} | OPTIONAL
   Submission patterns: {{xref:FORM-05}} | OPTIONAL
   Anti-abuse: {{xref:FORM-06}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:FE-01}} | OPTIONAL
Downstream: {{xref:FORM-02}}, {{xref:FORM-04}}, {{xref:FE-07}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN where form bindings are missing; do not invent form_ids.
intermediate: Required. Populate submit bindings and validation refs from inputs.
advanced: Required. Add multi-step/draft features and traceability pointers.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, route_id, multi_step, draft_support, related_feature_ids, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If form_id list is UNKNOWN → block Completeness Gate.
If submit_binding is UNKNOWN → block Completeness Gate.
If validation_schema_ref is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FORM
Pass conditions:
required_fields_present == true
all form_ids are unique
no new form_ids introduced
placeholder_resolution == true
no_unapproved_unknowns == true

FORM-02

FORM-02 — Field Spec (per field: type, rules, copy)
Header Block
template_id: FORM-02
title: Field Spec (per field: type, rules, copy)
type: form_field_spec
template_version: 1.0.0
output_path: 10_app/forms/FORM-02_Field_Spec.md
compliance_gate_id: TMP-05.PRIMARY.FORM
upstream_dependencies: ["FORM-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FORM-01", "DATA-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical per-field specification format for forms, including field types, validation rules, UX copy, default values, data mapping, and accessibility requirements. This template must be consistent with the forms inventory and data schemas and must not invent field IDs not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FORM-01 Forms Inventory: {{forms.inventory}}
DATA-06 Canonical Data Schemas: {{data.schemas}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
field_id (stable identifier)
form_id (binding to FORM-01)
field_name (human-readable)
field_type (text/number/select/date/file/etc.)
required (true/false)
validation_rules (min/max/pattern/enums)
ui_control (input/select/radio/textarea/etc.)
label copy
placeholder copy (optional)
help text (optional)
default value (optional)
data_binding (entity.field or schema pointer)
submission mapping (payload key)
error message policy (bind to FORM-03)
a11y attributes (aria-label/description)

Optional Fields
Conditional display rules | OPTIONAL
Formatting/masking rules | OPTIONAL
Autocomplete hints | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new field_ids; use only {{spec.fields_by_id}} as given.
Every field MUST bind to a form_id in FORM-01.
Validation rules MUST map cleanly to schema constraints when possible (DATA-06).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Field Entry (repeat per field_id)
   Field
   field_id: {{fields[0].field_id}}
   form_id: {{fields[0].form_id}}
   field_name: {{fields[0].field_name}}
   field_type: {{fields[0].field_type}}
   ui_control: {{fields[0].ui_control}}
   required: {{fields[0].required}}

copy:
label: {{fields[0].copy.label}}
placeholder: {{fields[0].copy.placeholder}} | OPTIONAL
help_text: {{fields[0].copy.help_text}} | OPTIONAL

validation:
rules: {{fields[0].validation.rules}}
messages: {{fields[0].validation.messages}} | OPTIONAL

data:
schema_ref: {{fields[0].data.schema_ref}} | OPTIONAL
entity_field: {{fields[0].data.entity_field}} | OPTIONAL
submission_key: {{fields[0].data.submission_key}}

a11y:
aria_label: {{fields[0].a11y.aria_label}} | OPTIONAL
aria_description: {{fields[0].a11y.aria_description}} | OPTIONAL

conditional_display: {{fields[0].conditional_display}} | OPTIONAL
formatting_mask: {{fields[0].formatting_mask}} | OPTIONAL
autocomplete_hint: {{fields[0].autocomplete_hint}} | OPTIONAL

open_questions:
{{fields[0].open_questions[0]}} | OPTIONAL

(Repeat the “Field” block for each field_id.)

2. References
   Forms inventory: {{xref:FORM-01}}
   Validation UX rules: {{xref:FORM-03}} | OPTIONAL
   Schema mapping: {{xref:FORM-04}} | OPTIONAL
   Data schemas: {{xref:DATA-06}} | OPTIONAL

Cross-References
Upstream: {{xref:FORM-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FORM-03}}, {{xref:FORM-04}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for optional copy; do not invent field_ids.
intermediate: Required. Bind validation to DATA-06 where possible and define submission keys.
advanced: Required. Add conditional display and formatting/masking specs.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, placeholder/help text, messages, schema_ref/entity_field, a11y attributes, conditional_display, formatting_mask, autocomplete_hint, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If field_id list is UNKNOWN → block Completeness Gate.
If form_id is UNKNOWN → block Completeness Gate.
If submission_key is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FORM
Pass conditions:
required_fields_present == true
all field_ids bind to existing form_ids (FORM-01)
no new field_ids introduced
placeholder_resolution == true
no_unapproved_unknowns == true

-

FORM-03

FORM-03 — Validation UX Rules (inline, focus, announce)
Header Block
template_id: FORM-03
title: Validation UX Rules (inline, focus, announce)
type: form_validation_ux_rules
template_version: 1.0.0
output_path: 10_app/forms/FORM-03_Validation_UX_Rules.md
compliance_gate_id: TMP-05.PRIMARY.FORM
upstream_dependencies: ["FORM-01", "FORM-02", "FE-05", "FE-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FORM-01", "FORM-02", "FE-05", "FE-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical UX rules for client-side and server-side form validation: when validation runs, how errors display (inline vs summary), focus management, screen reader announcements, and how server validation errors map to fields. This template must be consistent with accessibility and error UX rules and must not invent validation modes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FORM-01 Forms Inventory: {{forms.inventory}}
FORM-02 Field Specs: {{forms.field_specs}}
FE-05 Accessibility Notes: {{fe.a11y_notes}} | OPTIONAL
FE-07 Error Handling UX: {{fe.error_ux}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Validation timing rules (on blur/on submit/on change)
Client vs server validation precedence rules
Inline error display rules
Summary error display rules (top-of-form)
Focus first error rules
Announcement rules (aria-live)
Field-to-error mapping rules (server errors to field_id)
Error copy policy (tone, specificity)
Disable submit rules (when invalid/pending)
Success feedback rules (post-submit)

Optional Fields
Per-form overrides | OPTIONAL
Async field validation (unique checks) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Validation UX MUST meet accessibility rules from {{xref:FE-05}}.
Server-side validation errors MUST be mapped to fields when possible; otherwise show summary.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Validation Timing
   on_change: {{timing.on_change}}
   on_blur: {{timing.on_blur}}
   on_submit: {{timing.on_submit}}
   debounce_ms: {{timing.debounce_ms}} | OPTIONAL
2. Client vs Server Precedence
   client_first_rule: {{precedence.client_first_rule}}
   server_override_rule: {{precedence.server_override_rule}} | OPTIONAL
3. Inline Errors
   inline_display_rule: {{inline.display_rule}}
   inline_copy_policy: {{inline.copy_policy}}
   inline_spacing_rules: {{inline.spacing_rules}} | OPTIONAL
4. Summary Errors
   summary_display_rule: {{summary.display_rule}}
   summary_contents: {{summary.contents}} | OPTIONAL
   summary_link_to_fields: {{summary.link_to_fields}} | OPTIONAL
5. Focus Management
   focus_first_error: {{focus.first_error}}
   focus_on_submit_fail: {{focus.on_submit_fail}} | OPTIONAL
6. Screen Reader Announcements
   aria_live_policy: {{sr.aria_live_policy}}
   announce_on_submit_fail: {{sr.on_submit_fail}} | OPTIONAL
   announce_inline_changes: {{sr.inline_changes}} | OPTIONAL
7. Server Error → Field Mapping
   mapping_rule: {{server_map.mapping_rule}}
   unmapped_error_behavior: {{server_map.unmapped_behavior}}
   error_key_format: {{server_map.error_key_format}} | OPTIONAL
8. Submit Button Rules
   disable_when_invalid: {{submit.disable_when_invalid}}

disable_when_pending: {{submit.disable_when_pending}}
show_spinner_when_pending: {{submit.show_spinner}} | OPTIONAL

9. Success Feedback
   success_message_policy: {{success.message_policy}}
   post_submit_navigation: {{success.post_submit_navigation}} | OPTIONAL
10. References
    Field specs: {{xref:FORM-02}}
    Accessibility: {{xref:FE-05}} | OPTIONAL
    Error UX: {{xref:FE-07}} | OPTIONAL

Cross-References
Upstream: {{xref:FORM-01}}, {{xref:FORM-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FORM-05}}, {{xref:FORM-06}} | OPTIONAL
Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define on_submit + inline error rules; use UNKNOWN for debounce/async validation.
intermediate: Required. Define server-to-field mapping and focus/announcement policies.
advanced: Required. Add per-form override mechanisms and async validation guidance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, debounce_ms, server override rule, inline spacing, summary contents/links, focus submit fail, sr announce options, error key format, spinner, post submit nav, per-form overrides, async validation, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If timing.on_submit is UNKNOWN → block Completeness Gate.
If inline.display_rule is UNKNOWN → block Completeness Gate.
If server_map.mapping_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FORM
Pass conditions:
required_fields_present == true
validation_timing_defined == true
server_mapping_defined == true
a11y_rules_applied == true
placeholder_resolution == true
no_unapproved_unknowns == true

FORM-04

FORM-04 — Schema Mapping (forms ↔ DATA-06/DQV-03)
Header Block
template_id: FORM-04
title: Schema Mapping (forms ↔ DATA-06/DQV-03)
type: form_schema_mapping
template_version: 1.0.0
output_path: 10_app/forms/FORM-04_Schema_Mapping.md
compliance_gate_id: TMP-05.PRIMARY.FORM
upstream_dependencies: ["FORM-01", "FORM-02", "DATA-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FORM-01", "FORM-02", "DATA-06", "DQV-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical mapping between forms/fields and the underlying data schemas and validation rules, including how form payloads map to entities, how field constraints bind to schema constraints, and how server validation errors map back to field_ids. This template must be consistent with data schema definitions and must not invent schema IDs not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FORM-01 Forms Inventory: {{forms.inventory}}
FORM-02 Field Specs: {{forms.field_specs}}
DATA-06 Canonical Data Schemas: {{data.schemas}} | OPTIONAL
DQV-03 Data Validation Rules: {{data.validation_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Mapping registry (form_id → schema_ref/entity_ref)
Per-field mapping (field_id → schema path)
Submission payload mapping (field_id → payload key)
Constraint binding (field validation ↔ schema constraints)
Normalization/transforms (trim, parse, timezone, etc.)
Server error key mapping rules (schema path → field_id)
Unknown/unmapped behavior (what happens if no mapping)
Versioning notes (schema evolution effect)

Optional Fields
Derived fields (computed values) | OPTIONAL

Multi-entity submissions | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Every form MUST map to at least one schema/entity reference.
Every field MUST map to a schema path or be explicitly marked UNKNOWN and flagged.
Do not invent schema refs; use only those in {{xref:DATA-06}}/{{xref:DQV-03}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Form → Schema Mapping
   Mapping
   form_id: {{maps[0].form_id}}
   schema_ref: {{maps[0].schema_ref}}
   entity_ref: {{maps[0].entity_ref}} | OPTIONAL
   payload_root_key: {{maps[0].payload_root_key}} | OPTIONAL
   notes: {{maps[0].notes}} | OPTIONAL

(Repeat per form_id.)

2. Field → Schema Path Mapping
   Field Map
   field_id: {{field_maps[0].field_id}}
   form_id: {{field_maps[0].form_id}}
   schema_path: {{field_maps[0].schema_path}}
   payload_key: {{field_maps[0].payload_key}}
   transforms: {{field_maps[0].transforms}} | OPTIONAL
   constraints_bound: {{field_maps[0].constraints_bound}} | OPTIONAL
   open_questions:
   {{field_maps[0].open_questions[0]}} | OPTIONAL

(Repeat per field_id.)

3. Constraint Binding Rules
   binding_policy: {{constraints.binding_policy}}
   source_of_truth: {{constraints.source_of_truth}} (schema-first/form-first/UNKNOWN)
   mismatch_handling: {{constraints.mismatch_handling}} | OPTIONAL
4. Server Error Key Mapping
   error_key_format: {{errors.key_format}}
   schema_path_to_field_rule: {{errors.schema_path_to_field_rule}}
   unmapped_error_behavior: {{errors.unmapped_behavior}}

5. Unknown/Unmapped Handling
   unmapped_field_behavior: {{unknowns.unmapped_field_behavior}}
   unmapped_schema_behavior: {{unknowns.unmapped_schema_behavior}} | OPTIONAL
6. Versioning Notes
   schema_change_policy: {{versioning.schema_change_policy}}
   migration_notes: {{versioning.migration_notes}} | OPTIONAL
7. References
   Forms inventory: {{xref:FORM-01}}
   Field specs: {{xref:FORM-02}}
   Validation UX: {{xref:FORM-03}} | OPTIONAL
   Submission patterns: {{xref:FORM-05}} | OPTIONAL
   Data schemas: {{xref:DATA-06}} | OPTIONAL
   Data validation rules: {{xref:DQV-03}} | OPTIONAL

Cross-References
Upstream: {{xref:FORM-01}}, {{xref:FORM-02}}, {{xref:DATA-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FORM-05}}, {{xref:FORM-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN when schema refs/paths are missing; do not invent.
intermediate: Required. Provide per-field schema paths and transforms.
advanced: Required. Add schema evolution/migration policies and multi-entity mapping guidance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, entity_ref, payload_root_key, notes, transforms, constraints_bound, mismatch handling, versioning migration notes, multi-entity submissions, derived fields, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If schema_ref is UNKNOWN for any form → block Completeness Gate.
If schema_path_to_field_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FORM
Pass conditions:
required_fields_present == true
each form_id maps to DATA-06 schema_ref
each field_id maps to schema_path (or approved UNKNOWN)
server_error_mapping_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FO$RM-05

FORM-05 — Submission & Error Recovery Patterns
Header Block
template_id: FORM-05
title: Submission & Error Recovery Patterns
type: form_submission_error_recovery
template_version: 1.0.0
output_path: 10_app/forms/FORM-05_Submission_Error_Recovery_Patterns.md
compliance_gate_id: TMP-05.PRIMARY.FORM
upstream_dependencies: ["FORM-01", "FORM-03", "SMD-03", "FE-07", "API-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FORM-01", "FORM-03", "SMD-03", "FE-07", "API-03", "API-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical patterns for form submission, including request lifecycle, pending states, idempotency, server validation mapping, conflict handling, retries, and user recovery flows. This template must be consistent with mutation patterns and API error policy and must not invent submission flows not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FORM-01 Forms Inventory: {{forms.inventory}}
FORM-03 Validation UX Rules: {{forms.validation_ux}}
SMD-03 Mutation Patterns: {{smd.mutation_patterns}} | OPTIONAL
FE-07 Error Handling UX: {{fe.error_ux}} | OPTIONAL
API-03 Error Policy: {{api.error_policy}} | OPTIONAL
API-02 Endpoint Specs: {{api.endpoint_specs}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Submission lifecycle (idle → submitting → success/fail)
Pending UI rules (disable submit, spinner, prevent double submit)
Request construction rules (payload mapping ref to FORM-04)
Idempotency policy (client keys, dedupe)
Server validation error mapping (bind to FORM-03/FORM-04)
Conflict handling (409) rules
Retry policy (when allowed)
Timeout policy (client-side)
Recovery actions (edit & resubmit, retry, cancel)
Post-success behavior (navigate, toast, reset form)

Optional Fields
Multi-step submit rules | OPTIONAL
Autosave/draft submit rules | OPTIONAL
Offline queued submit binding | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Form submission MUST follow mutation patterns from {{xref:SMD-03}} when present.
Server field errors MUST map to field-level UI per {{xref:FORM-03}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Submission Lifecycle
   states: {{lifecycle.states}} (idle/submitting/success/error/UNKNOWN)
   transitions: {{lifecycle.transitions}} | OPTIONAL
2. Pending UI Rules
   disable_submit_while_pending: {{pending.disable_submit}}
   show_spinner: {{pending.show_spinner}} | OPTIONAL
   prevent_double_submit: {{pending.prevent_double_submit}}
   submit_lock_timeout_ms: {{pending.lock_timeout_ms}} | OPTIONAL
3. Request Construction
   payload_mapping_ref: {{request.payload_mapping_ref}} (expected: {{xref:FORM-04}})
   endpoint_binding_rule: {{request.endpoint_binding_rule}} (bind to API-02/FORM-01)
   headers_required: {{request.headers_required}} | OPTIONAL
   idempotency_key_rule: {{request.idempotency_key_rule}} | OPTIONAL
4. Server Validation Error Recovery
   field_mapping_ref: {{errors.field_mapping_ref}} (expected: {{xref:FORM-03}}/{{xref:FORM-04}})
   inline_errors_required: {{errors.inline_required}}
   summary_errors_when_unmapped: {{errors.summary_when_unmapped}}
   error_copy_policy: {{errors.copy_policy}} | OPTIONAL
5. Conflict Handling (409)
   on_conflict: {{conflict.on_conflict}}
   merge_or_prompt_policy: {{conflict.merge_or_prompt_policy}} | OPTIONAL
6. Retry Policy
   retry_allowed: {{retry.allowed}}
   retryable_error_classes: {{retry.retryable_error_classes}} | OPTIONAL
   backoff_policy: {{retry.backoff_policy}} | OPTIONAL
   max_attempts: {{retry.max_attempts}} | OPTIONAL
7. Timeout Policy
   client_timeout_ms: {{timeout.client_timeout_ms}}
   timeout_ui_behavior: {{timeout.ui_behavior}} | OPTIONAL

8. Recovery Actions
   edit_and_resubmit: {{recovery.edit_and_resubmit}}
   retry_action: {{recovery.retry_action}} | OPTIONAL
   cancel_action: {{recovery.cancel_action}} | OPTIONAL
9. Post-Success Behavior
   success_toast_policy: {{success.toast_policy}} | OPTIONAL
   navigate_after_success: {{success.navigate_after_success}} | OPTIONAL
   reset_form_after_success: {{success.reset_form_after_success}}
10. References
   Forms inventory: {{xref:FORM-01}}
   Validation UX: {{xref:FORM-03}}
   Schema mapping: {{xref:FORM-04}} | OPTIONAL
   Mutation patterns: {{xref:SMD-03}} | OPTIONAL
   Error UX: {{xref:FE-07}} | OPTIONAL
   API error policy: {{xref:API-03}} | OPTIONAL

Cross-References
Upstream: {{xref:FORM-01}}, {{xref:FORM-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FORM-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define lifecycle + pending UI; use UNKNOWN for idempotency if not specified.
intermediate: Required. Define server error mapping + conflict/retry behavior.
advanced: Required. Add offline/draft/multi-step rules if applicable and robust timeout handling.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, lifecycle transitions, spinner, lock timeout, headers required, idempotency key rule, field mapping ref, error copy policy, merge/prompt policy, retryable errors/backoff/max attempts, timeout UI behavior, retry/cancel actions, toast/nav policies, multi-step/draft/offline bindings, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If lifecycle.states is UNKNOWN → block Completeness Gate.
If pending.prevent_double_submit is UNKNOWN → block Completeness Gate.
If request.endpoint_binding_rule is UNKNOWN → block Completeness Gate.
If timeout.client_timeout_ms is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FORM
Pass conditions:
required_fields_present == true
submission_lifecycle_defined == true
pending_ui_rules_defined == true

error_recovery_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

FORM-06

FORM-06 — Anti-Abuse for Forms (spam, throttles, bot defense)
Header Block
template_id: FORM-06
title: Anti-Abuse for Forms (spam, throttles, bot defense)
type: form_anti_abuse_controls
template_version: 1.0.0
output_path: 10_app/forms/FORM-06_Anti_Abuse_For_Forms.md
compliance_gate_id: TMP-05.PRIMARY.FORM
upstream_dependencies: ["FORM-01", "RLIM-01", "RLIM-02", "RLIM-03", "RLIM-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FORM-01", "API-01", "API-02", "RLIM-01", "RLIM-02", "RLIM-03", "RLIM-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical anti-abuse controls for forms and form-backed endpoints, including rate limits, spam/bot defenses, validation hardening, abuse signal detection, and enforcement actions. This template must be consistent with rate limit and abuse policy and must not invent defenses not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FORM-01 Forms Inventory: {{forms.inventory}}
API-01 Endpoint Catalog: {{api.endpoint_catalog}} | OPTIONAL
API-02 Endpoint Specs: {{api.endpoint_specs}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}}
RLIM-02 Rate Limit Catalog: {{rlim.catalog}} | OPTIONAL
RLIM-03 Abuse Signals & Detection: {{rlim.abuse_signals}} | OPTIONAL
RLIM-04 Enforcement Actions: {{rlim.actions}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Threat model summary (spam, bots, credential stuffing if applicable)
Rate limit bindings (per form submit endpoints)
Bot defense model (captcha/challenge/UNKNOWN)
Honeypot/hidden field policy (optional)
Input validation hardening (length, encoding)
Abuse signal bindings (which rules apply)
Enforcement actions (throttle/ban/challenge)
User experience rules for anti-abuse (copy, retries)
Logging/audit fields for abuse events
Observability requirements (spam rate, challenge pass rate)

Optional Fields
Per-form stricter controls | OPTIONAL
Geo/IP restrictions | OPTIONAL
Email/phone verification gating | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Anti-abuse MUST bind to RLIM policy and action matrix.
Form submit endpoints MUST have explicit limits (or approved UNKNOWN).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Threat Model
   threats: {{threats.list}}
   notes: {{threats.notes}} | OPTIONAL
2. Rate Limit Bindings
   Binding
   form_id: {{bindings[0].form_id}}
   submit_endpoint_id: {{bindings[0].submit_endpoint_id}} | OPTIONAL
   limit_id_ref: {{bindings[0].limit_id_ref}} (expected: {{xref:RLIM-02}}) | OPTIONAL
   scope: {{bindings[0].scope}} | OPTIONAL
   notes: {{bindings[0].notes}} | OPTIONAL

(Repeat per form_id.)

3. Bot Defense Model
   captcha_supported: {{bot.captcha_supported}}
   challenge_type: {{bot.challenge_type}} (captcha/proof_of_work/UNKNOWN)
   trigger_rules: {{bot.trigger_rules}} | OPTIONAL
   fallback_behavior: {{bot.fallback_behavior}} | OPTIONAL
4. Honeypot Policy (Optional)
   honeypot_enabled: {{honeypot.enabled}} | OPTIONAL
   honeypot_field_rules: {{honeypot.field_rules}} | OPTIONAL
5. Validation Hardening
   max_field_lengths: {{hardening.max_field_lengths}}
   encoding_rules: {{hardening.encoding_rules}} | OPTIONAL
   payload_size_limits: {{hardening.payload_size_limits}} | OPTIONAL
6. Abuse Signals & Actions
   abuse_rule_refs: {{abuse.rule_refs}} (expected: {{xref:RLIM-03}}) | OPTIONAL
   action_refs: {{abuse.action_refs}} (expected: {{xref:RLIM-04}}) | OPTIONAL
   escalation_policy: {{abuse.escalation_policy}} | OPTIONAL

7. User Experience Rules
   challenge_copy_policy: {{ux.challenge_copy_policy}}
   retry_after_copy_policy: {{ux.retry_after_copy_policy}} | OPTIONAL
   lockout_copy_policy: {{ux.lockout_copy_policy}} | OPTIONAL
8. Logging / Audit
   abuse_event_fields: {{logs.abuse_event_fields}}
   pii_redaction: {{logs.pii_redaction}} | OPTIONAL
9. Observability Requirements
   metrics:
   submit_rate: {{obs.metrics.submit_rate}}
   blocked_rate: {{obs.metrics.blocked_rate}} | OPTIONAL
   challenge_rate: {{obs.metrics.challenge_rate}} | OPTIONAL
   challenge_pass_rate: {{obs.metrics.challenge_pass_rate}} | OPTIONAL
   dashboards: {{obs.dashboards}} | OPTIONAL
   alerts: {{obs.alerts}} | OPTIONAL
10. References
    Forms inventory: {{xref:FORM-01}}
    Rate limit policy: {{xref:RLIM-01}}
    Limits catalog: {{xref:RLIM-02}} | OPTIONAL
    Abuse signals: {{xref:RLIM-03}} | OPTIONAL
    Actions matrix: {{xref:RLIM-04}} | OPTIONAL

Cross-References
Upstream: {{xref:FORM-01}}, {{xref:RLIM-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define rate limit bindings skeleton; use UNKNOWN for captcha details.
intermediate: Required. Bind abuse rules/actions and define validation hardening.
advanced: Required. Add escalation policies, per-form stricter controls, and observability/alerts rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, submit_endpoint_id, limit_id_ref, scope, bot trigger/fallback rules, honeypot fields, encoding/payload limits, abuse refs, escalation policy, UX copy policies, pii redaction, dashboards, alerts, per-form controls, geo restrictions, verification gating, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If threats.list is UNKNOWN → block Completeness Gate.
If hardening.max_field_lengths is UNKNOWN → block Completeness Gate.
If logs.abuse_event_fields is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.FORM
Pass conditions:
required_fields_present == true
rate_limit_bindings_defined == true
validation_hardening_defined == true
abuse_actions_bound == true
placeholder_resolution == true
no_unapproved_unknowns == true

Client Routing & Deep Links (ROUTE)

Client Routing & Deep Links (ROUTE)
ROUTE-01 Route Contract (paths, params, types)
ROUTE-02 Navigation Map (flows, entry points)
ROUTE-03 Deep Link Map (URLs → screens/actions)
ROUTE-04 Guard Rules (auth gating, role gating)
ROUTE-05 Back/History Rules (expected behavior)
ROUTE-06 Link Validation & Unknown Handling (fallback routes)

ROUTE-01

ROUTE-01 — Route Contract (paths, params, types)
Header Block
template_id: ROUTE-01
title: Route Contract (paths, params, types)
type: client_route_contract
template_version: 1.0.0
output_path: 10_app/routing/ROUTE-01_Route_Contract.md
compliance_gate_id: TMP-05.PRIMARY.ROUTE
upstream_dependencies: ["SPEC_INDEX"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical route contract for the application: how routes are named, how paths are structured, how params and query params are typed/validated, and how route changes interact with guards. This template must be consistent with the Canonical Spec and must not invent route IDs not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Route identifier format (route_id naming rule)
Path format rules (leading slash, kebab-case, nesting)
Param naming rules (param keys)
Param typing rules (string/int/uuid)
Query param rules (allowed keys, typing)
Optional vs required params rules
Route versioning/alias rules (if paths change)
Validation rules (reject unknown params, sanitize)
Reserved routes (home, auth, not_found)
Unknown route handling policy binding (ROUTE-06)

Optional Fields
Locale prefixes | OPTIONAL
Multi-tenant prefixes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL

Routes MUST be deterministic and parseable.
Unknown routes MUST follow ROUTE-06 policy.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Route ID Rules
   route_id_format: {{ids.route_id_format}}
   examples:
   {{ids.examples[0]}}
   {{ids.examples[1]}} | OPTIONAL
2. Path Rules
   leading_slash: {{path.leading_slash}}
   case_style: {{path.case_style}} (kebab-case/UNKNOWN)
   nesting_rules: {{path.nesting_rules}} | OPTIONAL
   trailing_slash_policy: {{path.trailing_slash_policy}} | OPTIONAL
3. Param Rules
   param_key_style: {{params.key_style}}
   allowed_param_types: {{params.allowed_types}}
   required_param_rule: {{params.required_rule}}
   optional_param_rule: {{params.optional_rule}} | OPTIONAL
4. Query Param Rules
   allowed_query_keys_model: {{query.allowed_keys_model}} (allowlist/UNKNOWN)
   typing_rules: {{query.typing_rules}}
   unknown_query_key_policy: {{query.unknown_key_policy}} | OPTIONAL
5. Optional vs Required
   route_requires_params_when: {{optional.route_requires_params_when}}
   default_values_policy: {{optional.default_values_policy}} | OPTIONAL
6. Aliases / Versioning
   path_alias_supported: {{aliases.supported}}
   alias_rules: {{aliases.rules}} | OPTIONAL
   deprecation_policy: {{aliases.deprecation_policy}} | OPTIONAL
7. Validation
   reject_unknown_params: {{validation.reject_unknown_params}}
   sanitize_rules: {{validation.sanitize_rules}} | OPTIONAL
8. Reserved Routes
   home_route_id: {{reserved.home_route_id}}
   auth_route_id: {{reserved.auth_route_id}} | OPTIONAL
   not_found_route_id: {{reserved.not_found_route_id}}
   forbidden_route_id: {{reserved.forbidden_route_id}} | OPTIONAL
9. Unknown Route Handling
   unknown_route_policy_ref: {{unknown.policy_ref}} (expected: {{xref:ROUTE-06}})
10. References
    Navigation map: {{xref:ROUTE-02}} | OPTIONAL

Deep link map: {{xref:ROUTE-03}} | OPTIONAL
Guard rules: {{xref:ROUTE-04}} | OPTIONAL
Back/history rules: {{xref:ROUTE-05}} | OPTIONAL
Unknown handling: {{xref:ROUTE-06}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ROUTE-02}}, {{xref:ROUTE-03}}, {{xref:ROUTE-04}}, {{xref:ROUTE-06}} | OPTIONAL
Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for optional alias rules; define path/param rules.
intermediate: Required. Define query allowlist model and validation policies.
advanced: Required. Add versioning/deprecation and multi-tenant/locale prefix rules if applicable.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, ids examples, nesting/trailing slash rules, optional param rule, unknown query key policy, default values policy, alias rules/deprecation policy, sanitize rules, auth/forbidden route IDs, locale/tenant prefixes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If ids.route_id_format is UNKNOWN → block Completeness Gate.
If path.case_style is UNKNOWN → block Completeness Gate.
If unknown.policy_ref is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ROUTE
Pass conditions:
required_fields_present == true
route_id_and_path_rules_defined == true
param_and_query_rules_defined == true
unknown_route_policy_bound == true
placeholder_resolution == true
no_unapproved_unknowns == true

ROUTE-02

ROUTE-02 — Navigation Map (flows, entry points)
Header Block
template_id: ROUTE-02
title: Navigation Map (flows, entry points)
type: client_navigation_map
template_version: 1.0.0
output_path: 10_app/routing/ROUTE-02_Navigation_Map.md
compliance_gate_id: TMP-05.PRIMARY.ROUTE
upstream_dependencies: ["ROUTE-01", "SPEC_INDEX"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-01", "FE-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical navigation map of the app: primary entry points, user flows, navigation graph between screens/routes, and how navigation is structured (tabs/stacks/modals). This template must be consistent with the route contract and must not invent routes/screens not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-01 Route Contract: {{route.contract}}
FE-01 Route Map + Layout: {{fe.route_layout}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Entry points list (route_id list)
Primary navigation structure (tabs/sidebar/stacks)
Flow registry (flow_id list)
Flow steps (ordered route_id/screen_id sequence)
Transitions (push/replace/modal) per step
Back behavior expectations per flow
Role-based nav differences (if applicable)
Guarded transitions (auth required)
Deep link entry references (ROUTE-03)

Optional Fields
Alternative flows (error/recovery) | OPTIONAL
Cross-domain shortcuts | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new route_ids/screen_ids; use only those in SPEC_INDEX / FE-01.
Flow steps MUST be ordered and deterministic.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Entry Points
   entry_routes: {{entry.routes}}
   default_entry: {{entry.default_entry}}
   notes: {{entry.notes}} | OPTIONAL
2. Primary Navigation Structure
   nav_model: {{nav.model}} (tabs/sidebar/stack/UNKNOWN)
   nav_regions: {{nav.regions}} | OPTIONAL
   role_variants: {{nav.role_variants}} | OPTIONAL
3. Flow Registry (by flow_id)
   Flow
   flow_id: {{flows[0].flow_id}}
   name: {{flows[0].name}}
   primary_user: {{flows[0].primary_user}} | OPTIONAL
   entry_route_id: {{flows[0].entry_route_id}}
   exit_route_id: {{flows[0].exit_route_id}} | OPTIONAL
   steps (ordered):
- step_id: {{flows[0].steps[0].step_id}}
  route_id: {{flows[0].steps[0].route_id}}
  screen_id: {{flows[0].steps[0].screen_id}} | OPTIONAL
  transition: {{flows[0].steps[0].transition}}
  guard: {{flows[0].steps[0].guard}} | OPTIONAL
  back_behavior: {{flows[0].steps[0].back_behavior}} | OPTIONAL

(Repeat step blocks; repeat flow blocks.)

4. Back Behavior Rules
   global_back_rules: {{back.global_rules}}
   modal_back_rules: {{back.modal_rules}} | OPTIONAL
5. Guarded Transitions
   auth_gating_rules: {{guards.auth_gating_rules}}
   role_gating_rules: {{guards.role_gating_rules}} | OPTIONAL
6. Deep Link Entry References
   deep_link_ref: {{deep_links.ref}} (expected: {{xref:ROUTE-03}}) | OPTIONAL
7. References
   Route contract: {{xref:ROUTE-01}}
   Deep link map: {{xref:ROUTE-03}} | OPTIONAL

Guard rules: {{xref:ROUTE-04}} | OPTIONAL
Back/history rules: {{xref:ROUTE-05}} | OPTIONAL

Cross-References
Upstream: {{xref:ROUTE-01}}, {{xref:FE-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ROUTE-03}}, {{xref:ROUTE-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define entry points and at least the primary flows; use UNKNOWN if role variants unclear.
intermediate: Required. Define transitions, back behavior, and guard notes for each step.
advanced: Required. Add alternative/recovery flows and role-based navigation variants with traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, entry notes, nav regions, role variants, primary user, exit route, step screen_id, guard/back behavior, modal back rules, role gating rules, deep link ref, alternative flows, shortcuts, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If entry.routes is UNKNOWN → block Completeness Gate.
If flows registry is UNKNOWN → block Completeness Gate.
If any step route_id is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ROUTE
Pass conditions:
required_fields_present == true
entry_points_defined == true
flows_defined == true
all route_ids reference existing IDs
placeholder_resolution == true
no_unapproved_unknowns == true

ROUTE-03

ROUTE-03 — Deep Link Map (URLs → screens/actions)
Header Block
template_id: ROUTE-03
title: Deep Link Map (URLs → screens/actions)
type: client_deep_link_map
template_version: 1.0.0
output_path: 10_app/routing/ROUTE-03_Deep_Link_Map.md
compliance_gate_id: TMP-05.PRIMARY.ROUTE
upstream_dependencies: ["ROUTE-01", "ROUTE-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-01", "ROUTE-02", "MDL-01", "MDL-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical deep link mapping from external URLs/schemes into in-app routes/screens/actions, including parameter extraction, auth gating, safety validation, and fallback behavior. This template must be consistent with route contracts and mobile deep link rules (if applicable) and must not invent route IDs or URL patterns not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-01 Route Contract: {{route.contract}}
ROUTE-02 Navigation Map: {{route.nav_map}}
MDL-01 Link Scheme & Domains: {{mobile.links}} | OPTIONAL
MDL-02 Routing Rules: {{mobile.routing_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Deep link entry registry (link_id list)
URL pattern / scheme (https/app scheme)
Match rules (path params, query params)
Target route_id/screen_id/action
Param mapping (url → route params)
Auth gating behavior (logged out vs logged in)
Safety validation (allowlist, signature checks)
Fallback behavior (unknown/invalid/expired)
Telemetry requirements (link opens, failures)

Optional Fields
UTM/campaign param handling | OPTIONAL

Link expiry rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new route_ids; targets MUST exist in {{xref:ROUTE-01}}.
Deep link safety MUST follow MDL rules when present.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Deep Link Registry (by link_id)
   Link
   link_id: {{links[0].link_id}}
   url_pattern: {{links[0].url_pattern}}
   scheme: {{links[0].scheme}} | OPTIONAL
   target_route_id: {{links[0].target_route_id}}
   target_action: {{links[0].target_action}} | OPTIONAL
   param_mapping: {{links[0].param_mapping}}
   auth_gating: {{links[0].auth_gating}}
   safety_validation: {{links[0].safety_validation}}
   fallback_behavior: {{links[0].fallback_behavior}}
   telemetry: {{links[0].telemetry}} | OPTIONAL
   notes: {{links[0].notes}} | OPTIONAL
   open_questions:
   {{links[0].open_questions[0]}} | OPTIONAL

(Repeat per link_id.)

2. Global Safety Rules
   allowlisted_domains: {{safety.allowlisted_domains}} | OPTIONAL
   signed_links_supported: {{safety.signed_links_supported}} | OPTIONAL
   signature_rules: {{safety.signature_rules}} | OPTIONAL
3. Fallback Rules
   unknown_link_behavior: {{fallback.unknown_link_behavior}}
   invalid_param_behavior: {{fallback.invalid_param_behavior}} | OPTIONAL
4. Telemetry Requirements
   link_open_metric: {{telemetry.link_open_metric}}
   link_fail_metric: {{telemetry.link_fail_metric}} | OPTIONAL
5. References
   Route contract: {{xref:ROUTE-01}}
   Navigation map: {{xref:ROUTE-02}}
   Guard rules: {{xref:ROUTE-04}} | OPTIONAL

Link scheme/domains: {{xref:MDL-01}} | OPTIONAL
Routing rules: {{xref:MDL-02}} | OPTIONAL

Cross-References
Upstream: {{xref:ROUTE-01}}, {{xref:ROUTE-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ROUTE-04}}, {{xref:ROUTE-06}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define link registry skeleton; use UNKNOWN for signatures/telemetry specifics.
intermediate: Required. Define auth gating and safety validation and fallback behavior.
advanced: Required. Add signed link rules, expiry handling, and telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, scheme, target_action, telemetry, notes, allowlisted domains, signed links/signature rules, invalid param behavior, link expiry/utm handling, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If links registry is UNKNOWN → block Completeness Gate.
If any link has UNKNOWN url_pattern or target_route_id → block Completeness Gate.
If safety_validation is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ROUTE
Pass conditions:
required_fields_present == true
all target_route_ids exist in ROUTE-01
safety_validation_defined == true
fallback_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ROUTE-04

ROUTE-04 — Guard Rules (auth gating, role gating)
Header Block
template_id: ROUTE-04
title: Guard Rules (auth gating, role gating)
type: client_route_guard_rules
template_version: 1.0.0
output_path: 10_app/routing/ROUTE-04_Guard_Rules.md
compliance_gate_id: TMP-05.PRIMARY.ROUTE
upstream_dependencies: ["ROUTE-01", "API-04", "CSec-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-01", "API-04", "CSec-01", "CER-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical client-side route guard rules: authentication gating, role/permission gating, feature flag gating, and safe redirect behavior. This template must be consistent with backend AuthZ policies and must not invent roles/permissions not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-01 Route Contract: {{route.contract}}
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
CSec-01 Token Storage Policy: {{csec.token_policy}} | OPTIONAL
CER-04 Session Expiry Handling: {{cer.session_expiry}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Auth gating rules (logged-out redirects)
Role/permission gating rules (deny behavior)
Feature flag gating rules (if used)
Guard evaluation order (deterministic)
Redirect rules (where to send user)
Return-to behavior (after login)
Forbidden handling (403 vs hide route)
Session expiry handling binding (CER-04)
Telemetry requirements (guard denials)

Optional Fields
ABAC-style client hints (org context) | OPTIONAL
Per-route custom guards | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Client guards MUST NOT be the only security control; server remains authoritative.
Do not invent roles/permissions; bind to {{xref:API-04}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Evaluation Order
   order:
   {{order[0]}}
   {{order[1]}}
   {{order[2]}} | OPTIONAL
2. Auth Gating
   auth_required_rule: {{auth.required_rule}}
   redirect_to_login: {{auth.redirect_to_login}}
   return_to_param: {{auth.return_to_param}} | OPTIONAL
3. Role/Permission Gating
   gating_model: {{roles.model}} (role/permission/UNKNOWN)
   deny_behavior: {{roles.deny_behavior}} (403_route/404_hide/UNKNOWN)
   required_claims_source: {{roles.claims_source}} | OPTIONAL
4. Feature Flag Gating
   flag_gate_supported: {{flags.supported}}
   flag_gate_rule: {{flags.rule}} | OPTIONAL
5. Redirect Rules
   post_login_redirect: {{redirect.post_login}}
   forbidden_redirect: {{redirect.forbidden_redirect}} | OPTIONAL
   default_route: {{redirect.default_route}} | OPTIONAL
6. Session Expiry
   session_expiry_ref: {{session.expiry_ref}} (expected: {{xref:CER-04}})
   on_expiry_behavior: {{session.on_expiry_behavior}} | OPTIONAL
7. Telemetry
   guard_denial_metric: {{telemetry.guard_denial_metric}}
   guard_denial_log_fields: {{telemetry.guard_denial_log_fields}} | OPTIONAL
8. References
   Route contract: {{xref:ROUTE-01}}
   AuthZ rules: {{xref:API-04}} | OPTIONAL
   Token storage: {{xref:CSec-01}} | OPTIONAL
   Session expiry: {{xref:CER-04}} | OPTIONAL

Cross-References
Upstream: {{xref:ROUTE-01}}, {{xref:API-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ROUTE-06}} | OPTIONAL

Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define auth gating + deny behavior; use UNKNOWN for flags/telemetry details.
intermediate: Required. Define evaluation order, return-to behavior, and forbidden handling.
advanced: Required. Add per-route overrides and telemetry log fields with privacy constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, return_to_param, claims_source, flag gate rules, forbidden redirect, default route, expiry behavior, denial log fields, ABAC client hints, per-route custom guards, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If auth.redirect_to_login is UNKNOWN → block Completeness Gate.
If roles.deny_behavior is UNKNOWN → block Completeness Gate.
If telemetry.guard_denial_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ROUTE
Pass conditions:
required_fields_present == true
auth_and_role_gating_defined == true
evaluation_order_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ROUTE-05

ROUTE-05 — Back/History Rules (expected behavior)
Header Block
template_id: ROUTE-05
title: Back/History Rules (expected behavior)
type: client_back_history_rules
template_version: 1.0.0
output_path: 10_app/routing/ROUTE-05_Back_History_Rules.md
compliance_gate_id: TMP-05.PRIMARY.ROUTE
upstream_dependencies: ["ROUTE-02", "FE-01", "UICP-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-02", "FE-01", "UICP-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical back/history behavior across navigation stacks, tabs, and modals (web and mobile), including when back closes modals vs navigates, how deep links affect history, and how to handle edge cases like first screen back. This template must be consistent with navigation map and interaction patterns and must not invent navigation behaviors not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-02 Navigation Map: {{route.nav_map}}
FE-01 Route Map + Layout: {{fe.route_layout}} | OPTIONAL
UICP-05 Interaction Patterns: {{ui.interaction_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Global back behavior statement (principles)
Modal back behavior (close vs pop)
Tab back behavior (per-tab history vs reset)
Stack back behavior (pop rules)
Deep link entry back behavior
First-screen back behavior (exit app / confirm / go home)
Browser back behavior (web)
Android back behavior (mobile)
Unsaved changes handling (confirm on back)
Telemetry requirements (back events, exit events)

Optional Fields
Custom per-route back overrides | OPTIONAL

Gesture back rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Back behavior MUST be deterministic and consistent across shells.
Unsaved changes confirmation MUST prevent data loss.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Global Principles
   principles:
   {{principles[0]}}
   {{principles[1]}}
   {{principles[2]}}
2. Modal Back Behavior
   when_modal_open_back: {{modal.when_open_back}}
   modal_close_rule: {{modal.close_rule}}
   modal_stack_rule: {{modal.stack_rule}} | OPTIONAL
3. Tab Back Behavior
   per_tab_history: {{tabs.per_tab_history}}
   tab_reselect_behavior: {{tabs.reselect_behavior}} | OPTIONAL
   tab_reset_rule: {{tabs.reset_rule}} | OPTIONAL
4. Stack Back Behavior
   pop_rule: {{stack.pop_rule}}
   root_back_rule: {{stack.root_back_rule}} | OPTIONAL
5. Deep Link Entry Back Behavior
   deep_link_back_rule: {{deep.back_rule}}
   deep_link_initial_stack: {{deep.initial_stack}} | OPTIONAL
6. First-Screen Back
   first_screen_back_behavior: {{first.back_behavior}}
   (exit_app/confirm_exit/go_home/UNKNOWN)
   confirm_exit_copy: {{first.confirm_exit_copy}} | OPTIONAL
7. Browser Back (Web)
   browser_back_supported: {{web.supported}}
   browser_back_rules: {{web.rules}} | OPTIONAL
8. Android Back (Mobile)
   android_back_supported: {{android.supported}}
   android_back_rules: {{android.rules}} | OPTIONAL
9. Unsaved Changes
   unsaved_changes_detection: {{unsaved.detection}}
   confirm_on_back: {{unsaved.confirm_on_back}}
   confirm_copy_policy: {{unsaved.confirm_copy_policy}} | OPTIONAL

10. Telemetry
    back_event_metric: {{telemetry.back_event_metric}}
    exit_event_metric: {{telemetry.exit_event_metric}} | OPTIONAL
    fields: {{telemetry.fields}} | OPTIONAL
11. References
    Navigation map: {{xref:ROUTE-02}}
    Route/layout: {{xref:FE-01}} | OPTIONAL
    Interaction patterns: {{xref:UICP-05}} | OPTIONAL
    Deep links: {{xref:ROUTE-03}} | OPTIONAL

Cross-References
Upstream: {{xref:ROUTE-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ROUTE-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define modal/stack/tab basics; use UNKNOWN for platform-specific details if missing.
intermediate: Required. Define deep-link and first-screen back behavior and unsaved changes confirmation.
advanced: Required. Add per-route overrides, gesture rules, and telemetry fields.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, modal stack rule, tab reselect/reset rules, root back rule, deep initial stack, confirm copy, web/android rules, confirm copy policy, telemetry fields, overrides/gesture rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If first.back_behavior is UNKNOWN → block Completeness Gate.
If unsaved.confirm_on_back is UNKNOWN → block Completeness Gate.
If telemetry.back_event_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ROUTE
Pass conditions:
required_fields_present == true
global_principles_defined == true
modal_tab_stack_rules_defined == true
unsaved_changes_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ROUTE-06

ROUTE-06 — Link Validation & Unknown Handling (fallback routes)
Header Block
template_id: ROUTE-06
title: Link Validation & Unknown Handling (fallback routes)
type: client_link_validation_unknown_handling
template_version: 1.0.0
output_path: 10_app/routing/ROUTE-06_Link_Validation_Unknown_Handling.md
compliance_gate_id: TMP-05.PRIMARY.ROUTE
upstream_dependencies: ["ROUTE-01", "ROUTE-03", "ROUTE-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-01", "ROUTE-03", "ROUTE-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for validating links (internal + deep links) and handling unknown routes/invalid params safely, including fallback routing, safe error screens, and telemetry. This template must be consistent with route and guard rules and must not invent fallback routes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-01 Route Contract: {{route.contract}}
ROUTE-03 Deep Link Map: {{route.deep_link_map}} | OPTIONAL
ROUTE-04 Guard Rules: {{route.guard_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Internal link validation rules (route_id validity, params)
Deep link validation rules (scheme/domain allowlist, param parsing)
Unknown route handling behavior (not_found route)
Invalid param handling behavior (fallback or error)
Forbidden route handling behavior (403/404/hide)
Telemetry requirements (unknown route hits, invalid params)
Security constraints (no open redirects)
User copy policy (safe messaging)

Optional Fields
Recovery suggestions UX | OPTIONAL
Per-route custom fallback | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Unknown routes MUST be handled safely and deterministically.
Do not allow open redirects or arbitrary URL navigation.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Internal Link Validation
   route_id_check: {{internal.route_id_check}}
   params_validation: {{internal.params_validation}}
   query_validation: {{internal.query_validation}} | OPTIONAL
2. Deep Link Validation
   allowlisted_domains: {{deep.allowlisted_domains}} | OPTIONAL
   allowlisted_schemes: {{deep.allowlisted_schemes}} | OPTIONAL
   param_parsing_rules: {{deep.param_parsing_rules}}
   signature_validation: {{deep.signature_validation}} | OPTIONAL
3. Unknown Route Handling
   unknown_route_behavior: {{unknown.behavior}}
   not_found_route_id: {{unknown.not_found_route_id}}
4. Invalid Param Handling
   invalid_param_behavior: {{invalid.behavior}}
   fallback_route_id: {{invalid.fallback_route_id}} | OPTIONAL
5. Forbidden Handling
   forbidden_behavior: {{forbidden.behavior}} (403_route/404_hide/UNKNOWN)
   forbidden_route_id: {{forbidden.route_id}} | OPTIONAL
6. Telemetry
   unknown_route_metric: {{telemetry.unknown_route_metric}}
   invalid_param_metric: {{telemetry.invalid_param_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
7. Security Constraints
   no_open_redirects_rule: {{security.no_open_redirects_rule}}
   external_navigation_policy: {{security.external_navigation_policy}} | OPTIONAL
8. Copy Policy
   safe_copy_rules: {{copy.safe_copy_rules}}
   support_contact_policy: {{copy.support_contact_policy}} | OPTIONAL
9. References
   Route contract: {{xref:ROUTE-01}}
   Deep link map: {{xref:ROUTE-03}} | OPTIONAL
   Guard rules: {{xref:ROUTE-04}} | OPTIONAL
   Back/history: {{xref:ROUTE-05}} | OPTIONAL

Cross-References
Upstream: {{xref:ROUTE-01}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:FE-01}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define not_found behavior + telemetry; use UNKNOWN for signature rules.
intermediate: Required. Define deep link param parsing + forbidden handling.
advanced: Required. Add per-route fallbacks and robust security constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, query validation, allowlisted domains/schemes, signature validation, fallback route, forbidden route id, telemetry fields, external nav policy, support contact policy, recovery suggestions, per-route custom fallback, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If unknown.not_found_route_id is UNKNOWN → block Completeness Gate.
If telemetry.unknown_route_metric is UNKNOWN → block Completeness Gate.
If security.no_open_redirects_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ROUTE
Pass conditions:
required_fields_present == true
validation_rules_defined == true
unknown_and_invalid_handling_defined == true
telemetry_defined == true
security_constraints_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true